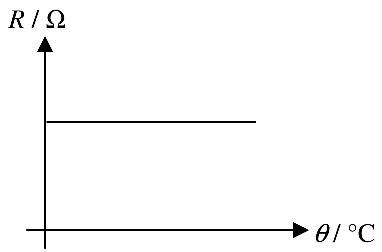
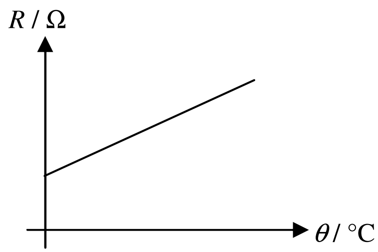
(2011-PP)1. The graphs below show how the electrical resistances R of three different circuit elements change with temperature θ. Which of the circuit elements can be used to measure temperature ?

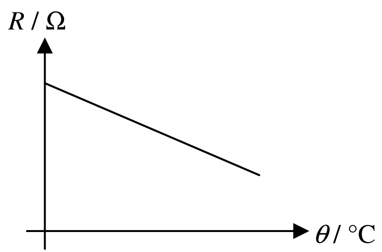
(1)



(2)

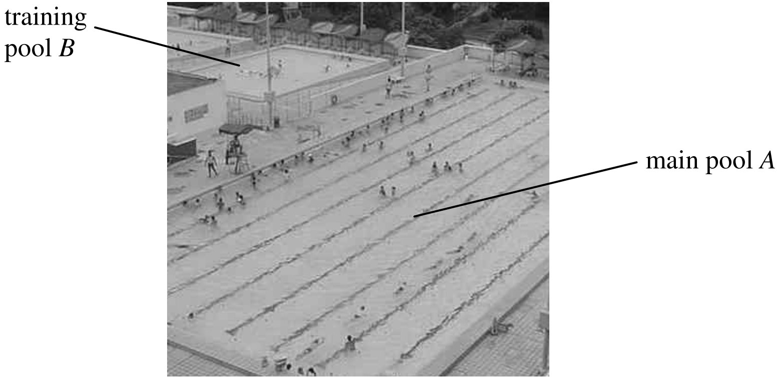


(3)



A. (1) only
B. (2) only
C. (1) and (3) only
D. (2) and (3) only

(2011-PP)2. In the figure below, a training pool *B* is located next to the main pool *A*. The training pool *B* has a smaller area and is shallower. If the pools are under the sunlight at the same time, which of the following statements about the rise in the water temperature of the two pools is correct ? Assume that the initial water temperatures of the pools are the same.



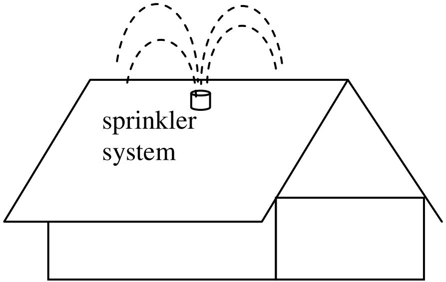
A. The water temperature of training pool *B* rises faster because it is shallower.
B. The water temperature of training pool *B* rises faster because it has a smaller surface area.
C. The water temperature of main pool *A* rises faster because it is deeper.
D. The water temperature of main pool *A* rises faster because it has a larger surface area.

(2011-PP)3. Peter adds 50 g of milk at 20°C to 350 g of tea at 80°C, what is the final temperature of the mixture ?

Given : Specific heat capacity of milk = 3800 J kg^{-1} $°C^{-1}$
Specific heat capacity of tea = 4200 J kg^{-1} $°C^{-1}$

A. 50.0°C
B. 72.5°C
C. 73.1°C
D. 77.4°C

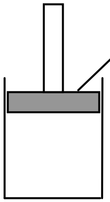
(2011-PP)4. The sprinkler system on a rooftop is able to spray small water droplets onto the rooftop which can lower the temperature of the rooftop on hot sunny days. Which of the following explanations about the sprinkler system is/are reasonable ?



(1) Water is a good conductor, which conducts heat quickly.
(2) Water has a high specific heat capacity, absorbing a lot of energy when its temperature rises.
(3) Water has a high specific latent heat of vaporization, absorbing a lot of energy when it evaporates.

A. (1) only
B. (2) only
C. (1) and (3) only
D. (2) and (3) only

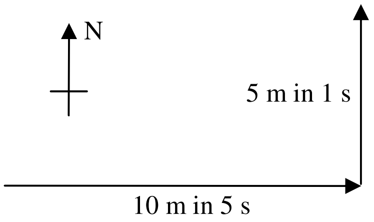
(2011-PP) *5. A fixed mass of an ideal gas is contained in a cylinder fitted with a frictionless piston as shown in the figure below. If the gas is cooled under constant pressure,



(1) the average separation of the gas molecules will decrease.
(2) the r.m.s. speed of the gas molecules will decrease.
(3) the number of collisions per second of the gas molecules on the piston will decrease.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

(2011-PP)6.



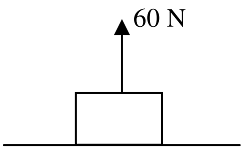
A toy car travelled due east for 10 m in 5 s, then immediately turned north and travelled 5 m for 1 s. What was the average speed of the car ?

A. 1.9 m s^{-1}
B. 2.2 m s^{-1}
C. 2.5 m s^{-1}
D. 3.5 m s^{-1}

(2011-PP)7. A stone falls from rest. Neglecting air resistance, the ratio of the distance travelled by the stone in the 1st second to that travelled in the 2nd second is

A. 1 : 1
B. 1 : 2
C. 1 : 3
D. 1 : 4

(2011-PP)8.

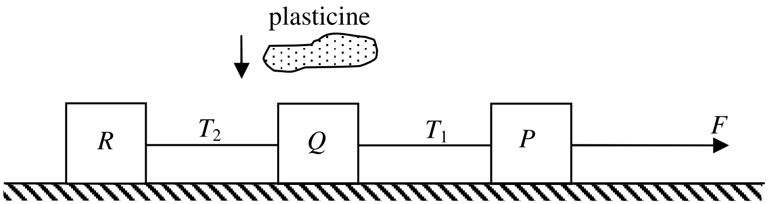


A block of weight 100 N is placed on a horizontal table and a vertical force of 60 N is exerted on the block as shown in the figure above. Which of the following statements is/are correct ?

(1) The weight of the block is balanced by the force exerted on the block by the table.
(2) The weight of the block and the force exerted on the table by the block are equal in magnitude.
(3) The force exerted on the table by the block and the force exerted on the block by the table are an action-reaction pair.

A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

(2011-PP)9. Blocks, P, Q and R, connected by light inextensible threads, are placed on a smooth horizontal surface as shown. A constant force F is applied to P so that the whole system travels to the right with acceleration.



A lump of plasticine is placed on Q and it moves together with Q. If the applied force F remains unchanged, how would the tensions T_1 and T_2 in the two threads change ?

	Tension T_1	Tension T_2
A.	increase	decrease
B.	increase	increase
C.	decrease	decrease
D.	decrease	increase

(2011-PP)10.

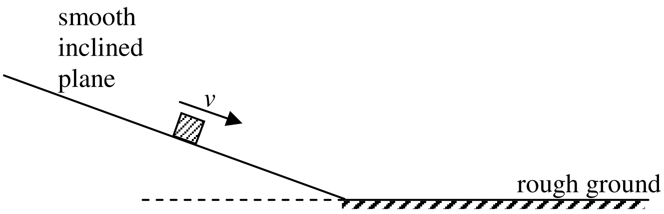


Figure (a)

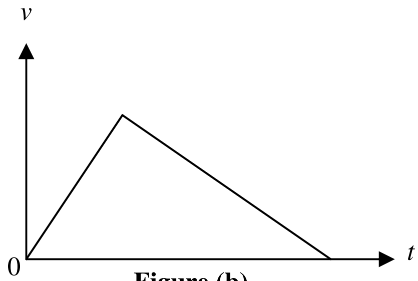
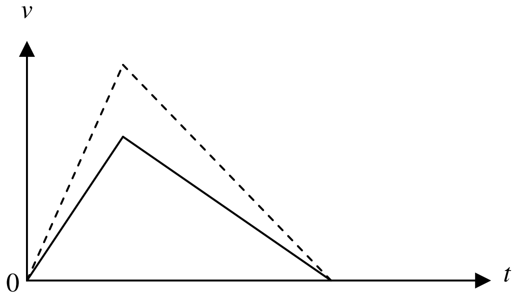


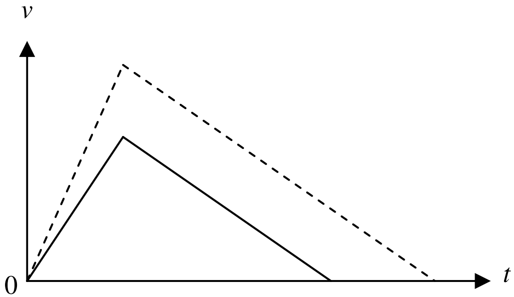
Figure (b)

As shown in Figure (a), a block slides down along a smooth inclined plane from rest. The corresponding speed-time graph of its motion is shown in Figure (b). Which of the following speed-time graphs (in dotted lines) best represents the motion of the block if it is released at a higher position on the plane instead ? Assume that the friction between the ground and the block remains unchanged.

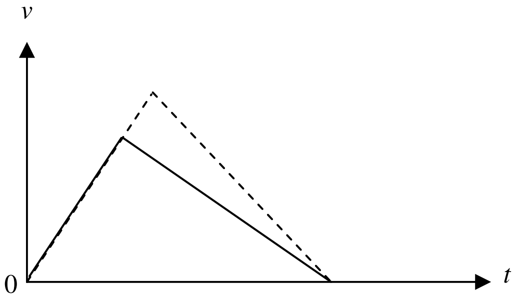
A.



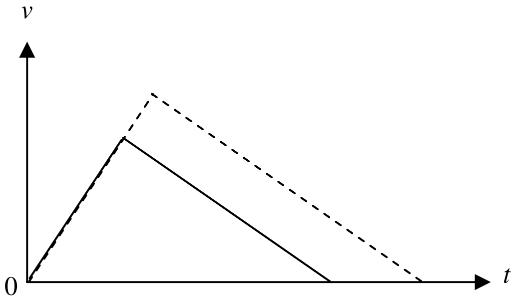
B.



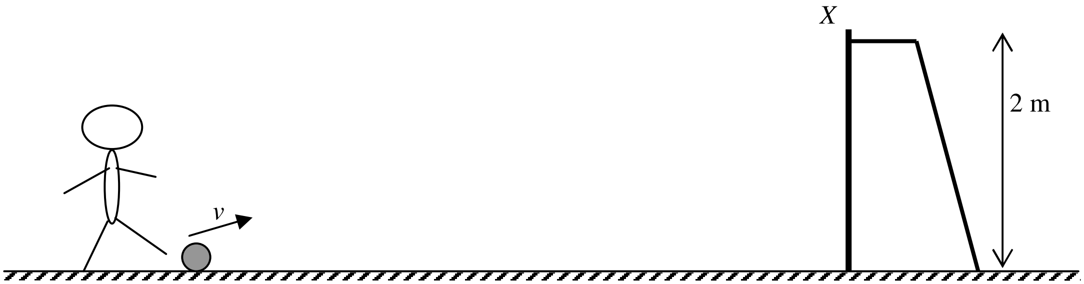
C.



D.



(2011-PP)11.

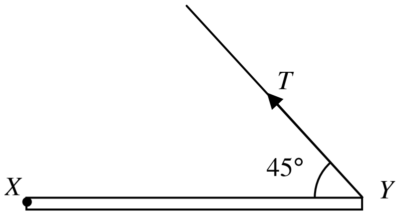


A football player kicks a ball on the ground. The ball leaves the ground with speed v and hits the bar at X with a speed of 17 m s^{-1}. X is 2 m above the ground. Neglecting air resistance, what is the value of v ?

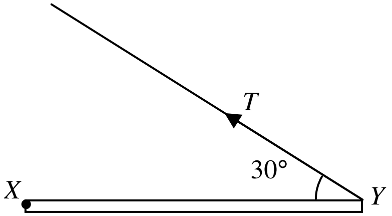
A. 15.8 m s^{-1}
B. 18.1 m s^{-1}
C. 19.0 m s^{-1}
D. 23.3 m s^{-1}

(2011-PP)12. A rod XY hinged at X is kept horizontal by a light string. M is the midpoint of XY. In which of the following arrangements will the tension T in the string be the smallest ?

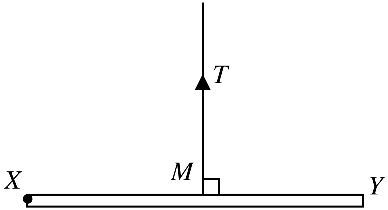
A.



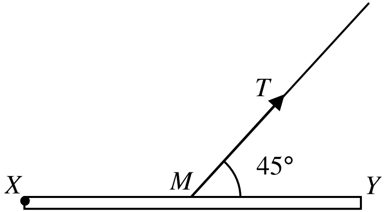
B.



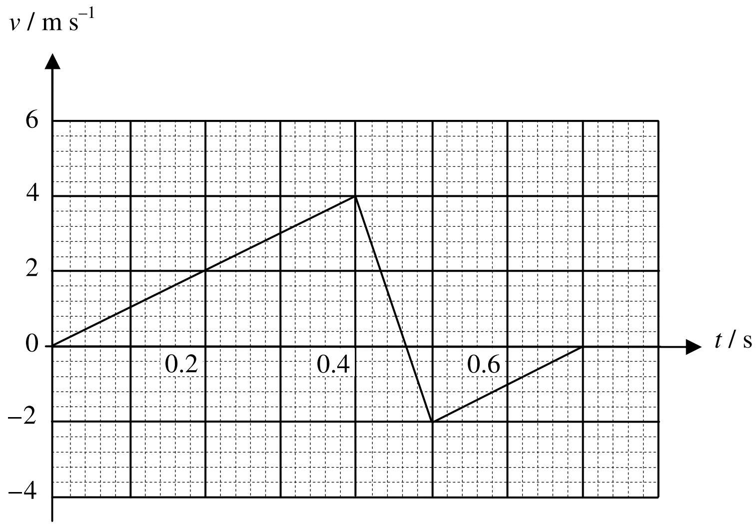
C.



D.



(2011-PP)13.

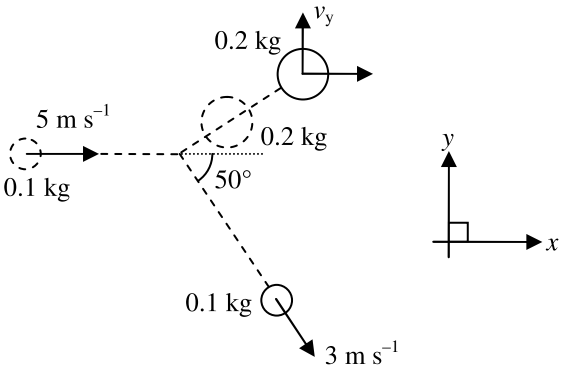


A ball of mass 0.2 kg is released from rest. It hits the ground and rebounds. The velocity-time graph of the ball is shown above. Which of the following statements are correct ?

(1) The magnitude of the change in momentum of the ball during the collision is 1.2 kg m s^{-1}.
(2) The magnitude of the average force acting on the ball by the ground during the collision is 12 N.
(3) There is mechanical energy loss during the collision.

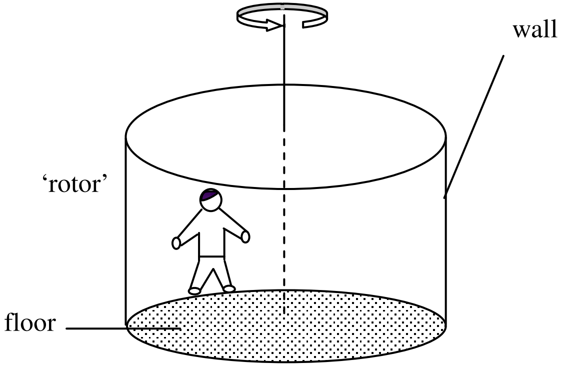
A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

(2011-PP)*14. A disc of mass 0.1 kg and velocity 5 m s^{-1} strikes a stationary disc of mass 0.2 kg on a smooth table. After the collision, the 0.1 kg disc moves with a speed of 3 m s^{-1} at 50° to the x direction. Find the component of the velocity of the 0.2 kg disc in y direction, v_y, after the collision.



A. 1.15 m s^{-1}
B. 1.54 m s^{-1}
C. 1.92 m s^{-1}
D. 2.01 m s^{-1}

(2011-PP)*15.



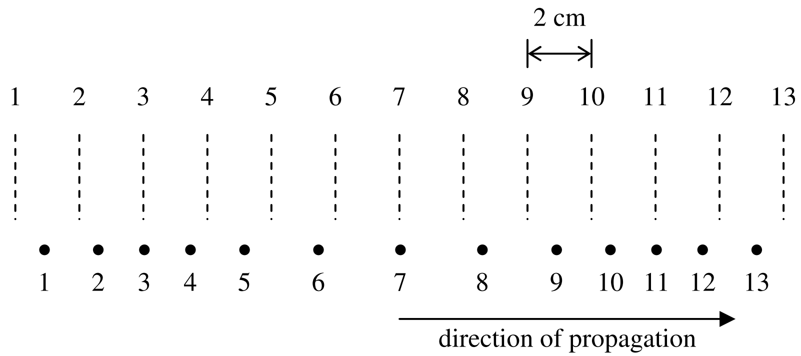
A man is rotating with constant speed inside a cylindrical 'rotor' and he remains pressed against the wall. The floor of the 'rotor' is smooth. Which of the following forces provides the centripetal force for the man ?

A. the weight of the man
B. the frictional force from the wall
C. the normal reaction from the wall
D. the supporting force from the floor

(2011-PP)16. Which of the following phenomena demonstrates that light is an electromagnetic wave ?

A. Light carries energy.
B. Light reflects when it meets a polished metal surface.
C. Light bends when it travels across a boundary from one medium into another.
D. Light can travel from the Sun to the Earth.

(2011-PP)17.

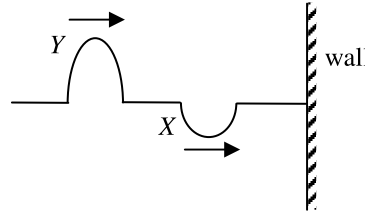


A longitudinal wave travels to the right through a medium containing a series of particles. The figure above shows the positions of the particles at a certain instant. The dotted lines indicate the equilibrium positions of the particles. Which of the following statements about the wave at the instant shown is/are correct ?

(1) The wavelength of the longitudinal wave is 16 cm.
(2) Particles 8 and 10 are moving in the same direction.
(3) Particle 3 is momentarily at rest.

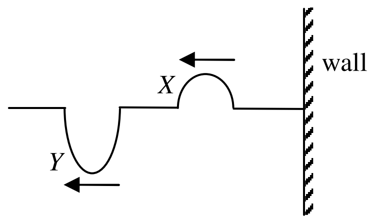
A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

(2011-PP)18.

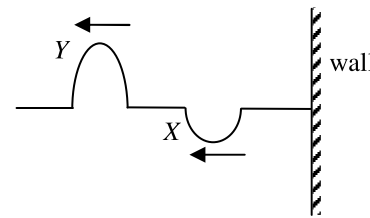


Two pulses, *X* and *Y*, are travelling along a string which is fixed at one end to the wall as shown in the figure above. Which of the following is a possible waveform of the string after the two pulses reflect ?

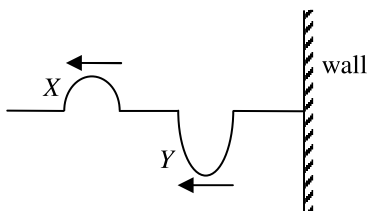
A.



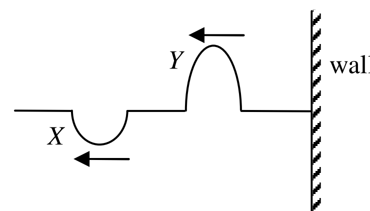
B.



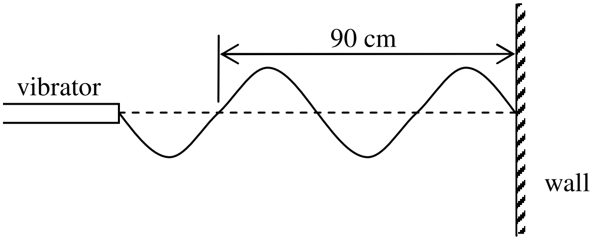
C.



D.



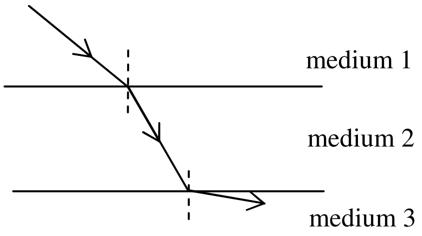
(2011-PP)19.



A stationary wave is set up along a string by a vibrator. The waveform at a certain instant is shown above. If the frequency of the vibrator is 50 Hz, what is the wave speed along the string ?

A. 15 m s^{-1}
B. 30 m s^{-1}
C. 45 m s^{-1}
D. 55 m s^{-1}

(2011-PP)20.



As shown above, a ray of light travels from medium 1 to medium 2, and then enters medium 3. The boundaries are parallel to each other. Arrange the speed of light, c, in the three media in **ascending** order.

A. $c_3 < c_2 < c_1$
B. $c_3 < c_1 < c_2$
C. $c_2 < c_3 < c_1$
D. $c_2 < c_1 < c_3$

(2011-PP)*21.

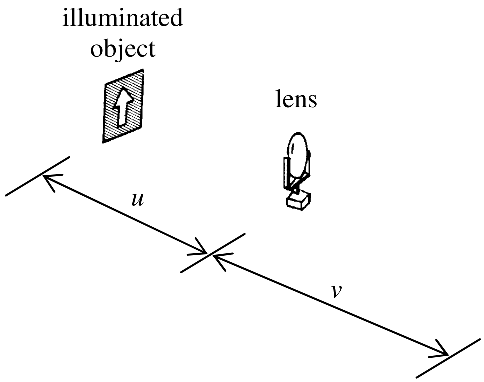


Figure (a)

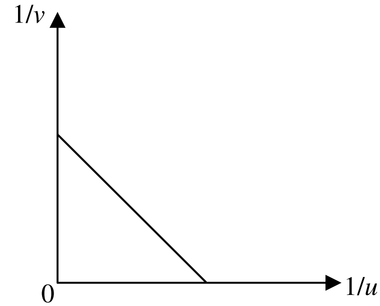
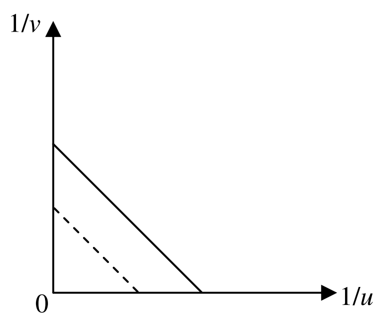


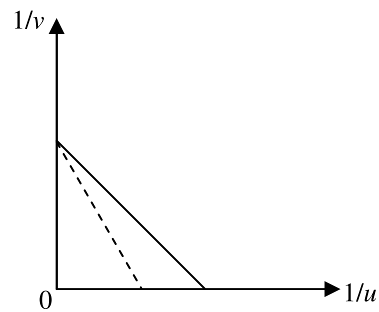
Figure (b)

A student uses the set-up in Figure (a) to study the relationship between the object distance u and the image distance v of a convex lens. A graph of $1/v$ against $1/u$ is plotted in Figure (b). If the lens is replaced by another convex lens of shorter focal length, which of the following graphs (in dotted lines) would be obtained ?

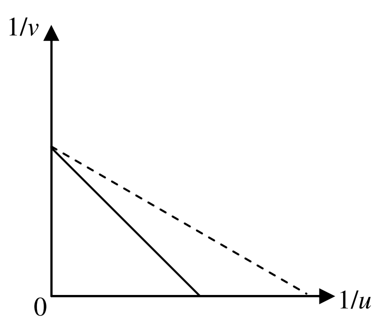
A.



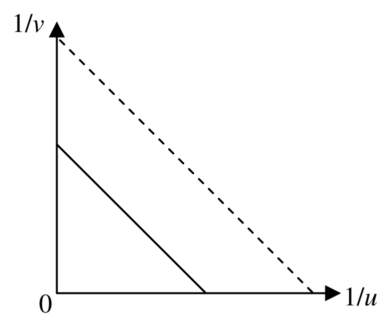
B.



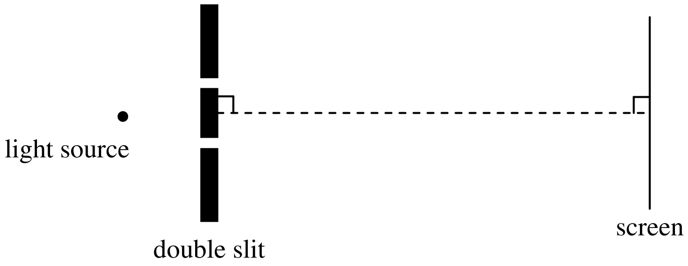
C.



D.



(2011-PP)*22.



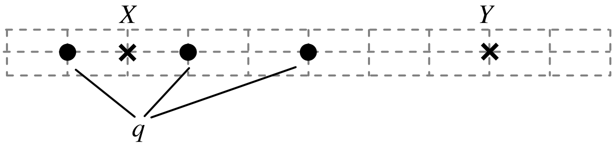
In a Young's double slit experiment, a monochromatic light source of wavelength 600 nm is used. The fringe separation is 5 mm on the screen. If the slit separation is halved and a monochromatic light source of wavelength 450 nm is used instead, what is the new fringe separation ?

A. 1.9 mm
B. 3.3 mm
C. 7.5 mm
D. 13.3 mm

(2011-PP)*23. Yellow light of wavelength 590 nm is incident normally on a diffraction grating with 400 lines per mm. Find the difference in angular positions for the third order and the fourth order bright fringes.

A. 13.7°
B. 25.7°
C. 45.1°
D. 70.7°

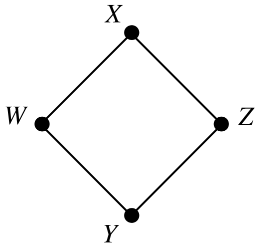
(2011-PP)24.



Three identical point charges q (represented by dots) are situated in the space as shown. Which of the following descriptions about the direction and magnitude of the electric field E at X and at Y is correct ?

	Direction	**Magnitude**
A.	Same	$E_X > E_Y$
B.	Same	$E_X < E_Y$
C.	Opposite	$E_X > E_Y$
D.	Opposite	$E_X < E_Y$

(2011-PP)*25.



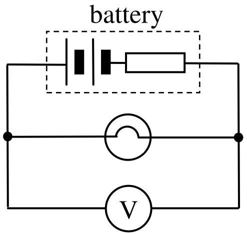
The figure above shows four points W, X, Y and Z in a uniform electric field. $WXZY$ is a square. The electric potential at W, X and Y are 1 V, 5 V and 5 V respectively. What is the electric potential at Z ?

A. 1 V
B. 6 V
C. 9 V
D. 11 V

(2011-PP)26. Two metal rods, X and Y, of uniform cross-sectional area are made of the same material and have the same volume. The length and resistance of X are l and R respectively. What is the resistance of Y if it has a length of $2l$?

A. $R/4$
B. $R/2$
C. $2R$
D. $4R$

(2011-PP)27. The figure below shows a battery of e.m.f. 3.0 V and internal resistance 2.0 Ω is connected to a light bulb of resistance 10.0 Ω. A voltmeter of internal resistance 10 kΩ is connected in parallel with the light bulb. What is the reading of the voltmeter ?



A. 2.4 V
B. 2.5 V
C. 2.9 V
D. 3.0 V

(2011-PP)28. In Figure (a), two identical resistors are connected in series to a cell of e.m.f. V and negligible internal resistance. The power dissipated by each resistor is P. If the two resistors are now connected in parallel as shown in Figure (b), what is the power dissipated by each resistor ?

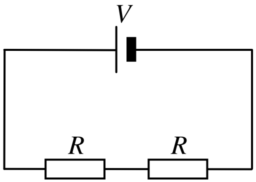


Figure (a)

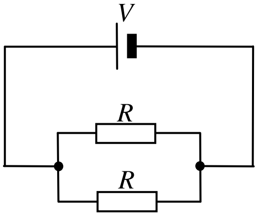
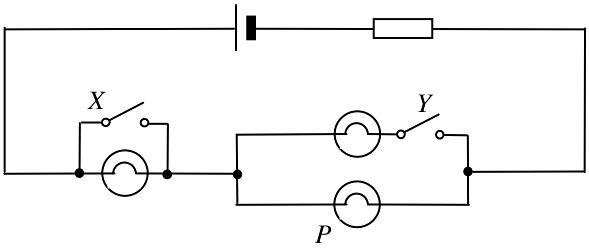


Figure (b)

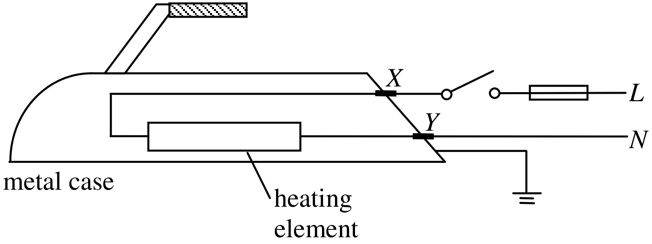
A. $2P$
B. $4P$
C. $8P$
D. $16P$

(2011-PP)29. In the circuit below, three identical light bulbs are connected to a cell. Under what conditions will light bulb *P* have the maximum brightness ?



	Switch *X*	**Switch *Y***
A.	closed	open
B.	closed	closed
C.	open	open
D.	open	closed

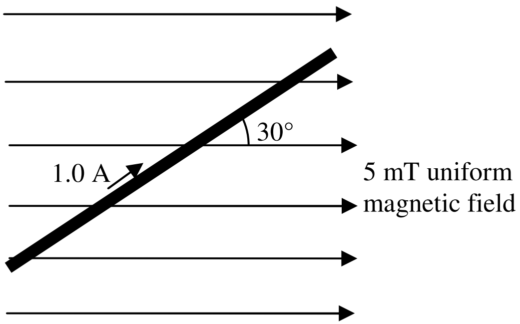
(2011-PP)30.



The figure above shows the main parts of an electric iron. In which of the following situations will the fuse blow when the switch is closed ?

A. The heating element is broken and becomes an open circuit.
B. The earth wire is worn out and becomes disconnected.
C. The insulation at contact point X is worn out so that the wire touches the metal case.
D. The insulation at contact point Y is worn out so that the wire touches the metal case.

(2011-PP) 31. The figure below shows a current of 1.0 A flowing in a metal rod of length 0.5 m. The rod is placed inside a region with a uniform magnetic field of strength 5 mT. What is the direction and the magnitude of the magnetic force acting on the rod ?

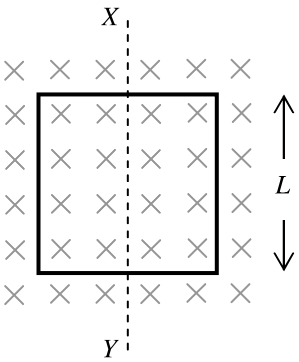


	Direction	**Magnitude**
A.	into the paper	1.25×10^{-3} N
B.	out of the paper	1.25×10^{-3} N
C.	into the paper	2.17×10^{-3} N
D.	out of the paper	2.17×10^{-3} N

(2011-PP)*32. A Hall probe is placed in a uniform magnetic field. The slice of semiconductor inside the Hall probe is 1.3×10^{-3} m thick and has 10^{25} charge carriers per cubic metre. When a steady current of 0.4 A passes through the slice, a Hall voltage of 2×10^{-5} V is set up. What is the magnetic field strength detected by the probe ? Assume that the magnitude of the charge of each charge carrier is 1.6×10^{-19} C.

A. 0.104 T
B. 0.962 T
C. 1.04 T
D. 9.62 T

(2011-PP)*33.



A square metal frame of side length L is placed inside a uniform magnetic field B as shown. What is the change in magnetic flux through the frame when it is rotated about the axis XY by 90° and 180° respectively ?

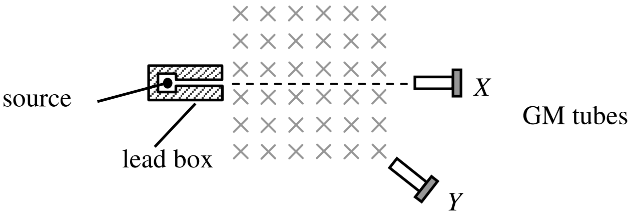
	90°	**180°**
A.	0	0
B.	0	$2BL^2$
C.	BL^2	0
D.	BL^2	$2BL^2$

(2011-PP)34. Which of the following statements about α and β particles is/are correct ?

(1) The mass of an α particle is greater than that of a β particle.
(2) α particles have a stronger penetrating power than β particles.
(3) An α source can discharge a positively charged metal sphere nearby.

A. (1) only
B. (2) only
C. (1) and (3) only
D. (2) and (3) only

(2011-PP)35.



A radioactive source is placed in front of a uniform magnetic field pointing into the paper as shown above. The count rates recorded by the GM tubes at X and Y are 101 counts per minute and 400 counts per minute respectively. Which of the following deductions must be correct ?

A. The source does not emit α radiation.
B. The source emits β radiations.
C. The source emits γ radiations.
D. The background count rate is about 100 counts per minute.

(2011-PP)*36. For the following nuclear reaction, state the type of reaction and determine the energy released.

$$^{2}_{1}\text{H} + {}^{3}_{1}\text{H} \rightarrow {}^{4}_{2}\text{He} + {}^{1}_{0}\text{n}$$

Given: mass of $^{2}_{1}\text{H} = 2.014$ u

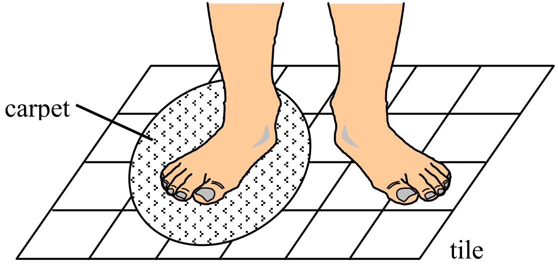
mass of $^{3}_{1}\text{H} = 3.016$ u

mass of $^{4}_{2}\text{He} = 4.003$ u

mass of $^{1}_{0}\text{n} = 1.009$ u

	Type of reaction	Energy released
A.	fusion	0.018 MeV
B.	fusion	16.76 MeV
C.	fission	0.018 MeV
D.	fission	16.76 MeV

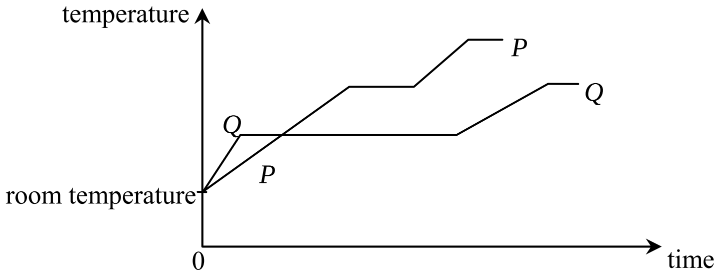
(2011-SP) 1.



Cynthia places a piece of carpet on a tiled floor. After a while, she stands in bare feet with one foot on the tiled floor and the other on the carpet as shown above. She feels that the tiled floor is colder than the carpet. Which of the following best explains this phenomenon ?

A. The tile is a better insulator of heat than the carpet.
B. The tile is at a lower temperature than the carpet.
C. The specific heat capacity of the tile is smaller than that of the carpet.
D. Energy transfers from Cynthia's foot to the tile at a greater rate than that to the carpet.

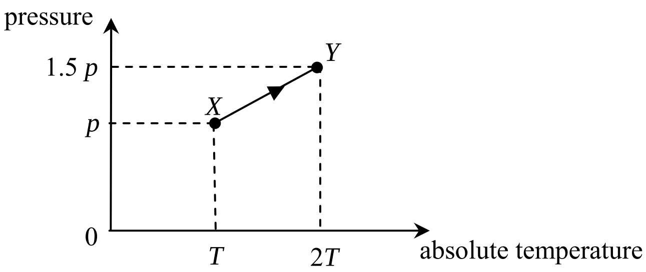
(2011-SP) 2.



The graph shows the variation in temperature of equal masses of two substances *P* and *Q* when they are separately heated by identical heaters. Which deduction is correct ?

A. The melting point of *P* is lower than that of *Q*.
B. The specific heat capacity of *P* in solid state is larger than that of *Q*.
C. The specific latent heat of fusion of *P* is larger than that of *Q*.
D. The energy required to raise the temperature of *P* from room temperature to boiling point is more than that of *Q*.

(2011-SP) *3.



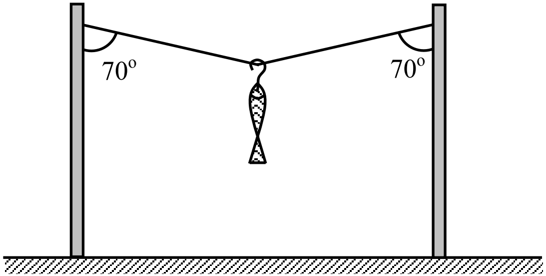
As the gas in a vessel of fixed volume is heated, it gradually leaks out. The gas in the vessel changes from state *X* to state *Y* along the path *XY* shown in the plot of pressure against absolute temperature. What percentage of the original mass of the gas leaks out from the vessel in this process ?

A. 10%
B. 20%
C. 25%
D. 50%

(2011-SP) *4. Two vessels contain hydrogen gas and oxygen gas respectively. Both gases have the same pressure and temperature and are assumed to be ideal. Which of the following physical quantities must be the same for the two gases ?

A. The volume of the gas
B. The mass per unit volume of the gas
C. The r.m.s. speed of the gas molecules
D. The number of gas molecules per unit volume

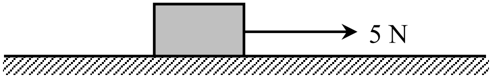
(2011-SP) 5.



A fish is hung on a light string as shown above. The tension in the string is 10 N. Find the total weight of the fish and the hook.

A. 20 sin 70° N
B. 20 cos 70° N
C. 10 sin 70° N
D. 10 cos 70° N

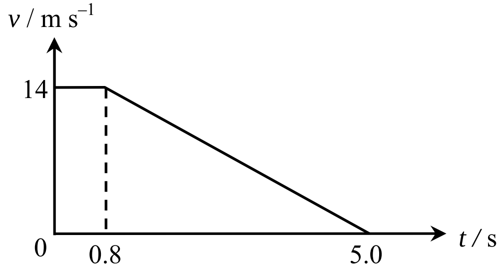
(2011-SP) 6.



A 1 kg block is pulled by a horizontal force of 5 N and moves with an acceleration of 2 m s^{-2} on a rough horizontal plane. Find the frictional force acting on the block.

A. zero
B. 2 N
C. 3 N
D. 7 N

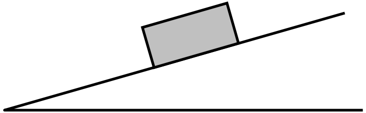
(2011-SP) 7. Patrick is driving along a straight horizontal road. At time t = 0, he observes that an accident has happened. He then applies the brakes to stop his car with uniform deceleration. The graph shows the variation of the speed of the car with time.



Find the distance travelled by the car from time $t = 0$ to 5.0 s.

A. 29.4 m
B. 40.6 m
C. 46.2 m
D. 81.2 m

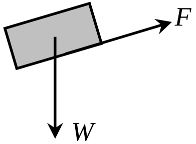
(2011-SP) 8.



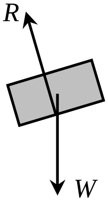
A block remains at rest on a rough inclined plane. Which diagram shows all the forces acting on the block ?

Note : W = gravitational force acting on the block,
R = normal reaction exerted by the inclined plane on the block, and
F = friction acting on the block.

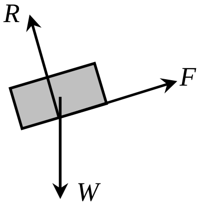
A.



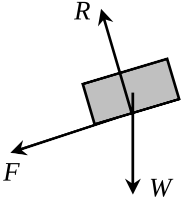
B.



C.



D.



(2011-SP) 9. Kelvin is standing on a balance inside a lift. The table shows the readings of the balance in three situations.

Motion of the lift	**Reading of the balance**
moving upwards with a uniform speed	R_1
moving downwards with a uniform speed	R_2
moving upwards with an acceleration	R_3

Which relationship is correct ?

A. $R_1 = R_2 > R_3$
B. $R_3 > R_1 = R_2$
C. $R_1 > R_2 > R_3$
D. $R_3 > R_1 > R_2$

(2011-SP) 10.

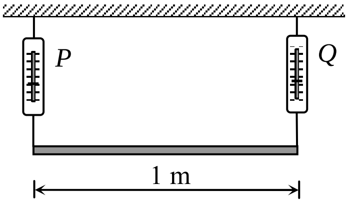


Figure (a)

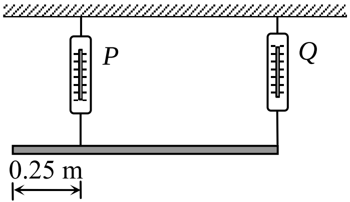


Figure (b)

Figure (a) shows a uniform plank supported by two spring balances *P* and *Q*. The readings of the two balances are both 150 N. *P* is now moved 0.25 m towards *Q* (see Figure (b)). Find the new readings of *P* and *Q*.

	Reading of *P*/N	**Reading of *Q*/N**
A.	100	200
B.	150	150
C.	200	100
D.	200	150

(2011-SP) 11. Which of the following pairs of forces is/are example(s) of action and reaction ?

(1) The centripetal force keeping a satellite in orbit round the earth and the weight of the satellite.
(2) The air resistance acting on an object falling through the air with terminal velocity and the weight of the object.
(3) The forces of attraction experienced by two parallel wires carrying currents in the same direction.

A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

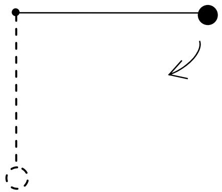
(2011-SP) 12. Two small identical objects P and Q are released from rest from the top of a building 80 m above the ground. Q is released 1 s after P. Neglecting air resistance, what is the maximum vertical separation between P and Q in the air ?

A. 5 m
B. 10 m
C. 35 m
D. 45 m

(2011-SP) 13. A car P of mass 1000 kg moves with a speed of 20 m s^{-1} and makes a head-on collision with a car Q of mass 1500 kg, which was moving with a speed of 10 m s^{-1} in the opposite direction before the collision. The two cars stick together after the collision. Find their common velocity immediately after the collision.

A. 2 m s^{-1} along the original direction of P
B. 2 m s^{-1} along the original direction of Q
C. 14 m s^{-1} along the original direction of P
D. 14 m s^{-1} along the original direction of Q

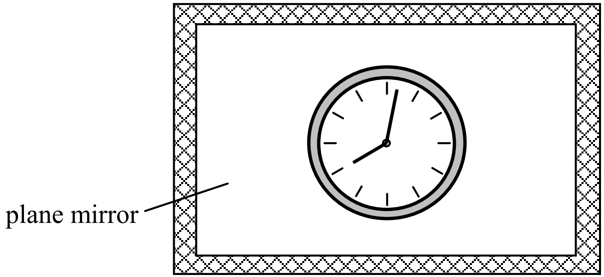
(2011-SP) *14.



A simple pendulum is held at rest in a horizontal position. It is then released with the string taut. Which statement about the tension in the string is **not correct** when the pendulum reaches its vertical position ?

A. The tension equals the weight of the pendulum bob in magnitude.
B. The tension attains its greatest value.
C. The tension does not depend on the length of the pendulum.
D. The tension depends on the mass of the pendulum bob.

(2011-SP) 15.



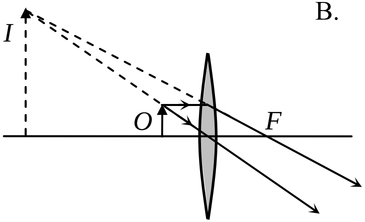
The diagram shows the image of a clock in a plane mirror. What is the time displayed by the clock ?

A. 3:58
B. 4:02
C. 7:58
D. 8:02

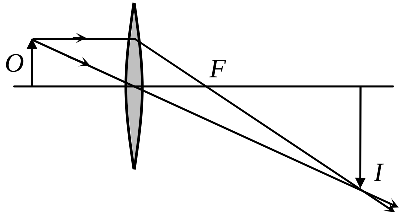
(2011-SP) 16.

Cecilia uses a magnifying glass to read some small print. Which diagram shows how the image of the print is formed ?

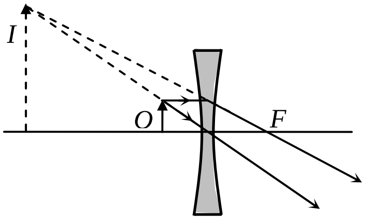
A.



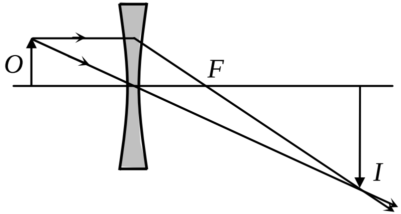
B.



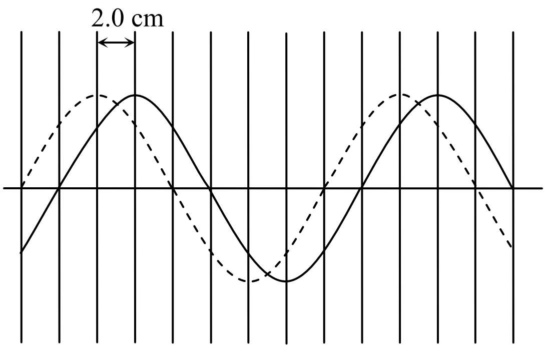
C.



D.



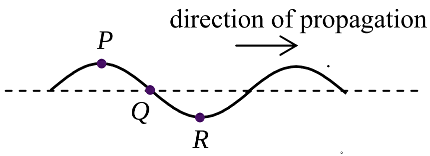
(2011-SP) 17.



The solid curve in the diagram shows a transverse wave at a certain instant. After 0.05 s, the wave has travelled a distance of 2.0 cm and is indicated by the dashed curve. Find the wavelength and frequency of the wave.

	Wavelength/cm	**Frequency/Hz**
A.	8	2.5
B.	16	2.5
C.	8	5
D.	16	5

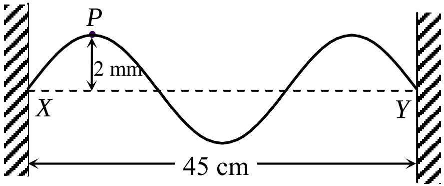
(2011-SP) 18.



The figure shows the shape of a transverse wave travelling along a string at a certain instant. Which statement about the motion of the particles *P*, *Q* and *R* on the string at this instant is correct ?

A. Particle *P* is moving downwards.
B. Particle *Q* is stationary.
C. Particle *R* attains its maximum acceleration.
D. *P* and *Q* are in phase.

(2011-SP) 19.



String XY is fixed at both ends. The distance between X and Y is 45 cm. Two identical sinusoidal waves travel along XY in opposite directions and form a stationary wave with an antinode at point P. The figure shows the string when P is 2 mm, its maximum displacement, from the equilibrium position. What is the amplitude and wavelength of each of the **travelling waves** on the string ?

	Amplitude	**Wavelength**
A.	1 mm	30 cm
B.	1 mm	15 cm
C.	2 mm	30 cm
D.	2 mm	15 cm

(2011-SP) 20. A Young's double-slit experiment was performed using a monochromatic light source. Which change would result in a greater fringe separation on the screen ?

(1) Using monochromatic light source of longer wavelength
(2) Using double slit with greater slit separation
(3) Using double slit with larger slit width

A. (1) only
B. (1) and (2) only
C. (2) and (3) only
D. (1), (2) and (3)

(2011-SP) 21. An object is placed at the focus of a concave lens of focal length 10 cm. What is the magnification of the image formed ?

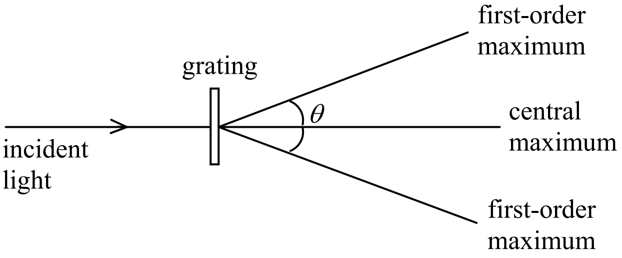
A. 0.5
B. 1.0
C. 2.0
D. infinite

(2011-SP) 22. Which of the following statements about sound waves is/are correct ?

(1) Sound waves are longitudinal waves.
(2) Sound waves are electromagnetic waves.
(3) Sound waves cannot travel in a vacuum.

A. (2) only
B. (3) only
C. (1) and (2) only
D. (1) and (3) only

(2011-SP) 23.

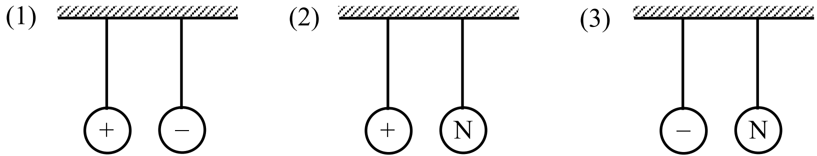


When monochromatic light is passed through a diffraction grating, a pattern of maxima and minima is observed as shown. Which combination would produce the largest angle θ between the first-order maxima ?

	Grating (lines per mm)	**Colour of light used**
A.	200	blue
B.	200	red
C.	400	blue
D.	400	red

(2011-SP) 24. Two conducting spheres are hanging freely in air by insulating threads. In which of the following will the two spheres attract each other ?

Note : ‘N’ denotes that the sphere is uncharged.



A. (1) only
B. (2) only
C. (3) only
D. (1), (2) and (3)

(2011-SP) 25. The table shows three electrical appliances which Clara used in a certain month :

Appliance	Rating	Duration
Air-conditioner	220 V, 1200 W	250 hours
television	220 V, 250 W	80 hours
computer	220 V, 150 W	60 hours

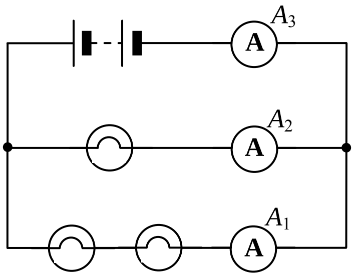
Calculate the cost of electricity used.
Note : 1 kW h of electricity costs $ 0.86.

A. $ 62.25
B. $ 73.79
C. $ 282.94
D. $ 536.64

(2011-SP) 26. If a 15 A fuse is installed in the plug of an electric kettle of rating ‘220 V, 900 W’, state what happens when the kettle is plugged in and switched on.

A. The kettle will not operate.
B. The kettle will be short-circuited.
C. The output power of the kettle will be increased.
D. The chance of the kettle being damaged by an excessive current will be increase

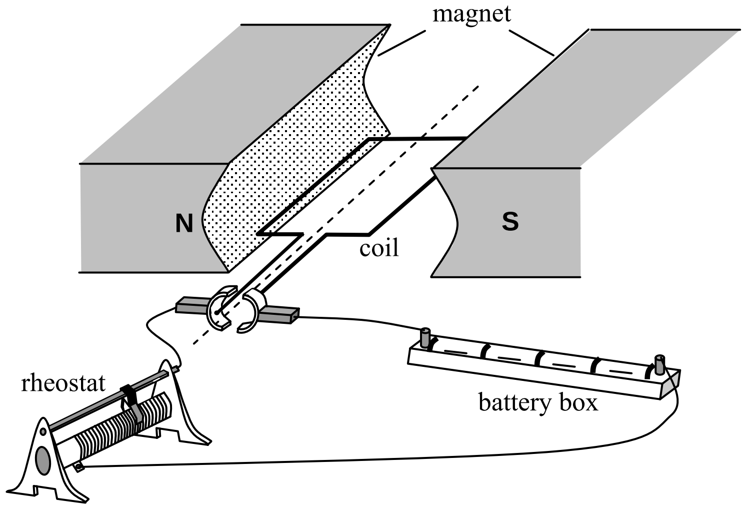
(2011-SP) 27.



In the above circuit, the bulbs are identical. The reading of ammeter A_1 is 1 A. Find the readings of ammeters A_2 and A_3.

	Reading of A_2	**Reading of A_3**
A.	2 A	2 A
B.	2 A	3 A
C.	0.5 A	1 A
D.	0.5 A	1.5 A

(2011-SP) 28.

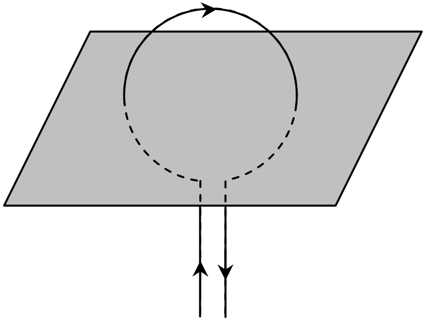


The figure shows a simple motor. Which of these changes would increase the turning effect of the coil ?

(1) using a stronger magnet
(2) reducing the resistance of the rheostat
(3) using a coil with a smaller number of turns

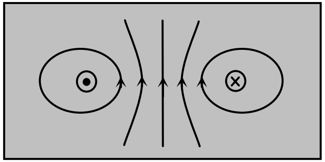
A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

(2011-SP) 29.

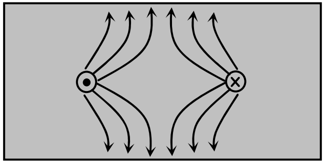


Which diagram shows the magnetic field pattern around a flat circular current-carrying coil, in the plane shown ?

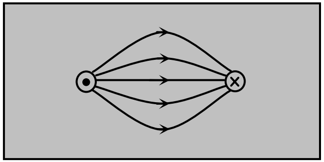
A.



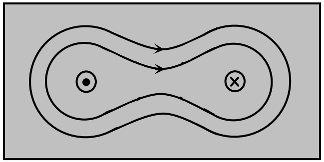
B.



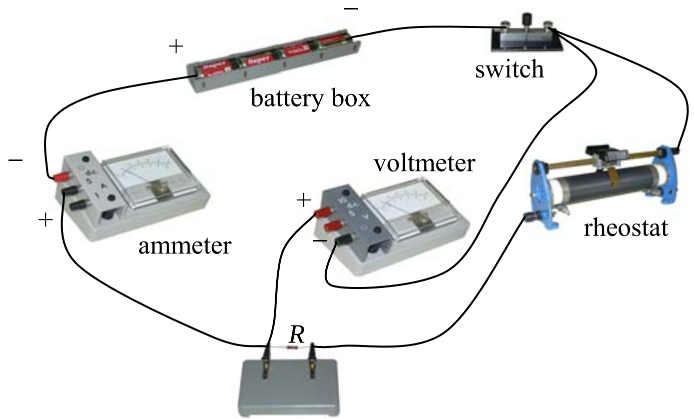
C.



D.



(2011-SP) 30.

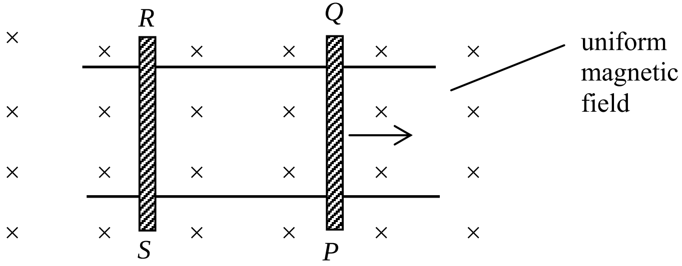


A student wants to measure the resistance of a resistor R and sets up the circuit shown. The student made which of these mistakes setting up the circuit ?

(1) The polarity of the ammeter was reversed.
(2) The polarity of the voltmeter was reversed.
(3) The voltmeter was connected across both R and the rheostat.

A. (1) only
B. (2) only
C. (1) and (3) only
D. (2) and (3) only

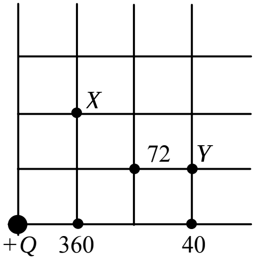
(2011-SP) 31.



The figure shows conducting rods PQ and RS placed on two smooth, parallel, horizontal conducting rails. A uniform magnetic field is directed into the plane of the paper. PQ is given an initial velocity to the right and left to roll. Which statement is **INCORRECT** ?

A. The induced current is in the direction $PQRS$.
B. The magnetic force acting on rod PQ is towards the left.
C. Rod RS starts moving towards the right.
D. Rod PQ moves with a uniform speed.

(2011-SP) 32.



The figure shows the location of an isolated charge of size +Q. The size (in an arbitrary unit) of the electric field strength is marked at certain points. What is the size (in the same arbitrary unit) of the electric field strength at *X* and *Y* ?

	electric field strength at *X*	**electric field strength at *Y***
A.	72	30
B.	72	36
C.	90	30
D.	90	36

(2011-SP) *33. Power is transmitted over long distances at high alternating voltages. Which statements are correct ?

(1) Alternating voltages can be stepped up or down efficiently by transformers.
(2) For a given transmitted power, the current will be reduced if a high voltage is adopted.
(3) The power loss in the transmission cables will be reduced if a high voltage is adopted.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

(2011-SP) 34. Which of these is a nuclear fusion reaction ?

A. $^{235}_{92}U + n \rightarrow \, ^{144}_{56}Ba + \, ^{90}_{36}Kr + 2n$

B. $^{2}_{1}H + \, ^{3}_{1}H \rightarrow \, ^{4}_{2}He + n$

C. $^{14}_{7}N + n \rightarrow \, ^{14}_{6}C + \, ^{1}_{1}H$

D. $^{238}_{92}U \rightarrow \, ^{234}_{90}Th + \alpha$

(2011-SP) *35. On which of the following does the activity of a radioactive source depend ?

(1) the nature of the nuclear radiation emitted by the source
(2) the half-life of the source
(3) the number of active nuclides in the source

A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

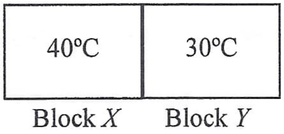
(2011-SP) 36. Different absorbers are placed in turn between a radioactive source and a Geiger-Muller tube. Three readings are taken for each absorber. The following data are obtained:

Absorber	**Count rate / s^{-1}**		
–	200	205	198
paper	197	202	206
5 mm aluminium	112	108	111
25 mm lead	60	62	58
50 mm lead	34	36	34

What type(s) of radiation does the source emit ?

A. β only
B. γ only
C. β and γ only
D. α , β and γ

(2012) 1. Two metal blocks X and Y of the same mass and of initial temperatures 40°C and 30°C respectively are in good thermal contact as shown. The specific heat capacity of X is greater than that of Y. Which statement is correct when a steady state is reached ? Assume no heat loss to the surroundings.

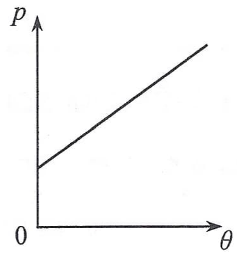


A. The temperature of block X is higher than that of block Y.
B. Their temperature becomes the same and is lower than 35°C.
C. Their temperature becomes the same and is higher than 35°C.
D. Their temperature becomes the same and is equal to 35°C.

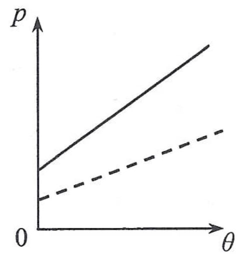
(2012) 2. When a patient's arm is wiped by a piece of cotton soaked with alcohol, the wiped area will feel cool as that patch of alcohol on the skin evaporates. Which statement explains this phenomenon ?

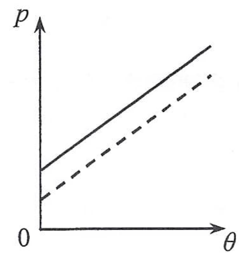
A. The evaporation of alcohol absorbs heat from the patient's arm.
B. The alcohol on the skin releases latent heat to the surrounding air.
C. The motion of all the molecules in the patch of alcohol slows down.
D. Air molecules remove heat from the patch of alcohol by conduction.

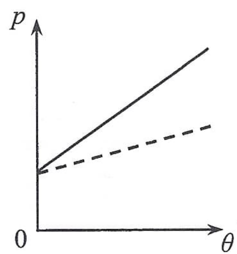
(2012) *3. An ideal gas is contained in a closed vessel of fixed volume. The graph below shows the variation of pressure p of the gas against its Celsius temperature θ.

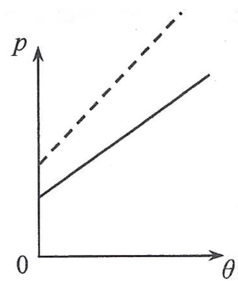


If the number of gas molecules in the vessel is halved, which graph of the dotted line best shows the relationship between p and θ ?

A.


B.


C.


D.


(2012) 4. Which of the following descriptions is correct ?

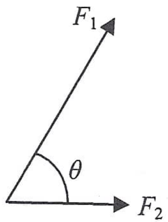
A. When water at 25°C is heated to 50°C, both the kinetic energy and potential energy of the water molecules increase.

B. When water at 25°C is heated to 50°C, only the potential energy of the water molecules increases.

C. When water boils at 100°C and turns into steam, the kinetic energy of the water molecules increases.

D. When water boils at 100°C and turns into steam, the potential energy of the water molecules increases.

(2012) 5.

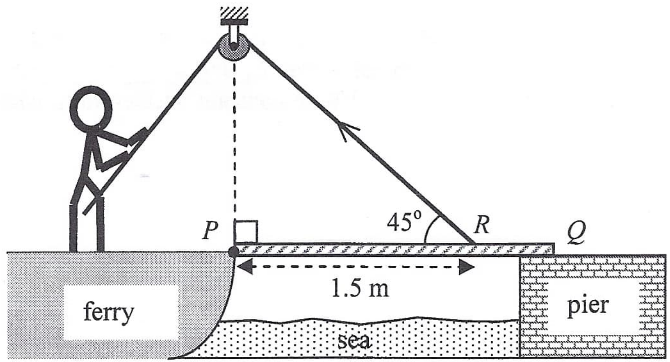


Two forces F_1 and F_2 of constant magnitudes act at the same point as shown. When the angle θ between F_1 and F_2 increases from 0° to 180°, the magnitude of the resultant force

A. decreases throughout.
B. increases throughout.
C. decreases and then increases.
D. increases and then decreases.

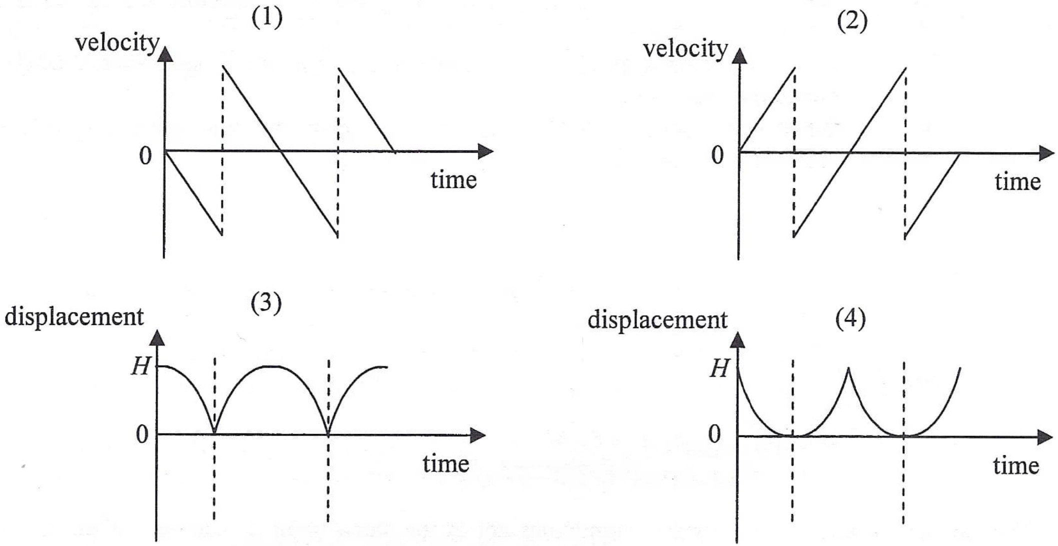
(2012)

6. A uniform gangplank *PQ* of a ferry smoothly hinged at end *P* initially rests horizontally on the pier. The gangplank has mass *M* and length 2 m. It is raised by a man on the ferry using a light rope passing a smooth fixed light pulley and connecting to *R* on the gangplank as shown. *R* is 1.5 m from end *P*. Which of the following correctly describes the force required to raise the gangplank steadily ?



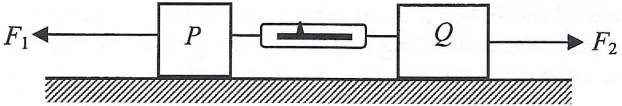
	initial force required to raise the gangplank when it is horizontal	*subsequent* force required to raise the gangplank
A.	0.67 *Mg*	greater than 0.67 *Mg*
B.	0.67 *Mg*	smaller than 0.67 *Mg*
C.	0.94 *Mg*	greater than 0.94 *Mg*
D.	0.94 *Mg*	smaller than 0.94 *Mg*

(2012) 7. Which of the following graphs (velocity-time and displacement-time) best represent the motion of a ball falling from rest under gravity at a height H and bouncing back from the ground two times ? Assume that the collision with the ground is perfectly elastic and neglect air resistance. (Downward measurement is taken to be negative.)



A. (1) and (3) only
B. (1) and (4) only
C. (2) and (3) only
D. (2) and (4) only

(2012) 8.



Blocks P and Q of mass m and $2m$ respectively are connected by a light spring balance and placed on a smooth horizontal surface as shown. If horizontal forces F_1 and F_2 (with $F_1 > F_2$) act on P and Q respectively and the whole system moves to the left with constant acceleration, what is the reading of the spring balance ?

A. $\dfrac{2F_1 - F_2}{3}$

B. $\dfrac{2(F_1 - F_2)}{3}$

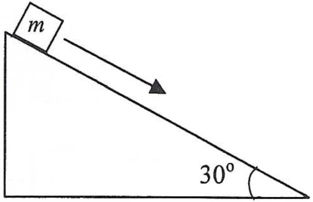
C. $\dfrac{2F_1 + F_2}{3}$

D. $\dfrac{F_1 + 2F_2}{3}$

(2012) 9. An object of mass 0.5 kg is raised vertically from the ground by a motor. The object is raised 2.5 m in 1.5 s with uniform speed. Estimate the output power of the motor. Neglect air resistance.
($g = 9.81\ \mathrm{m\ s^{-2}}$)

A. 5.5 W
B. 8.2 W
C. 11.0 W
D. 16.4 W

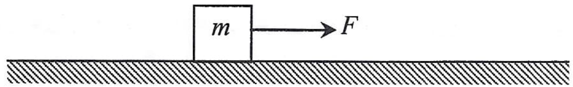
(2012) 10. A block of mass m resting on a 30° incline is given a slight push and slides down the incline with a uniform speed. Which of the following statements about the block's motion on the incline is/are correct ?



(1) There is no net force acting on the block.
(2) The frictional force acting on the block is 0.5 mg.
(3) If the block is given a greater initial speed, it will slide down the incline with acceleration.

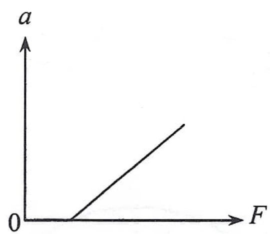
A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

(2012) 11.

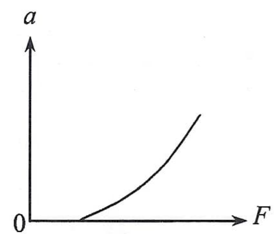


A block of mass m initially resting on a rough horizontal surface is pulled along the surface by a horizontal force F increasing from zero. If the frictional force is constant, which graph shows the relation between the acceleration of the block a and force F ?

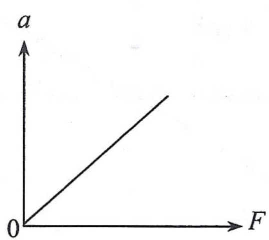
A.



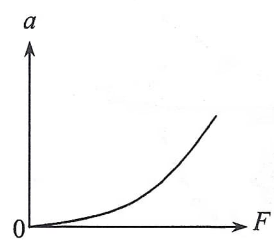
B.



C.



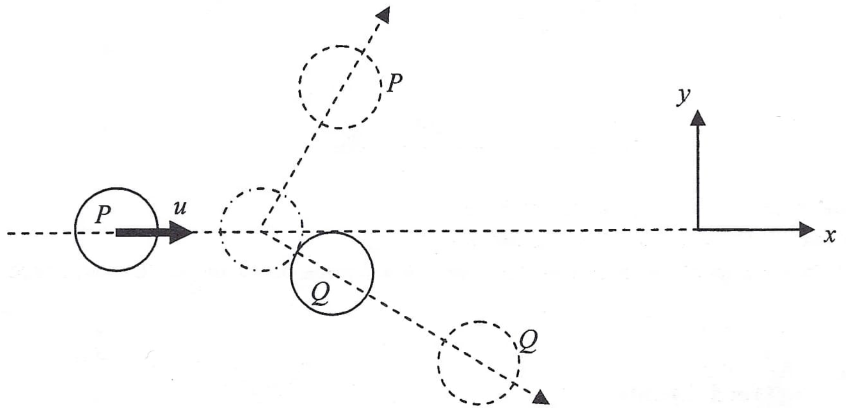
D.



(2012) *12. A bomber aircraft is 1 km above the ground and is flying horizontally at a speed of 200 m s^{-1}. The aircraft is going to release a bomb to destroy a target on the ground. How long before flying over the target should the bomb be released ? Assume that the bomber aircraft and the target are in the same vertical plane and neglect air resistance. ($g = 9.81$ m s^{-2})

A. 5.6 s
B. 10.1 s
C. 14.3 s
D. It cannot be calculated as the horizontal distance between the aircraft and the target is not known.

(2012) 13. On a smooth horizontal surface, a circular disk P moving at velocity u along the x direction collides obliquely with an identical disk Q initially at rest as shown below. The mass of each disk is m. Which statements about the collision is/are correct ?



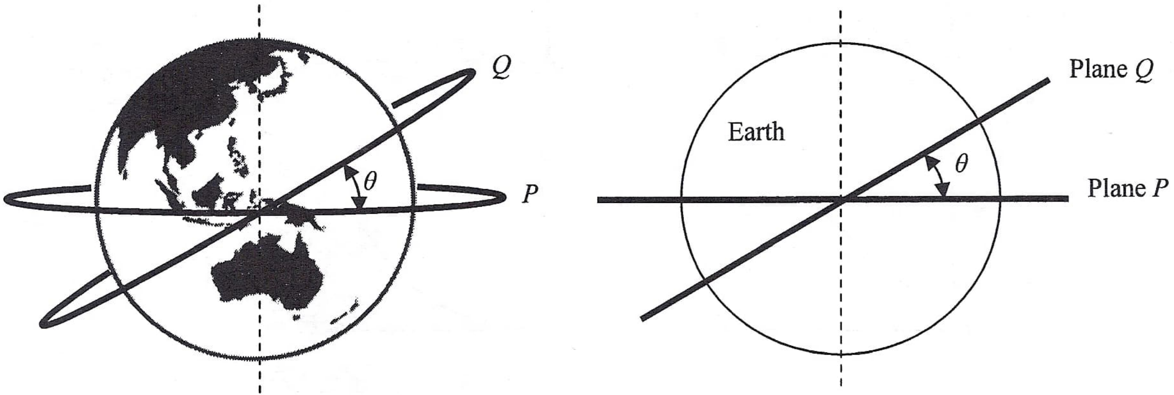
(1) Momentum of the system in the y direction is not conserved.

(2) The total kinetic energy of P and Q after collision is $\frac{1}{2}mu^2$ if the collision is perfectly elastic.

(3) Speed of Q after collision is less than u.

A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

(2012) *14. Two satellites move in circular orbits of the same radius R around the Earth (mass M). The orbits are in two different planes P and Q as shown. Plane P coincides with the Earth's equator while plane Q is inclined to the equator at θ. Which statement is **INCORRECT** ?



A. The speed of satellite P is $\sqrt{\frac{GM}{R}}$.

B. The centripetal force acting on satellite Q is pointing along the plane Q.

C. The acceleration of both satellites is the same in magnitude.

D. The period of satellite Q is longer than that of satellite P.

(2012) 15.

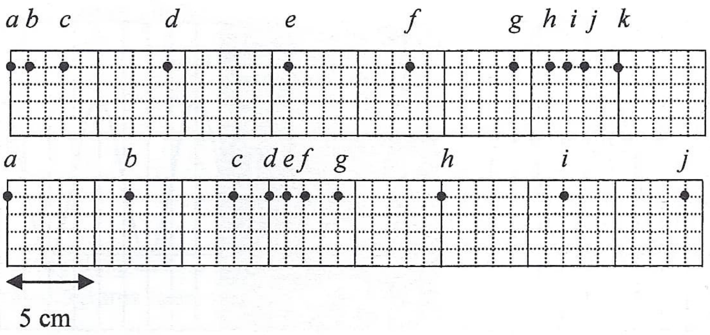


Figure (a)

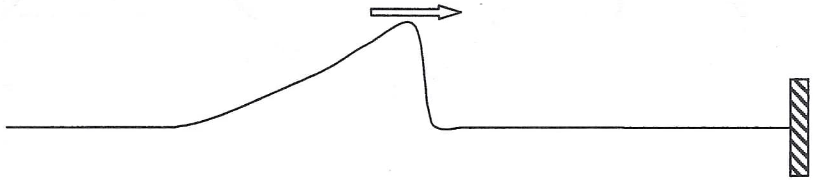
Figure (b)

A series of particles is uniformly distributed along a slinky spring initially. Figure (a) shows their positions at a certain instant when a travelling wave propagates along the slinky spring from left to right. Figure (b) shows their positions 0.1 s later. Which statement is correct ?

A. Particle *e* is always stationary.
B. Particles *a* and *i* are in phase.
C. The wavelength of the wave is 16 cm.
D. The frequency of the wave is 10 Hz.

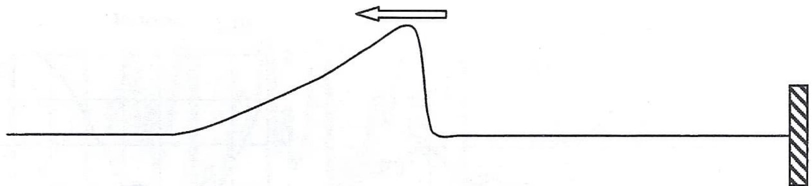
(2012)

16. A pulse on a string propagates towards the right end which is fixed.

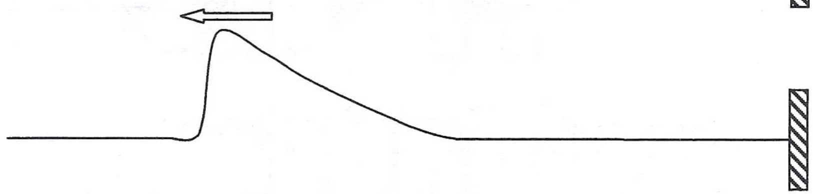


Which of the following represents the reflected pulse ?

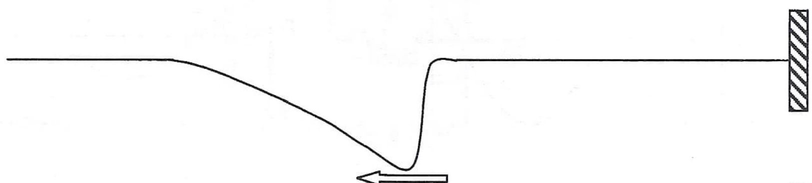
A.



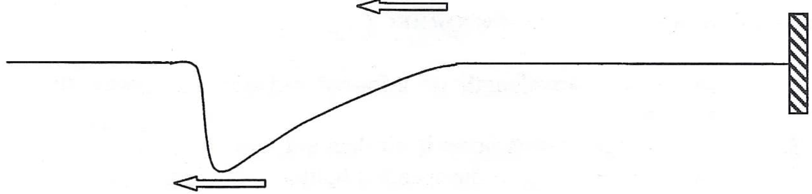
B.



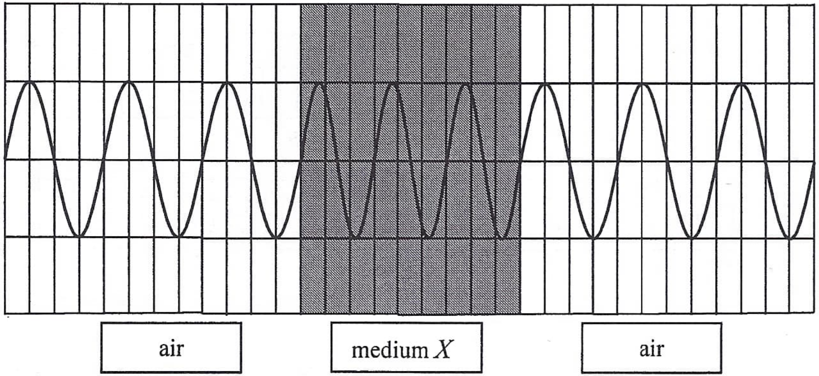
C.



D.

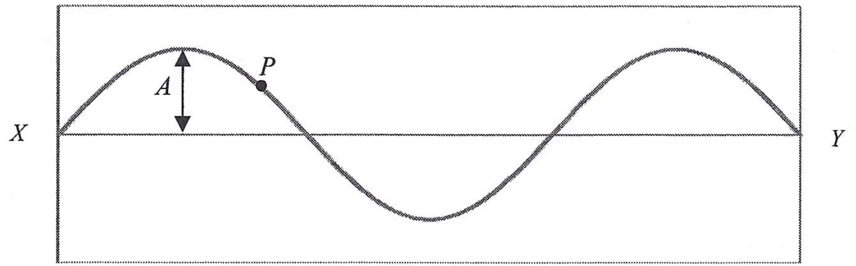


(2012) 17. A certain monochromatic light passes through medium X as shown below. What is the refractive index of medium X?



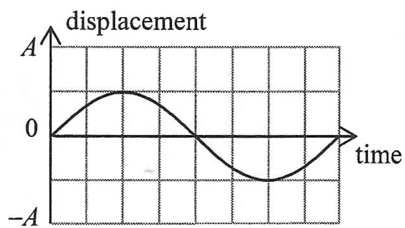
A. 1.25
B. 1.33
C. 1.50
D. 1.65

(2012) 18. A stationary wave is formed on a string fixed at both ends X and Y. The following is a snapshot of the string at time $t = 0$. The amplitude of vibration at an antinode is A.

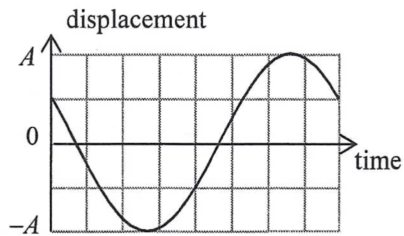


Which of the following shows the displacement-time graph of point P on the string for one period ? (Upward displacement is taken as positive.)

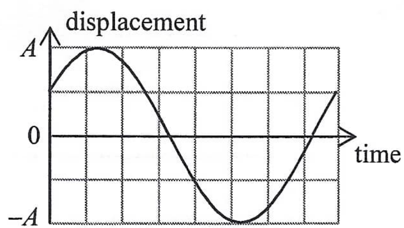
A.



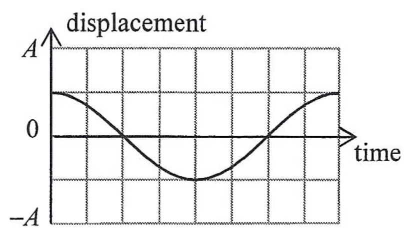
B.



C.



D.



(2012)

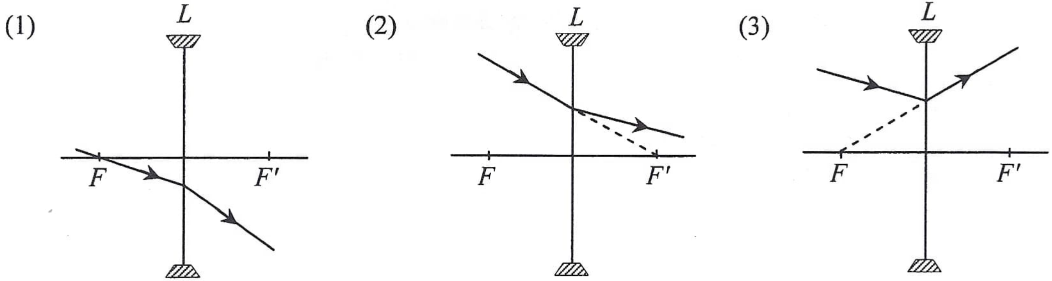
19. Which of the following statements is **INCORRECT** ?

A. In air, the wavelength of infra-red radiation is shorter than that of ultra-violet radiation.
B. Visible light travels faster in air than in glass.
C. Microwaves travel at the speed of light in a vacuum.
D. Both light and sound exhibit diffraction.

(2012) *20. For a diffraction grating of 600 lines per mm, the diffracted red light (657 nm) coincides with the diffracted violet light (438 nm) at angle of diffraction 52°. What are the respective orders of the diffracted red light and violet light ?

	red	violet
A.	2	3
B.	3	4
C.	3	2
D.	4	3

(2012) 21. In each of the following diagrams, L is a concave lens and its two principal foci are denoted by F and F'. Which of the ray diagrams is/are possible ?



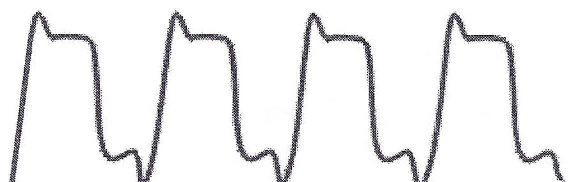
A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

(2012) 22. The figure shows the waveforms of sound notes generated by a violin, a piano and a tuning fork. The scale is the same in time and intensity axes for all three waveforms.

(I)

(II)

(III)



Which of the following about the sound notes are correct ?

(1) They all have the same pitch.
(2) The qualities of sound of (II) and (III) are different.
(3) (I) is generated by the tuning fork.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

(2012) 23. Which of the following about ultrasound is **INCORRECT** ?

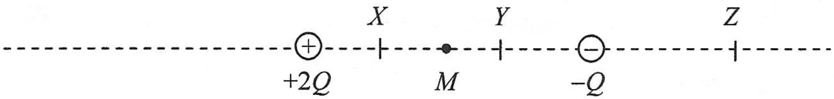
A. Ultrasound is a longitudinal wave.
B. The frequency of ultrasound is greater than 20000 Hz.
C. In air, the speed of ultrasound is faster than the speed of audible sound.
D. In air, the diffraction effect of ultrasound is less prominent than that of audible sound.

(2012) 24. P, Q, R, S are charged objects. When two of them are brought close to each other, P and Q repel, R and S also repel while Q and R attract each other. Which of the following descriptions about their charges is/are possible ?

(1) P and R are negatively charged.
(2) Q and S are positively charged.
(3) P is positively charged and S is negatively charged.

A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

(2012) *25.



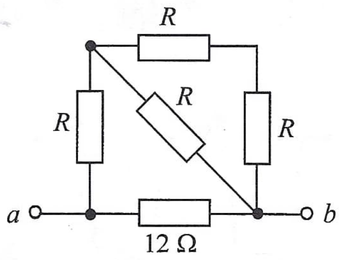
Two point charges $+2Q$ and $-Q$ are situated at fixed positions as shown. M is the mid-point between the charges. X, Y and Z are points marked on the line joining these two charges. At which point could

(1) the resultant electric field due to the two charges be zero ?
(2) the total electric potential due to the two charges be zero ?

	(1)	(2)
A.	Z	X
B.	Z	Y
C.	X	Z
D.	Y	Z

(2012)

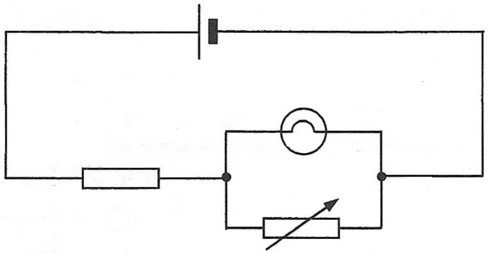
26.



In the above network, the resistance across terminals a and b is 6 Ω. If the 12 Ω resistor is replaced by a 6 Ω resistor, the resistance across terminals a and b

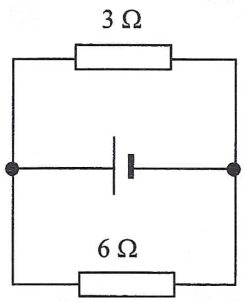
A. becomes 2 Ω.
B. becomes 4 Ω.
C. becomes 6 Ω.
D. cannot be found as the value of R is unknown.

(2012) 27. What will happen if the variable resistor is set to zero in the circuit below ?



A. The light bulb will burn out.
B. The light bulb will not light up.
C. The brightness of the light bulb will increase.
D. The brightness of the light bulb will remain unchanged.

(2012) 28.

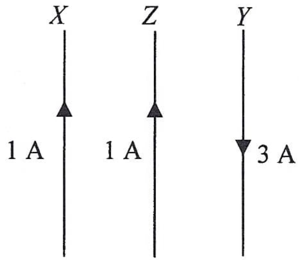


In the above circuit, the cell has e.m.f. 12 V and internal resistance 2 Ω. What is the current in the 6 Ω resistor ?

A. 0.5 A
B. 1.0 A
C. 1.5 A
D. 2.0 A

(2012)

29. In the figure below, X, Y and Z are three long straight parallel wires with Z placed midway between X and Y. X and Z carry currents of 1 A in the same direction while Y carries a current of 3 A in the opposite direction. The magnetic force per unit length experienced by wire X due to wire Z is of magnitude F.



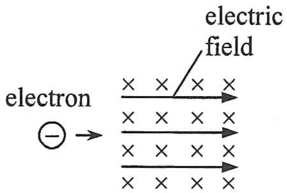
The magnetic force per unit length acting on wire Z due to both X and Y is

A. $2F$ to the right.
B. $2F$ to the left.
C. $4F$ to the right.
D. $4F$ to the left.

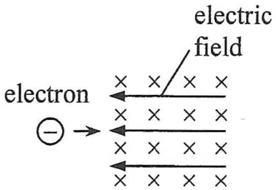
(2012)

30. An electron enters a region in which both a uniform electric field E and a uniform magnetic field B exist. The magnetic field B is pointing into the paper. In which direction should the electric field be applied so that the electron could be undeflected ?

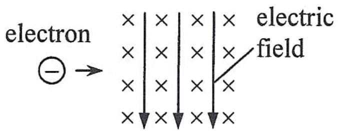
A.



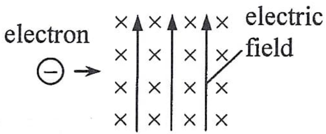
B.



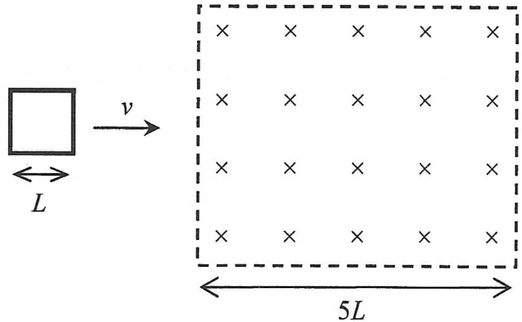
C.



D.



(2012) 31.



A square metal frame of length of side L moving with constant velocity v passes through a region of uniform magnetic field of width $5L$ as shown. What is the total time period during which a current is induced in the frame ?

A. $\dfrac{L}{v}$

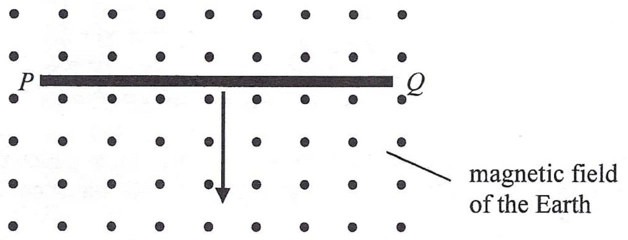
B. $\dfrac{2L}{v}$

C. $\dfrac{3L}{v}$

D. $\dfrac{4L}{v}$

(2012)

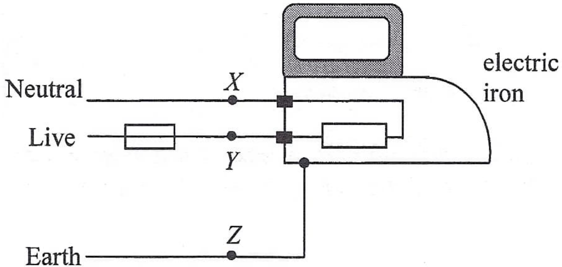
32. A copper rod PQ is placed horizontally as shown below. It is released and then falls vertically, cutting across the magnetic field of the Earth pointing out of the paper. Neglect air resistance. Which of the following statements is/are correct ?



(1) A voltage is induced across PQ.
(2) A steady induced current is generated in the rod.
(3) Due to the effect of the Earth's magnetic field, the copper rod falls with an acceleration less than the acceleration due to gravity.

A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

(2012) 33.



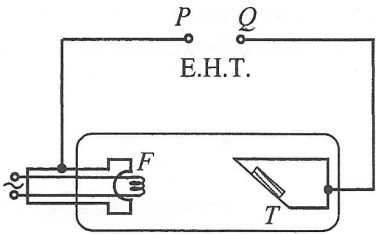
The figure shows a simple domestic circuit for an electric iron. The fuse will blow when which of the following points are short-circuited ?

(1) X and Y
(2) Y and Z
(3) X and Z

A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

(2012)

34.



The figure shows a schematic diagram of an X-ray tube in which the filament F and the metal target T are connected to terminals P and Q of an E.H.T. Which statement is correct ?

A. P is the positive terminal and X-rays are emitted from T.
B. P is the positive terminal and X-rays are emitted from F.
C. Q is the positive terminal and X-rays are emitted from T.
D. Q is the positive terminal and X-rays are emitted from F.

(2012) 35. A certain radioactive isotope X has a half-life of 20 hours. After a time interval of 10 hours, what is the approximate fraction (f) of a sample of the radioactive isotope X remaining ?

A. $\frac{1}{4} \leq f \leq \frac{1}{2}$

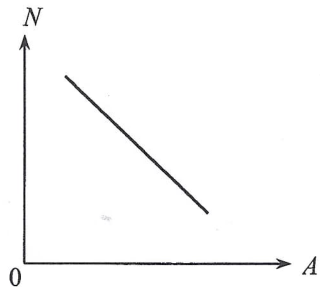
B. $f = \frac{1}{2}$

C. $\frac{3}{4} > f > \frac{1}{2}$

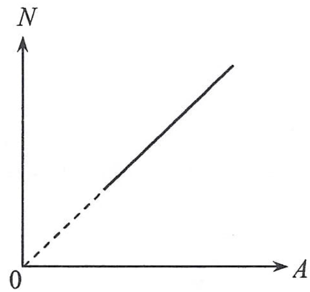
D. $f > \frac{3}{4}$

(2012) 36. Isotopes of an element have different mass number A and neutron number N. Which of the following N - A plots correctly shows the relationship of N and A for any given element ?

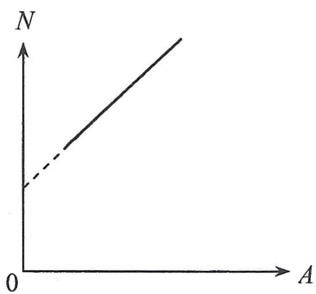
A.



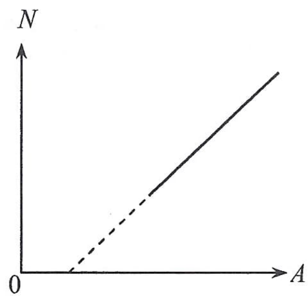
B.



C.



D.



(2013) 1. Which of the following statements about *boiling* and *evaporation* of a liquid is/are correct ?

(1) A liquid absorbs energy when it boils but does not absorb energy when it evaporates.
(2) Boiling occurs at a definite temperature while evaporation takes place above room temperature.
(3) Boiling occurs throughout the liquid while evaporation only takes place at the liquid's surface.

A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

(2013) 2. In an experiment to measure the specific latent heat of vaporization of water, a beaker of water is boiled off using an electric heater. Which of the following sources of error would lead to an experimental result smaller than the standard value ?

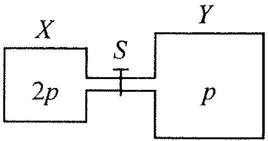
A. Energy is lost to the surroundings.
B. Water splashes out of the beaker.
C. Steam condenses on the cooler part of the heater and drops back to the beaker.
D. The heater is not completely immersed in water.

(2013) *3. In which of the following situations would the r.m.s. speed of the molecules of a fixed mass of an ideal gas increase ?

(1) The gas is heated under constant volume.
(2) The gas expands under constant pressure.
(3) The gas is compressed under constant temperature.

A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

(2013) *4.



Vessel X of volume V and vessel Y of volume $2V$ are connected by a short narrow tube as shown. Initially, tap S is closed and the same kind of ideal gas at the same temperature is contained in X and Y at pressure $2p$ and p respectively. The tap S is then opened and equilibrium state is finally reached with the temperature unchanged. Which statement is **INCORRECT** ?

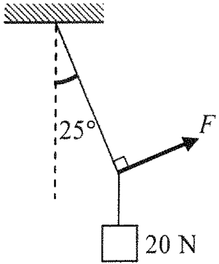
A. Before S is opened, both vessels contain the same number of gas molecules.

B. Before S is opened, the average kinetic energy of the gas molecules in both vessels is the same.

C. When S is opened, a net flow of gas from X to Y occurs.

D. When equilibrium is reached, the gas pressure becomes $\frac{3}{2}p$.

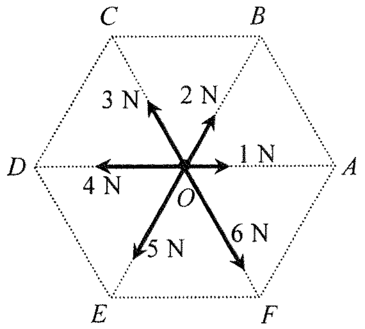
(2013) 5.



A block of weight 20 N is suspended by a light string from the ceiling. A force F is applied such that the block is displaced to one side with the string making an angle of 25° with the vertical as shown. Find the magnitude of F.

A. 8.5 N
B. 9.3 N
C. 18.1 N
D. 47.3 N

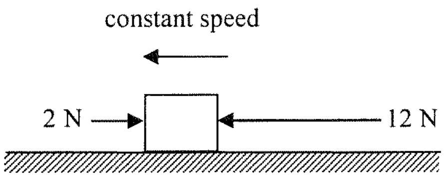
(2013) 6.



In the figure, O is the centre of a regular hexagon. A particle at O is subject to six forces with magnitudes indicated as shown. The resultant force acting on the particle is

A. 9 N along direction OE.
B. 8 N along direction OE.
C. 8 N along direction OF.
D. 6 N along direction OE.

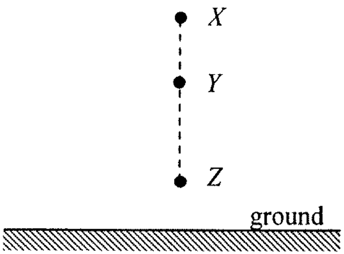
(2013) 7.



A block on a rough horizontal surface is moving to the left with constant speed under two horizontal forces 2 N and 12 N indicated as shown. If the force of 12 N is suddenly removed, what is the net force acting on the block at that instant ?

A. 12 N
B. 10 N
C. 8 N
D. 2 N

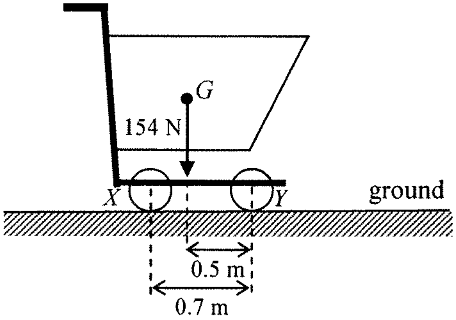
(2013) 8.



A particle is released from rest at X as shown. It takes time t_1 to fall from X to Y and time t_2 to fall from Y to Z. If $XY : YZ = 9 : 16$, find $t_1 : t_2$. Neglect air resistance.

A. 2 : 3
B. 3 : 4
C. 4 : 3
D. 3 : 2

(2013) 9.



The figure shows a supermarket trolley resting on the ground. The separation between cylindrical wheels X and Y is 0.7 m. When the trolley is loaded to a total weight of 154 N, its centre of gravity G is at a horizontal distance of 0.5 m from the wheel Y. What is the reaction acting on the wheel X from the ground ?

A. 44 N
B. 62 N
C. 92 N
D. 110 N

(2013) 10.

Two identical spheres are moving in opposite directions with speeds u and v (with $u > v$) respectively as shown. They make a head-on collision. Which of the following diagrams show(s) a *possible* situation of the spheres after collision ?

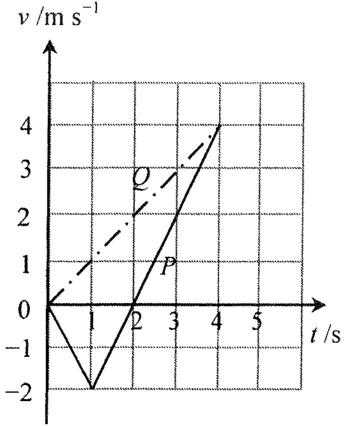
(1) 

(2) 

(3) 

A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

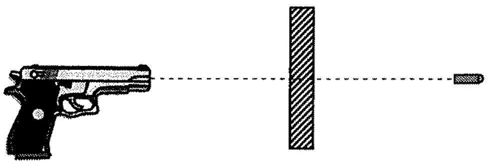
(2013) 11. Two particles P and Q start from the same position and travel along the same straight line. The figure shows the velocity-time (v-t) graph for P and Q. Which of the following descriptions about their motion is/are correct ?



(1) At $t = 1$ s, P changes its direction of motion.
(2) At $t = 2$ s, the separation between P and Q is 4 m.
(3) At $t = 4$ s, P and Q meet each other.

A. (1) only
B. (2) only
C. (1) and (3) only
D. (2) and (3) only

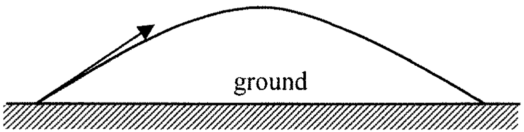
(2013) 12.



A bullet of mass 50 g is fired from a gun with a speed of 400 m s^{-1} and passes right through a fixed wooden block of 6 cm thickness as shown. Find the average resistive force acting on the bullet due to the block if it emerges with a speed of 250 m s^{-1}. Neglect air resistance and the effects of gravity.

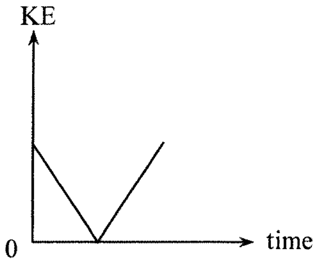
A. 4.06×10^4 N
B. 1.02×10^4 N
C. 125 N
D. Answer cannot be found as the time of travel of the bullet within the block is not known.

(2013) *13.

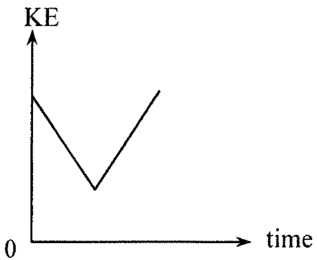


A particle is projected into the air at time $t = 0$ and it performs a parabolic motion before landing on the ground as shown. Which graph represents the variation of the kinetic energy (KE) of the particle with time before landing ? Neglect air resistance.

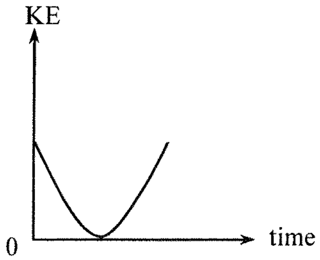
A.



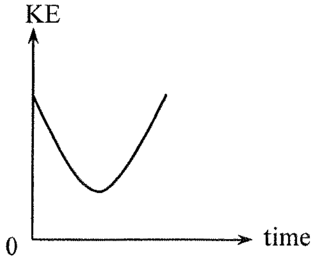
B.



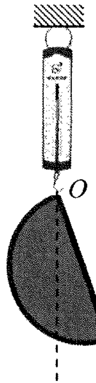
C.



D.



(2013) 14.



A semi-circular cardboard hangs from a spring balance from point O as shown. The reading of the spring balance is 5 N. Which statements are correct ?

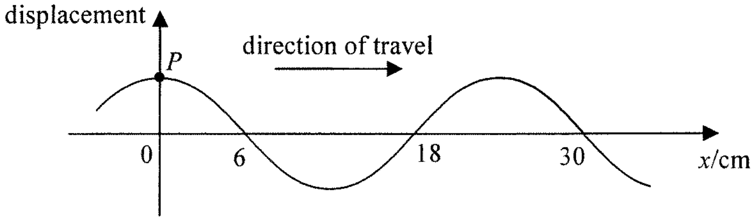
(1) The weight of the cardboard is 5 N.
(2) The centre of gravity of the cardboard is directly under O.
(3) The reading of the balance would become zero if the set-up is brought to the Moon's surface.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

(2013) *15. It is known that the mass of Mars is about $\frac{1}{10}$ of that of the Earth while its radius is about $\frac{1}{2}$ of the Earth's radius. In terms of the gravitational acceleration g on the Earth's surface, the approximate gravitational acceleration on the surface of Mars is

A. $0.2\,g$.
B. $0.4\,g$.
C. $2.5\,g$.
D. $4\,g$.

(2013) 16.



The figure shows a snapshot of a section of a continuous transverse wave travelling along the x-direction at time $t = 0$. At $t = 1.5$ s, the particle P just passes the equilibrium position for a *second* time at that moment. Find the wave speed.

A. 20 cm s^{-1}
B. 12 cm s^{-1}
C. 6 cm s^{-1}
D. 4 cm s^{-1}

(2013) 17.

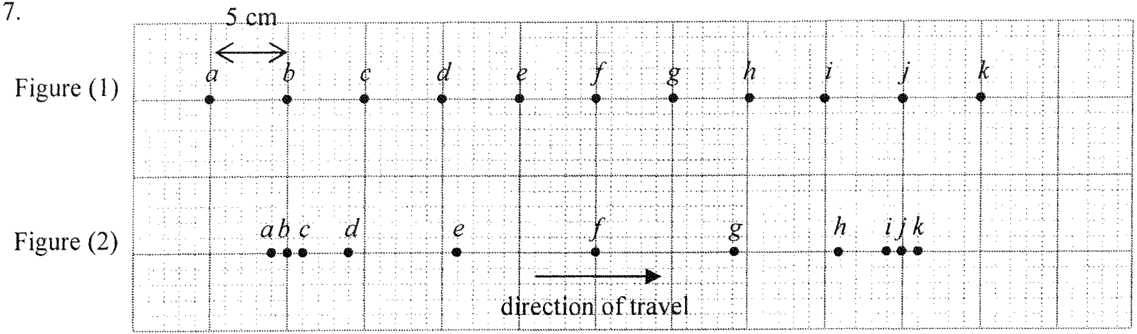
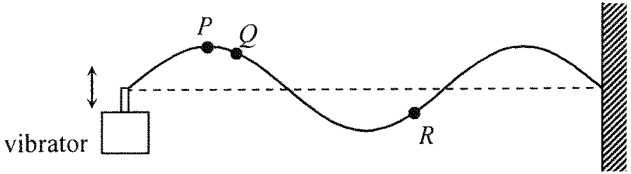


Figure (1) shows the equilibrium positions of particles *a* to *k* separated by 5 cm from each other in a medium. A longitudinal wave is travelling from left to right with a speed of 80 cm s^{-1}. At a certain instant, the positions of the particles are shown in Figure (2). Determine the amplitude and frequency of the wave.

	amplitude	frequency
A.	6 cm	2 Hz
B.	6 cm	4 Hz
C.	9 cm	2 Hz
D.	9 cm	4 Hz

(2013) 18.

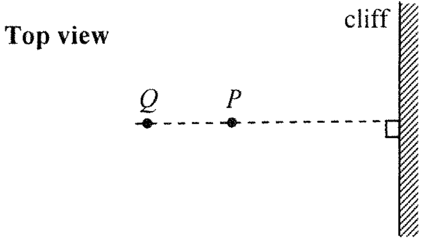


A vibrator generates a stationary wave on a string which is fixed at one end. The figure shows the appearance of the string at a certain instant. Which of the following descriptions about the motion of particles P, Q and R must be correct ?

(1) P and Q are momentarily at rest at this instant.
(2) Q and R take the same time to reach their respective equilibrium positions.
(3) P and R are always in antiphase.

A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

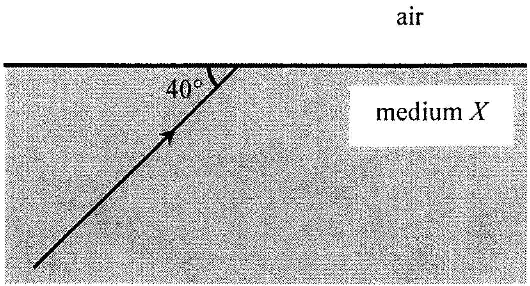
(2013) 19.



Astronauts P and Q stand at 400 m and 600 m respectively from a vertical cliff on the surface of a planet. The figure shows the top view. P claps his hands once and Q hears two clapping sounds separated by 4 s. What is the speed of sound in the atmosphere of this planet ?

A. 100 m s^{-1}
B. 150 m s^{-1}
C. 200 m s^{-1}
D. 250 m s^{-1}

(2013) 20.



A ray of light is travelling from a transparent medium X to air making an angle of 40° with the boundary plane as shown. If the angle between the refracted ray in air and the reflected ray in medium X is 70°, find the refractive index of medium X.

A. $\dfrac{\sin 40°}{\sin 30°}$

B. $\dfrac{\sin 30°}{\sin 40°}$

C. $\dfrac{\sin 60°}{\sin 50°}$

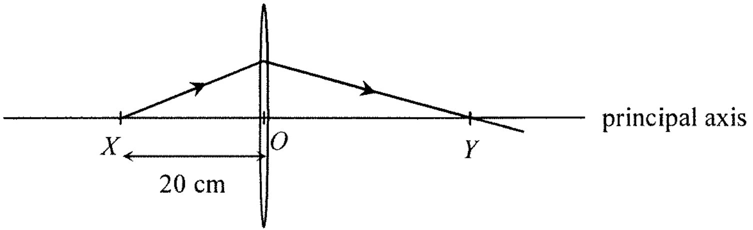
D. $\dfrac{\sin 50°}{\sin 60°}$

(2013) 21. White light can be resolved into component colours by using a glass prism. Which of the following statements is/are correct ?

(1) The refractive indices of glass for different component colours are not the same.
(2) Red light travels faster than violet light in a vacuum.
(3) The frequencies of all the component colours are reduced when entering the prism.

A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

(2013) 22.

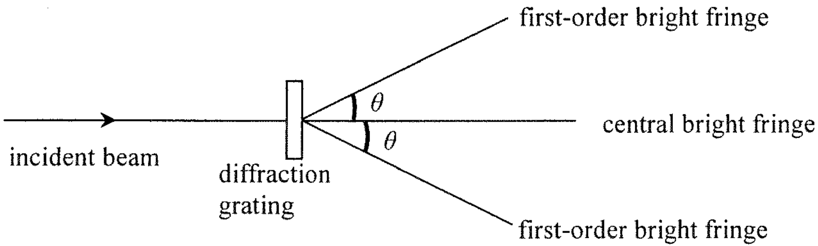


A point light source at X on the principal axis of a thin convex lens emits a ray of light. The ray passes through the lens and reaches the principal axis at point Y as shown. O is the optical centre of the lens such that $OX = 20$ cm and $OY > OX$. Which of the following statements is/are correct ?

(1) The focal length of the lens is shorter than 20 cm.
(2) If the point light source is shifted away from the lens, separation OY would increase.
(3) An object placed at Y would give a diminished image at X.

A. (1) only
B. (2) only
C. (1) and (3) only
D. (2) and (3) only

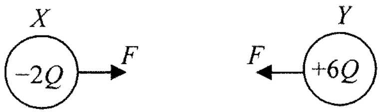
(2013) *23.



When monochromatic light passes through a diffraction grating, a pattern of bright fringes is formed. Which arrangement would produce the greatest angle θ between the central and first-order bright fringes ?

	grating (lines per mm)	colour of light
A.	400	green
B.	400	blue
C.	200	green
D.	200	blue

(2013) 24. X and Y are two small identical metal spheres carrying charges $-2Q$ and $+6Q$ respectively. When X and Y are separated by a certain distance, the magnitude of the electrostatic force between them is F.



The spheres are brought to touch each other and then placed back to their original positions. The electrostatic force between them becomes

A. $\frac{1}{4}F$, attractive.

B. $\frac{1}{4}F$, repulsive.

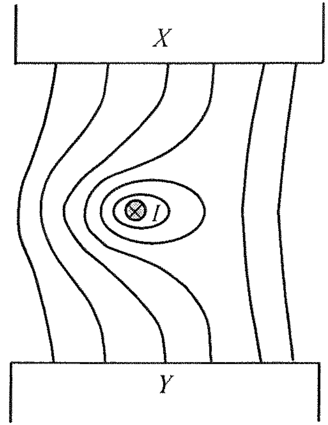
C. $\frac{1}{3}F$, attractive.

D. $\frac{1}{3}F$, repulsive.

(2013) *25. Lightning flash may occur when the strength of the electric field (assumed uniform) between a thundercloud and the ground reaches 3×10^6 N C^{-1}. A lightning flash on average discharges about 20 C of charge. If a thundercloud is at a height of 500 m above the ground, estimate the order of magnitude of the energy released in a lightning flash.

A. 10^6 J
B. 10^8 J
C. 10^{10} J
D. 10^{12} J

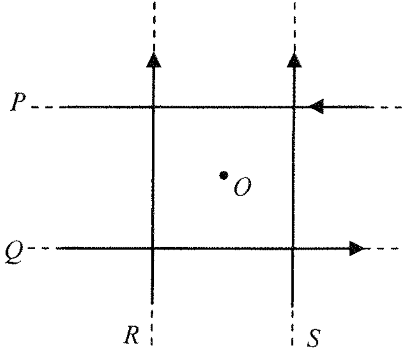
(2013) 26.



A straight wire carrying current I pointing into the paper is placed in a magnetic field between pole pieces X and Y. The figure shows the resultant field line pattern. What is the polarity of pole piece X and in what direction is the magnetic force acting on the wire ? Ignore the effect of the Earth's magnetic field.

	polarity of *X*	**direction of magnetic force**
A.	**N**	to right
B.	**N**	to left
C.	**S**	to right
D.	**S**	to left

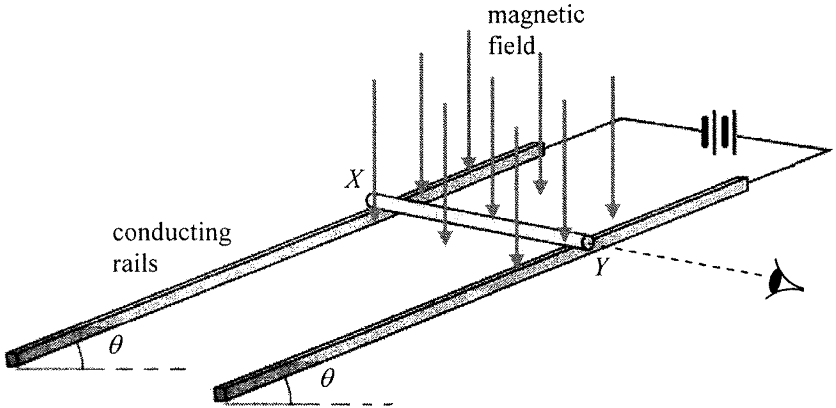
(2013) 27.



In the figure, four long straight wires P, Q, R and S in the same plane carry equal currents in the directions shown. The wires are insulated from each other. O is a point on the same plane and is equidistant from each wire. Removing which wire would increase the magnetic field strength at O ?

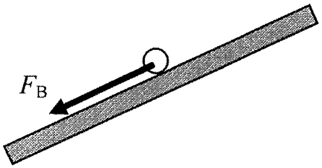
A. wire P
B. wire Q
C. wire R
D. wire S

(2013) 28.

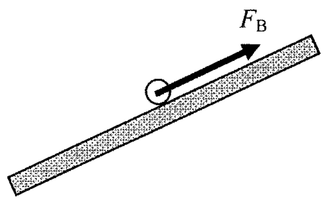


A copper rod XY is placed on a pair of smooth inclined conducting rails which are located in a magnetic field applied vertically downward. The rails make an angle θ to the horizontal and a battery is connected to the rails as shown above. Which diagram shown below represents the magnetic force F_B acting on the rod when viewed from end Y?

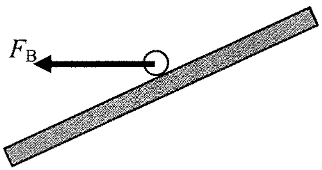
A.



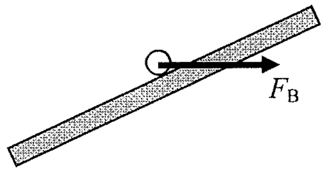
B.



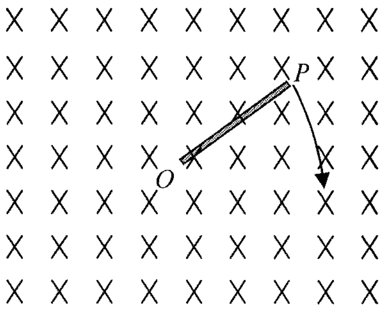
C.



D.



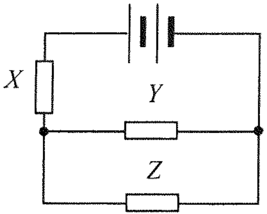
(2013) 29.



A metal rod OP is rotated about O in a clockwise direction in the plane of the paper with a uniform magnetic field pointing into the paper. Which statement is correct ?

A. An induced current flows in the rod from O to P.
B. An induced current flows in the rod from P to O.
C. E.m.f. is induced in the rod with end O at a higher electric potential.
D. E.m.f. is induced in the rod with end P at a higher electric potential.

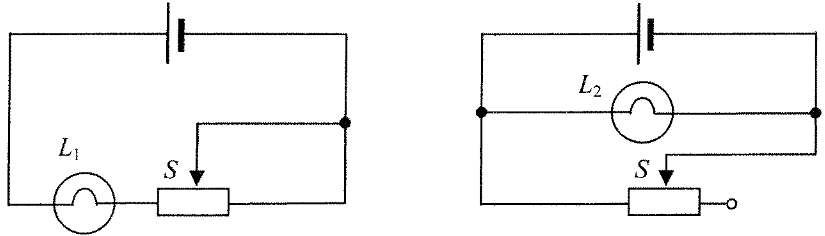
(2013) 30.



Resistors X, Y and Z in the above circuit are identical while the battery of negligible internal resistance supplies a total power of 24 W. What is the power dissipated in resistor Z ?

A. 3 W
B. 4 W
C. 6 W
D. 8 W

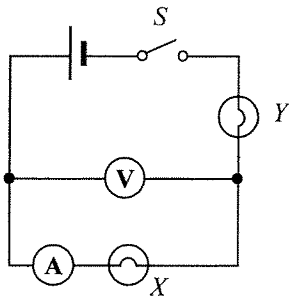
(2013) 31.



In each of the above circuits, the cell has constant e.m.f. and negligible internal resistance. When the sliding contact S of each rheostat shifts from the mid-position to the right, how would the brightness of each bulb change ?

	bulb L_1	bulb L_2
A.	becomes dimmer	remains unchanged
B.	becomes dimmer	becomes brighter
C.	remains unchanged	becomes dimmer
D.	becomes brighter	remains unchanged

(2013) 32.



In the above circuit, the cell has negligible internal resistance. When switch S is closed, both bulbs are not lit. The voltmeter has a reading but the ammeter reads zero. If only one fault has been developed in the circuit, which of the following is possible ?

A. Bulb X has been shorted accidentally.
B. Bulb Y has been shorted accidentally.
C. Bulb X is burnt out and becomes open circuit.
D. Bulb Y is burnt out and becomes open circuit.

(2013) 33. Which of the following domestic electrical appliances consumes a power close to 1 kW in normal working conditions ?

A. an electric fan
B. a microwave oven
C. a fluorescent lamp
D. a TV set

(2013) 34. $^{238}_{92}U$ undergoes α-β-β-α decay and becomes a nuclide X. What are the atomic number and mass number of X?

	atomic number	**mass number**
A.	90	230
B.	90	234
C.	88	230
D.	88	234

(2013) *35. Polonium-210 is a pure α-emitter with a half-life of 140 days and it will decay into lead, which is stable. Initially there is a sample containing 420 mg of pure polonium-210. Estimate the mass of polonium-210 left after 70 days.

A. 315 mg
B. 297 mg
C. 210 mg
D. 105 mg

(2013) *36. The sun releases huge amount of energy through thermonuclear fusion while at the same time its mass decreases. The average power released by the sun is about 3.8×10^{26} W. Estimate the decrease in mass of the sun in one second.

A. 4.2×10^{6} kg
B. 4.2×10^{9} kg
C. 1.3×10^{15} kg
D. 1.3×10^{18} kg

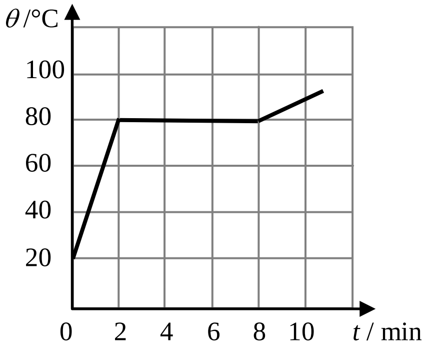
(2014)

1.

Two identical scoops of ice-cream are transferred from a refrigerator into paper cup *X* and vacuum flask *Y* shown above. Under room temperature, the time required for the ice-cream in the containers to melt completely is t_X and t_Y respectively. What is the expected result and explanation ?

A. $t_X > t_Y$ as the vacuum flask reduces heat loss to the surroundings.
B. $t_X > t_Y$ as the vacuum flask retains the heat.
C. $t_Y > t_X$ as the vacuum flask keeps things cold by releasing heat into the surroundings.
D. $t_Y > t_X$ as the vacuum flask reduces the rate of heat gain from the surroundings.

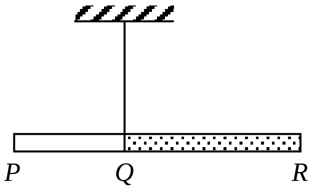
(2014) 2.



An electric heater of constant power is used to heat a solid substance X which is insulated from the surroundings. The variation of its temperature θ with time t is shown above. X has a specific heat capacity of 800 J kg^{-1} °C^{-1} in its solid state. What is the specific latent heat of fusion of X ?

A. 144 kJ kg^{-1}
B. 192 kJ kg^{-1}
C. 202 kJ kg^{-1}
D. Answer cannot be found as both the mass of X and the power of the heater are not known.

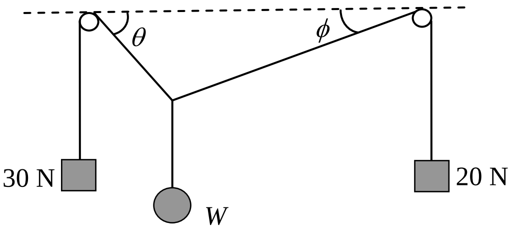
(2014) 3.



PQR is a composite rod having a uniform cross-section but with portions *PQ* and *QR* made of different materials, each of uniform density. The ratio of the length of *PQ* to that of *QR* is 2 : 3. When the rod is suspended at *Q*, it remains horizontal as shown. What is the ratio of the mass of *PQ* to that of *QR* ?

A. 2 : 3
B. 1 : 1
C. 3 : 2
D. Answer cannot be found as the ratio of their densities is not given.

(2014) 4.



The figure shows a weight W attached to two light strings which pass over two smooth pegs at the same height and with weights 30 N and 20 N attached to the respective ends of each string. The system is at equilibrium. Which deduction about W is correct ?

A. W is less than 50 N.
B. W is equal to 50 N.
C. W is greater than 50 N.
D. No information about W can be obtained as angles θ and ϕ are not known.

(2014) 5. A particle is moving along a straight line with uniform acceleration. It takes 4 s to travel a distance of 36 m and then 2 s to travel the next 36 m. What is its acceleration ?

A. 2.5 m s^{-2}
B. 3.0 m s^{-2}
C. 4.0 m s^{-2}
D. 4.5 m s^{-2}

(2014) 6. Two small identical blocks slide down from rest on smooth incline planes from the same height H as shown in Figure (1) and Figure (2) below. Their respective speeds at the bottom of the incline planes are denoted by v_1 and v_2 and the respective times taken to reach the bottom are t_1 and t_2. Which of the following is correct ? Neglect air resistance.

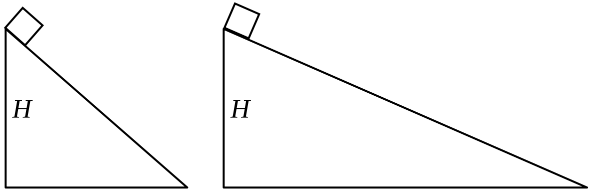
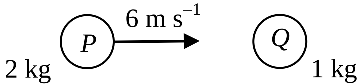


Figure (1) Figure (2)

A. $v_1 > v_2$ and $t_1 = t_2$
B. $v_1 > v_2$ and $t_1 < t_2$
C. $v_1 = v_2$ and $t_1 = t_2$
D. $v_1 = v_2$ and $t_1 < t_2$

(2014) 7.



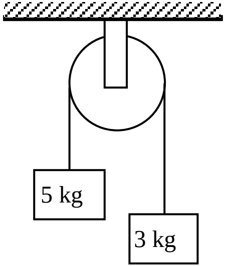
A sphere P of mass 2 kg makes a head-on collision with another sphere Q of mass 1 kg which is initially at rest. The speed of P just before collision is 6 m s^{-1}. If the two spheres move in the same direction after collision, which of the following could be the speed(s) of Q just after collision ?

(1) 2 m s^{-1}
(2) 4 m s^{-1}
(3) 6 m s^{-1}

A. (1) only
B. (1) and (2) only
C. (2) and (3) only
D. (1), (2) and (3)

(2014)

8. Two blocks of masses 5 kg and 3 kg respectively are connected by a light string passing through a frictionless fixed light pulley. Find the magnitude of the acceleration of the blocks in terms of the acceleration due to gravity g when they are released. Neglect air resistance.



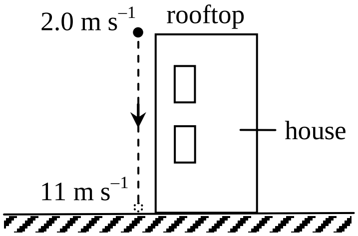
A. g

B. $\frac{g}{2}$

C. $\frac{g}{4}$

D. $\frac{g}{8}$

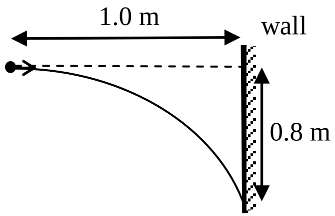
(2014) 9.



A particle is projected vertically downward with an initial speed of 2.0 m s^{-1} from the rooftop of a house. The particle reaches the ground with a speed of 11 m s^{-1} as shown. Estimate the height of the house. Neglect air resistance. (g = 9.81 m s^{-2})

A. 3.3 m
B. 6.0 m
C. 6.5 m
D. 12 m

(2014) *10.



A particle is projected horizontally towards a vertical wall 1.0 m away. It hits the wall at a position 0.8 m vertically below its point of projection. At what speed is it projected ? Neglect air resistance. (g = 9.81 m s^{-2})

A. 2.0 m s^{-1}
B. 2.5 m s^{-1}
C. 5.0 m s^{-1}
D. 6.3 m s^{-1}

(2014) *11. An astronaut inside a spacecraft moving in a circular orbit around the Earth is apparently weightless because

A. the astronaut is too far from the Earth to feel the Earth's gravitational force.
B. the astronaut and the spacecraft are both moving with the same acceleration towards the Earth.
C. the Earth's gravitational force on the astronaut is balanced by the reaction force of the spacecraft's floor.
D. the Earth's gravitational force on the astronaut is balanced by the centripetal force.

(2014) *12. An artificial satellite revolves in a circular orbit above the Earth's surface at a height equal to the radius of the Earth. Find the acceleration of the satellite in terms of the acceleration due to gravity g on the Earth's surface.

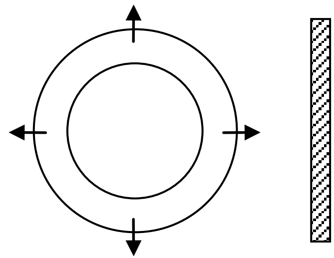
A. $\dfrac{g}{8}$

B. $\dfrac{g}{4}$

C. $\dfrac{g}{2}$

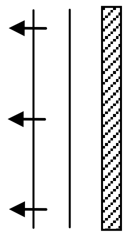
D. g

(2014) 13.

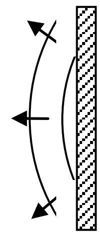


The above figure shows two circular pulses produced by drops of water falling on a ripple tank. The pulses are then reflected by a straight barrier. Which diagram best shows the reflected pulses ?

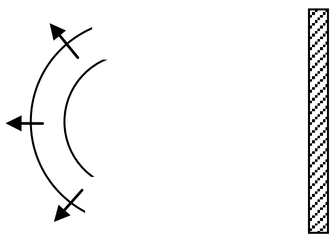
A.



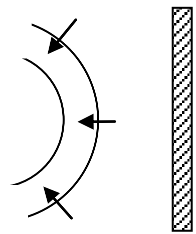
B.



C.

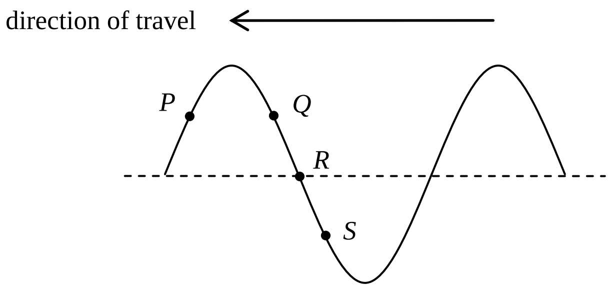


D.



(2014)

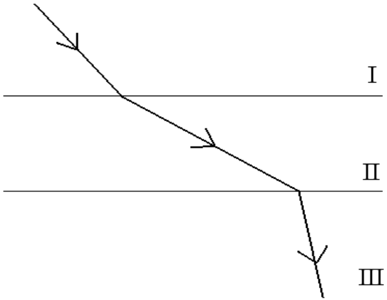
14. A transverse wave travels towards the left on a long string. *P*, *Q*, *R* and *S* are particles on the string. Which of the following statements correctly describe(s) their motions at the instant shown ?



(1) *P* is moving upwards.
(2) *Q* and *S* are moving in opposite directions.
(3) *R* is momentarily at rest.

A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

(2014) 15. The figure shows the path of a light ray travelling from medium I to medium III separated by parallel boundaries. Arrange in **ascending order** the speed of light in the respective media.



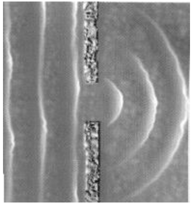
A. I < III < II

B. II < III < I

C. III < I < II

D. III < II < I

(2014) 16.



The photograph shows a series of plane sea waves travelling through a gap in a sea wall which exhibits diffraction. Assuming that the frequency of the waves remains unchanged, which of the following will increase the degree of diffraction ?

(1) The gap in the sea wall becomes narrower.
(2) The wavelength of the waves increases.
(3) The amplitude of the waves becomes larger.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

(2014) 17.

Figure (1)

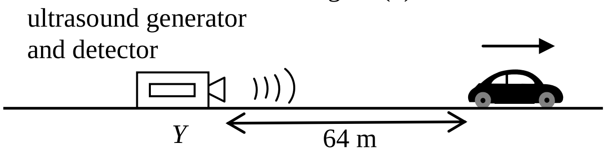


Figure (2)

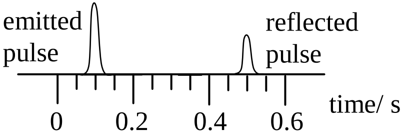
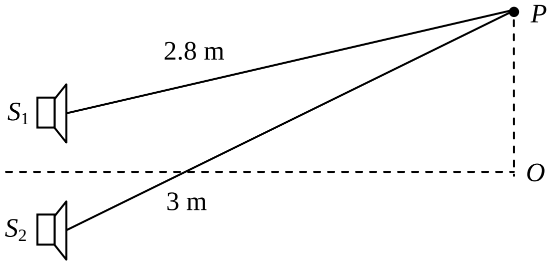


Figure (1) shows a car travelling with a uniform speed along a straight road away from a stationary ultrasound generator and detector at Y. When the car is 64 m from Y, the generator emits an ultrasound pulse towards the car. The pulse is then reflected back to the detector at Y and displayed on a CRO as shown in Figure (2). Estimate the speed of the car. Given: speed of ultrasound in air is 340 m s^{-1}

A. 16 m s^{-1}
B. 20 m s^{-1}
C. 24 m s^{-1}
D. 32 m s^{-1}

(2014) 18.



S_1 and S_2 are two loudspeakers connected to a signal generator but the sound waves produced by them are in anti-phase. Point O is equidistant from the loudspeakers while point P is at the distances shown in the figure from the loudspeakers. What type of interference occurs at O and P if the wavelength of the sound waves is 10 cm ?

	O	***P***
A.	destructive	constructive
B.	constructive	constructive
C.	destructive	destructive
D.	constructive	destructive

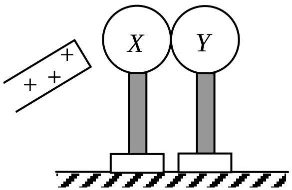
(2014)

19. Which of the following statements about sound waves is/are correct ?

(1) Sound waves are electromagnetic waves.
(2) Sound waves cannot travel in a vacuum.
(3) Sound waves cannot form stationary waves.

A. (2) only
B. (3) only
C. (1) and (2) only
D. (1) and (3) only

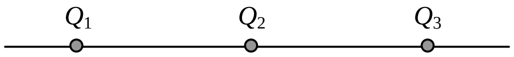
(2014) 20.



Two insulated uncharged metal spheres X and Y are placed in contact. A positively-charged rod is brought near X as shown. X is then touched by a finger momentarily and the two spheres are then separated by removing Y. The charged rod is removed afterwards. Which of the following describes the charges on X and Y ?

	sphere *X*	**sphere *Y***
A.	uncharged	uncharged
B.	uncharged	positive
C.	negative	uncharged
D.	negative	negative

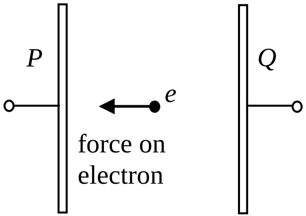
(2014) 21.



Three point charges Q_1, Q_2 and Q_3 are fixed on a straight line with Q_2 at the mid-point of Q_1 and Q_3. The resultant electrostatic force on each charge is zero. Which of the following can be the sign and magnitude (in the same arbitrary units) of Q_1, Q_2 and Q_3 ?

	Q_1	Q_2	Q_3
A.	+2	+1	+2
B.	+2	−1	+2
C.	−4	+1	+4
D.	−4	+1	−4

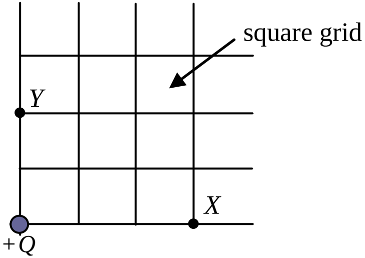
(2014) 22.



Two parallel metal plates P and Q are maintained at a certain p.d. by a battery (not shown in the figure). An electron placed between the plates would experience an electrostatic force of 8.0×10^{-18} N towards P. Which of the following descriptions about the electric field E between the plates is correct ?

A. $E = 0.02\ \text{N C}^{-1}$ from Q to P.
B. $E = 0.02\ \text{N C}^{-1}$ from P to Q.
C. $E = 50\ \text{N C}^{-1}$ from Q to P.
D. $E = 50\ \text{N C}^{-1}$ from P to Q.

(2014) *23.



The figure shows the location of an isolated point charge $+Q$. If the electric potential at X is V, what is the electric potential at Y?

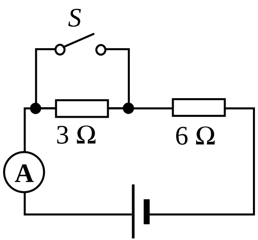
A. $\frac{2}{3}V$

B. $\frac{3}{2}V$

C. $\frac{4}{9}V$

D. $\frac{9}{4}V$

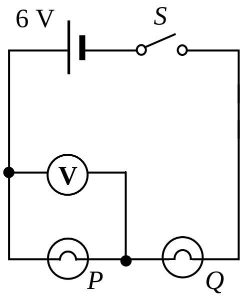
(2014) 24.



In the above circuit, the cell has constant e.m.f. and a fixed internal resistance. When S is closed, the ammeter reads 3.0 A. When S is open, which of the following is a possible reading of the ammeter ?

A. 1.6 A
B. 2.0 A
C. 2.4 A
D. 3.2 A

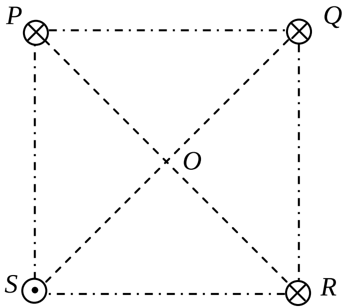
(2014) 25.



The figure shows two light bulbs P and Q connected to a cell of e.m.f. 6 V and negligible internal resistance. The voltmeter reads 6 V when the switch S is closed. Which of the following is possible ?

A. Both P and Q are short-circuited.
B. Both P and Q are burnt out and become open circuit.
C. P is short-circuited or Q is burnt out and becomes open circuit.
D. P is burnt out and becomes open circuit or Q is short-circuited.

(2014) 26.



Four long straight parallel wires P, Q, R and S carrying currents of equal magnitude are situated at the vertices of a square as shown. P, Q and R each carries a current directed into the paper while S carries a current directed out of the paper. The direction of the resultant magnetic field at the centre O of the square is along

A. OP.
B. OQ.
C. OR.
D. OS.

(2014) 27.

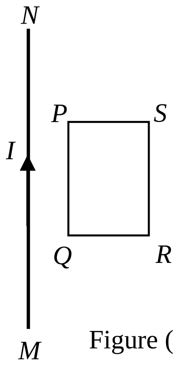


Figure (1)

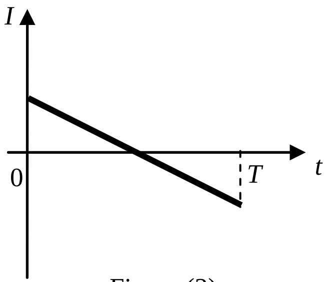
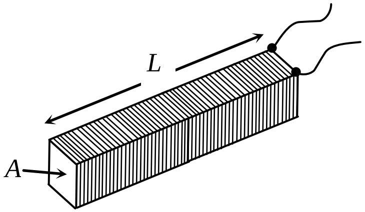


Figure (2)

A long straight current-carrying wire *MN* and a rectangular coil *PQRS* are fixed in the same plane as shown in Figure (1). The current *I* is taken as positive when it flows from *M* to *N* and it varies with time *t* as shown in Figure (2). The direction of the induced current in the coil during the time interval $0 - T$ is

A. first anti-clockwise and then clockwise.
B. first clockwise and then anti-clockwise.
C. anti-clockwise throughout.
D. clockwise throughout.

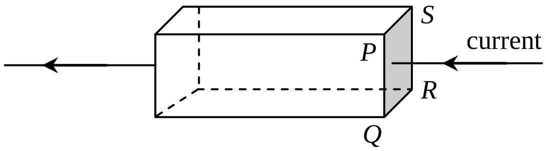
(2014) 28.



The figure shows a closely packed long solenoid of cross-sectional area A and length L having a total of N turns. If the solenoid carries a constant direct current throughout, which of the following changes can increase the magnetic flux density B at its central cross-section ?

	length	**cross-sectional area**	**total number of turns**
A.	$2L$	$2A$	$2N$
B.	L	$2A$	N
C.	$2L$	A	N
D.	L	A	$2N$

(2014) *29.



The figure shows a current flowing from right to left in a metal block with cross-section *PQRS*. When a uniform magnetic field is applied to the block, the electric potential of the side *PQ* is found to be higher than that of the side *SR*. In which direction can the magnetic field be applied ?

A. from *P* to *Q*
B. from *Q* to *P*
C. from *P* to *S*
D. from *S* to *P*

(2014) *30. When a heater is connected to a d.c. voltage of 10 V, the power dissipated is P. If the heater is connected to a sinusoidal a.c., the power dissipated becomes $\frac{1}{2}P$. What is the **r.m.s. voltage** of this a.c. source ? Assume that the resistance of the heater is constant.

A. 5 V
B. $5\sqrt{2}$ V
C. 10 V
D. $10\sqrt{2}$ V

(2014)

31. Nucleus W decays to nucleus Z as shown below:

$$W \xrightarrow{\alpha} X \xrightarrow{\beta} Y \xrightarrow{\beta} Z$$

Which of the following statements is/are correct ?

(1) Nucleus X has 1 more proton than nucleus Y.
(2) Nucleus W has 2 more neutrons than nucleus X.
(3) W and Z are isotopes of the same element.

A. (1) only
B. (2) only
C. (1) and (3) only
D. (2) and (3) only

(2014) 32. A GM counter is placed close to and in front of a radioactive source which emits both α and γ radiations. The count rate recorded is 450 counts per minute while the background count rate is 50 counts per minute. Three different materials are placed in turn between the source and the counter. The following results are obtained.

Material	**Recorded count rate / counts per minute**
(Nil)	450
cardboard	x
1 mm of aluminium	y
2 mm of lead	z

Which of the following is the most suitable set of values for x, y and z ?

	x	y	z
A.	300	300	100
B.	300	100	50
C.	100	100	0
D.	100	50	50

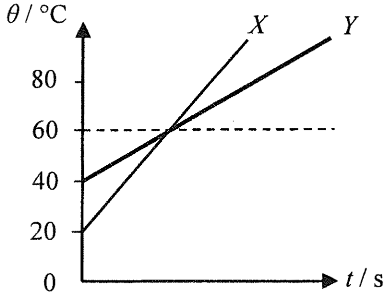
(2014) *33. A radium nucleus decays to a radon nucleus by emitting an α-particle. The energy released in the process is 4.9 MeV. Compared to the mass of a radium nucleus, the total mass of a radon nucleus and an α-particle is

A. 5.4×10^{-11} kg less.
B. 5.4×10^{-11} kg more.
C. 8.7×10^{-30} kg less.
D. 8.7×10^{-30} kg more.

(2015) 1. A driver turns off the engine after parking his car outdoors under the sun. Two hours later when getting into the car, he feels that the inside of the car is far hotter than outside. The best explanation is

A. the car's engine is still generating heat after the engine has been switched off.
B. the car's metal parts absorb infra-red radiation at a faster rate than the surroundings.
C. the glass windows of the car trap infra-red radiation and a greenhouse effect results.
D. the surrounding air is a good insulator of heat which reduces heat loss by conduction.

(2015) 2.



Two objects X and Y are made of the same material. They are heated separately by heaters of the same power. The graph shows the variation of temperature θ of X and Y with time t. What is the ratio of mass of X to that of Y ?

A. 3 : 1
B. 2 : 1
C. 1 : 2
D. 2 : 3

(2015) 3. When an object P is in contact with another object Q, heat flows from P to Q. P must have a higher

(1) temperature.
(2) internal energy.
(3) specific heat capacity.

A. (1) only
B. (3) only
C. (1) and (2) only
D. (1) and (3) only

(2015) 4. A particle travels at 2.0 m s^{-1} due east for 1.5 s and then travels at 4.0 m s^{-1} due north for 1.0 s. What is the magnitude of its average velocity for the whole journey ?

A. 2.0 m s^{-1}
B. 2.8 m s^{-1}
C. 3.0 m s^{-1}
D. 5.0 m s^{-1}

(2015) 5. A constant net force acting on an object of mass m_1 produces an acceleration a_1 while the same force acting on another object of mass m_2 produces an acceleration a_2. If this net force acts on an object of mass (m_1+m_2), what would be the acceleration produced ?

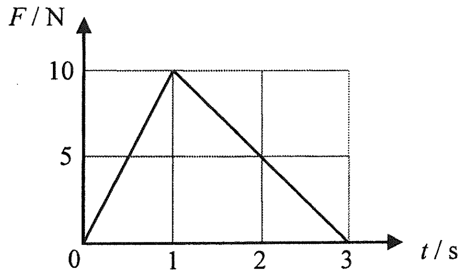
A. $a_1 + a_2$

B. $\dfrac{a_1 + a_2}{2}$

C. $\dfrac{a_1 a_2}{a_1 + a_2}$

D. $\dfrac{2a_1 a_2}{a_1 + a_2}$

(2015) 6.

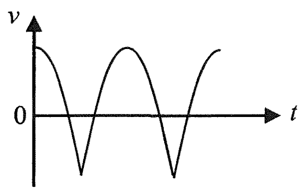


An object of mass 3 kg is initially at rest on a smooth horizontal ground. A force F is applied horizontally to the object such that the magnitude of F varies with time t as shown. What is the speed of the object at $t = 3$ s ? Neglect air resistance.

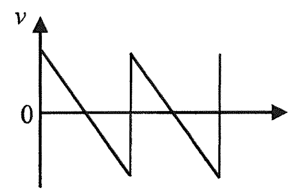
A. $2.5\ \mathrm{m\ s^{-1}}$
B. $5\ \mathrm{m\ s^{-1}}$
C. $10\ \mathrm{m\ s^{-1}}$
D. $15\ \mathrm{m\ s^{-1}}$

(2015) 7. A rubber ball bounces vertically up and down from the ground. Which graph best shows the variation of its velocity v with time t if the collisions are elastic ? Neglect air resistance.

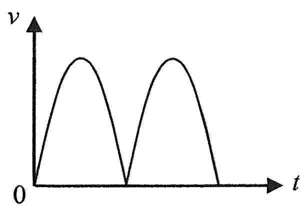
A.



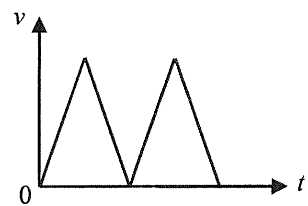
B.



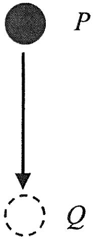
C.



D.



(2015) 8. An object falls from P to Q as shown below. Throughout the motion, air resistance increases with the speed of the object.

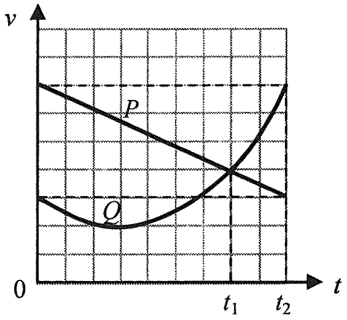


Which of the following descriptions is/are correct ?

(1) The net force acting on the object is constant throughout the motion.
(2) The magnitude of the object's acceleration decreases from P to Q.
(3) The kinetic energy gained by the object from P to Q is equal to its loss in gravitational potential energy.

A. (1) only
B. (2) only
C. (1) and (3) only
D. (2) and (3) only

(2015) 9.

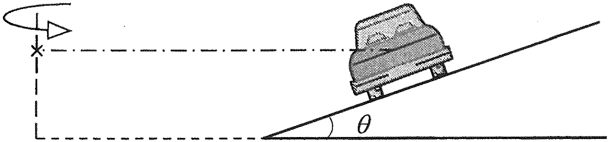


The figure shows the velocity-time (v-t) graph of two cars P and Q travelling along the same straight road. At $t = 0$, the cars are at the same position. Which deductions about the cars between $t = 0$ and $t = t_2$ are correct ?

(1) P and Q are always travelling in the same direction.
(2) At $t = t_1$, the separation between P and Q is at a maximum.
(3) At $t = t_2$, Q lags behind P.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

(2015) *10.



The figure shows the rear view of a car of mass m which travels along a circular road banked with an angle θ to the horizontal. The car moves at a certain speed such that it experiences **no frictional force along the inclined surface**. Which of the following represents the centripetal force on the car ?

A. $mg\sin\theta$

B. $mg\sin\theta\cos\theta$

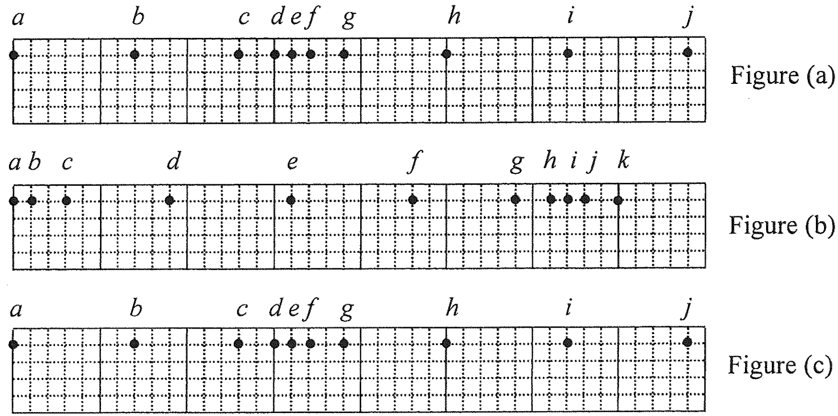
C. $\dfrac{mg\cos\theta}{\sin\theta}$

D. $\dfrac{mg\sin\theta}{\cos\theta}$

(2015) 11. The gravitational force exerted on the Earth by the Sun is F_0. The gravitational force exerted on the Sun by the Earth is

A. equal to F_0 and in the same direction.
B. equal to F_0 and in the opposite direction.
C. much smaller than F_0 and in the same direction.
D. much smaller than F_0 and in the opposite direction.

(2015) 12.

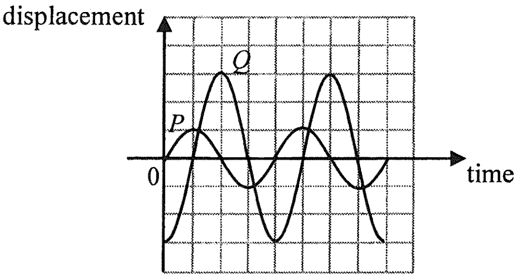


A series of particles is uniformly distributed along a slinky spring initially. When a travelling wave propagates along the slinky spring from left to right, Figure (a) shows the positions of the particles at a certain instant. Figures (b) and (c) respectively show their positions 0.05 s and 0.1 s later. Which of the following is/are a possible frequency of the wave ?

(1) 10 Hz
(2) 20 Hz
(3) 40 Hz

A. (1) only
B. (2) only
C. (3) only
D. (1), (2) and (3)

(2015) 13.

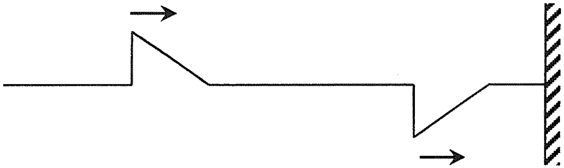


Two waves P and Q travel in the same direction and meet at a point. The above graph shows the variation of the displacement of each wave with time at that point. Which of the following statements is/are correct ?

(1) P and Q have the same frequency.
(2) The oscillation due to P is in anti-phase with that due to Q.
(3) The amplitude of the resultant wave at that point is four times the amplitude of P.

A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

(2015) 14.



Two pulses of the same shape travel along a stretched string with one end fixed to the wall as shown above. Which of the following can be the resultant waveform at different instants later ?

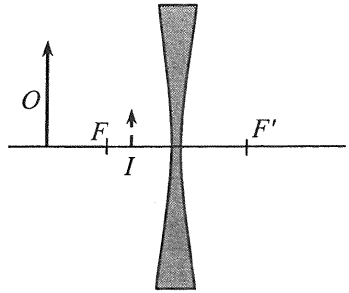
(1)

(2)

(3)

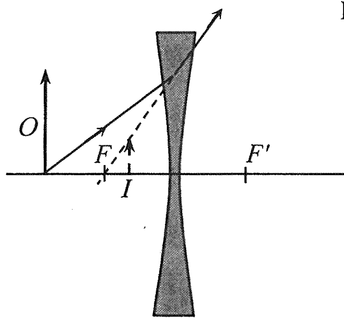
A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

(2015) 15.

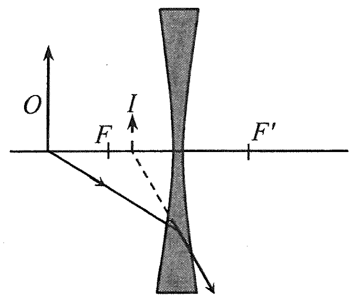


An object O placed in front of a concave lens forms an image I as shown. F and F' are the foci of the lens. Which ray diagram is correct ?

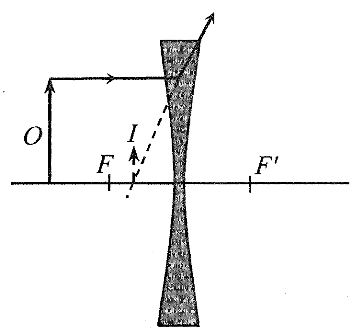
A.



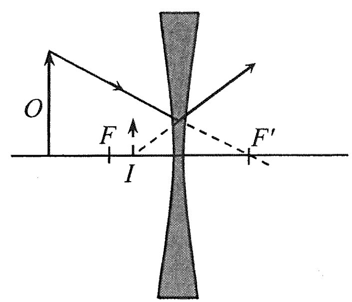
B.



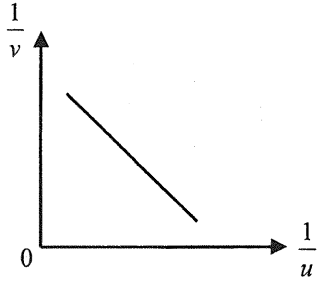
C.



D.

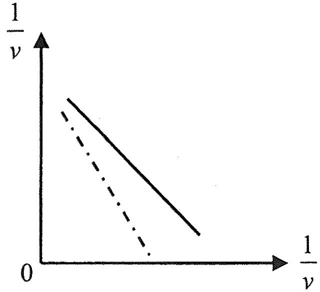


(2015) *16.

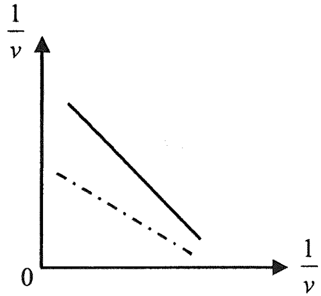


A student uses a convex lens to investigate the variation of image distance v with object distance u for real images. The graph of $\frac{1}{v}$ plotted against $\frac{1}{u}$ is shown above. If a convex lens of longer focal length is used, what would be the expected result (in dotted lines) ?

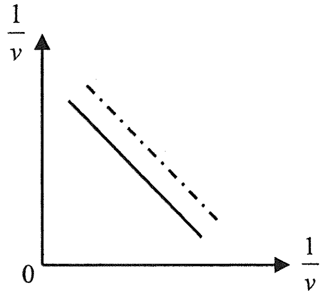
A.



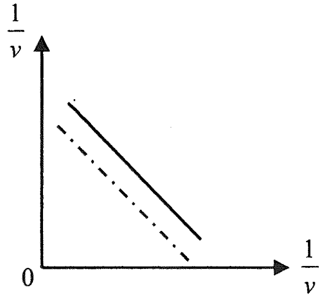
B.



C.

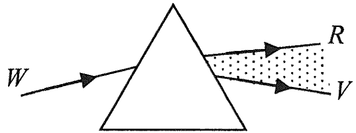


D.

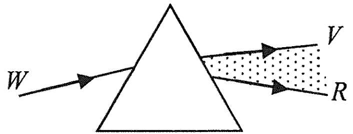


(2015) 17. Which diagrams below correctly show the spectra formed from white light by a glass prism and a diffraction grating respectively ? It is known that red light travels faster than violet light in glass.
(R = red, V = violet, W = white)

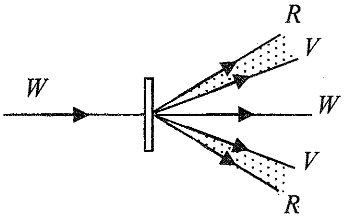
(1)



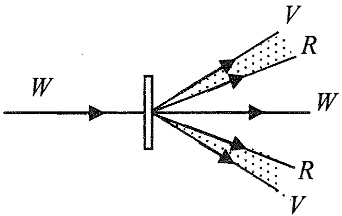
(2)



(3)

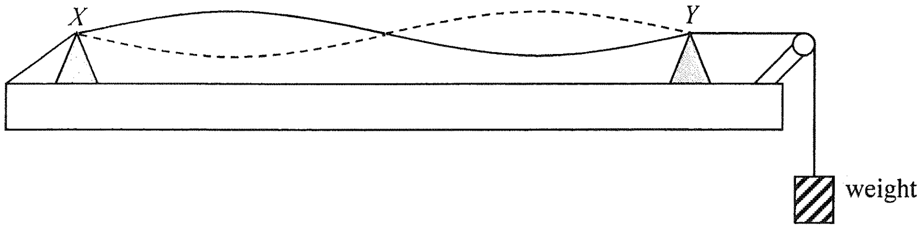


(4)

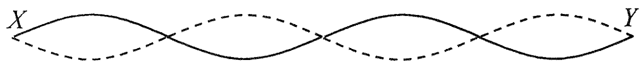


A. (1) and (3) only
B. (1) and (4) only
C. (2) and (3) only
D. (2) and (4) only

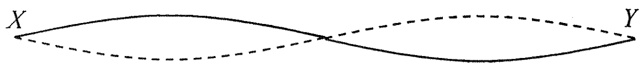
(2015) 18.

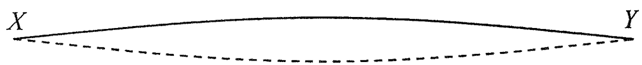


A string is set to vibrate at frequency f such that a standing wave is formed between two fixed supports X and Y as shown. If the tension in the string is increased by adding weight gradually while the frequency is kept at f, which of the following is a possible mode of vibration at a steady state ?

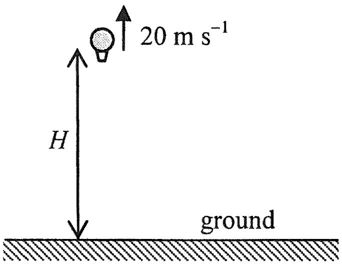
A. 

B. 

C. 

D. 

(2015) 19. A balloon is rising at a uniform speed of 20 m s^{-1}. When the balloon is at an altitude H as shown, it sends a sound signal towards the ground. After 5 s, the balloon receives the echo of the signal. Estimate H. Given: speed of sound in air = 340 m s^{-1}



A. 1600 m
B. 850 m
C. 800 m
D. 750 m

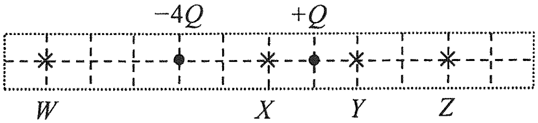
(2015) 20. Which of the following gives the order of magnitude of the wavelengths of ultra-violet radiation and microwave in a vacuum ?

	ultra-violet radiation	**microwave**
A.	10^{-8} m	10^{-2} m
B.	10^{-8} m	10^{-5} m
C.	10^{-10} m	10^{-2} m
D.	10^{-10} m	10^{-5} m

(2015) 21. Three conducting balls are suspended by insulating threads. **Any two of them** are found to attract each other if brought close to each other. Which conclusion below is correct ?

A. Only one ball is uncharged while the other two carry like charges.
B. Only one ball is uncharged while the other two carry unlike charges.
C. Only one ball is charged.
D. All three balls are charged.

(2015) 22.



Two point charges $-4Q$ and $+Q$ are fixed as shown. At which point indicated in the figure is the resultant electric field due to these two charges zero ?

A. W
B. X
C. Y
D. Z

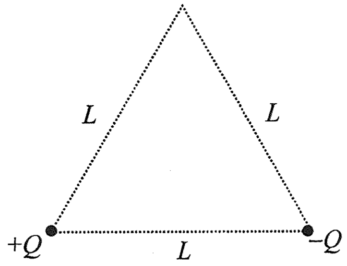
(2015) *23. Point charges $+Q$ and $-Q$ are fixed respectively at two vertices of an equilateral triangle with sides of length L as shown. What is the minimum energy required for bringing another point charge $+Q$ from infinity to the third vertex ?

A. 0

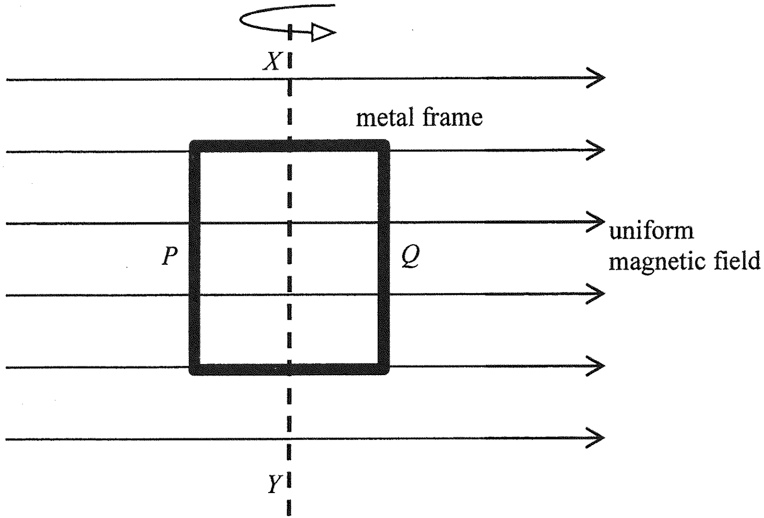
B. $\frac{1}{4\pi\varepsilon_0}\left(\frac{Q^2}{L}\right)$

C. $\frac{1}{4\pi\varepsilon_0}\left(\frac{2Q^2}{L}\right)$

D. $\frac{1}{4\pi\varepsilon_0}\left(\frac{3Q^2}{L}\right)$



(2015) 24.

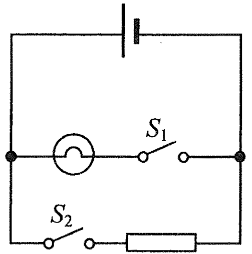


A rectangular metal frame is made to rotate steadily about its axis XY in a uniform magnetic field. At the instant shown, the frame is in the plane of the paper and side P is moving out of the paper while side Q is moving into the paper. Which statement is **INCORRECT at this instant** ?

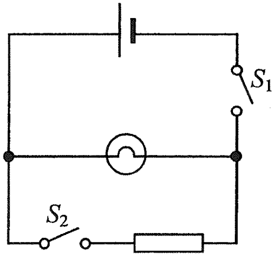
A. The induced e.m.f. in the frame is at a maximum.
B. The induced current produced in the frame is flowing in anti-clockwise direction.
C. The magnetic force acting on side P is in a direction pointing into the paper.
D. The magnetic forces acting on the frame produce a moment opposing the frame's rotation.

(2015) 25. For safety purposes, the driver seat of a car is equipped with a seat belt warning light. When the driver seat is occupied, the switch S_1 under his seat will close. If the seat belt is not yet fastened, switch S_2 will remain open and the warning light will light up. If the seat belt is fastened, the switch S_2 will close and the warning light will shut off. Which circuit design below is the best design ?

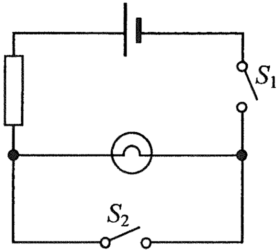
A.



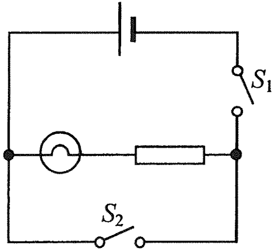
B.



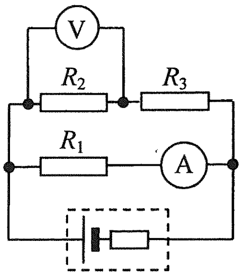
C.



D.



(2015) 26. In the following circuit, the cell has a finite internal resistance and both meters are ideal.



In which situation below will the readings of the ammeter and the voltmeter suddenly increase ?

A. R_1 is faulty and becomes a short circuit.
B. R_2 is faulty and becomes a short circuit.
C. R_3 is faulty and becomes a short circuit.
D. R_2 is faulty and becomes an open circuit.

(2015) *27. A copper wire of uniform cross-section carries a current of 0.5 A. Each metre of the wire contains 10^{22} free electrons. Find the magnitude of the drift velocity of the electrons in the wire.

A. 2.5×10^{-3} m s^{-1}
B. 7.8×10^{-3} m s^{-1}
C. 3.1×10^{-4} m s^{-1}
D. 9.7×10^{-4} m s^{-1}

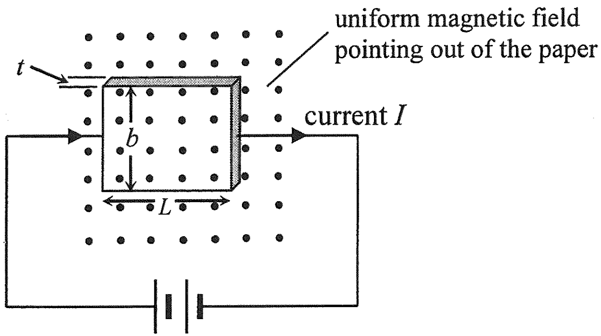
(2015) 28. Which statement is **NOT** a reason why mains sockets at home are connected in parallel instead of a series circuit ?

A. Electrical appliances connected to different sockets can be switched on or off independently.
B. Voltage supplied to each socket is fixed and all electrical appliances can operate at their rated voltages.
C. The current supplied can be reduced and thinner cables can then be used.
D. When an electrical appliance breaks down and becomes an open circuit, other appliances can still work normally.

(2015) 29. An electric iron of 1800 W sold in Hong Kong (220 V 50 Hz) is connected to a 110 V 60 Hz mains socket in another country. How does its performance compare on the same ironing setting ?

A. The electric iron does not work because the a.c. supply is 60 Hz instead of 50 Hz.
B. The electric iron is as hot as when it is used in Hong Kong.
C. The electric iron is hotter than when it is used in Hong Kong.
D. The electric iron is colder than when it is used in Hong Kong.

(2015) *30.



A slice of conductor (with a certain resistivity) of thickness t, breadth b and length L is connected to a battery of constant e.m.f. and negligible internal resistance. A steady current I passes through it as shown. When a uniform magnetic field is applied normally to the slice, a Hall voltage V is set up between two of its opposite faces. If the thickness and the breadth of the slice are reduced to ½ t and ½ b respectively, what would the following quantities be ?

	current passing through the slice	**Hall voltage developed**
A.	$\frac{I}{4}$	$\frac{V}{4}$
B.	I	$\frac{V}{4}$
C.	$\frac{I}{4}$	$\frac{V}{2}$
D.	I	$\frac{V}{2}$

(2015) 31. Which of the following nuclear reactions is/are **spontaneous** reaction(s) ?

(1) $^{24}_{11}\text{Na} \rightarrow {}^{24}_{12}\text{Mg} + {}^{0}_{-1}\text{e}$

(2) $^{10}_{5}\text{B} + {}^{1}_{0}\text{n} \rightarrow {}^{7}_{3}\text{Li} + {}^{4}_{2}\text{He}$

(3) $^{2}_{1}\text{H} + {}^{3}_{1}\text{H} \rightarrow {}^{4}_{2}\text{He} + {}^{1}_{0}\text{n}$

A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

(2015) 32. Workers in nuclear power plants wear clothes with film badges to measure the dosage of radiation received over a period of time. Which type of radiation below **CANNOT** be monitored by the film badges ?

A. α-radiation
B. β-radiation
C. γ-radiation
D. X-rays

(2015) *33. A piece of wood found in an archaeological site is dated using carbon-14 dating method. It registers a corrected count rate of 11.0 counts per minute while a fresh wood sample cut from the same kind of trees gives a corrected count rate of 15.6 counts per minute. What is the approximate age of the wood found in the archaeological site ? Given: half-life of carbon-14 is 5730 years.

A. 890 years
B. 1300 years
C. 2000 years
D. 2900 years

(2016) 1. Some icy cold liquid is kept cold inside a vacuum flask. Which statements are correct ?

(1) The flask's cork stopper reduces heat gain from the surroundings.
(2) The silver coating on the inner surface of the glass wall is a good reflector of infra-red.
(3) The vacuum between the double glass walls reduces heat gain by radiation.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

(2016) 2. 0.3 kg of water at temperature 50 °C is mixed with 0.2 kg of ice at temperature 0 °C in an insulated container of negligible heat capacity. What is the final temperature of the mixture ?

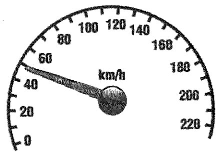
Given: specific heat capacity of water = 4200 J kg^{-1} $°C^{-1}$
specific latent heat of fusion of ice = 3.34×10^5 J kg^{-1}

A. −1.8 °C
B. 0 °C
C. 1.8 °C
D. 3.0 °C

(2016) *3. When an ideal gas is heated from 25 °C to 50 °C, the average kinetic energy of the gas molecules will

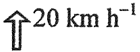
A. double.
B. increase by 41 %.
C. increase by 8.4%.
D. increase by 4.1%.

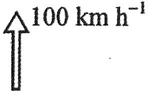
(2016) 4. The speedometer of a car shown below indicates the car's

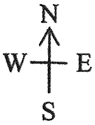


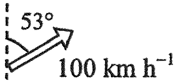
A. instantaneous speed.
B. instantaneous velocity.
C. average speed of the whole journey.
D. average velocity of the whole journey.

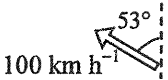
(2016) 5. A car travelling at 80 km h^{-1} due east changes direction and travels at 60 km h^{-1} due north. Which diagram represents the change in velocity of the car ?

A. 

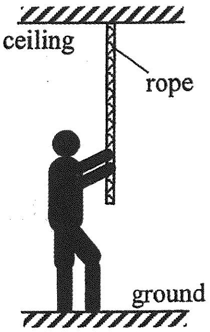
B. 



C. 

D. 

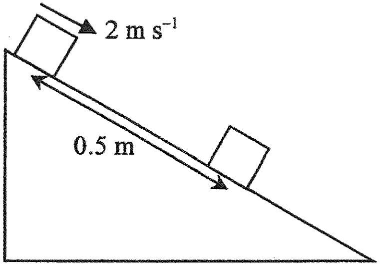
(2016) 6. A boy of weight W exerts a downward pulling force F on a rope of weight G hung vertically from the ceiling. He stands still on the ground as shown. Which of the following gives the magnitude of the force exerted by



(1) the boy on the ground;
(2) the rope on the ceiling ?

	(1)	(2)
A.	W	$G-F$
B.	W	$G+F$
C.	$W-F$	$G-F$
D.	$W-F$	$G+F$

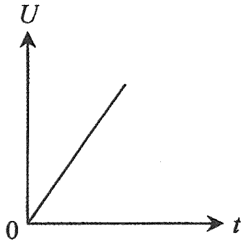
(2016) 7. A block with initial speed 2 m s^{-1} slides down a rough inclined plane and stops after travelling a distance of 0.5 m. What is the deceleration of the block ?



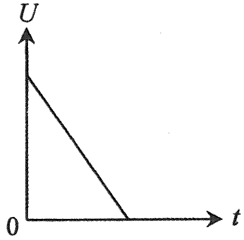
A. 1 m s^{-2}
B. 2 m s^{-2}
C. 4 m s^{-2}
D. Answer cannot be found as the angle of inclination of the plane is not given.

(2016) 8. An object at a certain height falls freely from rest under gravity. Which graph correctly shows the variation of its gravitational potential energy U with time t ? Neglect air resistance and take $U = 0$ at the ground.

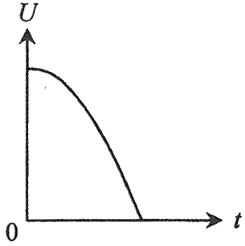
A.



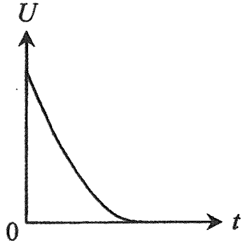
B.



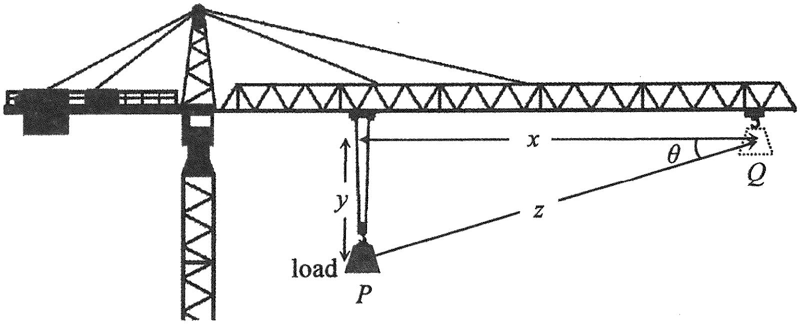
C.



D.



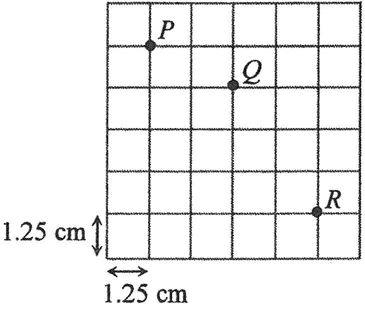
(2016) 9. A crane moves a load of weight W steadily from point P to point Q as shown.



The work done on the load by the crane is

A. Wy.
B. $W(x+y)$.
C. Wz.
D. $Wz\cos\theta$.

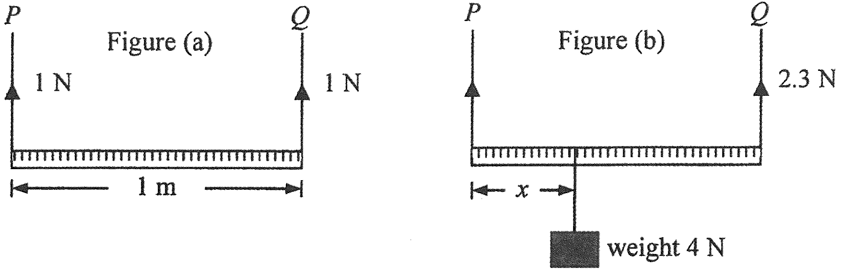
(2016) *10.



The above stroboscopic picture shows a particle projected horizontally at position P into the air in a vertical plane. Subsequently the particle reaches positions Q and R such that the time interval between P and Q is equal to that between Q and R. Each square of the grid measures 1.25 cm × 1.25 cm. Find the particle's speed of projection at P. Neglect air resistance. ($g = 9.81$ m s^{-2})

A. 0.3 m s^{-1}
B. 0.4 m s^{-1}
C. 0.5 m s^{-1}
D. 0.6 m s^{-1}

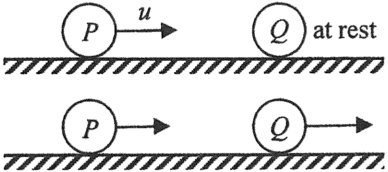
(2016) 11.



A uniform metre rule is supported by vertical wires *P* and *Q* and remains at rest horizontally as shown in Figure (a). The tension in each wire is 1 N. When a weight of 4 N is hung from the metre rule at a certain position as shown in Figure (b), the tension in *Q* becomes 2.3 N while the metre rule remains horizontal. Find the distance *x* shown.

A. 32.5 cm
B. 57.5 cm
C. 67.5 cm
D. Answer cannot be found as the tension in *P* is not known.

(2016) 12. On a smooth horizontal surface, a marble P moving with speed u collides head-on with another marble Q, which is at rest. After collision, P and Q move with different speeds as shown.

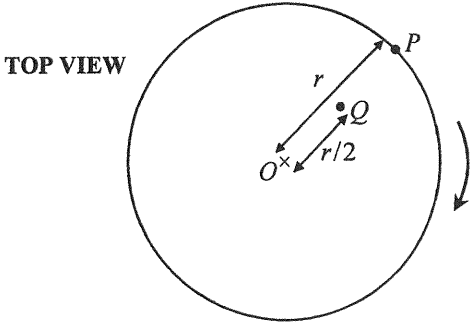


Which of the following statements about this collision is/are correct ?

(1) During collision, the force acting on Q by P is equal and opposite to that acting on P by Q.
(2) The total momentum of the two marbles is conserved only when the collision is perfectly elastic.
(3) The kinetic energy lost by P must be equal to that gained by Q.

A. (1) only
B. (2) only
C. (1) and (3) only
D. (2) and (3) only

(2016) *13. Particles P and Q are fixed at distances r and $r/2$ respectively from the centre O of a horizontal circular platform which is rotating uniformly as shown.



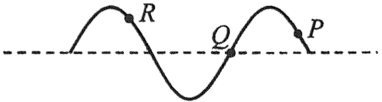
The ratio of the acceleration of P to that of Q is

A. 1 : 2.
B. 2 : 1.
C. 1 : 4.
D. 4 : 1.

(2016) *14. A satellite orbits the Earth in a circular path of radius 7.2×10^6 m. What is the period of the satellite ?
Given : mass of the Earth = 6.0×10^{24} kg

A. 1.4 hours
B. 1.7 hours
C. 1 day
D. Answer cannot be found as the mass of the satellite is not known.

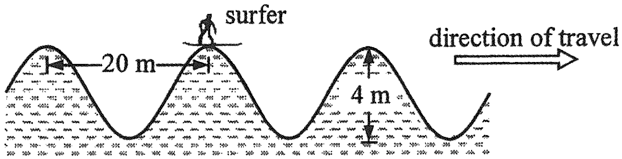
(2016) 15.



The above figure shows a snapshot of a transverse wave which travels along a string. Which statement is correct ?

A. The wave is travelling to the left if particle P is moving upwards at this instant.
B. Particles P and R are moving in the same direction at this instant.
C. Particle Q is at rest at this instant.
D. Particle R vibrates with an amplitude larger than that of particle Q.

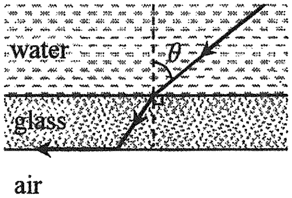
(2016) 16.



The surfer in the figure reaches a crest at the moment shown. The crests of the water wave are 20 m apart and the surfer descends a vertical distance of 4 m from a crest to a trough in a time interval of 2 s. What is the speed of the wave ?

A. $1\ \mathrm{m\ s^{-1}}$
B. $2\ \mathrm{m\ s^{-1}}$
C. $5\ \mathrm{m\ s^{-1}}$
D. $10\ \mathrm{m\ s^{-1}}$

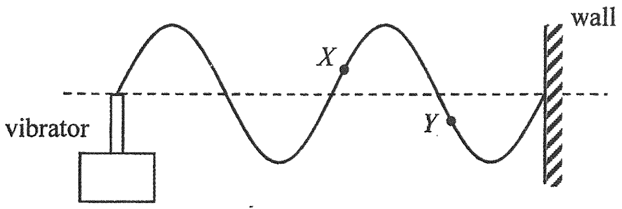
(2016) 17.



A parallel-sided glass sheet separates water from air. A ray of light in water is incident at an angle θ on the glass sheet and finally emerges into air along the glass-air interface as shown. Find θ.
Given: refractive index of water is 1.33.

A. 41.2°
B. 48.8°
C. 53.1°
D. It depends on the refractive index of glass.

(2016) 18. A string is tied to a vibrator while the other end is fixed to a wall. A stationary wave is formed as shown.



Which statement is correct when the frequency of the vibrator doubles ?

A. The wavelength will double.
B. The wave speed will double.
C. The amplitude will be halved.
D. Particles X and Y will become vibrating in phase.

(2016) 19. Diffraction will occur when light

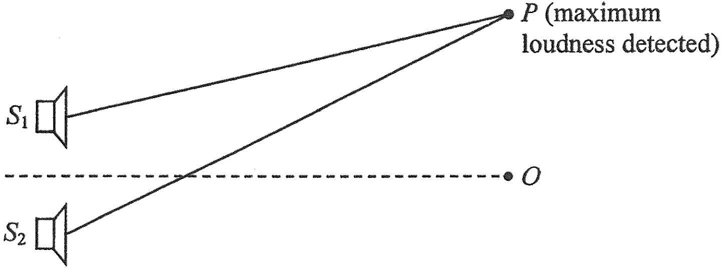
(1) passes through a pinhole.
(2) passes by a sharp edge.
(3) passes through a slit.

A. (1) only
B. (2) only
C. (3) only
D. (1), (2) and (3)

(2016) 20. A beam of white light is separated into different colours after entering a glass prism because lights of different colours

A. are diffracted to different extents by the prism.
B. undergo total internal reflection at different angles inside the prism.
C. travel at different speeds in vacuum.
D. travel at different speeds in glass.

(2016) 21.



Loudspeakers S_1 and S_2 connected to a signal generator emit sound waves which are in phase. Point O is equidistant from the loudspeakers while at point P maximum loudness is detected. The wavelength of the sound waves is λ. Which statement is **INCORRECT** ?

A. Both PS_1 and PS_2 must be integral multiples of wavelength λ.

B. The definite value of the path difference $PS_2 - PS_1$ cannot be determined from the information given.

C. At least one point of minimum loudness can be detected between O and P.

D. Minimum loudness will be detected at P if the sound waves from S_1 and S_2 are in antiphase.

(2016) 22. An object is moving at constant speed towards a convex lens of focal length 10 cm. At the moment when it is at 100 cm from the lens, which of the following descriptions of the image is correct ?

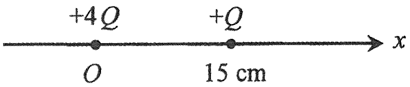
	direction of image movement	speed of the image
A.	away from the lens	faster than that of the object
B.	towards the lens	faster than that of the object
C.	away from the lens	slower than that of the object
D.	towards the lens	slower than that of the object

(2016) 23. Which of the following are applications of ultrasound ?

(1) sterilizing drinking water
(2) detecting cracks in railway tracks
(3) breaking up kidney stones

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

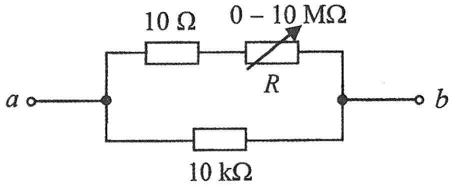
(2016) *24.



Point charges $+4Q$ and $+Q$ are fixed on the x-axis with $+4Q$ at the origin O and $+Q$ at $x = 15$ cm as shown. The respective electric fields due to the two charges are equal at

A. $x = 10$ cm.
B. $x = 12$ cm.
C. $x = 20$ cm.
D. $x = 30$ cm.

(2016) 25.



In the above circuit, the variable resistor R can be adjusted over its full range from 0 to 10 MΩ. What is the approximate range of resistance between a and b ?

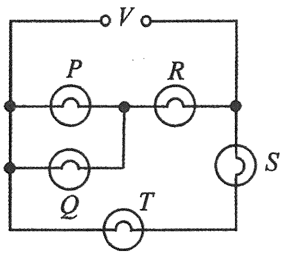
A. 0 to 10 kΩ
B. 10 Ω to 10 kΩ
C. 10 Ω to 10 MΩ
D. 10 kΩ to 10 MΩ

(2016) 26. Two filament light bulbs X and Y are connected in parallel to a dry cell. X is brighter than Y. Which statements are correct ?

(1) In 1 s, the number of charges flowing through X is greater than that flowing through Y.
(2) In 1 s, the electrical energy dissipated by X is greater than that dissipated by Y.
(3) For every unit charge passing, the electrical energy dissipated by X is equal to that dissipated by Y.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

(2016) 27.



In the above circuit, all the bulbs are identical. If the voltage V gradually increases, which bulb(s) will burn out first ?

A. P and Q
B. R
C. S
D. T

(2016) 28. A television set in stand-by mode consumes 1.5 W. If it is in this mode for 16 hours a day, estimate the carbon dioxide (CO_2) emission due to the electricity consumed in stand-by mode in a 30-day month. Given: 1 kW h of electricity consumed corresponds to 0.8 kg CO_2 emission from the power station.

A. 0.576 kg
B. 0.720 kg
C. 576 kg
D. 720 kg

(2016) *29. A student uses a search coil to study the strength of the magnetic field inside a long solenoid which is connected to an a.c. signal generator set at a certain frequency. Which of the following can improve the accuracy of this experiment ?

(1) Ensure the plane of the search coil is perpendicular to the field lines.
(2) Increase the signal generator's frequency and use the same current as before.
(3) Set the axis of the solenoid along an east-west direction to avoid the effects of the Earth's magnetic field.

A. (1) only
B. (1) and (2) only
C. (2) and (3) only
D. (1), (2) and (3)

(2016) *30. A sinusoidal a.c. of a certain frequency delivers a r.m.s. voltage $V_{\text{r.m.s.}}$. If its frequency is doubled and its peak voltage is halved, what would be the r.m.s. voltage ?

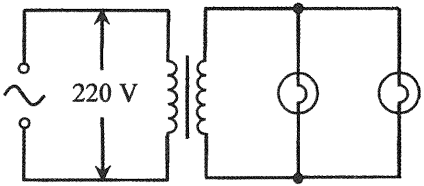
A. $\frac{1}{2} V_{\text{r.m.s.}}$

B. $\frac{1}{\sqrt{2}} V_{\text{r.m.s.}}$

C. $\frac{1}{2\sqrt{2}} V_{\text{r.m.s.}}$

D. $V_{\text{r.m.s.}}$

(2016) *31.



In the above circuit, each light bulb works at its rated value '22 W, 11 V'. The current in the primary coil is 0.25 A. Find the efficiency of the transformer.

A. 20%
B. 40%
C. 64%
D. 80%

(2016) 32. Which of the following statements about ionizing radiations is/are correct ?

(1) The ionizing power of α-particles is much stronger than that of β-particles.
(2) γ-radiation can be completely shielded by a 10 cm thick concrete wall.
(3) Ionizing radiations α, β and γ all undergo deflection in an electric field.

A. (1) only
B. (1) and (2) only
C. (1) and (3) only
D. (2) and (3) only

(2016) 33. Two radionuclides X and Y are of half-lives 3 hours and 4 hours respectively and initially there are N_X and N_Y undecayed nuclei respectively. After 24 hours, the number of undecayed nuclei of both nuclides becomes the same. Find $N_X : N_Y$.

A. 8 : 1
B. 4 : 3
C. 4 : 1
D. 2 : 1

(2017)

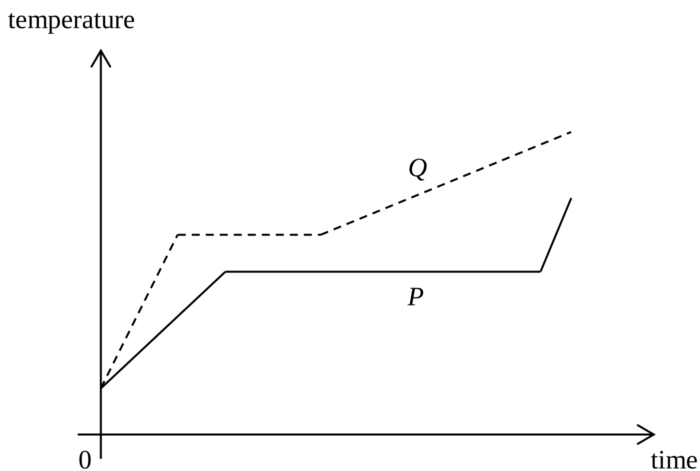
1. 30 g of milk at 10°C is added to 120 g of coffee at 80°C. Assuming there is no heat loss to the surroundings, what is the final temperature of the mixture ?

Given: specific heat capacity of milk = 3800 J kg^{-1} $°C^{-1}$
specific heat capacity of coffee = 4200 J kg^{-1} $°C^{-1}$

A. 64.8°C
B. 65.2°C
C. 66.0°C
D. 67.1°C

(2017)

2. Same mass of solids P and Q are heated at the same rate. The temperature-time graphs of the two substances are shown below.



Which of the following comparisons about their melting points and specific latent heats of fusion is correct ?

	higher melting point	**larger specific latent heat of fusion**
A.	P	P
B.	P	Q
C.	Q	P
D.	Q	Q

(2017)

3. Which of the following statements about the internal energy of a substance are correct ?

(1) When a solid melts, the latent heat of fusion absorbed becomes potential energy of the molecules in the substance.
(2) When a vapour condenses, its internal energy decreases.
(3) When a liquid evaporates, the internal energy of the remaining liquid increases.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

(2017)

*4. The pressure of a fixed mass of an ideal gas at 10°C is 2×10^5 N m^{-2}. If the volume of the gas is reduced to half of its original volume and its temperature is increased to 100°C, what would the pressure be ?

A. 1.00×10^5 N m^{-2}
B. 1.32×10^5 N m^{-2}
C. 4.00×10^5 N m^{-2}
D. 5.27×10^5 N m^{-2}

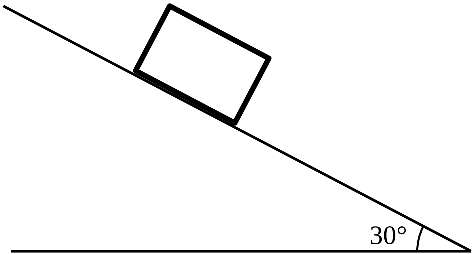
(2017)

5. Which of the following statements about the motion of any two objects is correct ?

A. The object that takes a shorter time to complete the same path must have greater average speed.
B. The object that travels a greater distance in 1 s must have greater average velocity.
C. The object with greater velocity must have greater acceleration.
D. If the two objects have the same acceleration, they must be moving in the same direction.

(2017)

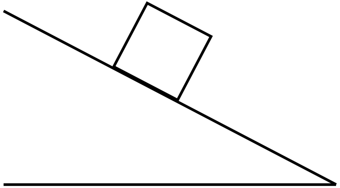
6. A block is released from rest on an inclined plane as shown. The inclined plane makes an angle of 30° to the horizontal. The block moves with uniform acceleration, and travels a distance of 1 m in the first 3 s. Determine the acceleration of the block.



A. $0.22\ \mathrm{m\ s^{-2}}$
B. $0.33\ \mathrm{m\ s^{-2}}$
C. $4.91\ \mathrm{m\ s^{-2}}$
D. Cannot be determined as the frictional force acting on the block is unknown.

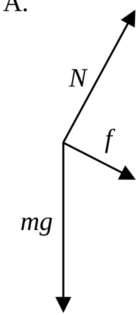
(2017)

7. A block of mass m stays at rest on a rough inclined plane as shown.

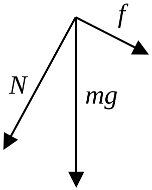


Which of the following diagrams correctly shows the forces acting on the block ?
(N is the normal reaction from the inclined plane, and f is the frictional force between the block and the plane.)

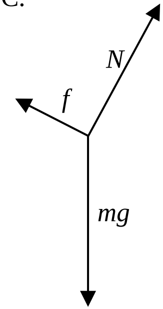
A.



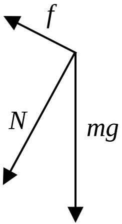
B.



C.

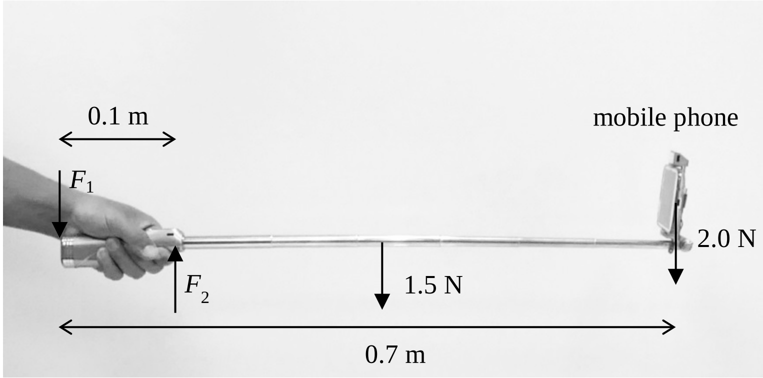


D.



(2017)

8. Selfie sticks are popular nowadays. A uniform selfie stick of length 0.7 m is held horizontally as shown. Assume that the forces required to hold the selfie stick by the hand are represented by F_1 and F_2 , and F_1 and F_2 are perpendicular to the stick.

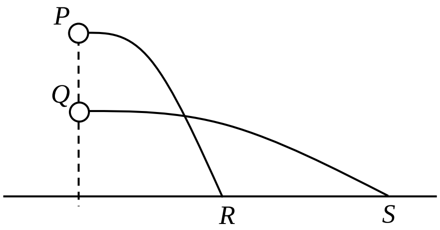


It is given that the weight of the selfie stick and the mobile phone are 1.5 N and 2.0 N respectively. Taking the mobile phone as a point mass, estimate the magnitude of F_2 .

A. 3.5 N
B. 19.3 N
C. 35 N
D. Cannot be determined as F_1 is unknown.

(2017)

*9. Marbles P and Q of the same mass are shot horizontally. They hit the horizontal ground at points R and S respectively as shown. Neglect air resistance.

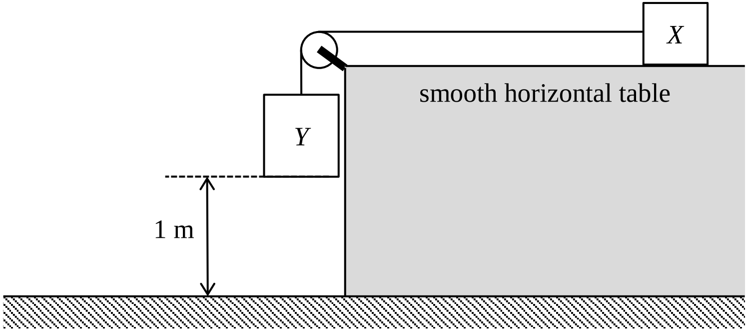


Which of the following statements is **INCORRECT** ?

A. The initial speed of marble P is smaller than that of marble Q.
B. The time of flight of marble P is shorter than that of marble Q.
C. The potential energy loss of marble P is greater than that of marble Q.
D. The acceleration of marbles P and Q is the same during the flight.

(2017)

10. Blocks X and Y are connected by a light inextensible string passing over a fixed frictionless light pulley as shown. The mass of X and Y are 0.5 kg and 1 kg respectively. Initially, Y is 1 m above the ground and the string is taut. The system is then released from rest.



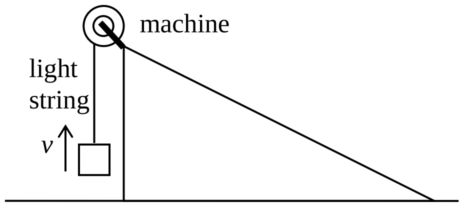
What is the speed of Y just before it reaches the ground ? (Take $g = 9.81\ \text{m s}^{-2}$)

A. $3.62\ \text{m s}^{-1}$
B. $4.43\ \text{m s}^{-1}$
C. $6.26\ \text{m s}^{-1}$
D. $9.81\ \text{m s}^{-1}$

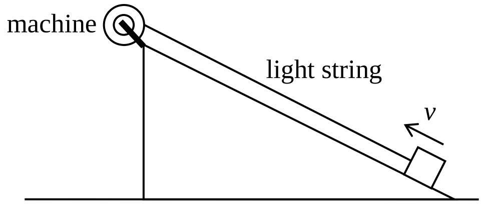
(2017)

11. A machine is fixed at the top of a smooth inclined plane. Two methods, (I) and (II), are used to lift a block from the ground to the top of the inclined plane by the machine.

(I) Pull the block vertically upward at a uniform speed *v*.
(II) Pull the block up along the inclined plane at the same uniform speed *v*.



(I)



(II)

Which of the following statements correctly compare(s) the two methods ?

(1) The tension in the string is the same.
(2) The average output power of the machine is the same.
(3) The work done by the machine on the block is the same.

A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

(2017) 12. Players of ‘bubble soccer’ wear air-filled plastic ‘bubbles’ as shown.

Which of the following statements best explains why the bubble can reduce the chance of injury during a collision ?

A. The bubble increases the mass of the player, thus the momentum of the player increases.
B. The bubble increases the air resistance acting on the player.
C. The bubble lengthens the impact time during a collision.
D. Like a balloon, the bubble provides a lifting force to the player.

(2017) *13. A small object is released from rest at a point very far away from a planet *X*. The object then starts moving towards *X*. *X* does not have an atmosphere. Neglect the effect of other celestial bodies.

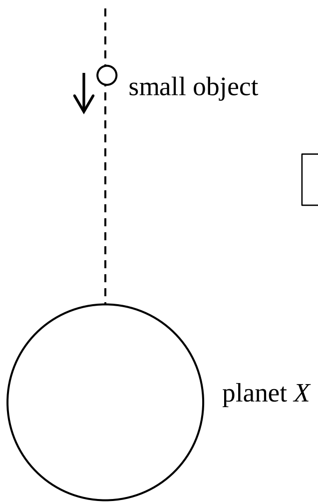
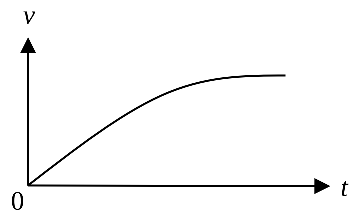


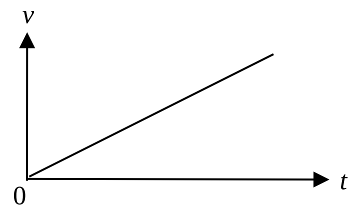
Diagram **NOT** drawn to scale

Which of the following graphs best shows the variation of the velocity *v* of the object with time *t* before it hits *X* ?

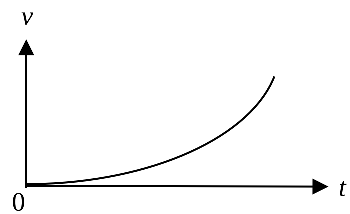
A.



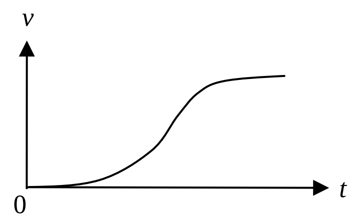
B.



C.

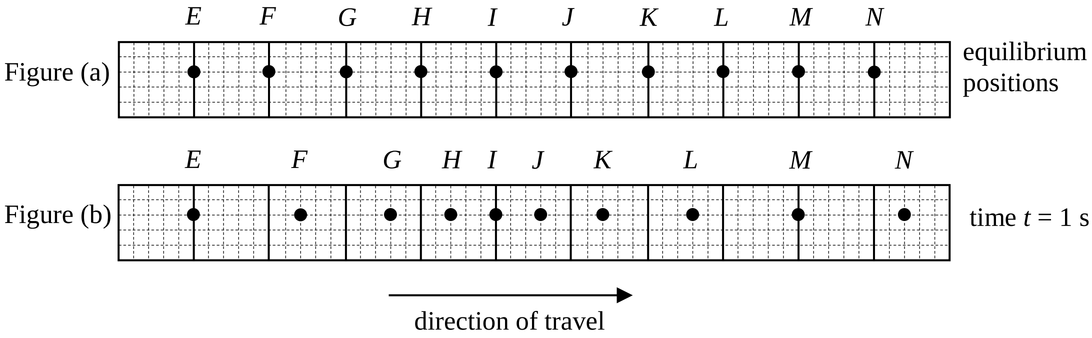


D.



(2017)

14. Figure (a) shows the equilibrium positions of particles E to N in a medium. At time t = 0, a longitudinal wave starts travelling from left to right. At time t = 1 s, the positions of the particles are shown in Figure (b).

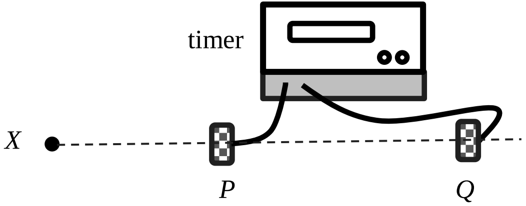


Which of the following statements **MUST BE** correct ?

A. The distance between particles F and N is equal to the wavelength of the wave.
B. The period of the wave is 1 s.
C. Particle E is always at rest.
D. Particle I is momentarily at rest at t = 1 s.

(2017)

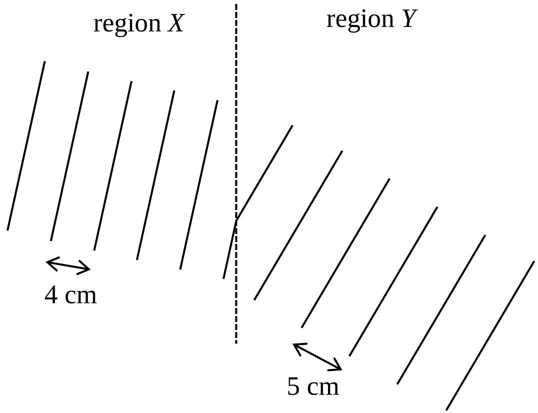
15. An experiment is set up to measure the speed of sound in air as shown. *P* and *Q* are two microphones connected to a timer. A sound is produced at *X*. The timer starts when *P* receives the sound, and stops when *Q* receives the sound. The timer shows the time taken for the sound to travel from *P* to *Q*. The distance *PQ* and the time shown can be used to calculate the speed of sound.



Which of the following statements is **INCORRECT** ?

A. *X*, *P* and *Q* must be along the same straight line.
B. The percentage error in the time measured will increase if the distance *PQ* is reduced.
C. The speed of sound determined should be independent of the distance between *X* and *P*.
D. The distance *PQ* must be equal to an integral multiple of the wavelength of the sound produced at *X*.

(2017) 16. The figure shows plane water waves travelling from region X to region Y. The wavelengths of the water waves in regions X and Y are 4 cm and 5 cm respectively.



Which of the following statements is correct ?

A. The speed of the water waves in region X is higher than that in region Y.
B. The direction of travel of the water waves bends towards the normal as they enter region Y.
C. The frequency of the water waves is the same in both regions.
D. If plane water waves of wavelength 5 cm travel from region Y to region X, the wavelength becomes 6 cm after the waves enter region X.

(2017)

17. In which of the following situations **MUST** the direction of travel of a wave change ?

(1) when a wave is reflected by a barrier
(2) when a wave enters from one medium to another medium
(3) when a wave travels through a gap smaller than its wavelength

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

(2017)

18. Two musical notes of the same pitch and loudness are produced by two different musical instruments. They sound different to the human ears because they have different

A. amplitudes.
B. phases.
C. wave speeds.
D. waveforms.

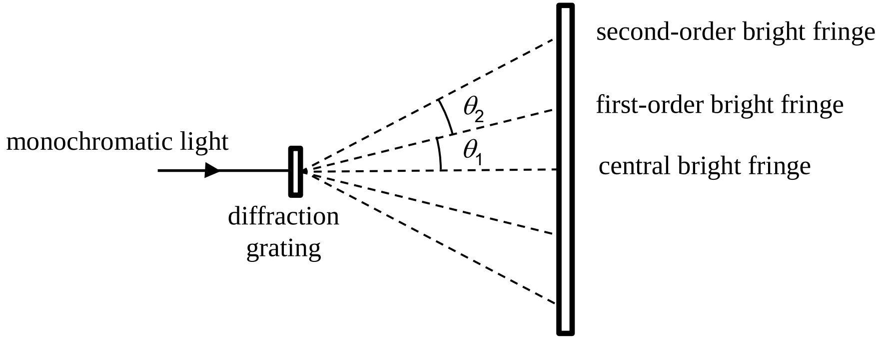
(2017)

*19. When an object is placed 30 cm in front of a concave lens, an image is formed 20 cm away from the lens. If the concave lens is replaced by a convex lens of the same focal length and the object distance remains unchanged, which of the following descriptions about the image formed is correct ?

	nature of the image	**image distance**
A.	real	20 cm
B.	real	60 cm
C.	virtual	20 cm
D.	virtual	60 cm

(2017)

*20. The figure below shows some of the bright fringes formed when monochromatic light passes through a diffraction grating.



Which of the following is/are correct ?

(1) $\theta_1 = \theta_2$

(2) The maximum order of bright fringe is 4 if $\theta_1 = 20°$.

(3) θ_1 will decrease if the experiment is performed in water but not in air.

A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

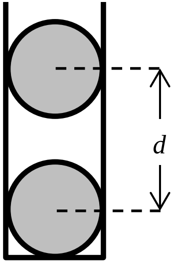
(2017)

21. If the speed of sound in water is x and the speed of light in water is y, which of the following is correct ?

	speed of sound in air	speed of light in air
A.	$> x$	$> y$
B.	$> x$	$< y$
C.	$< x$	$> y$
D.	$< x$	$< y$

(2017)

22. In the figure, two charged conducting spheres of the same mass *m* are put in a vertical plastic cylinder. The inner wall of the cylinder is smooth. The spheres are separated by a distance *d* and remain in equilibrium.

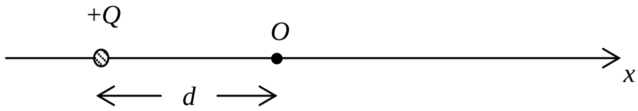


Which of the following statements **MUST BE** correct ?

(1) Both spheres carry positive charges.
(2) The amount of charges on the two spheres are the same.
(3) The separation *d* depends on *m*.

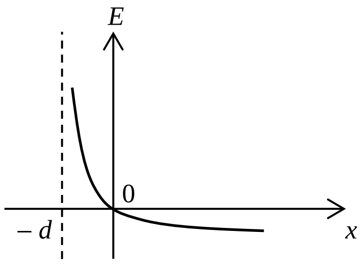
A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

(2017) *23. A point charge $+Q$ is fixed at a distance d away from the origin O as shown.

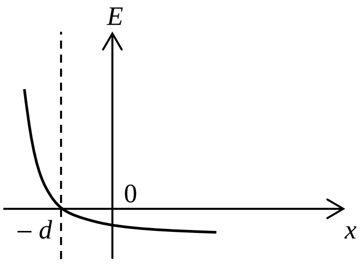


Which of the following graphs best represents the variation of the electric field strength E along the x-axis ? (Take the electric field pointing to the right as positive.)

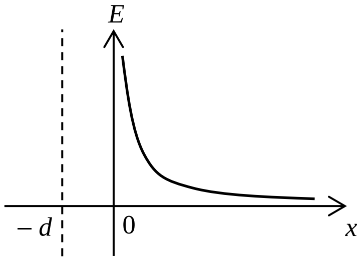
A.



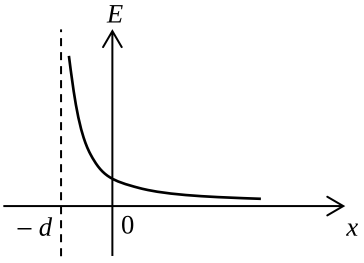
B.



C.

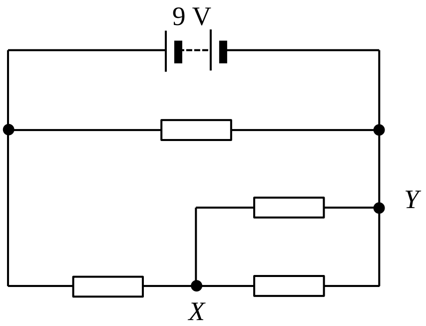


D.



(2017)

24. In the circuit, all resistors are identical. The internal resistance of the battery can be neglected.



What is the potential difference between *X* and *Y* ?

A. 1.5 V
B. 3.0 V
C. 4.5 V
D. 6.0 V

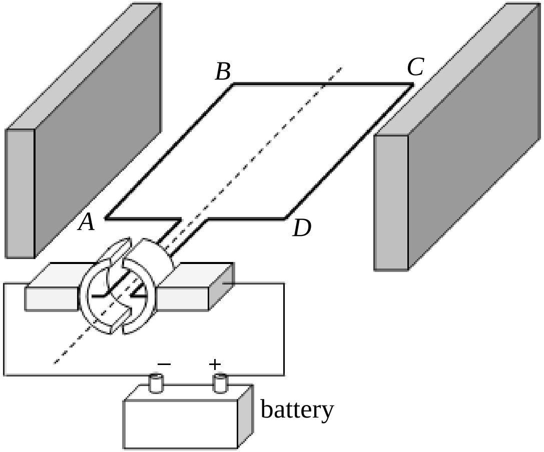
(2017)

25. Which of the following statements about the use of a fuse is correct ?

A. A fuse should be installed in the neutral wire.
B. A fuse is not required in an electrical appliance with double insulation.
C. A 5A fuse is suitable for a heater of rating ‘220 V, 1500 W’.
D. The melting point of a fuse should be lower than that of copper.

(2017)

26. The figure shows a simple d.c. motor, the coil *ABCD* is mounted between the poles of two slab-shaped magnets.

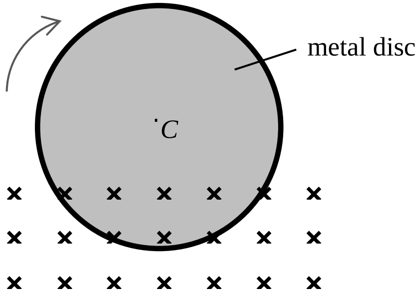


Which of the following statements is correct ?

A. The turning effect is zero when the coil is vertical.
B. The magnetic force acting on *BC* is the greatest when the coil is horizontal.
C. The direction of the magnetic force acting on *AB* remains constant.
D. The direction of the current in the coil remains unchanged.

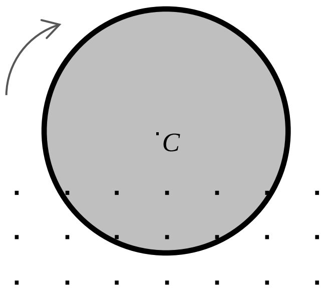
(2017)

27. A metal disc is rotating about its centre C with constant speed. Part of the metal disc is inside a uniform magnetic field pointing into the paper as shown. An eddy current flows in the metal disc.

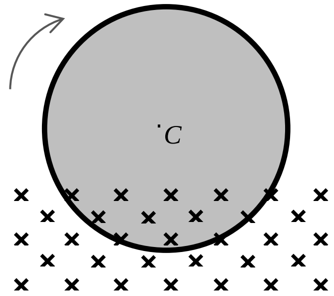


After which of the following changes will the eddy current increase ?

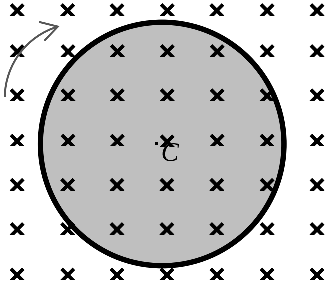
A. Reverse the direction of the magnetic field



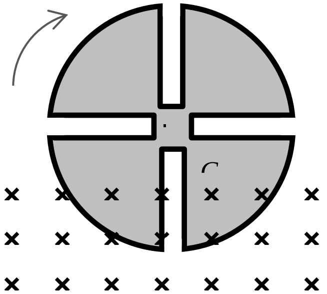
B. Increase the strength of the magnetic field



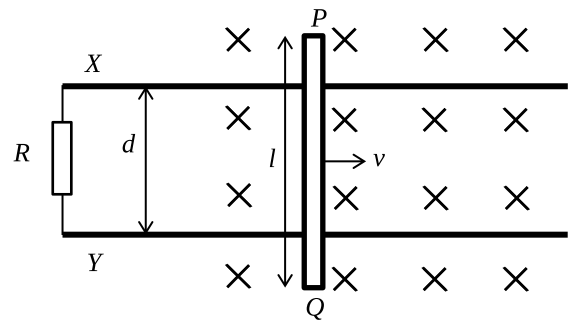
C. Apply the magnetic field over the whole metal disc



D. Cut several slits from the metal disc



(2017) *28. A metal rod PQ of length l is moving along smooth horizontal metal rails X and Y with constant speed v in a uniform magnetic field of magnetic field strength B pointing into the paper. The metal rails X and Y are separated by a distance of d and are connected to a resistor of resistance R as shown.



Which of the following descriptions about the induced current is correct ?

	magnitude	**direction**
A.	$\frac{Blv}{R}$	from X to Y through R
B.	$\frac{Blv}{R}$	from Y to X through R
C.	$\frac{Bdv}{R}$	from X to Y through R
D.	$\frac{Bdv}{R}$	from Y to X through R

(2017)

*29. A heater of resistance 100 Ω is connected to the mains supply. The r.m.s. voltage of the mains supply is 110 V. Which of the following statements are correct ?

(1) The peak voltage across the heater is 156 V.
(2) The power dissipated by the heater is 121 W.
(3) The power dissipated by the heater will be doubled if the r.m.s. voltage of the mains supply doubles.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

(2017)

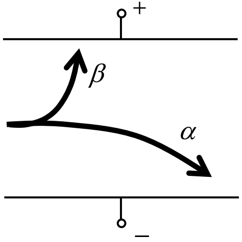
*30. The input terminal of a transformer is connected to the 220 V mains supply. Ten identical light bulbs are connected in parallel to the output terminal of the transformer. All the light bulbs are working at their rated values of '3 V, 1.5 W'. If the efficiency of the transformer is 70%, what is the current drawn from the mains supply ?

A. 0.007 A
B. 0.048 A
C. 0.068 A
D. 0.097 A

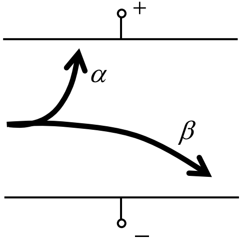
(2017)

31. Which of the following diagrams best shows the deflection of α and β particles in an uniform electric field in vacuum ?

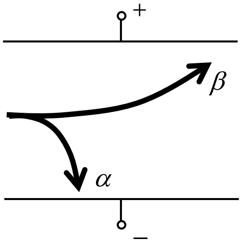
A.



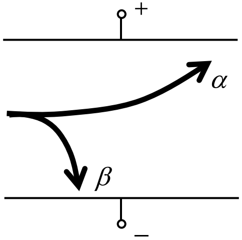
B.



C.



D.



(2017)

32. Which of the following statements about β particles and γ rays is correct ?

A. Only β particles can ionize air particles.
B. Only γ rays can travel through vacuum.
C. Both of them can be detected by a photographic film.
D. Both of them carry charge.

(2017)

*33. The following shows the decay of uranium-238 ($^{238}_{92}U$).

$$^{238}_{92}U \rightarrow {}^{234}_{90}Th + {}^{4}_{2}He$$

Given that : mass of $^{238}_{92}U$ = 238.05079 u

mass of $^{234}_{90}Th$ = 234.04363 u

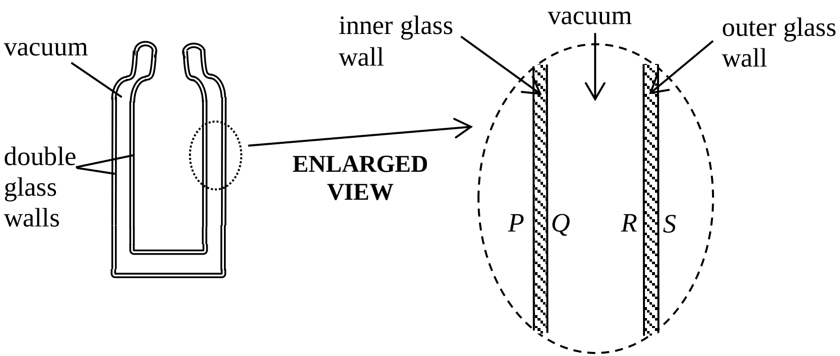
mass of $^{4}_{2}He$ = 4.00260 u

Which of the following statements is/are correct ?

(1) The temperature required to start the decay is about 10^7 K.
(2) The energy released in the decay of one uranium-238 nucleus is 4.25 MeV.
(3) All the energy released in the decay becomes the kinetic energy of $^{4}_{2}He$.

A. (1) only
B. (2) only
C. (1) and (3) only
D. (2) and (3) only

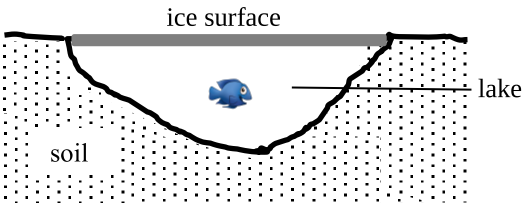
(2018) 1.



The figure shows a vacuum flask with double glass walls for keeping liquids cold. *P*, *Q* and *R*, *S* are the glass surfaces of the inner and outer glass walls respectively. Which two surfaces are usually coated with silver ?

A. *P* and *R*
B. *Q* and *R*
C. *P* and *S*
D. *R* and *S*

(2018) 2. In some countries, the outdoor temperature can drop below 0 ℃ in winter such that the lake surface becomes a thick layer of ice. However the water beneath the ice surface does not easily turn into ice and most aquatic life can survive during winter time.



Which statement below best explains this phenomenon ?

A. The layer of ice provides good heat insulation.
B. The freezing point of water below the ice surface is much lower than 0 ℃.
C. There is thermal energy transferred from the soil to the water in the lake.
D. Ice releases latent heat when it melts.

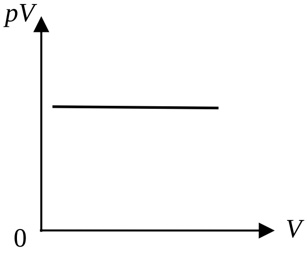
(2018) 3. Milk at 5 °C is added to a cup of tea at 25 °C. Which statements below are correct ? Neglect the heat capacity of the cup and assume that there is no heat exchange with the surroundings.

(1) The average kinetic energy of the water molecules in the tea decreases.
(2) The average potential energy of the water molecules in the tea remains unchanged.
(3) The energy lost by the tea equals the energy gained by the milk.

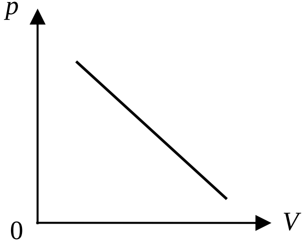
A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

(2018) *4. From which graph below can one deduce that the pressure p of a fixed mass of an ideal gas is inversely proportional to its volume V when the temperature of the gas is kept constant ?

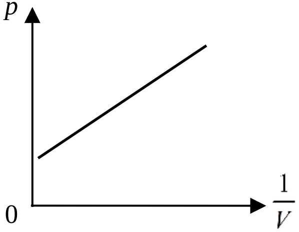
A.



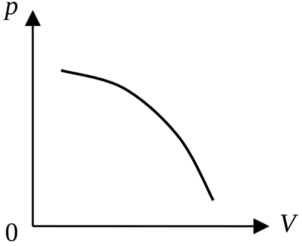
B.



C.

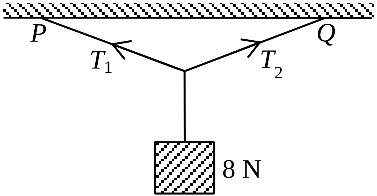


D.



(2018)

5. A block of weight 8 N is suspended from a horizontal ceiling by light inextensible strings to two different points P and Q as shown. The strings are equal in length.

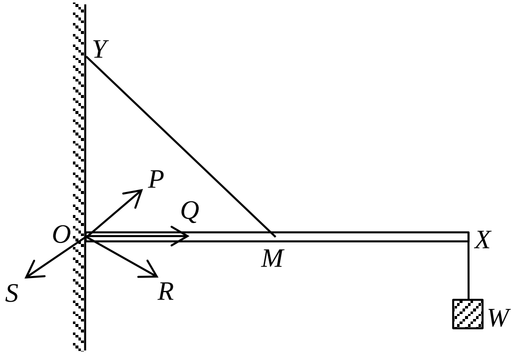


Which of the following descriptions about the tensions T_1 and T_2 in the two strings is/are correct ?

(1) The magnitude of T_1 must be greater than 4 N.
(2) The maximum value of T_2 would not exceed 8 N.
(3) The resultant force of T_1 and T_2 is zero.

A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

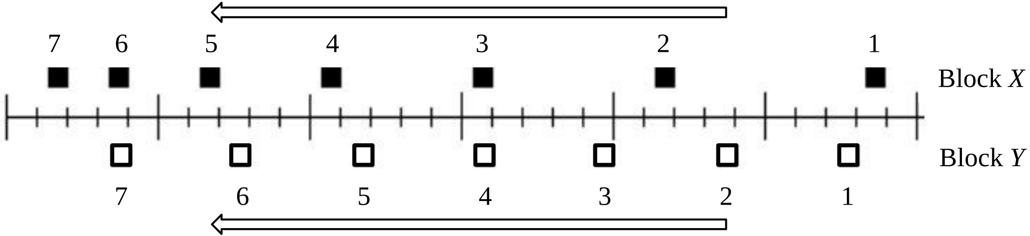
(2018) 6.



A uniform light rigid rod OX is hinged smoothly to a wall at one end O. Its mid-point M is connected by a light inextensible string to a point Y directly above O while a weight W is suspended from the other end X of the rod as shown. Rod OX remains horizontal. The reaction force acting on the rod due to the wall is along the direction

A. OP.
B. OQ.
C. OR.
D. OS.

(2018) 7. Two blocks X and Y are moving towards the left. Their positions at successive instants (indicated by the numbers) of equal time intervals are shown below.

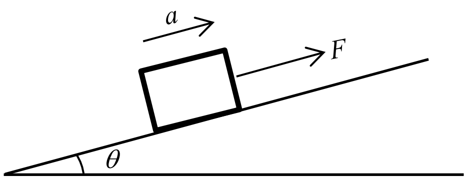


Do these two blocks ever have the same instantaneous speed ?

A. Yes, at instant 3.
B. Yes, at a moment between instants 4 and 5.
C. Yes, at instant 6.
D. No.

(2018)

8. A block of mass m is placed on a smooth incline making an angle θ with the horizontal as shown. When a force of magnitude F parallel to the incline is applied to the block, it travels up the incline with an acceleration a. If the applied force becomes $2F$, what would the magnitude of the acceleration be ?



A. greater than $2a$
B. equal to $2a$
C. between a and $2a$
D. whether it is greater than $2a$, equal to $2a$ or between a and $2a$ depends on the value of θ

(2018)

9. A particle moving along a straight line with speed 0.5 m s^{-1} changes its direction of motion within a period of 0.2 s and subsequently travels with the same speed in the opposite direction. Find the magnitude of its average acceleration during this 0.2 s.

A. 5 m s^{-2}
B. 2.5 m s^{-2}
C. 0 m s^{-2}
D. its magnitude depends on the mass of the particle

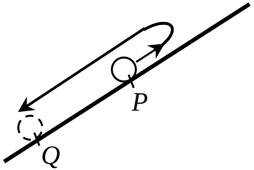
(2018)

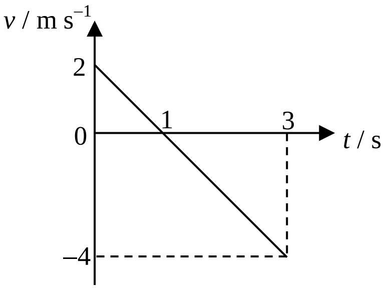
10. A train departs from one station and stops at the next station after travelling 1.2 km along a straight line. The maximum acceleration and deceleration of the train are both 5 m s^{-2} and the highest speed of the train is limited to 20 m s^{-1}. Find the shortest time taken for this trip.

A. 56 s
B. 58 s
C. 62 s
D. 64 s

(2018)

11. A bead is projected up along a smooth incline at point P at time $t = 0$. After reaching the highest point, the bead then travels downwards and passes point Q at $t = 3$ s as shown. The velocity-time (v-t) relationship of the bead is shown in the graph below. Find the separation PQ along the incline.





A. 2 m
B. 3 m
C. 4 m
D. 5 m

(2018)

12. At a certain instant, an object flying horizontally to the right at 1 m s^{-1} suddenly explodes into two fragments of mass ratio 1 : 2. If just after the explosion the more massive fragment flies to the right at 3 m s^{-1}, what would happen to the other fragment just after the explosion ?

A. It would fly to the left at 3 m s^{-1}.
B. It would fly to the left at 4 m s^{-1}.
C. It would be momentarily at rest.
D. It would fly to the right at 1 m s^{-1}.

(2018) *13. A satellite of mass m moves around a planet of mass M in a circular orbit of radius r. What does the angular velocity of the satellite depend on ?

(1) r
(2) m
(3) M

A. (1) only
B. (2) only
C. (1) and (3) only
D. (2) and (3) only

(2018)

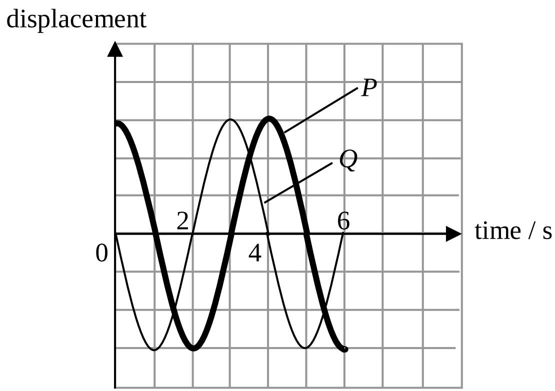
14. Which of the following statements about waves is/are correct ?

(1) Longitudinal waves can transmit energy from one place to another but transverse waves cannot.
(2) Sound waves propagate faster in water than in air.
(3) Infra-red radiation is a kind of electromagnetic wave.

A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

(2018)

15. The figure below shows the displacement-time graph of particles P and Q on the same transverse travelling wave of wavelength λ.

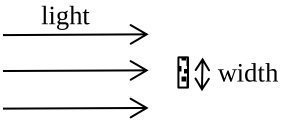


Which of the following statements **MUST BE** correct ? Upward displacement is taken to be positive.

(1) At time t = 2 s, P is momentarily at rest.
(2) At time t = 4 s, Q is moving downwards.
(3) The separation between the equilibrium positions of P and Q is 0.25λ.

A. (2) only
B. (3) only
C. (1) and (2) only
D. (1) and (3) only

(2018) 16. Light undergoes diffraction round an obstacle.



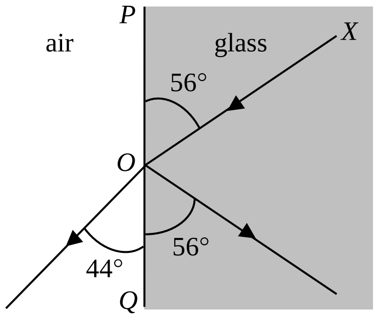
The angle of diffraction would increase when

(1) the amplitude of the incident light is increased.
(2) the width of the obstacle is increased.
(3) the wavelength of the incident light is increased.

A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

(2018)

17.



In the figure above, *XO* is a light ray incident on the glass-air boundary plane *PQ*. Which of the following gives the refractive index of glass ?

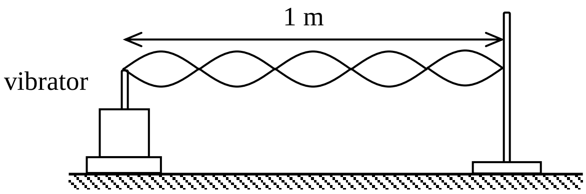
A. $\frac{\sin 56^{o}}{\sin 44^{o}}$

B. $\frac{\sin 44^{o}}{\sin 34^{o}}$

C. $\frac{\sin 56^{o}}{\sin 46^{o}}$

D. $\frac{\sin 46^{o}}{\sin 34^{o}}$

(2018) 18. The figure shows a string with one end fixed and the other end tied to a vibrator. A stationary wave is formed as shown at a certain frequency.



If the speed of the wave along the string is 7 m s^{-1}, what is the frequency of the wave ?

A. 2.8 Hz
B. 7 Hz
C. 17.5 Hz
D. 35 Hz

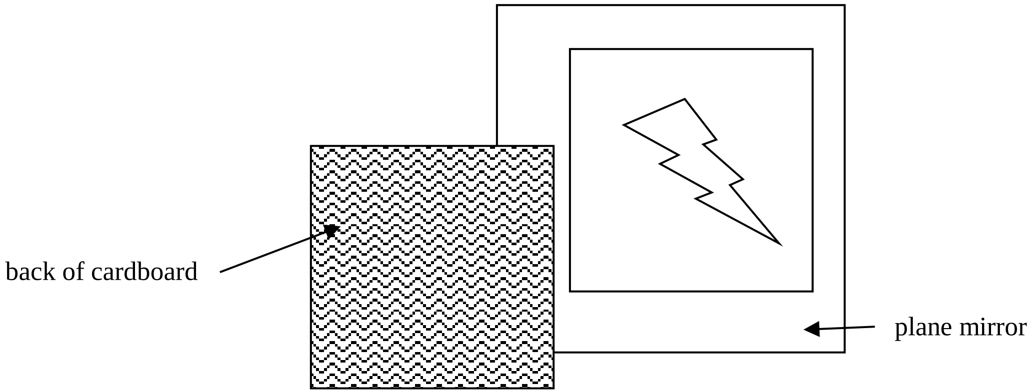
(2018)

*19. An object placed 25.0 cm in front of a lens forms a virtual image at a distance 11.1 cm from the lens. The lens is a

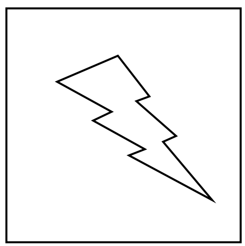
A. concave lens of focal length 7.7 cm.
B. concave lens of focal length 20 cm.
C. convex lens of focal length 7.7 cm.
D. convex lens of focal length 20 cm.

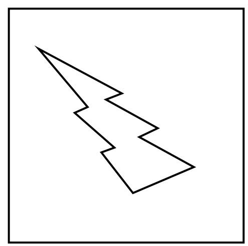
(2018)

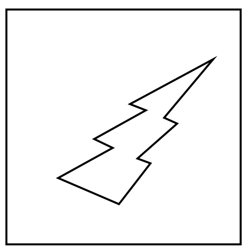
20. The figure shows the image seen when a plane mirror is placed in front of a cardboard with a design on its front surface.

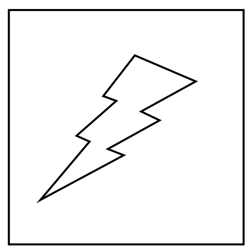


Which diagram below shows the design on the cardboard ?

A.


B.


C.


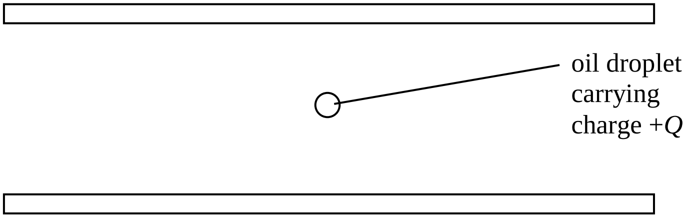
D.


(2018)

21. Which of the following is **NOT** a typical sound intensity level that occurs in daily life ?

A. 130 dB : when an airplane take-off
B. 110 dB : at a rock concert
C. 80 dB : having a normal conversation
D. 30 dB : inside a library

(2018) 22.

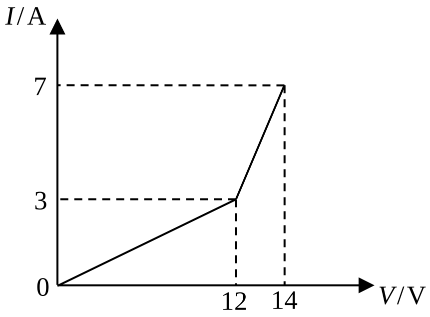


A charged oil droplet of mass *m* is found floating between two horizontal parallel metal plates with an electric field of constant strength in between. The oil droplet is carrying charge $+Q$. What are the direction and strength of the electric field ?

	direction	**strength**
A.	upward	$\dfrac{mg}{Q}$
B.	upward	$\dfrac{Q}{mg}$
C.	downward	$\dfrac{mg}{Q}$
D.	downward	$\dfrac{Q}{mg}$

(2018)

23. The graph below shows the current-voltage (I-V) relationship of a conductor.

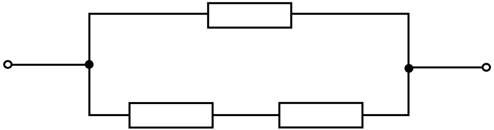


Which statement below is **INCORRECT** ?

A. When the voltage across the conductor is less than 12 V, it obeys Ohm's law.
B. When the voltage across the conductor exceeds 12 V, its resistance begins to decrease.
C. The resistance of the conductor is 0.5 Ω when the current flowing through it is 5 A.
D. The resistance of the conductor is 2 Ω when the voltage across it is 14 V.

(2018)

24.



Three identical resistors are arranged as shown. The rated power of each resistor is 12 W. If no resistor exceeds its rated power, what is the maximum power dissipation in such an arrangement ?

A. 16 W
B. 18 W
C. 20 W
D. 24 W

(2018)

25. Two wires X and Y of the same length are made from different materials. The radius of X is half that of Y. When the two wires are connected in parallel to the same power supply, the current flowing through each wire is the same. What is the ratio of the resistivity of the material of X to that of Y?

A. 1 : 4
B. 4 : 1
C. 1 : 2
D. 2 : 1

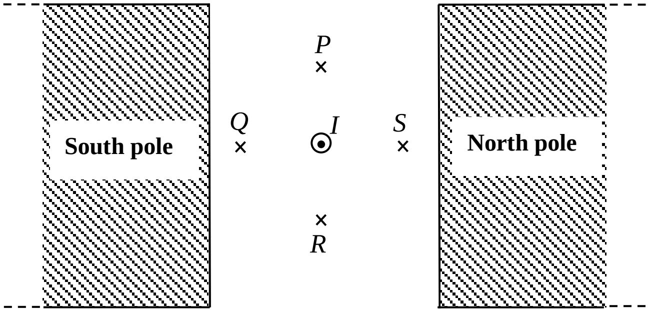
(2018)

26. A mobile phone battery labelled 2800 mA h capacity is fully charged initially. What percentage of capacity remains after it has delivered a current of 200 mA for 3 hours ?

A. 7.1%
B. 21.4%
C. 78.6%
D. 92.9%

(2018)

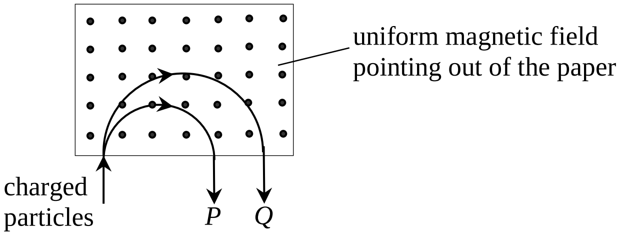
27. A straight wire carrying a current I pointing out of the paper is placed in a uniform magnetic field between two pole pieces as shown. At which point, P, Q, R or S, can the resultant magnetic field be zero ? Neglect the effects of the Earth's magnetic field.



A. P
B. Q
C. R
D. S

(2018)

*28.

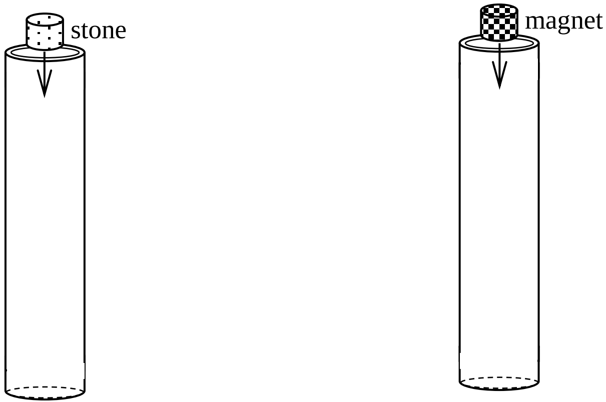


P and *Q* are two particles carrying the same amount of charge but of different masses. They travel with the same speed and enter a uniform magnetic field pointing out of the paper as shown. Semi-circular paths with different radii are described before they emerge from the field. Which descriptions below are correct ?

(1) Both *P* and *Q* are positively charged.
(2) *P* and *Q* emerge from the field with the same speed.
(3) The mass of *Q* is greater than that of *P*.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

(2018) 29.

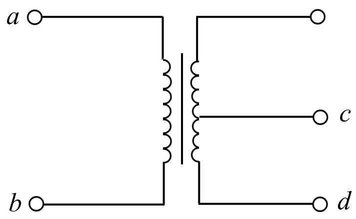


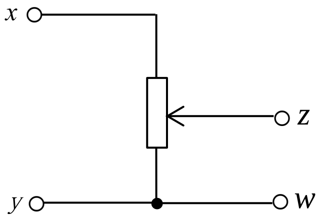
A stone and a strong magnet of the same size and shape are released from rest into a hollow aluminium tubing respectively. Which of the following is correct ? Neglect air resistance.

	drops slower	**reason**
A.	stone	the stone is more massive
B.	magnet	the stone is more massive
C.	stone	the magnet induces eddy current in the aluminium tubing
D.	magnet	the magnet induces eddy current in the aluminium tubing

(2018)

*30. In the circuits below, if a 12 V sinusoidal a.c. is applied across *ab* and across *xy* respectively, the voltages across *cd* and *zw* are both 6 V. Now if a 6 V sinusoidal a.c. is applied across *cd* and across *zw* respectively, what would be the voltages across *ab* and *xy* respectively ?





	voltage across *ab*	**voltage across *xy***
A.	12 V	12V
B.	12 V	6 V
C.	6 V	6 V
D.	12 V	0 V

(2018)

31. Which of the following nuclear reactions is/are possible for a chain reaction to occur ?

(1) $^{2}_{1}H + ^{3}_{1}H \rightarrow ^{1}_{0}n + ^{4}_{2}He$

(2) $^{235}_{92}U + ^{1}_{0}n \rightarrow ^{141}_{56}Ba + ^{92}_{36}Kr + 3\,^{1}_{0}n$

(3) $^{239}_{94}Pu + ^{1}_{0}n \rightarrow ^{148}_{58}Ce + ^{89}_{36}Kr + 3\,^{1}_{0}n$

A. (1) only
B. (2) only
C. (1) and (3) only
D. (2) and (3) only

(2018)

*32. X and Y are two radioactive nuclides. The ratio of the mass of an atom of X to that of an atom of Y is 1 : 2. The half-lives of X and Y are T and $2T$ respectively. If two samples consisting of purely X and Y respectively have the same initial mass, find the ratio of the number of undecayed nuclei of X to that of Y after a period of $4T$.

A. 1 : 4
B. 1 : 2
C. 1 : 1
D. 2 : 1

(2018)

*33. Given: mass of proton = 1.007276 u
mass of neutron = 1.008665 u
mass of ${}^{3}_{2}He$ nucleus = 3.016030 u
1 u = 931 MeV

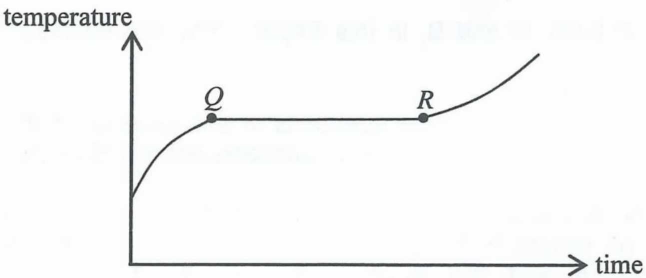
When a ${}^{3}_{2}He$ nucleus is formed from 2 protons and 1 neutron,

A. 6.7 MeV energy is released.
B. 6.7 MeV energy is required.
C. 8.0 MeV energy is released.
D. 8.0 MeV energy is required.

(2019) 1. A block measuring 80 °C is put into water at a temperature of 40 °C. The final temperature of the mixture is 70 °C. Which of the following deductions must be correct ? Assume that there is no heat loss to the surroundings.

A. The energy gained by the water is greater than the energy lost by the block.
B. The mass of the water is greater than the mass of the block.
C. The specific heat capacity of water is smaller than that of the block's material.
D. The heat capacity of the water is smaller than that of the block.

(2019) 2.

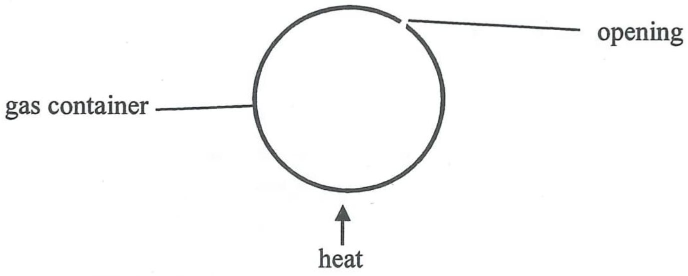


A substance undergoes the fusion process. The figure shows how the substance's temperature varies with time. Its temperature remains constant during the period from Q to R. Which of the following deductions **within this period** is/are correct ?

(1) The substance does not absorb heat.
(2) The mass ratio of the substance in solid and liquid states remains constant throughout.
(3) The average potential energy of the molecules of the substance increases with time.

A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

(2019) *3. The figure shows an inexpansible container with an opening.

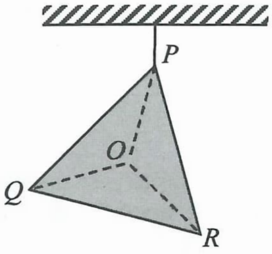


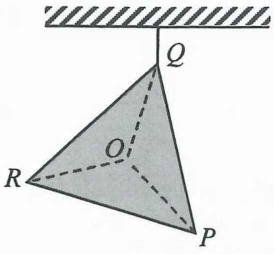
When the gas inside the container is being heated slowly by a heater, which statements about the gas molecules in the container are correct ?

(1) The number of molecules decreases.
(2) The average kinetic energy of molecules increases.
(3) The average separation between molecules remains unchanged.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

(2019) 4.

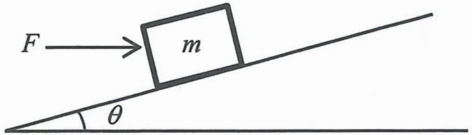




O is the centre of a metal plate PQR in the form of an equilateral triangle with **non-uniform** mass distribution. The plate is suspended from the ceiling at P and then at Q as shown. The centre of gravity of the metal plate is

A. at O.
B. within the region POQ.
C. within the region ROQ.
D. within the region POR.

(2019) 5.



In the above figure, a horizontal force F is applied to a block of mass m so as to keep it at rest on a smooth incline making an angle θ with the horizontal. Find the magnitude of F.

A. $\dfrac{mg\sin\theta}{\cos\theta}$

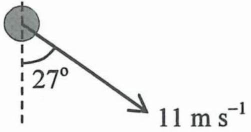
B. $mg\sin\theta\cos\theta$

C. $\dfrac{mg\cos\theta}{\sin\theta}$

D. $mg\sin\theta$

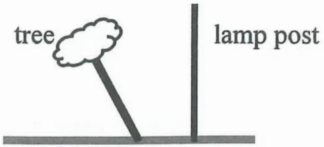
(2019)

*6. A small ball after projection moves under the effect of gravity only. Its velocity at a certain instant is shown below. What is the speed of the ball **1 s before** ? Neglect air resistance. ($g = 9.81$ m s^{-2})



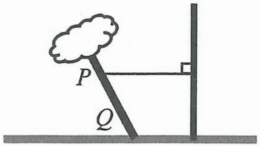
A. 19.1 m s^{-1}
B. 9.8 m s^{-1}
C. 5.0 m s^{-1}
D. 0.2 m s^{-1}

(2019) 7.

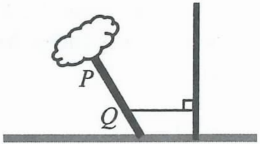


A tree blown by strong wind is found leaning to one side. In order to give the tree some support, a rope is wrapped around its trunk and tied to a fixed lamp post nearby. In which of the following arrangements would the rope have the highest chance of breaking ?

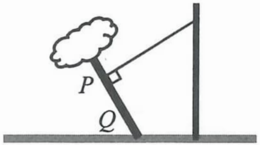
A.



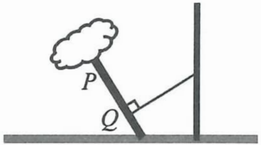
B.



C.

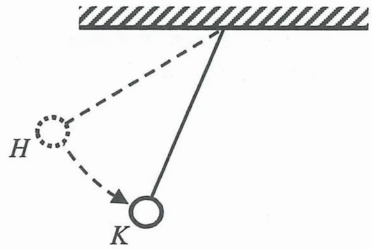


D.



(2019)

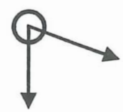
8.



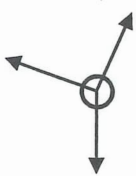
A small sphere suspended by a light inextensible string is released from point *H* as shown. The string remains taut when the sphere swings downward. Which free-body diagram below best shows all the forces acting on the sphere at *K* ? Neglect air resistance.

A.

B.

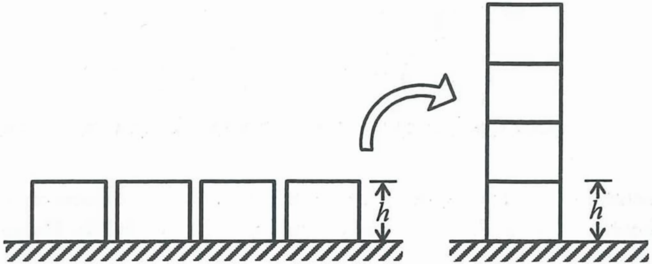


C.



D.

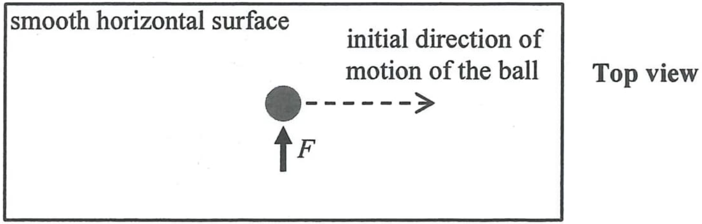
(2019) 9.



Four identical uniform blocks, each of mass m and height h, are first placed on a horizontal table. If the blocks are stacked on top of one another as shown, what is the minimum work done ?

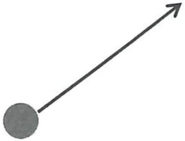
A. $8\,mgh$
B. $6\,mgh$
C. $4\,mgh$
D. $3\,mgh$

(2019) 10.



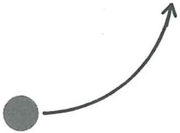
The above figure shows a ball moving with a constant speed along a straight line on a smooth horizontal surface. At a certain instant, the ball is acted on by a force F for a very short time as shown above. Which **subsequent** path below would the ball most closely follow ?

A.

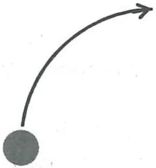


B.

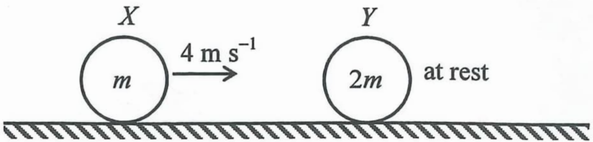
C.



D.



(2019) 11.



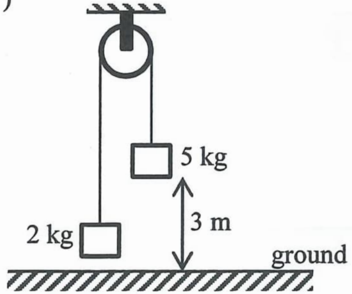
On a smooth horizontal surface, sphere X of mass m travels with speed 4 m s^{-1}. It collides head-on with another sphere Y of mass $2m$, which is at rest initially. Which of the following can be the speed of Y just after collision ?

(1) 1 m s^{-1} (2) 2 m s^{-1} (3) 3 m s^{-1}

A. (1) only
B. (2) only
C. (1) and (2) only
D. (2) and (3) only

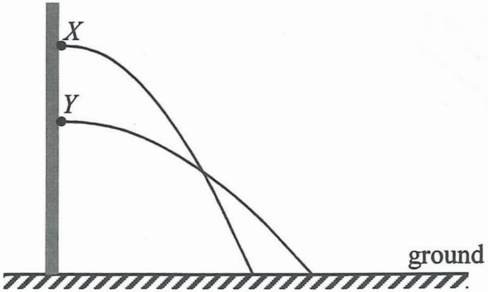
(2019)

12. Two blocks of respective masses 2 kg and 5 kg are connected by a light inextensible string which passes over a smooth fixed light pulley as shown. The system is released from rest when the 5-kg block is 3 m above the ground. What is the speed of the 5-kg block just when reaching the ground ? Neglect air resistance. ($g = 9.81 \text{ m s}^{-2}$)



A. 5.0 m s^{-1}
B. 6.0 m s^{-1}
C. 6.5 m s^{-1}
D. 7.7 m s^{-1}

(2019) *13.

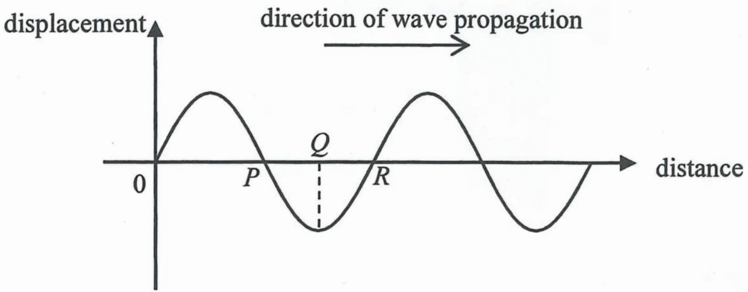


Particles X and Y are projected horizontally from a vertical wall and the figure shows their paths in air before reaching the ground. Which statements below are correct ? Neglect air resistance.

(1) The time of flight of Y is longer.
(2) The projection speed of Y is greater.
(3) X and Y can have the same landing speed.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

(2019) 14. The figure shows the displacement-distance graph at a certain instant of a longitudinal wave which travels to the right. Displacement to the right is taken to be positive.

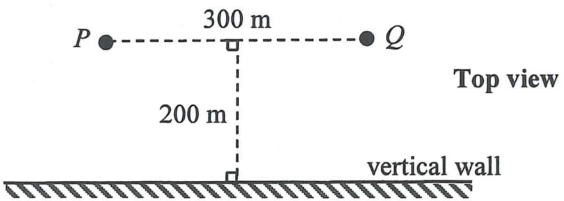


At the instant shown, which of the following statements is/are correct ?

(1) P is a centre of compression.
(2) A particle with its equilibrium position at Q is at rest.
(3) A particle with its equilibrium position at R is moving downwards.

A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

(2019) 15.

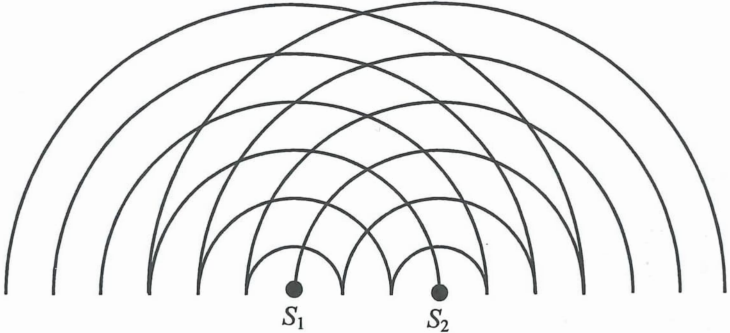


Boys P and Q are 300 m apart and are both at a distance of 200 m from a vertical wall as shown. When P shouts once, two sounds are heard by Q. Which description below is correct ?
Given : speed of sound in air = 340 m s^{-1}

A. The first sound is louder and 0.59 s later the second sound is heard.
B. The first sound is louder and 0.29 s later the second sound is heard.
C. The second sound is louder and 0.59 s earlier the first sound is heard.
D. The second sound is louder and 0.29 s earlier the first sound is heard.

(2019)

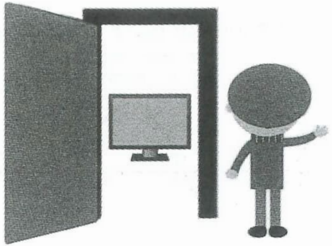
16.



The figure shows the circular water waves generated by two dippers S_1 and S_2 vibrating in phase. The lines represent wave crests. What is the number of nodal lines (i.e. minimum amplitude) formed ?

A. 3
B. 4
C. 6
D. 7

(2019) 17. Peter is standing next to the door of a room. He can hear the sound emitted by the television inside the room but cannot see the television pictures. Which of the following is/are the possible reason(s) ?



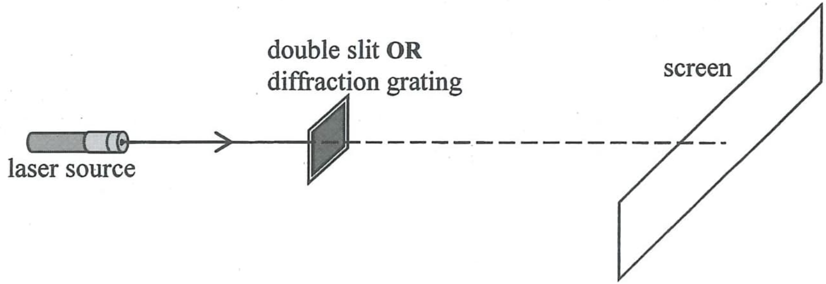
(1) Sound wave diffracts while light wave does not.
(2) Sound wave is mechanical in nature while light wave is electromagnetic.
(3) Sound wave has a much longer wavelength than visible light.

A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

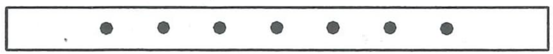
(2019) 18. How does the speed of propagation of waves along a stretched string change if the tension in the string is increased or the string is replaced by a more massive one of the same length and tension ?

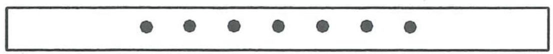
	tension increased	using a more massive string of the same length and tension
A.	speed increases	speed decreases
B.	speed increases	speed increases
C.	speed decreases	speed decreases
D.	speed decreases	speed increases

(2019) 19.



A double slit and a diffraction grating are used in turns in the above set-up such that red and green laser lights are directed one after the other on each of them. The four resulting patterns of bright spots obtained on the screen are shown below. Which one belongs to the pattern formed by **green light** incident on the **diffraction grating** ?

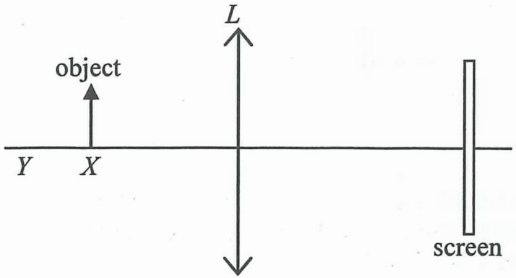
A. 

B. 

C. 

D. 

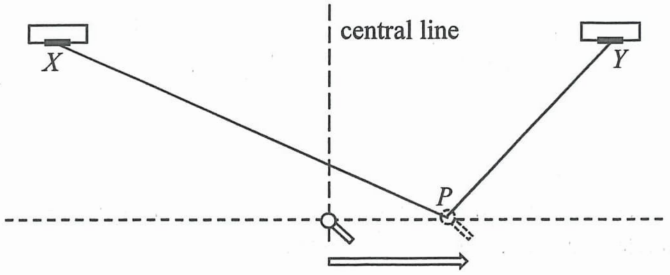
(2019) 20.



An object is placed at point X in front of a convex lens L as shown. A sharp image is captured by the screen. The object is then shifted to point Y. Which adjustment below may give a sharp image on the screen again ?

A. replacing L with another convex lens of longer focal length
B. replacing L with another convex lens of the same shape but made from material of a larger refractive index
C. replacing L with a concave lens
D. moving the screen to the right

(2019) 21.



Two loudspeakers X and Y emit sound waves of frequency 500 Hz. A microphone is moved steadily along a line perpendicular to the central line as shown. It detects sound waves of maximum amplitude at the central line and the next maximum amplitude is detected at point P. Find $PX - PY$.
Given: speed of sound in air = 340 m s^{-1}

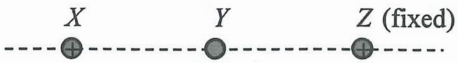
A. 0.17 m
B. 0.34 m
C. 0.51 m
D. 0.68 m

(2019) 22. Which of the following statements about infra-red radiation is/are correct ?

(1) It bends towards the normal when it travels from air to water.
(2) It travels faster in water than in air.
(3) It is used for satellite communication.

A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

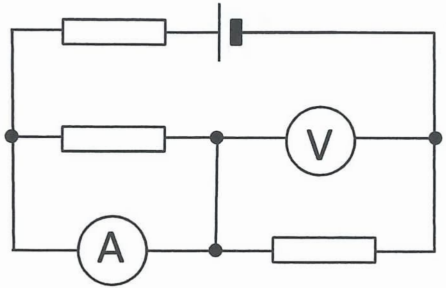
(2019) 23.



In the above figure, point charge Y is placed in the middle of two identical positive point charges X and Z, with Z being fixed. Both X and Y are in equilibrium and at rest initially. What would happen to X if Y is slightly pushed towards Z?

A. It moves towards the left.
B. It moves towards the right.
C. It remains at rest.
D. It cannot be determined as the sign of Y is not known.

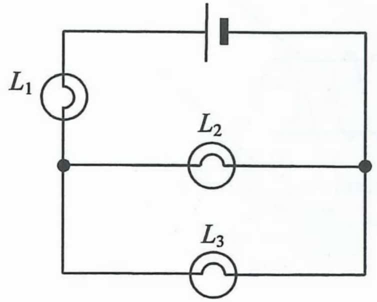
(2019) 24.



The figure shows a 6 V cell of negligible internal resistance connected to three identical resistors. Both ammeter and voltmeter are ideal. Find the voltmeter reading.

A. 6 V
B. 4 V
C. 3 V
D. 2 V

(2019) 25.



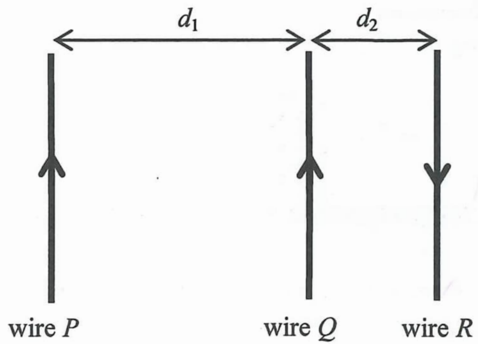
In the above circuit, L_1, L_2 and L_3 are three light bulbs and the cell has negligible internal resistance. Which of the following changes will make L_3 brighter ?

(1) L_1 is faulty and becomes a short circuit.
(2) L_2 is faulty and becomes a short circuit.
(3) L_2 is faulty and becomes an open circuit.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

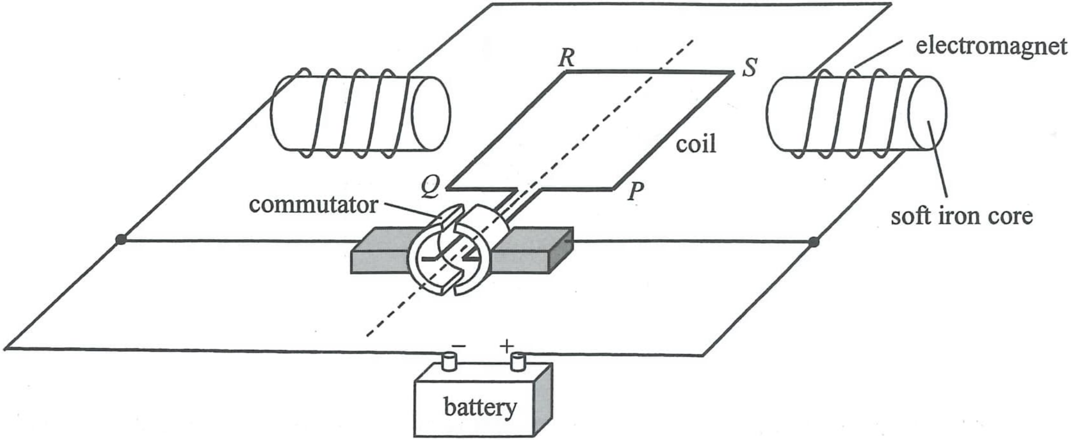
(2019)

26. Three parallel wires P, Q and R are arranged with separations d_1 and d_2 (with $d_1 > d_2$) as shown. Each wire carries the same magnitude of current flowing in the directions indicated. If the magnitude of the magnetic force per unit length exerted on Q by P is F, what are the direction and magnitude of the **resultant magnetic force** per unit length exerted on Q ?



	direction of resultant magnetic force exerted on Q	**magnitude of resultant magnetic force per unit length exerted on Q**
A.	to the right	greater than $2F$
B.	to the right	smaller than F
C.	to the left	greater than $2F$
D.	to the left	smaller than F

(2019) 27.

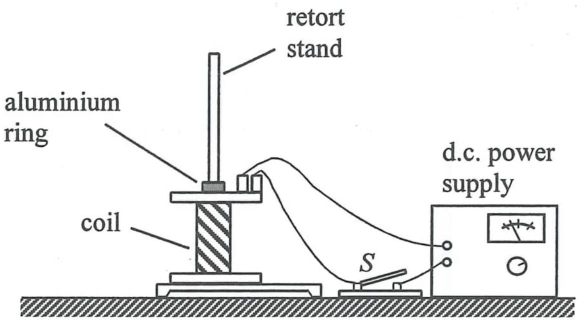


The figure shows the structure of a motor. The coil *PQRS* and the two electromagnets are connected to a battery so that the coil rotates continuously. If a sinusoidal a.c. source of frequency 50 Hz is used instead of a battery, the coil will

A. remain at rest.
B. oscillate at a frequency 50 Hz.
C. rotate to a vertical position and then stop.
D. rotate continuously.

(2019)

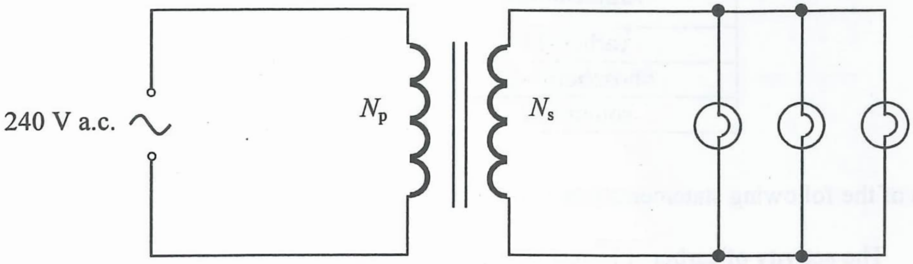
28.



A retort stand and a coil connecting to a d.c. power supply are arranged as shown. An aluminium ring threaded through the stand is placed on top of the coil. When the switch *S* is closed, the aluminium ring jumps up momentarily and then falls back down. Which of the following modifications would enable the ring to rise up and float in the air ?

A. using a ring made from a lighter material
B. using a ring made from a metal of smaller resistivity
C. using a coil with the number of turns doubled
D. using an a.c. power supply instead of a d.c. power supply

(2019) *29. In the circuit below, each light bulb works at rated power '12 V 24 W'. What should be the turns ratio (N_p:N_s) of the transformer ?



A. 40 : 1
B. 30 : 1
C. 20 : 1
D. 10 : 1

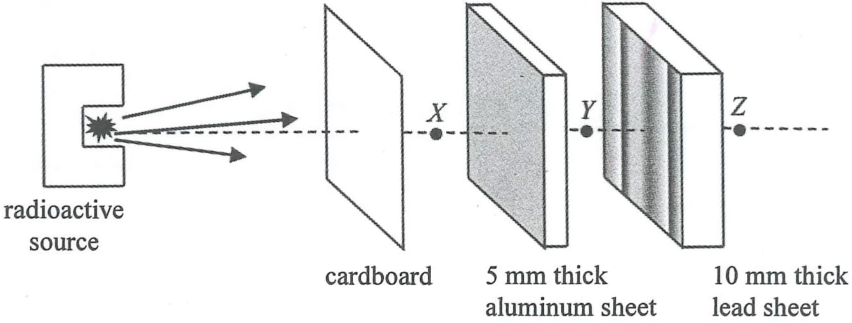
(2019)

*30. The power consumption of the heating element of an electric heater connected to an a.c. mains can be increased by

(1) increasing the electrical resistance of the heating element.
(2) increasing the frequency of the a.c. voltage.
(3) increasing the r.m.s. value of the a.c. voltage.

A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

(2019) 31. A radioactive source emits α, β and γ radiations.



Which statement about the radiation(s) detected at positions *X*, *Y*, *Z* indicated in the figure is correct ?

A. No radiation from the radioactive source is detected at *Z*.
B. Both β and γ radiations can be detected at *Y*.
C. α radiation can only be detected at *X* but not at *Y* and *Z*.
D. β radiation can only be detected at *X* but not at *Y* and *Z*.

(2019) *32. The half-lives of some radioisotopes are tabulated below.

radioisotope	half-life
carbon-11	20.3 minutes
phosphorus-32	14.3 days
sodium-22	2.60 years

Which of the following statements is/are correct ?

(1) The activity of carbon-11 must be the highest.
(2) The decay constant of phosphorus-32 is larger than that of carbon-11.
(3) If the initial activity of sodium-22 is 1520 Bq, its activity would be lower than 380 Bq after 6 years.

A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

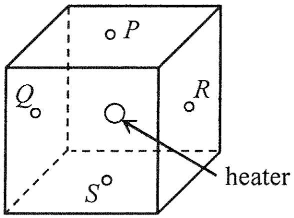
(2019)

*33. Given: mass of a neutron = 16749×10^{-31} kg
mass of a proton = 16726×10^{-31} kg
mass of an electron = 9×10^{-31} kg

In a nuclear reaction, a neutron becomes a proton and a β particle. Estimate the energy released in this process.

A. 1.8 MeV
B. 1.3 MeV
C. 0.79 MeV
D. 0.51 MeV

(2020) 1. A heater is installed at the centre of a fully filled cubic water tank. Temperature sensors P, Q, R and S are fixed at the respective centres of the top, left, right and bottom surfaces of the tank.



After the heater is switched on for **a short duration**, which pair of sensors below would indicate the largest temperature difference ?

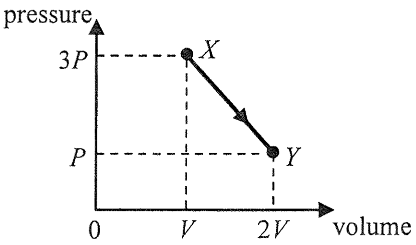
A. Q and R
B. R and S
C. Q and S
D. P and R

(2020) 2. An electric kettle which contains 1 kg of water at room temperature takes 168 s to heat up the water to boiling point. The kettle's rated value is '220 V, 2000 W'. Assume that all the electrical energy consumed by the kettle is transferred to the water. Which of the following statements is/are correct ?
Given: specific heat capacity of water = 4200 J kg^{-1} $°C^{-1}$

(1) The initial temperature of the water is 20 °C.
(2) The resistance of the kettle's heating element is about 24 Ω.
(3) If the electric kettle is operated with 110 V, the time taken to heat up the water to boiling point will be doubled.

A. (1) only
B. (3) only
C. (1) and (2) only
D. (1), (2) and (3)

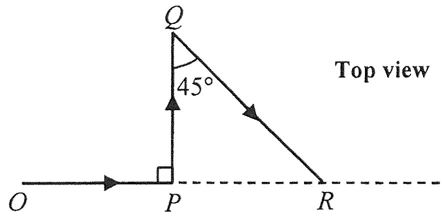
(2020) *3. A fixed mass of an ideal gas expands from state X to state Y through a process as represented in the pressure-volume graph below.



If the temperature of the gas at state Y is 25 °C, what is its temperature at state X?

A. −74.3 °C
B. 16.7 °C
C. 37.5 °C
D. 174 °C

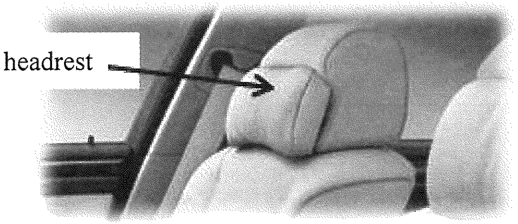
(2020) 4.



A car takes 8 minutes to travel along a path $OPQR$ on a horizontal surface as shown. Given that $OP = PQ = 2$ km, find the magnitude of the average velocity of the car in this journey.

A. 30 km h^{-1}
B. 36 km h^{-1}
C. 41 km h^{-1}
D. 51 km h^{-1}

(2020) 5.

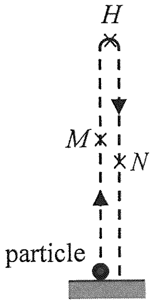


For a car travelling on a highway, which of the following statements about the safety design of the headrest is/are correct ?

(1) As the headrest is soft, it can reduce the force exerted on the passenger's head during impact.
(2) It can minimise injury of the passenger when the car is struck by another one from behind.
(3) It can minimise injury of the passenger when the car brakes suddenly.

A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

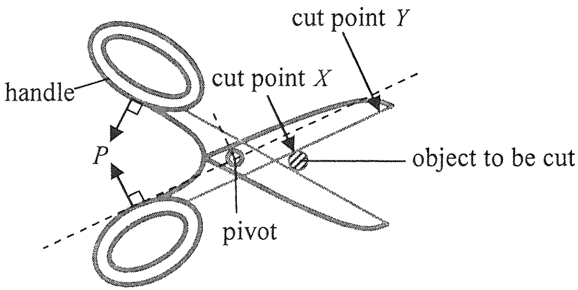
(2020) 6. A particle is thrown vertically upward and its path is shown below. H is the highest point that the particle reached. Which of the following statements about the particle is/are correct ? Neglect air resistance.



(1) Its acceleration at M is upward.
(2) Its acceleration at H is zero.
(3) Its acceleration at N is downward.

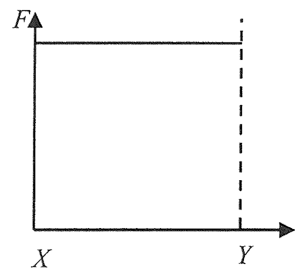
A. (1) only
B. (3) only
C. (2) and (3) only
D. (1), (2) and (3)

(2020) 7. The figure shows that a pair of forces P of constant magnitude is applied at right angles to the handles of a pair of scissors in order to cut an object.

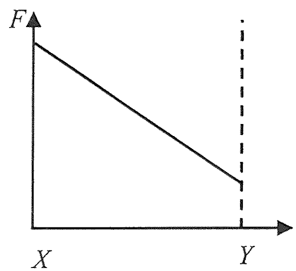


Which graph below best shows the variation of force F produced at cut point from X to Y when the pair of scissors is closed ?

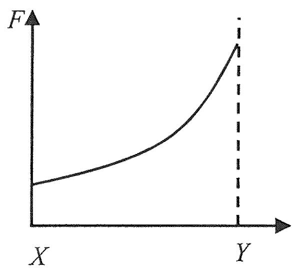
A.



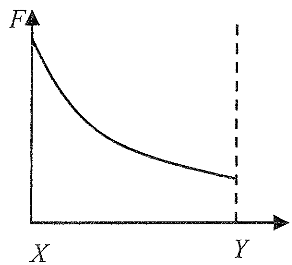
B.



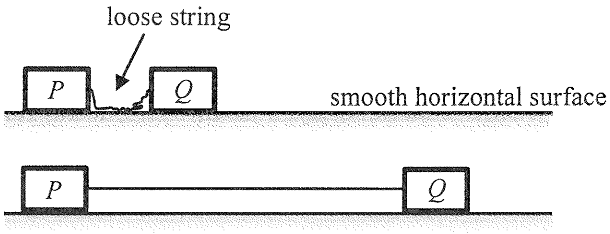
C.



D.



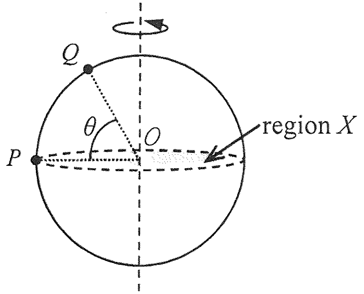
(2020) 8. On a smooth horizontal surface, two identical blocks P and Q are connected by a light inextensible string. The blocks are at rest and the string is loose initially.



Q is given a speed of $4\ \mathrm{m\ s^{-1}}$ and moves to the right. Find the speeds of the blocks when the string just becomes taut and P starts to move.

	block P	block Q
A.	$1\ \mathrm{m\ s^{-1}}$	$1\ \mathrm{m\ s^{-1}}$
B.	$2\ \mathrm{m\ s^{-1}}$	$1\ \mathrm{m\ s^{-1}}$
C.	$2\ \mathrm{m\ s^{-1}}$	$2\ \mathrm{m\ s^{-1}}$
D.	$4\ \mathrm{m\ s^{-1}}$	$2\ \mathrm{m\ s^{-1}}$

(2020) *9. Particles P and Q are fixed on the surface of a sphere rotating about a vertical axis passing through the centre O of the sphere as shown. The horizontal shaded region X divides the sphere into two halves. P is at the edge of region X while Q is at an angle of elevation θ above region X.



Find the ratio of the centripetal acceleration of P to that of Q.

A. $1 : \cos\theta$
B. $1 : \sin\theta$
C. $\cos\theta : 1$
D. $\sin\theta : 1$

(2020) *10. The diameter of Neptune is about 4 times that of the Earth and its mass is about 17 times that of the Earth. Estimate the acceleration due to gravity on Neptune's surface.
Given: acceleration due to gravity on Earth's surface $g = 9.81$ m s^{-2}

A. 2.3 m s^{-2}
B. 9.2 m s^{-2}
C. 10.4 m s^{-2}
D. 41.7 m s^{-2}

(2020) 11.

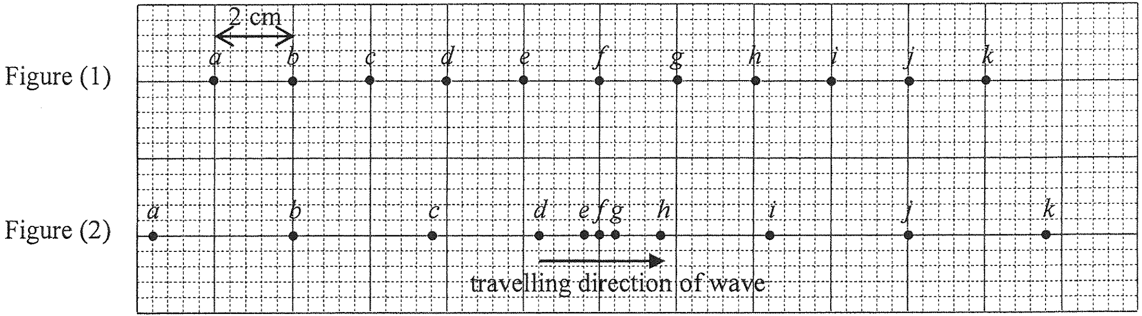
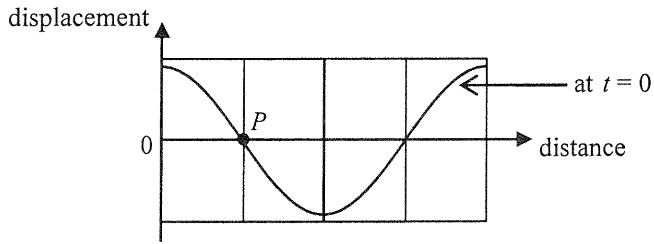


Figure (1) shows the equilibrium positions of particles *a* to *k* in a medium. The particles are separated by 2 cm from each other. A longitudinal wave of frequency 5 Hz is travelling from left to right. At a certain instant, the positions of the particles are shown in Figure (2). Determine the amplitude and speed of the wave.

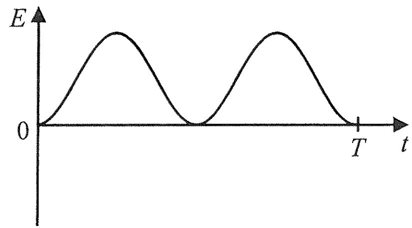
	amplitude	**speed**
A.	3.6 cm	40 cm s^{-1}
B.	3.6 cm	80 cm s^{-1}
C.	2.4 cm	40 cm s^{-1}
D.	2.4 cm	80 cm s^{-1}

(2020) 12. The figure shows part of the displacement-distance graph of a travelling wave of period T at time $t = 0$. P is a particle on the wave.

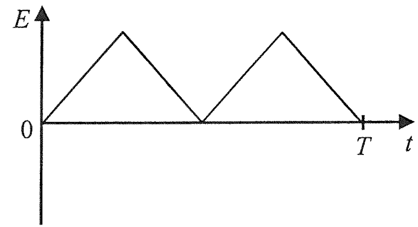


Which graph below correctly shows the variation of the particle's kinetic energy E within a period starting from $t = 0$?

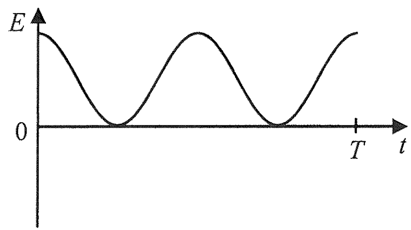
A.



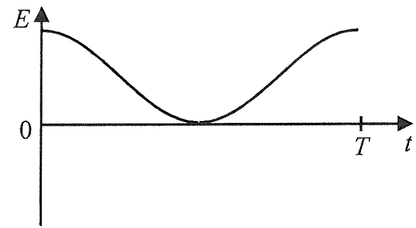
B.



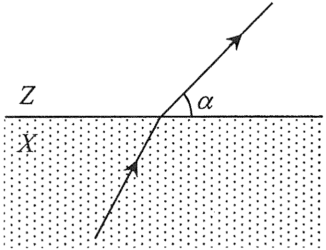
C.

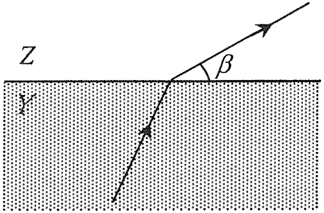


D.



(2020) 13. Monochromatic light travels with the same incident angle from media X and Y respectively to another medium Z as shown.





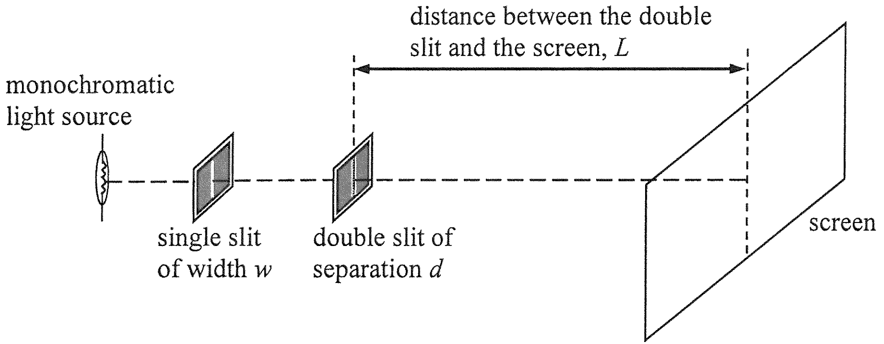
The corresponding refracted rays in Z make angles α and β respectively with the boundary plane (with $\alpha > \beta$). Which medium, X or Y, has a greater refractive index ? In which medium, X or Y, does light travel faster ?

	medium with a greater refractive index	**medium in which light travels faster**
A.	X	X
B.	X	Y
C.	Y	X
D.	Y	Y

(2020) *14. A light beam consisting of wavelengths λ_1 and λ_2 is incident normally on a diffraction grating. The third-order diffraction of wavelength λ_1 coincides with the fourth-order diffraction of wavelength λ_2 in the resulting pattern. If λ_1 is 680 nm, find λ_2.

A. 510 nm
B. 680 nm
C. 907 nm
D. It cannot be determined because the grating spacing is unknown.

(2020) 15. The figure shows a typical set-up of Young's double slit experiment.

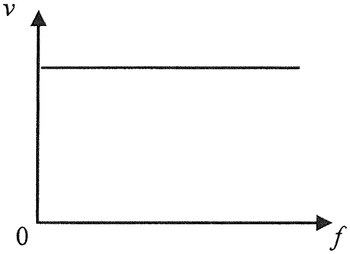


Which combination below is the best setting for displaying an observable fringe pattern on the screen ?

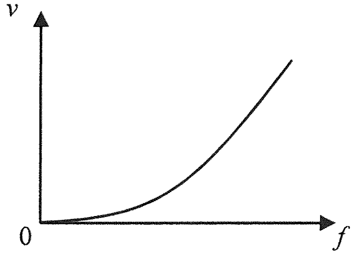
	w	d	L
A.	0.1 mm	1 mm	10 m
B.	0.1 mm	1 mm	1 m
C.	1 mm	0.1 mm	1 m
D.	1 mm	0.1 mm	0.1 m

(2020) 16. A transverse wave propagates along a stretched string. Which graph below correctly shows the variation of the speed v of the wave with its frequency f?

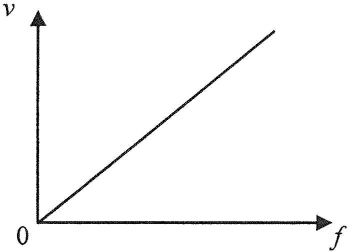
A.



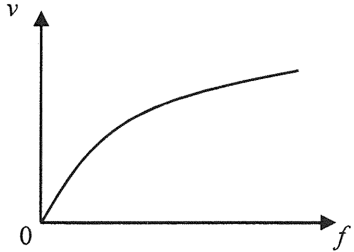
B.



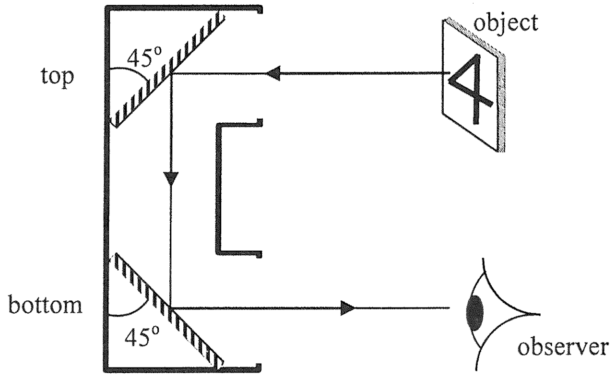
C.



D.

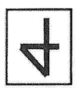


(2020) 17. The figure shows a periscope designed by a student. An object is observed via the periscope.

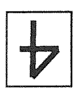


Which image will the observer see ?

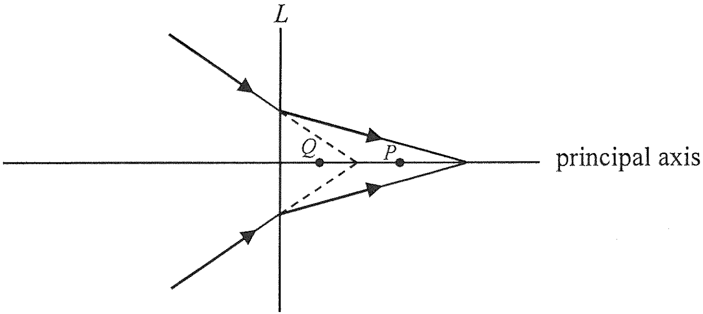
A. 

B. 

C. 

D. 

(2020) 18.



Referring to the above ray diagram, what kind of lens is represented by L ? Which point, P or Q, can be its focus ?

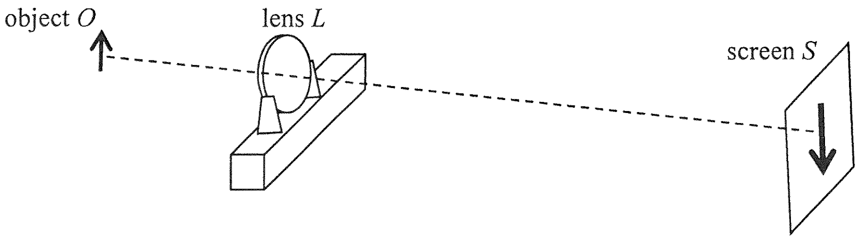
	lens L	focus
A.	concave	P
B.	convex	P
C.	concave	Q
D.	convex	Q

(2020) 19. Which of the following phenomena provides conclusive evidence that sound is a wave ?

(1) reflection of sound from a wall
(2) refraction of sound at the boundary between two media
(3) interference of sound

A. (2) only
B. (3) only
C. (1) and (2) only
D. (1) and (3) only

(2020) 20. The figure shows an enlarged sharp image of an object O formed on a screen S by a convex lens L.



Which of the following can give a diminished sharp image on the screen ?

(1) Keeping the positions of O and L unchanged, move S suitably closer to L.
(2) Keeping the positions of L and S unchanged, move O suitably farther away from L.
(3) Keeping the positions of O and S unchanged, move L suitably closer to S.

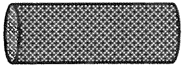
A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

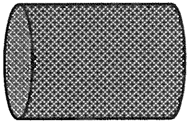
(2020) 21. Which of the following statements about ultrasound is/are correct ?

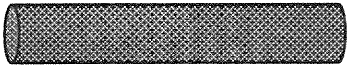
(1) Ultrasound has a shorter wavelength than audible sound.
(2) Ultrasound cannot be produced by vibrating objects.
(3) Ultrasound cannot be heard as it cannot travel through air.

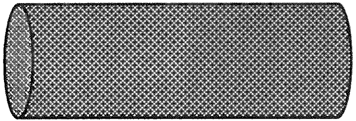
A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

(2020) 22. The cylindrical resistors below are made from the same metal. Which one would produce the most power when the same voltage is applied in turns across the two ends of each resistor ?

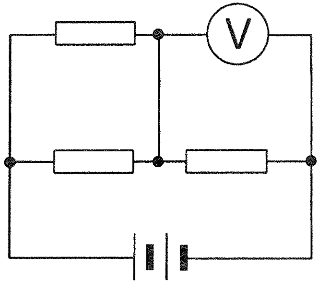
A. 

B. 

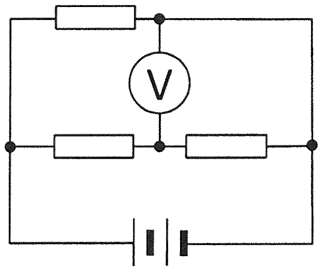
C. 

D. 

(2020) 23. Three identical resistors, a battery of negligible internal resistance and an ideal voltmeter are connected to form Circuits (a) and (b) respectively.



Circuit (a)

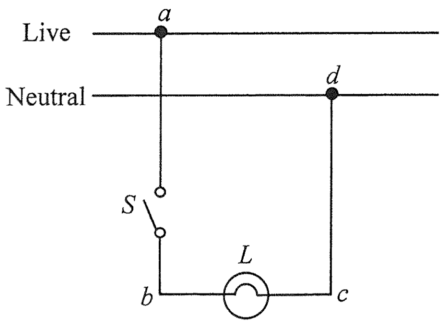


Circuit (b)

Given that the voltmeter reading is 8 V in Circuit (a), what is the voltmeter reading in Circuit (b) ?

A. 4 V
B. 6 V
C. 8 V
D. 12 V

(2020) 24. The figure shows part of a domestic lighting circuit in which the bulb L does not light up when switch S is closed.



The circuit is then checked with switch S closed. Using a voltage tester to touch points b and c in turns, the tester indicates that both points are at high voltage. When touching points a and d in turns, the tester indicates only point a is at high voltage. Which of the following can be a reason for the fault ?

A. The switch S has been damaged.
B. The filament of bulb L has been burnt out and becomes an open circuit.
C. There is a short circuit between a and d.
D. There is an open circuit between c and d.

(2020) 25.

The battery shown has a capacity of 1100 mA h. How much energy is delivered when the battery operates normally at a current of 250 mA for one hour ? Assume that the battery's operating voltage remains at 3.7 V during that period.

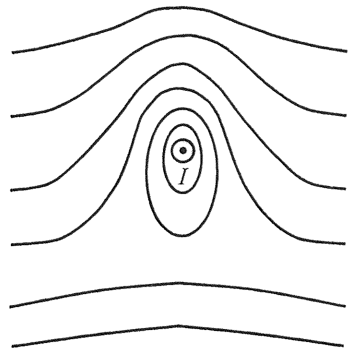
A. $(3.7 \times \frac{250}{1000} \times 3600)$ J

B. $(3.7 \times \frac{1100}{1000} \times 3600)$ J

C. $(3.7 \times \frac{250}{1000} \times 1)$ J

D. $(3.7 \times \frac{1100}{1000} \times 1)$ J

(2020) 26. The figure below shows the magnetic field pattern on a horizontal surface around a long vertical straight wire carrying a steady current I pointing out of the paper. The Earth's magnetic field is **NOT** neglected.

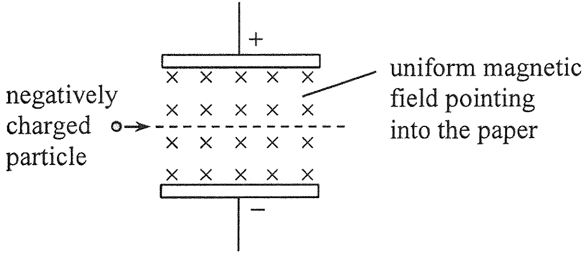


Top view

What are the directions of the following ?

	the horizontal component of the Earth's magnetic field	**the magnetic force experienced by the current-carrying wire**
A.	$\leftarrow$	$\downarrow$
B.	$\leftarrow$	$\uparrow$
C.	$\rightarrow$	$\downarrow$
D.	$\rightarrow$	$\uparrow$

(2020) *27.

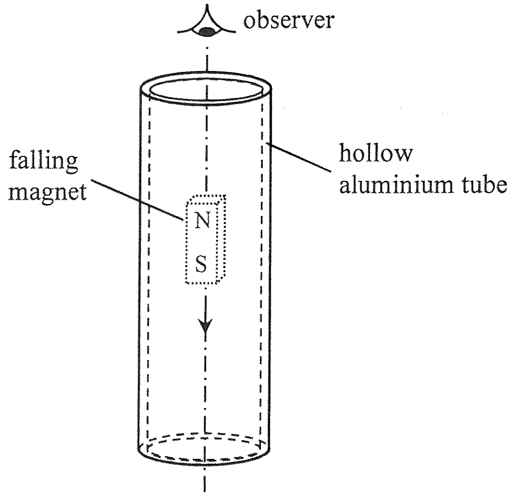


A negatively charged particle goes undeflected through a region in which a uniform electric field and a uniform magnetic field are set up as shown. The electric field is set up by the potential difference across the two parallel metal plates. Which of the following changes may cause the charged particle to deflect downward ? Neglect the effects of gravity.

(1) increasing the potential difference across the plates
(2) increasing the magnitude of the charge on the particle
(3) increasing the particle's speed entering the region

A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

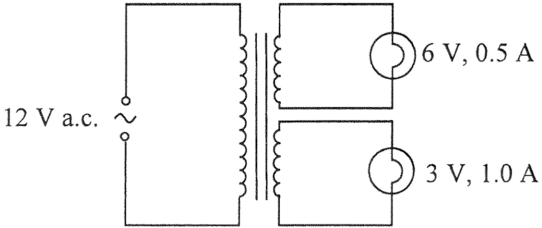
(2020) 28.



When a small strong magnet falls through a hollow aluminium tube as shown, eddy currents are induced. Which of the following correctly describes the direction of current induced in the tube when viewed by an observer from above ?

A. clockwise both above and below the magnet
B. anti-clockwise both above and below the magnet
C. clockwise above the magnet and anti-clockwise below the magnet
D. anti-clockwise above the magnet and clockwise below the magnet

(2020) *29.



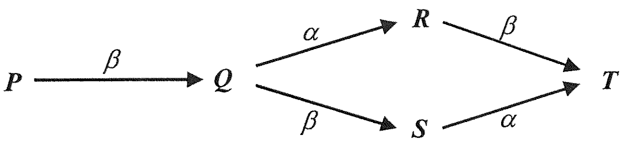
The figure shows an ideal transformer with two secondary coils connected to two light bulbs marked '6 V, 0.5 A' and '3 V, 1.0 A' respectively. When a 12 V a.c. supply is connected to the primary coil, the bulbs work at their respective rated values. Estimate the current in the primary coil.

A. 0.25 A
B. 0.50 A
C. 0.75 A
D. 1.0 A

(2020) 30. The background count rate in an experiment is determined using a GM counter. Four readings of the count rate in each minute are taken. Which set of readings below is the most probable ?

	1st minute	2nd minute	3rd minute	4th minute
A.	5	62	8	69
B.	40	40	40	40
C.	60	50	30	20
D.	29	26	31	35

(2020) 31.



Nuclide P can decay into nuclide T through either process $P - Q - R - T$ or process $P - Q - S - T$ as shown. Which deductions below are correct ?

(1) P and T are isotopes of the same element.
(2) Q and S have the same number of protons.
(3) S has one more neutron than R.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

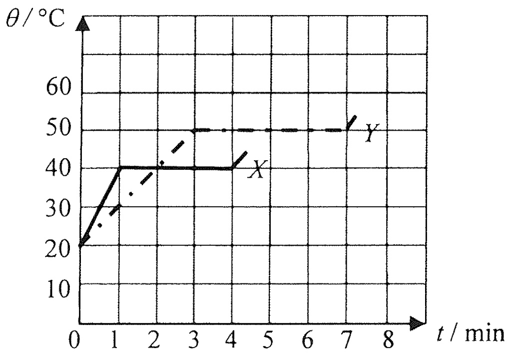
(2020) *32. The decay constant of a radioisotope of an element

A. is random.
B. depends on pressure and temperature.
C. is directly proportional to the number of nucleons in the isotope.
D. is an identifying characteristic of that isotope.

(2020) 33. Two radioactive samples P and Q are freshly prepared. It is found that when $\frac{15}{16}$ of all the nuclei of P have decayed, $\frac{63}{64}$ of all the nuclei of Q have also decayed. Find the ratio $\frac{\text{half-life of } P}{\text{half-life of } Q}$.

A. 1 : 4
B. 2 : 3
C. 3 : 2
D. 4 : 1

(2021) 1. Solid substances X and Y of equal mass are heated by heaters of power $2P$ and P respectively. The graph shows how the temperature θ of each substance varies with the heating time t.



What is the ratio of the specific latent heat of fusion of X to that of Y ?

A. 3 : 2
B. 3 : 4
C. 4 : 3
D. 2 : 3

(2021) 2. Metal blocks X and Y are identical in size and shape. X is of a higher temperature than Y. Which of the following statements must be correct ?

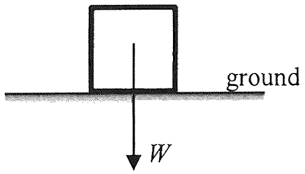
(1) Energy will flow from X to Y if they are in thermal contact.
(2) X is a better conductor of heat compared to Y.
(3) The total internal energy of X is higher than that of Y.

A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

(2021) *3. For an ideal gas, kinetic theory deduces that $pV = \frac{1}{3} Nm\overline{c^2}$. Which physical quantity below can be represented by $\frac{3p}{\overline{c^2}}$?

A. the total mass of the gas
B. the volume of one mole of the gas
C. the density of the gas
D. the number of gas molecules per unit volume

(2021) 4. A block of weight W is at rest on a horizontal ground as shown.



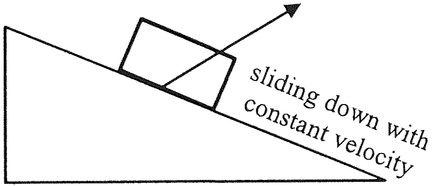
The force acting on the block by the ground is R. Which of the following statements is/are correct ?

(1) R and W are opposite in direction.
(2) R and W are equal in magnitude.
(3) R and W is an action-and-reaction pair.

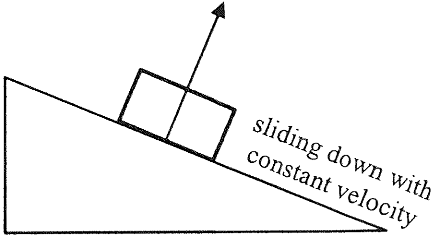
A. (1) only
B. (1) and (2) only
C. (2) and (3) only
D. (1), (2) and (3)

(2021) 5. A block is sliding down a rough incline with constant velocity as shown. Which arrow indicates the direction of **the resultant force acting on the block by the incline** ? Neglect air resistance.

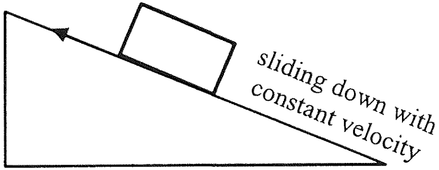
A.



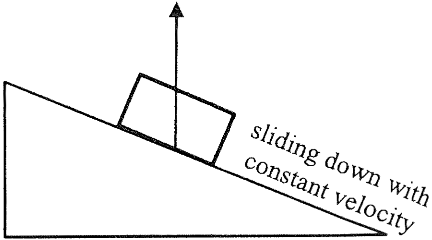
B.



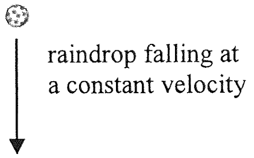
C.



D.

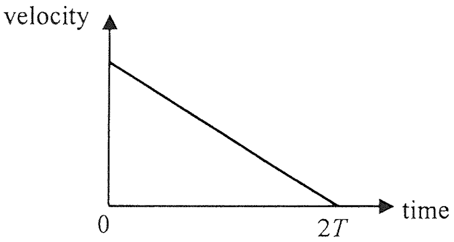


(2021) 6. Which statement below about a raindrop falling at a constant terminal velocity is correct ?



A. No work is done on the raindrop by the gravitational force.
B. As the raindrop falls, all its gravitational potential energy loss is converted into kinetic energy gain.
C. The only force acting on the raindrop is its weight.
D. No net force is acting on the raindrop.

(2021) 7. At time $t = 0$, a small sphere is projected up along a smooth incline with a certain initial velocity. It travels a distance L and becomes momentarily at rest after a time $2T$. The corresponding velocity-time graph is shown below.



What is the distance travelled by the sphere from $t = 0$ to $t = T$?

A. $\frac{1}{4}L$

B. $\frac{1}{2}L$

C. $\frac{3}{4}L$

D. $\frac{4}{5}L$

(2021) 8. A stationary uranium nucleus ${}^{238}_{92}\mathrm{U}$ decays to give a thorium nucleus ${}^{234}_{90}\mathrm{Th}$ and an α particle ${}^{4}_{2}\mathrm{He}$.

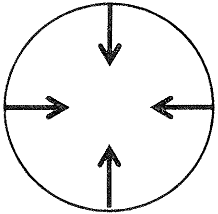
$$ {}^{238}_{92}\mathrm{U} \longrightarrow {}^{234}_{90}\mathrm{Th} + {}^{4}_{2}\mathrm{He} $$

Which of the following correctly describes the situation about the ${}^{234}_{90}\mathrm{Th}$ nucleus and the α particle just after the decay ?

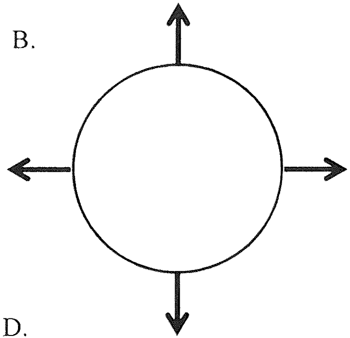
	magnitude of momentum p	kinetic energy KE
A.	$p(\mathrm{Th}) = p(\alpha)$	$\mathrm{KE(Th)} < \mathrm{KE}(\alpha)$
B.	$p(\mathrm{Th}) > p(\alpha)$	$\mathrm{KE(Th)} > \mathrm{KE}(\alpha)$
C.	$p(\mathrm{Th}) = p(\alpha)$	$\mathrm{KE(Th)} > \mathrm{KE}(\alpha)$
D.	$p(\mathrm{Th}) = p(\alpha)$	$\mathrm{KE(Th)} = \mathrm{KE}(\alpha)$

(2021) *9. A particle is moving clockwise (top view) in a horizontal circle with uniform speed. Which top view diagram below correctly shows the net force acting on the particle at various positions ?

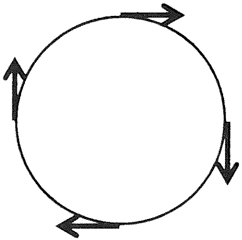
A.



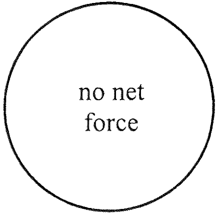
B.



C.



D.

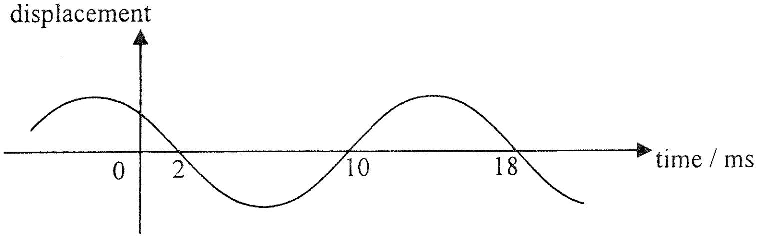


(2021) 10. Which of the following can be transferred by mechanical waves from one place to another along the direction of propagation ?

(1) mass
(2) momentum
(3) energy

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

(2021) 11.



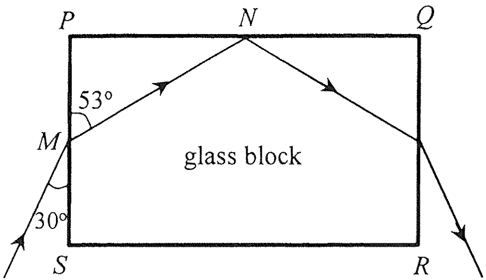
The displacement-time graph of a particle on a travelling wave is as shown. Find the frequency of the wave.

A. 55.6 Hz
B. 62.5 Hz
C. 111 Hz
D. 125 Hz

(2021) 12. Earthquakes propagate in the form of waves. The quake centre produces both longitudinal waves (P-wave) and transverse waves (S-wave) which travel with speeds 9.6 km s^{-1} and 6.4 km s^{-1} respectively on the Earth's crust. In an earthquake, a monitoring station detects the arrival of the P-wave pulse at 7:02 a.m. while the S-wave pulse arrives 2 minutes later at 7:04 a.m. Estimate the time that this earthquake occurs.

A. 6:53 a.m.
B. 6:56 a.m.
C. 6:58 a.m.
D. 6:59 a.m.

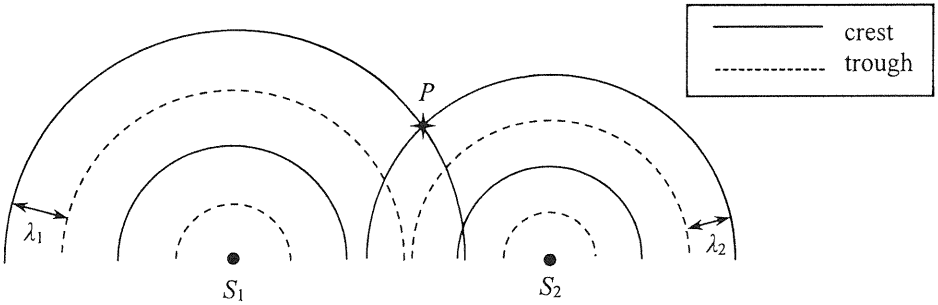
(2021) 13.



The figure shows the cross-section of a rectangular glass block *PQRS*. A light ray is incident from air at *M* on face *PS* and the refracted ray strikes face *PQ* at *N*. Find the critical angle for the glass-air interface.

A. 37°
B. 44°
C. 53°
D. 60°

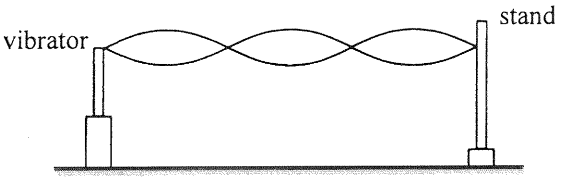
(2021) 14. In a ripple tank, circular water waves of wavelengths λ_1 and λ_2 ($\lambda_1 > \lambda_2$) are produced by two vibrators S_1 and S_2 respectively. The figure represents the two circular waves propagating on the water surface at a certain moment.



Which of the following statements is correct ?

A. The particle at P is always at crest position.
B. At P, the two waves always reinforce to give a larger amplitude.
C. The principle of superposition cannot be applied at P as $\lambda_1 \neq \lambda_2$.
D. The principle of superposition can be applied at P but the two waves do not always reinforce at that location.

(2021) 15. In the set-up below, different stationary wave patterns are formed on an elastic string by adjusting the frequency f of the vibrator.

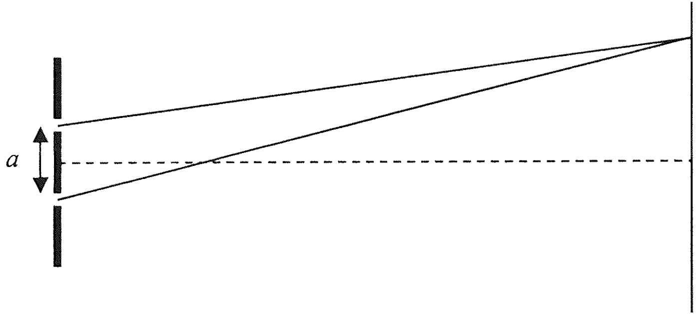


Which statements are correct when f increases ?

(1) The number of antinodes increases.
(2) The speed of the waves on the string increases.
(3) The frequency of the waves produced in air by the string increases.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

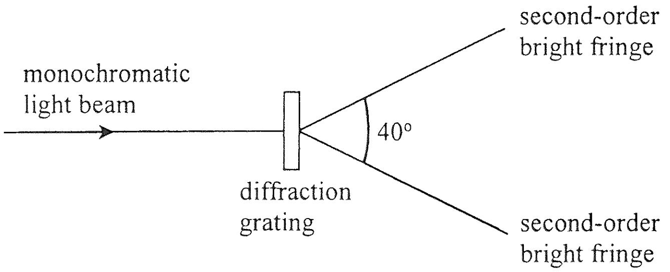
(2021) 16. In Young's double slit experiment employing monochromatic light, how will the interference pattern change if the separation a of the double slit is reduced ?



(1) The pattern will become brighter.
(2) The number of fringes that can be observed will increase.
(3) The fringe separation of the pattern will become larger.

A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

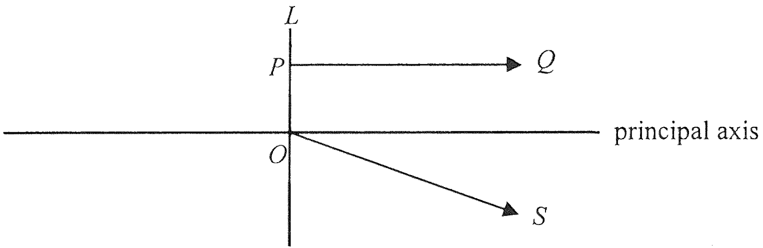
(2021) *17.



When a monochromatic light beam is incident normally on a diffraction grating with 300 lines per mm, a pattern of bright fringes is formed. The angle between the two second-order bright fringes is 40°. Find the frequency of the light.

A. 1.4×10^{14} Hz
B. 2.6×10^{14} Hz
C. 2.8×10^{14} Hz
D. 5.3×10^{14} Hz

(2021) 18. In the figure below, PQ and OS are light rays refracted from a lens L. Both light rays come from a point object situated on the left of L.



Which of the following deductions is/are correct ?

(1) The image of the object must be virtual.
(2) The object must lie along the straight line containing OS.
(3) L must be a concave lens.

A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

(2021) 19. Typical wavelengths of X-rays and microwaves are in the ratio $10^n : 1$. The value of n could be

A. -10.
B. -4.
C. $+4$.
D. $+10$.

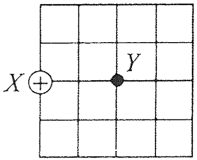
(2021) 20. Submarines employ ultrasound instead of microwaves to detect obstacles in the sea. This is because

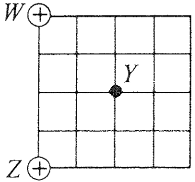
A. wavelengths of ultrasound are shorter than those of microwaves.
B. ultrasound travels faster than microwaves in the sea.
C. microwaves are easily absorbed by sea water.
D. microwaves diffract too much in the sea.

(2021) 21. Three isolated identical metal spheres X, Y and Z carry charges $+2Q$, $-4Q$ and $+5Q$ respectively. Y is first moved to touch X and then Y is brought in contact with Z. When Y and Z are separated, find the charge on each sphere.

	X	Y	Z
A.	0	$+1.5Q$	$+1.5Q$
B.	$-Q$	$+0.5Q$	$+0.5Q$
C.	$+Q$	$+Q$	$+Q$
D.	$-Q$	$+2Q$	$+2Q$

(2021) *22. When a point charge $+Q$ is placed at X as shown, the strength of the electric field at Y is E_0.



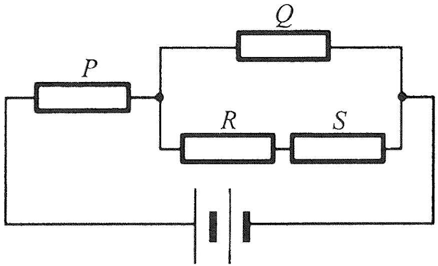


If W and Z are each placed with a point charge of $+Q$, what will be the electric field strength at Y ?

Note: $\sin 45° = \cos 45° = \frac{\sqrt{2}}{2}$

A. $\frac{\sqrt{2}}{2} E_0$
B. E_0
C. $\sqrt{2}\, E_0$
D. $2\, E_0$

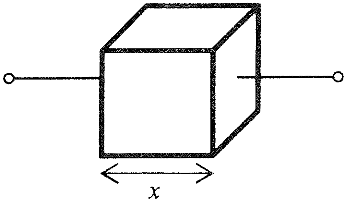
(2021) 23. Four identical resistors P, Q, R and S are connected to a battery of negligible internal resistance as shown.



If the power dissipated by R is 1 W, find the total power output of the battery.

A. 11 W
B. 15 W
C. 19 W
D. 21 W

(2021) 24. The figure shows a metallic cube of side length x. How is its resistance R between any two opposite faces related to x ?



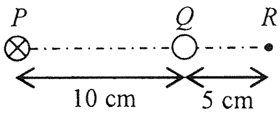
A. $R \propto \frac{1}{x}$

B. $R \propto x$

C. $R \propto x^2$

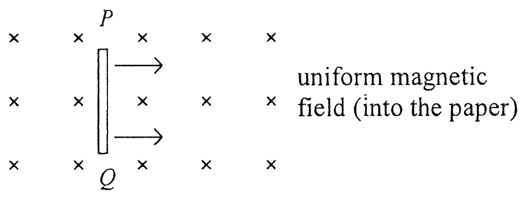
D. $R \propto \frac{1}{x^2}$

(2021) 25. In the figure below, PQR is a straight line with $PQ = 10$ cm and $QR = 5$ cm. A long straight wire carrying a current of 0.3 A (directed into the paper) is placed at P. When another long straight wire carrying a current I is placed at Q, the resultant magnetic field at R becomes zero. Determine the direction and magnitude of I.



	direction of I	**magnitude of I**
A.	into the paper	0.1 A
B.	into the paper	0.9 A
C.	out of the paper	0.1 A
D.	out of the paper	0.9 A

(2021) 26. When a copper rod PQ moves with a constant velocity across a uniform magnetic field as shown, an e.m.f. is induced across the rod.

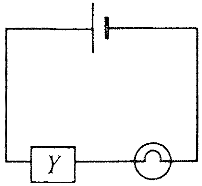


Which of the following statements is/are correct ?

(1) The magnitude of the induced e.m.f. depends on the length of the rod.
(2) Rod PQ acts like a cell providing an e.m.f. with P being its positive terminal.
(3) There is a force acting on the rod to oppose its motion.

A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

(2021) 27. A light bulb is connected in series with a device Y and a cell as shown. Assume that the internal resistance of the cell is negligible and its e.m.f. remains unchanged.

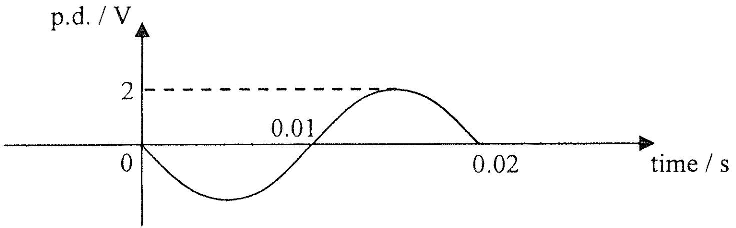


It is found that the brightness of the light bulb decreases with time. Which deductions must be correct ?

(1) The current in Y decreases with time.
(2) The voltage across Y decreases with time.
(3) The power supplied by the cell decreases with time.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

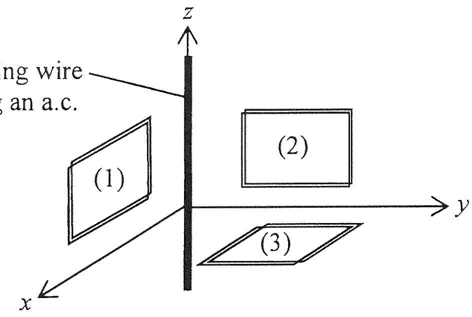
(2021) *28. The graph shows the waveform of a sinusoidal alternating p.d. applied across a 10 Ω resistor.



Find the root-mean-square current in this 10 Ω resistor and the average power dissipated by it.

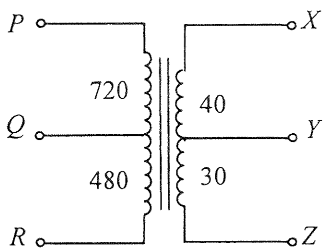
	root-mean-square current / A	average power / W
A.	0.14	0.2
B.	0.14	0.4
C.	0.2	0.2
D.	0.2	0.4

(2021) 29. The figure shows three mutually perpendicular coils (1), (2) and (3) placed near a conducting wire carrying an a.c. along the z-axis direction. In which coil(s) would e.m.f. be induced ?



A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

(2021) *30.



The above figure represents a multi-tapped transformer. The number of turns between the various 'tapping points' are indicated as shown. Which connections should be used for stepping down a voltage from 240 V to 6 V ?

	primary coil	**secondary coil**
A.	*PQ*	*XY*
B.	*PQ*	*YZ*
C.	*PR*	*XY*
D.	*PR*	*YZ*

(2021) 31. A radioactive nuclide plutonium-239 ($^{239}_{94}Pu$) becomes a stable lead-207 isotope ($^{207}_{82}Pb$) after a series of α and β decays. Find the number of β decays in the process.

A. 3
B. 4
C. 5
D. 6

(2021) 32. The activity of a radioactive sample is measured to be 18 MBq. What is its activity 3 half-lives before ?

A. 6 MBq
B. 54 MBq
C. 72 MBq
D. 144 MBq

(2021) 33. Which of the following may contain sources of ionizing radiations ?

(1) sea water
(2) a rock sample
(3) human body

A. (1) only
B. (2) only
C. (2) and (3) only
D. (1), (2) and (3)

(2022) 1. A well-insulated vessel with negligible heat capacity contains warm water at 45 °C. The temperature of warm water drops by 5 °C after mixing with 50 g water at temperature 0 °C added to the vessel. Find the original mass of warm water in the vessel.

A. 400 g
B. 450 g
C. 500 g
D. 550 g

(2022) 2. When ice melts into water at 0 °C, what happens to the molecules during the melting process ?

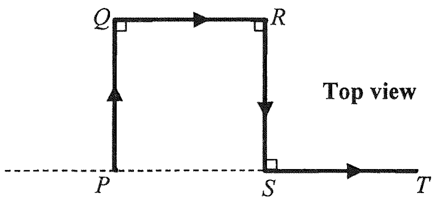
(1) their average separation increases
(2) their average kinetic energy increases
(3) their average potential energy increases

A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

(2022) *3. A weather balloon filled with 21 kg of helium gas has a volume of 120 m^3 at 27 °C. Find the gas pressure in the balloon. Given: mass of one mole of helium gas = 4×10^{-3} kg

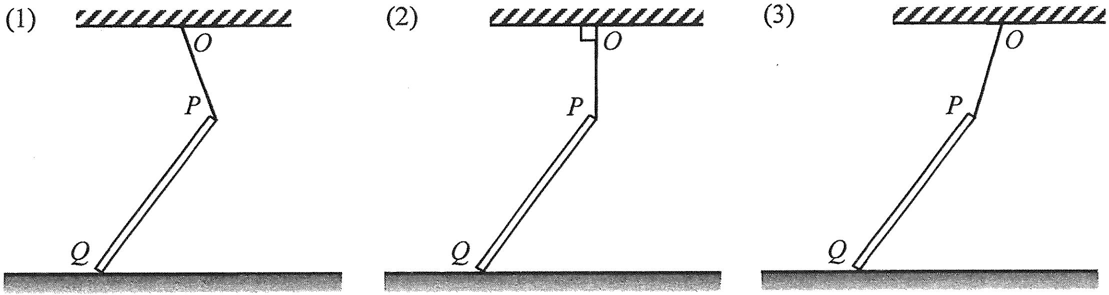
A. 9.93×10^4 Pa
B. 1.00×10^5 Pa
C. 1.05×10^5 Pa
D. 1.09×10^5 Pa

(2022) 4. A car travels with constant speed v along a horizontal road $PQRST$ with four sections of equal length as shown. Determine v if the car's average velocity for the journey $PQRST$ is 20 km h^{-1} in magnitude.



A. 10 km h^{-1}
B. 20 km h^{-1}
C. 40 km h^{-1}
D. It cannot be found as the length of each section is unknown.

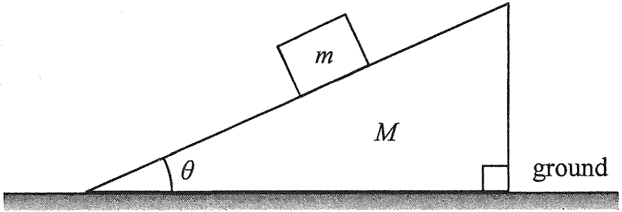
(2022) 5. In each figure below, a uniform rod PQ has its upper end P attached to point O on the ceiling by a light inextensible string. The rod's lower end Q rests on a rough horizontal ground as shown.



In which figure is the frictional force acting on the rod due to the ground towards the right ?

A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

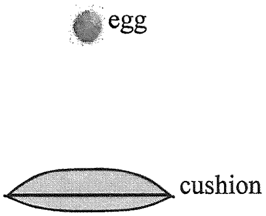
(2022) 6.



A block of mass m is placed on a wedge of mass M as shown. The system remains at rest. What is the normal force exerted on the wedge by the ground ?

A. Mg
B. $(M+m)g$
C. $Mg+mg\cos\theta$
D. $Mg+mg\tan\theta$

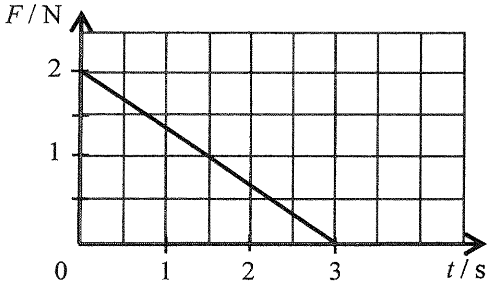
(2022) 7. An egg is released from a height and falls towards a cushion without breaking.



Which of the following is the most probable explanation ?

A. The cushion makes the egg stop at a shorter distance.
B. The cushion helps lengthen the time of impact.
C. The cushion helps reduce part of the gravitational force acting on the egg.
D. The cushion increases the reaction force acting on the egg during impact.

(2022) 8. An object of mass 2 kg moves with velocity 1 m s^{-1} initially. It is acted upon by a force F which varies with time t as shown in the graph. The force and the object's velocity are in the same direction. Find the speed of the object at $t = 3$ s.



A. 1.5 m s^{-1}
B. 2.5 m s^{-1}
C. 3.5 m s^{-1}
D. 4.5 m s^{-1}

(2022) *9. A trolley with a spring-loaded launcher in Figure (1) can project a ball vertically upward. The trolley travels on the ground at a constant speed in a horizontal circle (with centre O) as shown in Figure (2).

Figure (2)

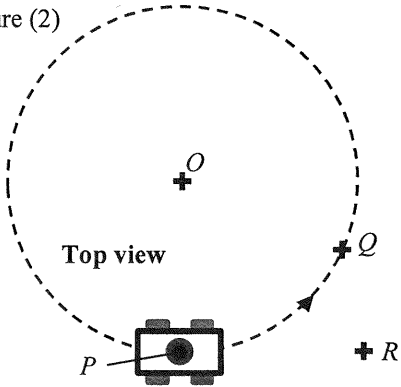
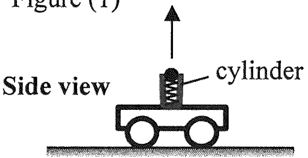


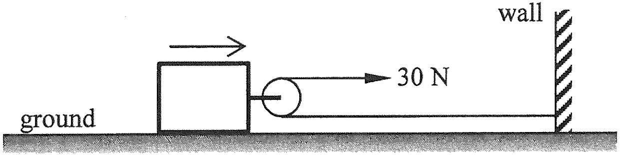
Figure (1)



The ball is projected when the trolley is at point P. After some time, the ball falls back to the ground as the trolley just reaches point Q. Neglecting air resistance, where would the ball land ?

A. O
B. P
C. Q
D. R

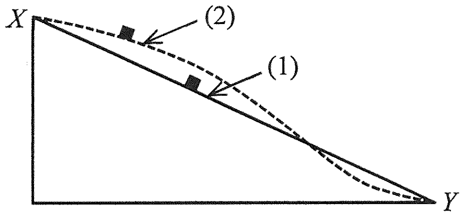
(2022) 10.



On a horizontal ground there is a block attached with a smooth light pulley as shown. A light and inextensible horizontal string, with one end fixed to a wall, passes the pulley. A man exerts a horizontal force of 30 N at the other end of the string and pulls a distance of 4 m. Find the work done by the man if the ground exerts a frictional force of 10 N on the block.

A. 20 J
B. 80 J
C. 100 J
D. 120 J

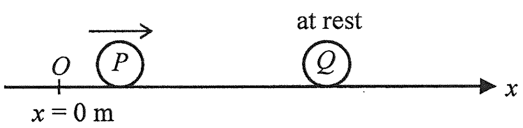
(2022) 11. In the figure below, the straight path (1) and the curved path (2) in a vertical plane are smooth. A small block slides from rest at *X* along the respective paths to *Y*.

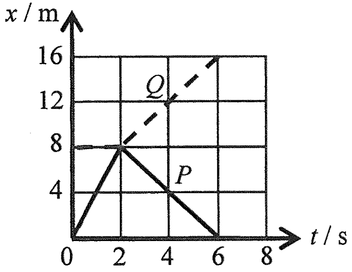


Which of the following about the speed of the block at *Y* and the time taken to reach *Y* is correct ? Neglect air resistance.

	speed at *Y*	time taken to reach *Y*
A.	the same	different
B.	the same	the same
C.	different	the same
D.	different	different

(2022) 12.

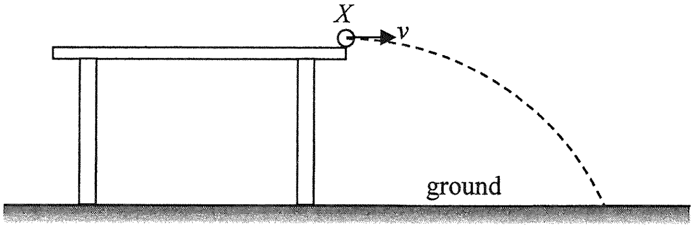




On a smooth horizontal surface, sphere P travelling along the x-axis collides head-on with another sphere Q, which is initially at rest at $x = 8$ m. The graph shows the displacement-time (x-t) relationship of P (solid line) and Q (dotted line). The collision takes place at $t = 2$ s and the collision time is negligible. Which statement is correct ?

A. The collision is perfectly inelastic.
B. The mass of P is larger than that of Q.
C. P still moves along $+x$ direction after collision.
D. P and Q move with the same speed after collision.

(2022) *13. A marble leaves the edge of a table at point X with a horizontal speed v and hits the ground at a certain point as shown.

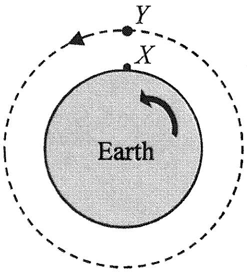


If the marble leaves the table at a higher speed, which of the following would remain unchanged ? Neglect air resistance.

(1) the time of flight of the marble in the air
(2) the acceleration of the marble during its flight in air
(3) the horizontal distance between X and the landing point

A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

(2022) *14. In the figure, X is an object resting on the Earth's equator. Y is a geostationary satellite moving in a circular orbit above the equator such that it always appears stationary to an observer on Earth.



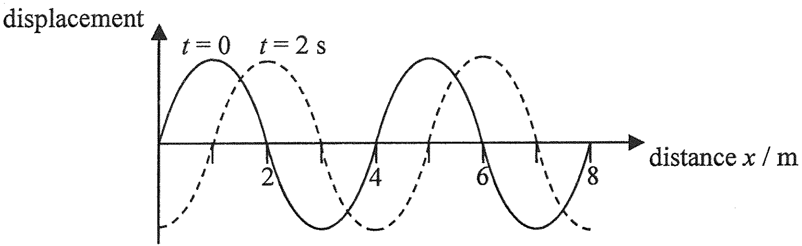
NOT drawn to scale

Which descriptions about the motion of X and Y are correct ?

(1) The motion of X and Y takes the same period.
(2) X is moving with a slower speed.
(3) X has a greater acceleration.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

(2022) 15. The figure shows the displacement-distance graph at time $t = 0$ and $t = 2$ s respectively of a wave travelling to the right.

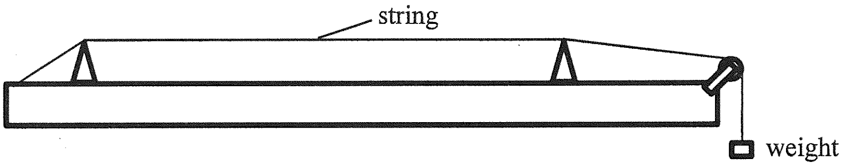


Which statements about the wave are correct ?

(1) The wavelength is 4 m.
(2) The period is 4 s.
(3) The speed of the wave can be 2.5 m s^{-1}.

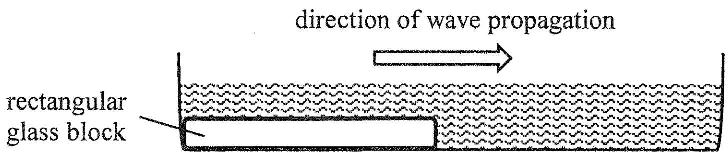
A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

(2022) 16. Referring to the set-up below, which of the following combinations would give the highest speed of wave propagation along the string when the string is plucked ?



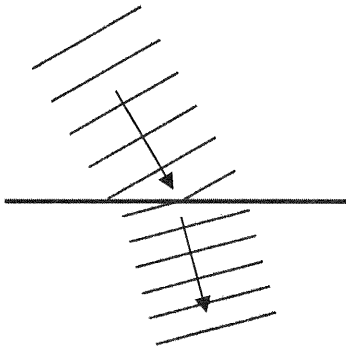
	tension of the string	radius of the cross-section of the string
A.	T	r
B.	T	$2r$
C.	$2T$	r
D.	$2T$	$2r$

(2022) 17. Plane water waves travel from shallow water to deep water in a ripple tank as shown below.

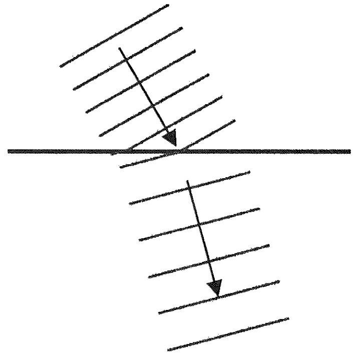


Which diagram correctly shows the top view of the wavefronts in the ripple tank ?

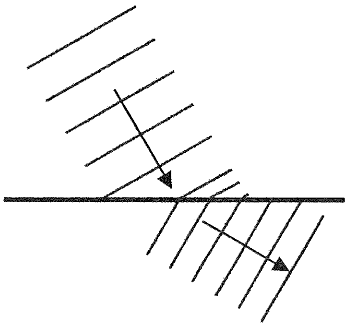
A.



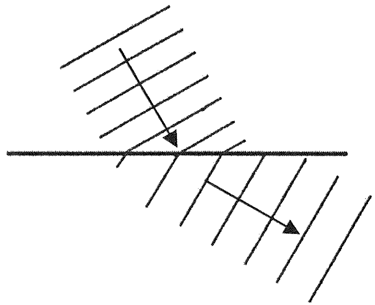
B.



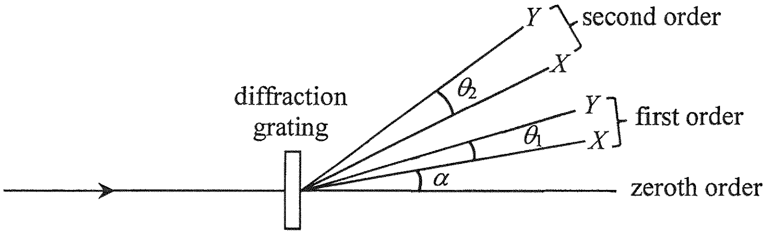
C.



D.



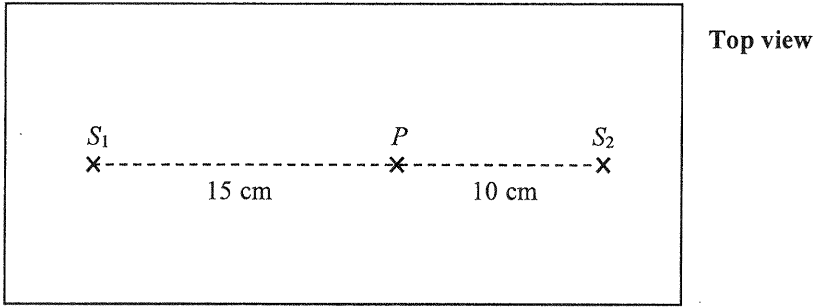
(2022) *18.



A light beam which consists of monochromatic lights X and Y is incident normally on a diffraction grating. The figure shows the spectra of the first two orders. The diffraction angle of X in the first order is α. The angular separation between X and Y in the first order is θ_1 and that in the second order is θ_2. Which statement **must be** correct ?

A. The wavelength of Y is shorter than that of X.
B. α would be larger if the grating spacing is smaller.
C. θ_1 is independent of the grating spacing.
D. $\theta_2 = 2\theta_1$

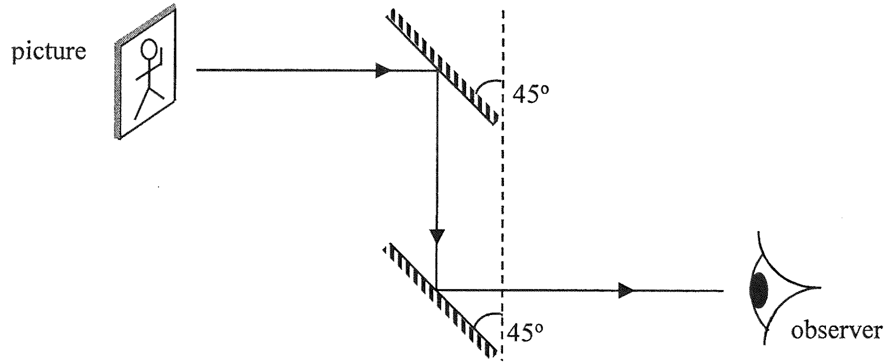
(2022) 19. In the ripple tank shown below, dippers S_1 and S_2 are vibrating in phase with frequency f and generate two sets of waves towards each other at a speed of 20 cm s^{-1}. Destructive interference occurs at point P on the straight line S_1S_2.



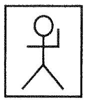
Which of the following is a possible value of f ?

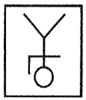
A. 24 Hz
B. 20 Hz
C. 18 Hz
D. 16 Hz

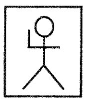
(2022) 20. A periscope is formed by two plane mirrors as shown.

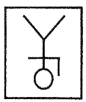


Which image will the observer see ?

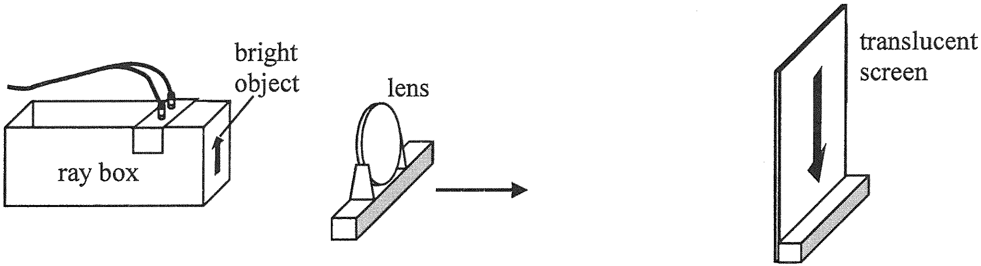
A. 

B. 

C. 

D. 

(2022) 21. In the set-up below, the separation between the bright object and the translucent screen is fixed. A lens is moved from the object towards the screen as shown.



The first sharp image formed is inverted as shown and its length is 9 cm. When the lens is moved further towards the screen, a second sharp image of length 1 cm is observed. Which of the following statements is/are correct ?

(1) The second image is erect.
(2) The length of the object is 3 cm.
(3) There are at most two positions of the lens that can give a sharp image on the screen when the lens is moved.

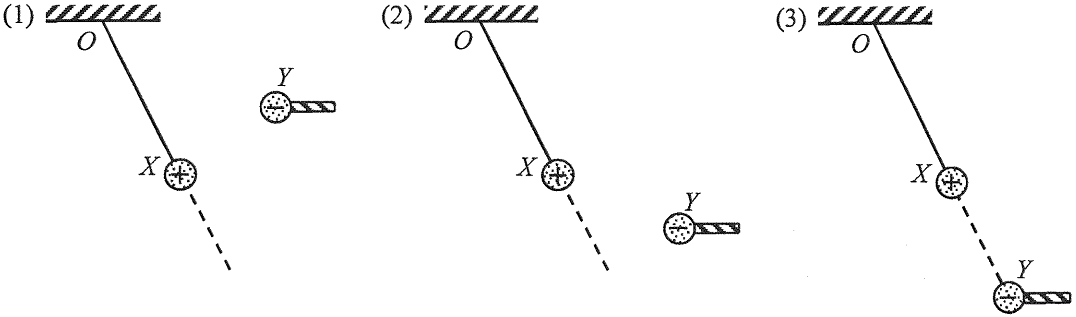
A. (1) only
B. (2) only
C. (2) and (3) only
D. (1), (2) and (3)

(2022) 22. Which statements about ultrasound are correct ?

(1) Ultrasound is longitudinal waves.
(2) Ultrasound needs media to propagate.
(3) The speed of ultrasound in glass is higher than that in air.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

(2022) 23. A positively charged sphere X of mass m is suspended from a fixed point O by a nylon thread. Another negatively charged sphere Y at the end of an insulating rod is held in various positions as shown. O, X and Y are in the same vertical plane.

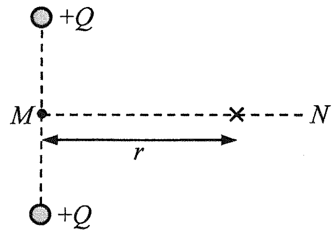


In which situation(s) could X be kept at rest as shown ?

A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

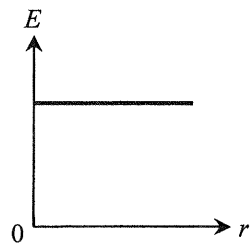
(2022)

*24.

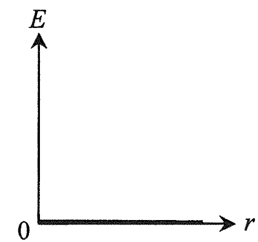


Two positive point charges $+Q$ are fixed as shown above. MN is the perpendicular bisector of the line joining the charges. Which graph correctly shows the variation of the electric field strength E on line MN with distance r from M ?

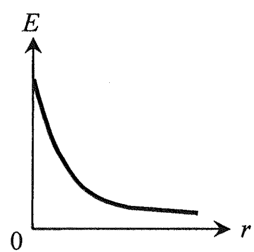
A.



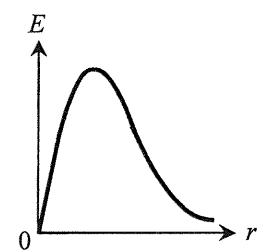
B.



C.



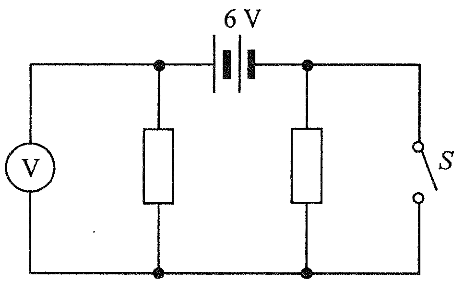
D.



(2022) 25. An electrical device is charged with a steady d.c. of 60 mA for 30 minutes. Find the **number of electrons** that pass through the device during charging.

A. 108
B. 1800
C. 3.75×10^{17}
D. 6.75×10^{20}

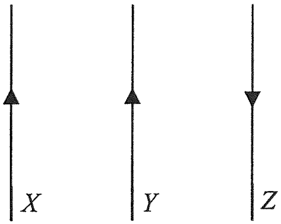
(2022) 26.



In the above circuit, the resistors are identical and the 6 V battery has negligible internal resistance. Which of the following is the set of readings on the voltmeter when (1) S is open; (2) S is closed ?

	***S* open**	***S* closed**
A.	0 V	6 V
B.	3 V	6 V
C.	6 V	0 V
D.	6 V	3 V

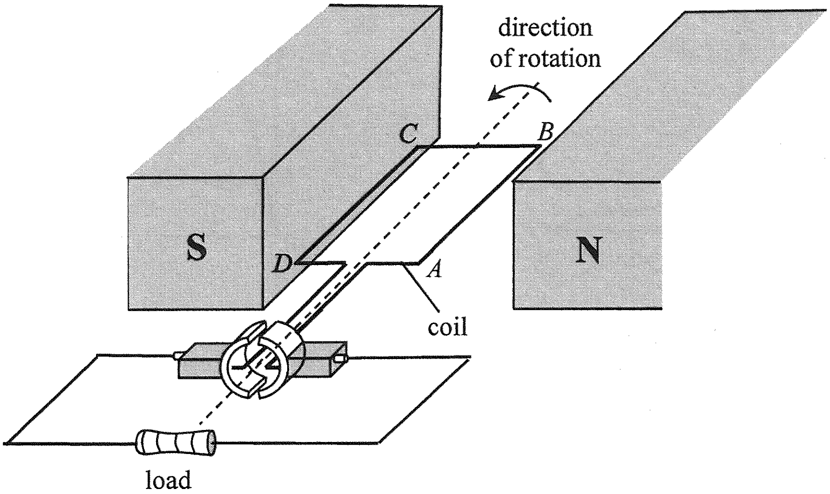
(2022) 27. The figure below shows three parallel wires carrying currents in the directions shown.



If one of the wires experiences zero resultant magnetic force, that wire

A. must be X.
B. must be Y.
C. must be Z.
D. may be Y or Z.

(2022) 28.

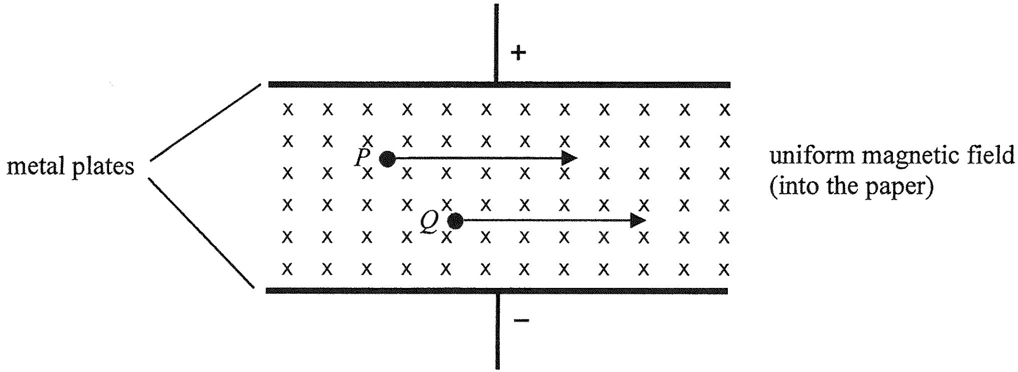


The figure shows the structure of a simple generator. Which of the following statements is/are correct ?

(1) The magnetic force acting on side *AB* of the coil is upward at the instant shown.
(2) The commutator would reverse the direction of current in the coil whenever the coil passes its vertical position.
(3) The current flowing through the load is an unsteady d.c.

A. (1) only
B. (2) only
C. (3) only
D. (1) and (3) only

(2022) *29. Charged particles P and Q are moving in a region with mutually perpendicular uniform electric and magnetic fields as shown.



If both particles are not deflected by the fields, which of the following statements **must be** correct ? Neglect the effects of gravity.

(1) They are positively charged.
(2) They are moving with the same velocity.
(3) They have the same charge to mass ratio.

A. (1) only
B. (2) only
C. (1) and (3) only
D. (2) and (3) only

(2022) *30. Arrange the following in ascending order of power delivered when each of them is connected to the same resistor.

(1) a 100 Hz sinusoidal a.c. of peak voltage 2 V
(2) a 50 Hz sinusoidal a.c. of r.m.s. voltage 2 V
(3) a steady d.c. of voltage 1.5 V

A. (1) (3) (2)
B. (2) (3) (1)
C. (1) (2) (3)
D. (2) (1) (3)

(2022) 31. Which of the following statements about X-rays is **INCORRECT** ?

A. X-rays can be produced by high-speed electrons hitting a metal target.
B. X-rays are a kind of electromagnetic waves.
C. X-rays can be deflected by an electric field.
D. Although X-rays do not carry charges, they can cause ionization.

(2022) *32. Radioactive sources X and Y have initial activities 100 kBq and 200 kBq respectively. One day later, their activities are found to be equal. After another day, the activity of X becomes 80 kBq. Deduce the corresponding activity of Y.

A. 40 kBq
B. 50 kBq
C. 89 kBq
D. 160 kBq

(2022) 33. What is the value of x for the fission reaction below ?

$$^{235}_{92}U + ^{1}_{0}n \rightarrow ^{92}_{36}Kr + ^{141}_{56}Ba + x^{1}_{0}n$$

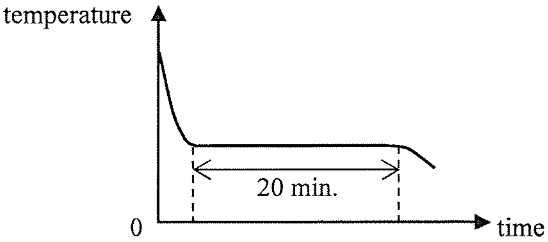
A. 1
B. 2
C. 3
D. 4

(2023) 1. Which of the following statements involving heat transfer via radiation is/are correct ?

(1) Objects hotter than the surroundings do not absorb heat by radiation.
(2) The silvery surfaces of survival blankets help the body absorb heat from the surroundings by radiation.
(3) The cooling fins of a car engine should preferably be black in colour.

A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

(2023) 2.

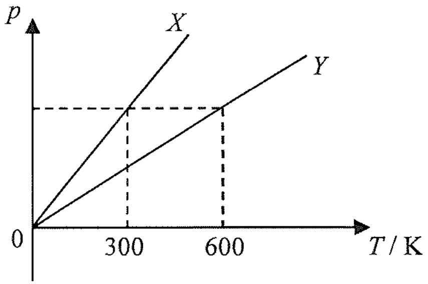


A hot liquid substance of mass 0.3 kg and with heat capacity 600 J °C^{-1} is left to cool in air. Its temperature falls at a rate of 2 °C per minute just before it begins to solidify. Then its temperature remains steady for 20 minutes until all the liquid just solidifies. Estimate the specific latent heat of fusion of the substance.

A. 20000 J kg^{-1}
B. 24000 J kg^{-1}
C. 48000 J kg^{-1}
D. 80000 J kg^{-1}

(2023)

*3. A fixed mass of an ideal gas contained in a closed vessel X of volume V is heated. Line X in the figure shows the variation of pressure p with temperature T of the gas. The experiment is repeated with another closed vessel Y containing the same amount of this ideal gas and line Y represents the result.



Find the volume of vessel Y.

A. 0.25 V
B. 0.5 V
C. 2V
D. 4V

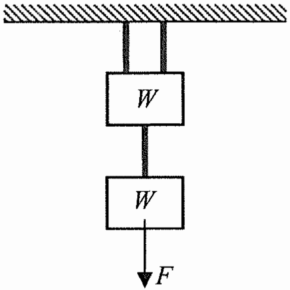
(2023) 4. Which of the following are vector quantities ?

(1) acceleration
(2) momentum
(3) work done

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

(2023)

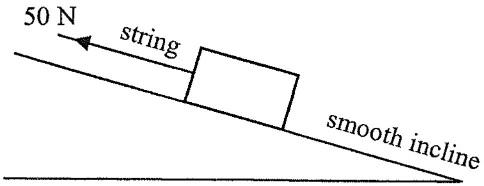
5. Two blocks, each of weight W, are hung by three identical light strings as shown. Each string has a limiting tension of $2W$, i.e. it will break for a tension larger than this value.



Force F acts vertically on the lower block. What is the largest value of F without breaking any of the strings ?

A. $0.5W$
B. W
C. $1.5W$
D. $3W$

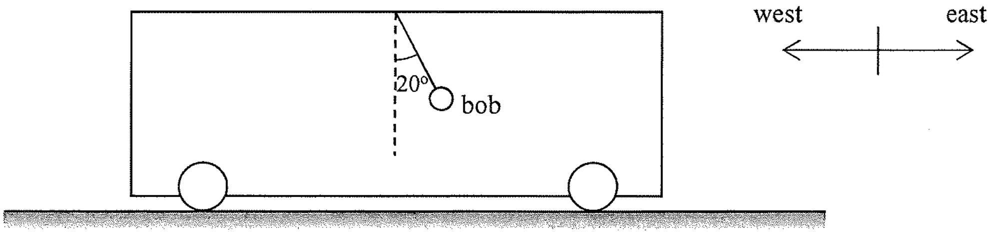
(2023) 6. On a smooth incline, a block of weight 120 N is kept stationary by a string with tension 50 N parallel to the incline.



The string is then cut and the block accelerates down the incline. What is the **net force acting on the block** before and after the string is cut ?

	before the string is cut	after the string is cut
A.	50 N	70 N
B.	50 N	50 N
C.	0 N	70 N
D.	0 N	50 N

(2023) 7. A bob is suspended from the ceiling of a train moving along a horizontal east-west rail.



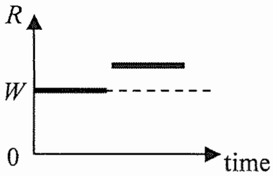
The bob slants towards east and makes an angle of 20° with the vertical as shown. What is the direction and magnitude of the acceleration of the train ? ($g = 9.81\ \text{m s}^{-2}$)

	direction	**magnitude**
A.	due west	$3.36\ \text{m s}^{-2}$
B.	due west	$3.57\ \text{m s}^{-2}$
C.	due east	$3.36\ \text{m s}^{-2}$
D.	due east	$3.57\ \text{m s}^{-2}$

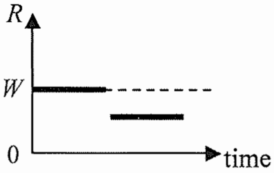
(2023)

8. In an amusement park ride, a boy of weight W undergoes free fall vertically in a chair at first and then decelerates uniformly. Which graph below correctly shows the time variation of the reaction force R acting on the boy by the chair ? Neglect air resistance.

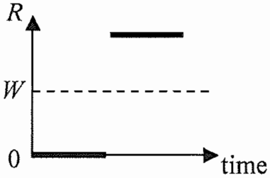
A.



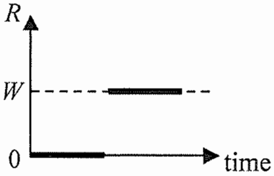
B.



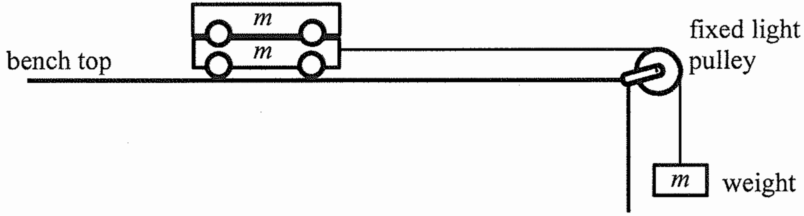
C.



D.



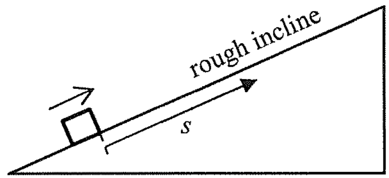
(2023) 9. The figure shows a set-up in which a weight of mass m is attached to two trolleys stacked together, each trolley of mass m, via a light inextensible string.



After the weight is released from rest, which of the following is correct ? Assume that all contact surfaces are smooth and neglect air resistance. g is the acceleration due to gravity.

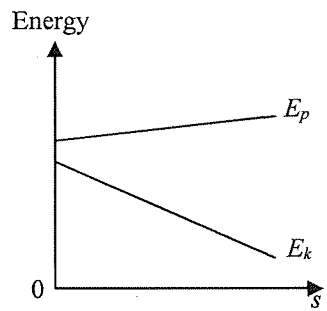
	magnitude of the acceleration	**tension in the string**
A.	$g/2$	smaller than mg
B.	$g/2$	equal to mg
C.	$g/3$	smaller than mg
D.	$g/3$	equal to mg

(2023) 10. A block is projected upwards along a **rough** incline which has a constant frictional force acting on the block.

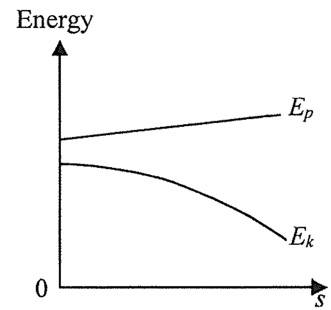


Which of the following correctly shows the variation of potential energy E_p and kinetic energy E_k with distance s that the block travelled up the incline ? Neglect air resistance.

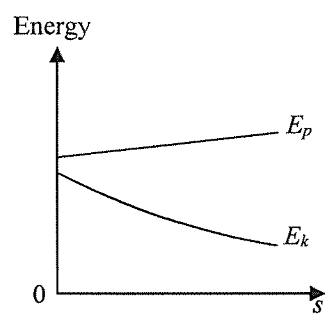
A.



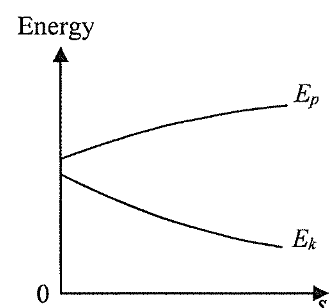
B.



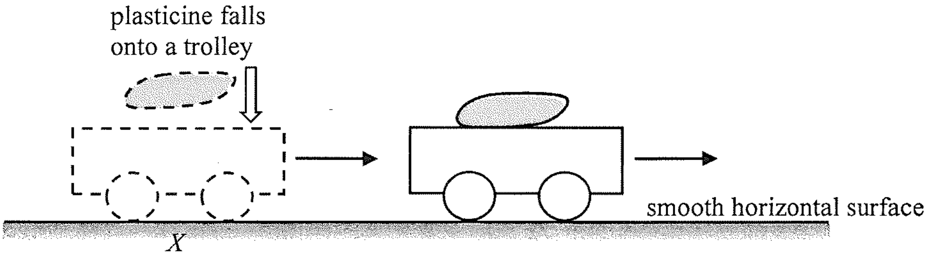
C.



D.



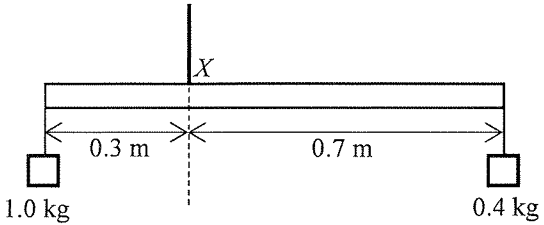
(2023) 11. A trolley moves with a constant velocity along a smooth horizontal surface. When the trolley reaches point X, a plasticine falls vertically onto it. After the collision, they stick together and continue to move forward.



Which description about the **total linear momentum of the trolley and the plasticine** just before and just after collision is correct ?

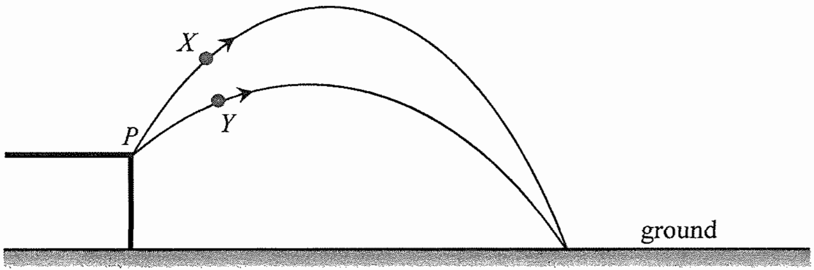
	along the horizontal direction	**along the vertical direction**
A.	conserved	conserved
B.	conserved	not conserved
C.	not conserved	conserved
D.	not conserved	not conserved

(2023) 12. A uniform metre rule of a certain weight is hung by a string at X such that it is kept in equilibrium with weights hanging at both ends as shown. If the 1.0-kg weight is shifted 0.1 m towards X, what distance should the 0.4-kg weight be shifted towards X in order to restore equilibrium ?



A. 0.1 m
B. 0.2 m
C. 0.25 m
D. 0.45 m

(2023) *13. Two identical particles X and Y are projected from point P **simultaneously with the same initial speed** but with different angles as shown. They finally hit the ground at the same point via different paths.

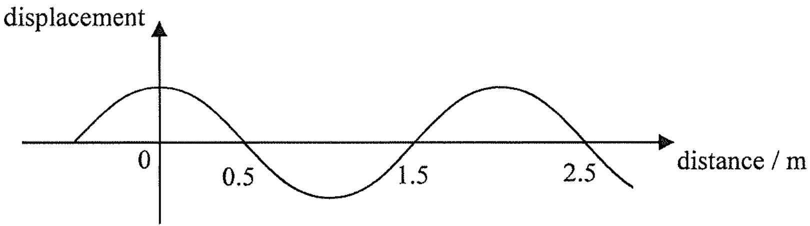


Which of the following statements is/are correct ? Neglect air resistance.

(1) They hit the ground at the same time.
(2) They hit the ground with the same speed.
(3) The kinetic energy of Y is greater than that of X at their respective maximum heights.

A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

(2023) 14.

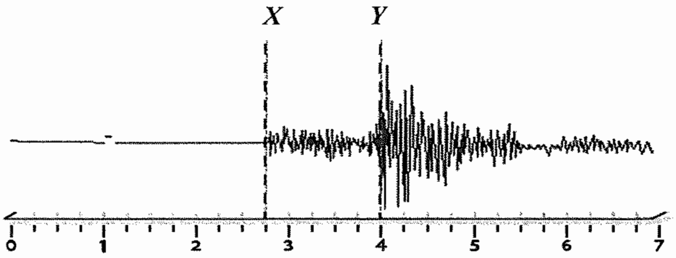


The above figure shows the displacement-distance graph of a wave travelling at a speed of 10 m s^{-1}. What is the period of this wave ?

A. 0.10 s
B. 0.15 s
C. 0.20 s
D. 0.25 s

(2023)

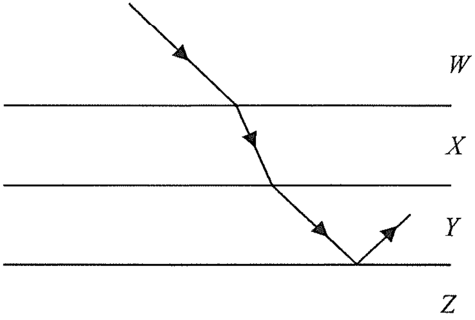
15. Earthquakes produce P-waves (longitudinal) and S-waves (transverse) which travel at 7.0 km s^{-1} and 4.0 km s^{-1} respectively. On a given day, a seismometer detected the P- and S- waves from a shallow earthquake (see the figure below) with its quake centre located at a distance D from the seismometer.



Which deduction below is correct ?

	Y in the figure denotes the arrival of	D / km
A.	P-waves	225
B.	P-waves	700
C.	S-waves	225
D.	S-waves	700

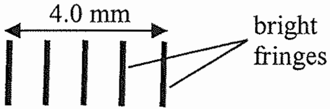
(2023) 16. The figure shows a light ray passing through the horizontal boundaries between media W, X, Y and Z. The light ray in W is parallel to that entered into Y from X. Total internal reflection occurs at the boundary between Y and Z.



Which medium has the highest refractive index ?

A. W
B. X
C. Y
D. Z

(2023) *17.

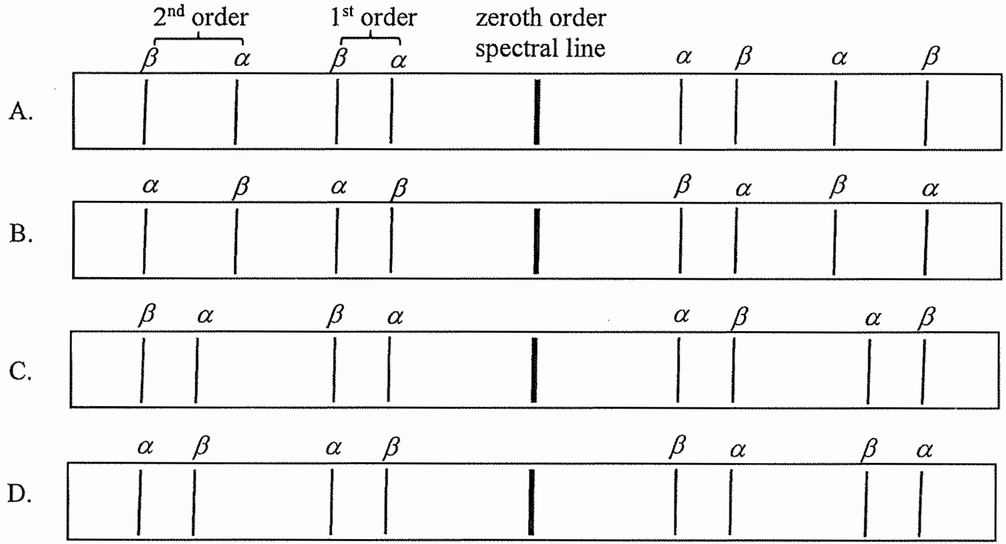


In Young's double-slit experiment, monochromatic light of wavelength λ is used and the slit separation is a. On a screen placed at a distance of 1.8 m from the double slits, the separation between five consecutive bright fringes is 4.0 mm. Which relationship below is correct ?

A. $a = 1800\ \lambda$
B. $a = 2250\ \lambda$
C. $a = 7200\ \lambda$
D. $a = 11250\ \lambda$

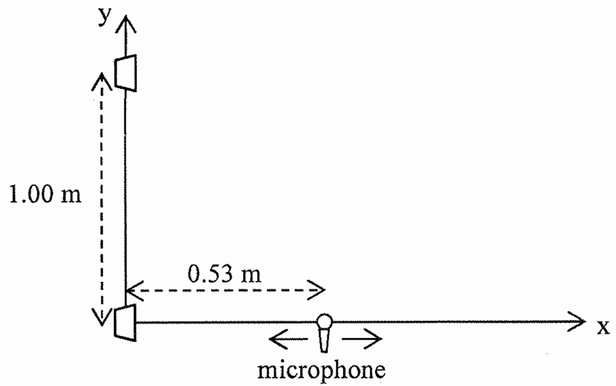
(2023)

18. A diffraction grating is used to observe the α (656 nm) and β (486 nm) lines emitted from a gas discharge tube. Which of the following best represents the pattern observed ?



(2023)

19. Two loudspeakers separated by 1.00 m along the y-axis emit coherent sound waves which are in phase. When a microphone is moved along the x-axis, only one maximum is detected at a distance of 0.53 m as shown. Find the wavelength of the sound waves.



A. 0.60 m
B. 0.75 m
C. 1.00 m
D. 1.20 m

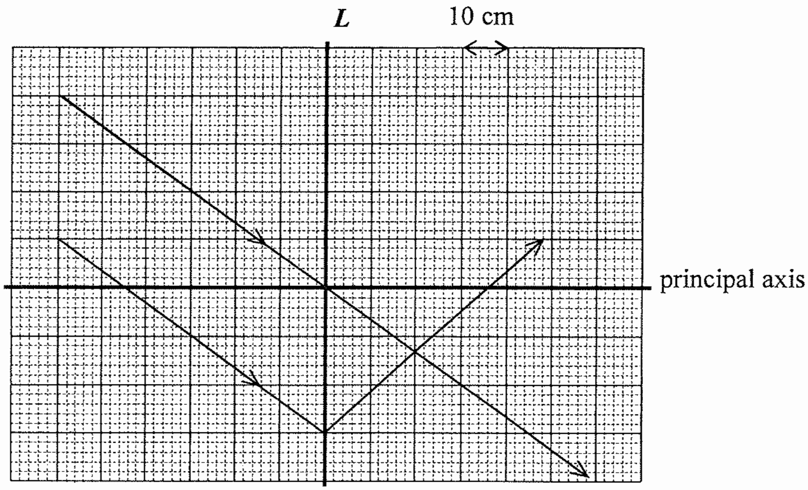
(2023)

20. Virtual images of an object formed by a single lens are always

(1) erect.
(2) on the opposite side of the lens to the object.
(3) smaller than the object.

A. (1) only
B. (3) only
C. (2) and (3) only
D. (1), (2) and (3)

(2023) 21.

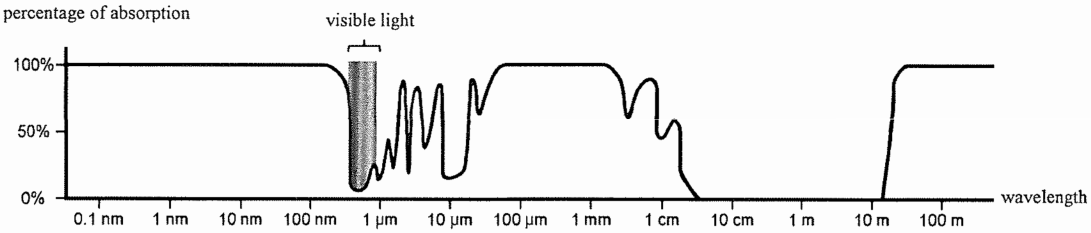


The ray diagram shows the refraction of two parallel light rays through a lens *L*. What is the type and the focal length of *L* ?

	type	focal length
A.	convex lens	20 cm
B.	convex lens	36 cm
C.	concave lens	20 cm
D.	concave lens	36 cm

(2023)

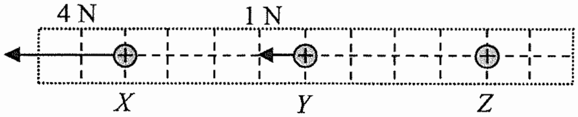
22. When measuring electromagnetic waves from outer space on the Earth's surface, the percentage of absorption by the atmosphere for various wavelengths of electromagnetic waves is shown below.



Referring to this absorption graph, determine which statement below is correct.

A. Gamma rays from galaxies can be observed on Earth's surface.
B. Nearly all ultra-violet radiation can reach Earth's surface.
C. Infra-red radiation from stars can reach Earth's surface without absorption.
D. GHz radio waves (wavelength about 0.5 m) are the best for communications between ground station and space station in the upper atmosphere.

(2023) 23. Positive point charges X, Y and Z are placed along a straight line with Y at the mid-point between X and Z. Assume that the only interaction among them is the electrostatic force.

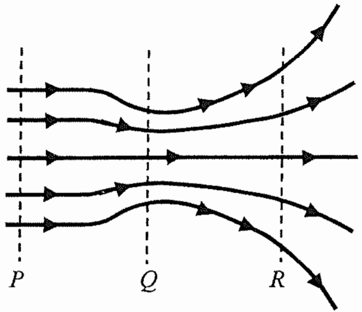


At the instant shown, the net forces acting on X and Y respectively are 4 N and 1 N both towards the left. At that instant, which statements about Z are correct ?

(1) The net force acting on Z is towards the right.
(2) The net force acting on Z is 5 N in magnitude.
(3) The charge of Z must be larger than that of X.

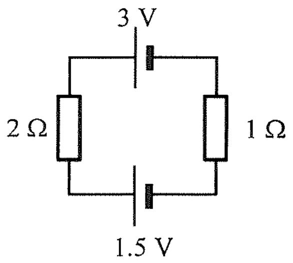
A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

(2023) 24. The figure shows the field lines of an electric field. Arrange the respective electric field strength E_P, E_Q and E_R of regions P, Q and R in ascending order.



A. $E_P < E_Q < E_R$
B. $E_Q < E_R < E_P$
C. $E_R < E_P < E_Q$
D. $E_Q < E_P < E_R$

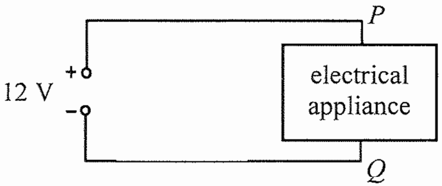
(2023) 25. Two cells of negligible internal resistance are connected to two resistors as shown.



Which of the following is correct ?

	direction of current	current in the circuit
A.	anticlockwise	0.5 A
B.	clockwise	0.5 A
C.	anticlockwise	1.5 A
D.	clockwise	1.5 A

(2023) 26. The figure shows an electrical appliance connected to a potential difference of 12 V.

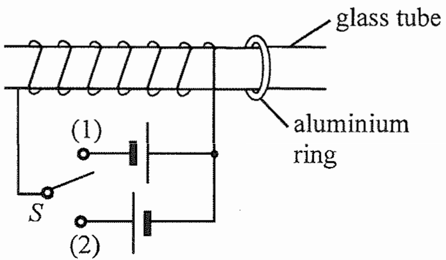


When a charge of +2 C passes from P to Q in the circuit, how much electrical energy is supplied to the appliance ?

A. 2 J
B. 6 J
C. 12 J
D. 24 J

(2023)

27. A solenoid is tightly wound around a smooth horizontal glass tube as shown. A movable aluminium ring is threaded on the right side of the tube.



Initially the ring is at rest and the 2-way switch *S* is in open circuit. In what direction would the ring move at the moment when *S* is connected in turns to (1) and (2) ?

	when *S* is connected to (1)	**when *S* is connected to (2)**
A.	to the left	to the left
B.	to the right	to the right
C.	to the right	to the left
D.	to the left	to the right

(2023) 28. Which of the following statements about domestic circuits is/are correct ?

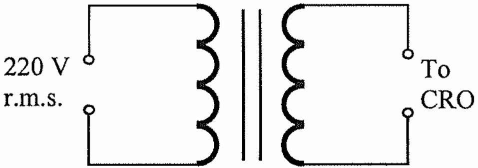
(1) The live (L) wire is sometimes positive and sometimes negative with respect to the neutral (N) wire.

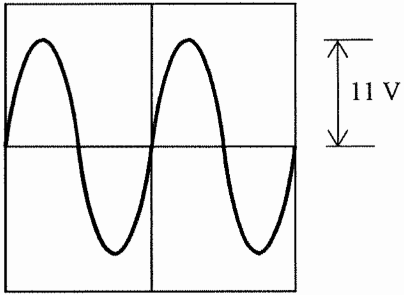
(2) The kilowatt-hour meter measures the total power consumed by the domestic electrical appliances.

(3) With more electrical appliances connected, the total resistance of the domestic circuit becomes smaller.

A. (1) only
B. (2) only
C. (1) and (3) only
D. (2) and (3) only

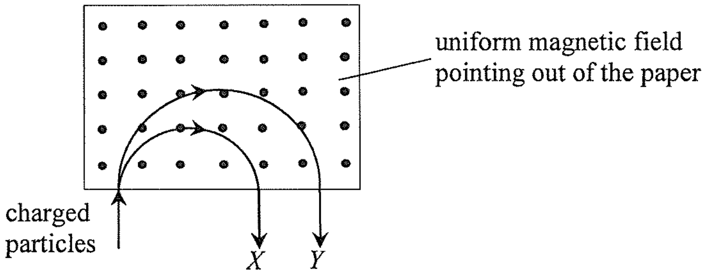
(2023) *29. A transformer is connected to a.c. mains of 220 V r.m.s. value. The peak value of the output voltage displayed on a CRO is found to be 11 V. If the transformer has 2000 turns in its primary coil, find the number of turns in the secondary coil.





A. 71
B. 100
C. 141
D. 200

(2023) *30. The figure shows two charged particles X (mass m_X) and Y (mass m_Y) which enter a region of uniform magnetic field pointing out of the paper. X and Y, having the **same speed and charge**, move along the paths in the plane of the paper as shown below.



X and Y are

A. positively charged and $m_X > m_Y$.
B. positively charged and $m_X < m_Y$.
C. negatively charged and $m_X > m_Y$.
D. negatively charged and $m_X < m_Y$.

(2023)

31. A radioactive source contains a radioactive nuclide X which decays to become a stable nuclide Y. Besides X there is no other radioactive nuclides in this source. Which of the following statements **must be correct** after a time of one half-life of X has elapsed ?

(1) The number of nuclides of X decreases to about half of its initial value.
(2) The number of nuclides of Y increases to about twice of its initial value.
(3) The activity of the radioactive source falls to about half of its initial value.

A. (1) only
B. (2) only
C. (1) and (3) only
D. (2) and (3) only

(2023)

*32. When a proton (^1_1H) bombards a lithium nucleus (^7_3Li), two α particles (^4_2He) are produced.

$$^1_1H + ^7_3Li \rightarrow 2\ ^4_2He$$

The energy released in this nuclear reaction is 17.32 MeV. Find the mass of a lithium nucleus.

Given: mass of ^1_1H = 1.0078 u

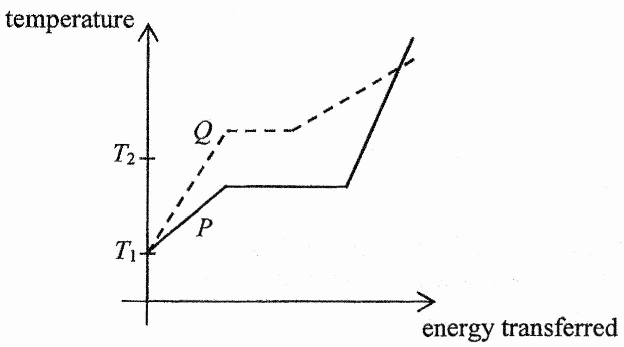
mass of ^4_2He = 4.0026 u

A. 6.9788 u
B. 6.9974 u
C. 7.0017 u
D. 7.0160 u

(2023) 33. Which statement about nuclear fission is correct ?

A. It involves two nuclei combined together.
B. All nuclear fission reactions are spontaneous.
C. Rate of fission reaction is temperature dependent.
D. Fission of heavy nuclei yields products that are more stable in terms of energy.

(2024) 1. Substance P and Q are of the same mass and both are solids at temperature T_1. The graph below shows the variation of temperature of the substances with energy transferred.



Which statement below is correct ?

A. P has a higher melting point than Q.
B. The specific latent heat of fusion of P is smaller than that of Q.
C. The specific heat capacity of solid P is greater than that of solid Q.
D. At temperature T_2, P is gas while Q is solid.

(2024)

2. If 0.5 kg of water at 50 °C is mixed with 2.0 kg of ice at 0 °C, what is the final temperature of the mixture ?

Given: specific heat capacity of water = 4200 J kg^{-1} °C^{-1}

specific latent heat of fusion of ice = 3.34×10^5 J kg^{-1}

A. −54 °C
B. 0 °C
C. 10 °C
D. 25 °C

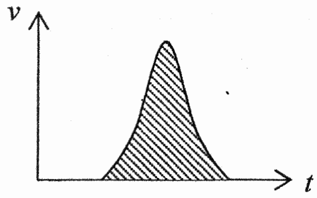
(2024)

*3. The best vacuum attained in the laboratory has a pressure of about 10^{-8} Pa. Estimate the order of magnitude of the number of air molecules in 1 cm^3 of such 'vacuum' at room temperature.

A. 10^4
B. 10^6
C. 10^8
D. 10^{12}

(2024)

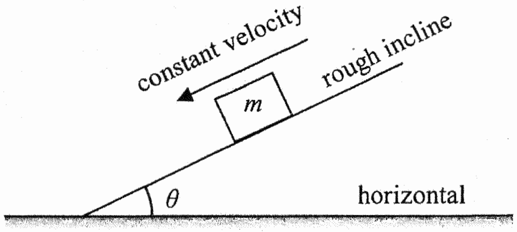
4.



What physical quantity does the shaded area represent in the above velocity-time (v-t) graph ?

A. acceleration
B. change of velocity
C. momentum
D. displacement

(2024) 5.

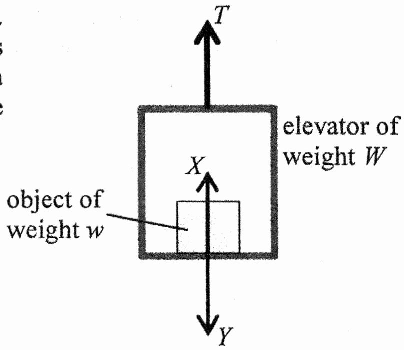


A block of mass m slides down along a fixed rough incline with constant velocity as shown. The incline makes an angle θ with the horizontal. What is the magnitude of the force exerted on the incline by the block ?

A. zero
B. mg
C. $mg \sin\theta$
D. $mg \cos\theta$

(2024)

6. An object of weight w is placed inside an elevator of weight W. A force T pulls the elevator upward. A **supporting force** X acts on the object while the **floor of the elevator experiences a force** Y due to the presence of the object. Which of the following is an action and reaction pair ?



A. T and $(W + w)$
B. w and X
C. X and Y
D. It cannot be determined as the direction of acceleration of the elevator is unknown.

(2024) 7. In an experiment, a horizontal force F acts on a block placed on a rough horizontal surface as shown in Figure (a). Figure (b) shows the variation of the block's kinetic energy E_k with the distance s it travelled. Neglect air resistance and assume frictional force unchanged.

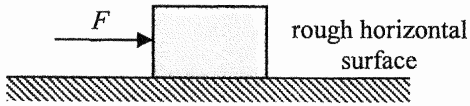


Figure (a)

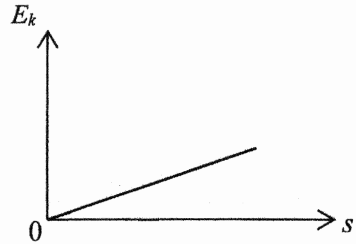
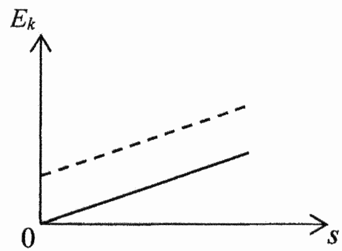


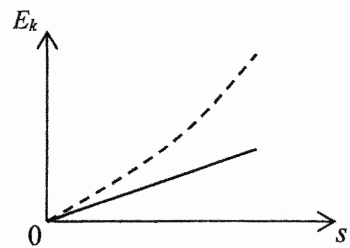
Figure (b)

If the experiment is repeated with force F doubled, which of the following graphs represents the expected result (in dotted line) ?

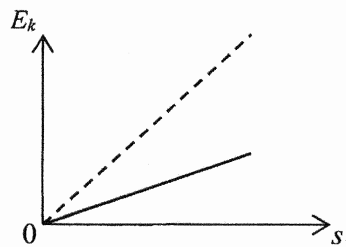
A.



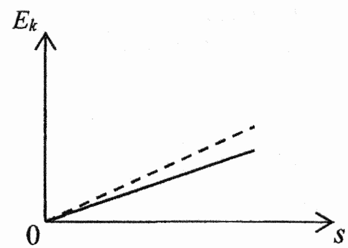
B.



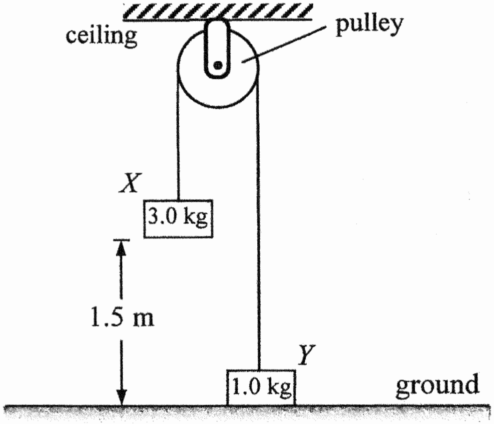
C.



D.



(2024) 8. Block X and Y of mass 3.0 kg and 1.0 kg respectively are connected by a light inextensible string passing over a smooth light pulley fixed to the ceiling as shown. The blocks are released from rest. Find the time taken for block X to reach the ground. Neglect air resistance.

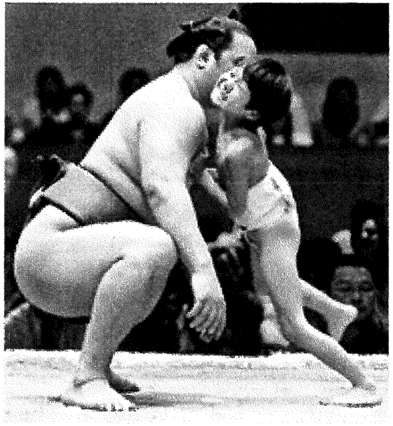


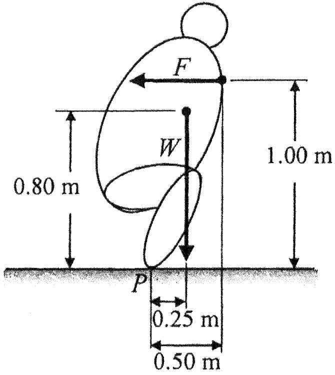
A. 0.55 s
B. 0.64 s
C. 0.68 s
D. 0.78 s

(2024) 9. A person thought that when he pulls his hair with a large enough force, eventually he will be able to pull himself off the ground. This is a misconception because

A. the pulling force is acting internally within that person but it is not an external force acting on the person.
B. it is impossible to generate a pulling force larger than a person's weight.
C. the pulling force is balanced by the person's weight.
D. it is impossible for the pulling force to overcome the force due to atmospheric pressure.

(2024) 10. A child tries to push a 150 kg (sumo) wrestler in order to topple him backwards in a game. The simplified diagram indicates the weight W of the wrestler and the horizontal pushing force F acting on the wrestler by the child. P is the wrestler's contact point on the ground.



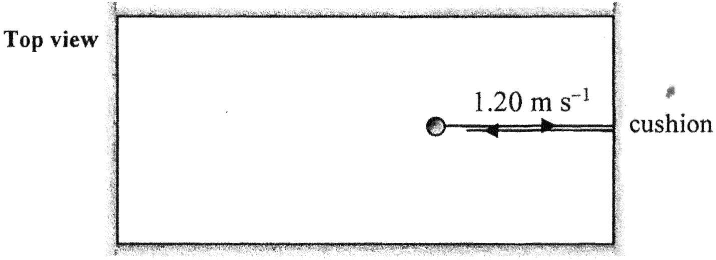


Find the minimum value of F required.

A. 368 N
B. 736 N
C. 1177 N
D. 2354 N

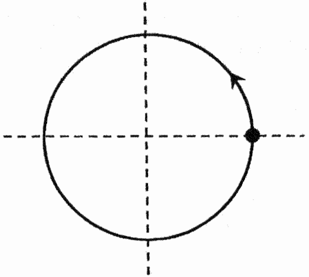
(2024)

11. On a billiard table, a billiard ball of mass 0.16 kg and moving with a uniform speed of 1.20 m s^{-1} collides elastically with the cushion. It rebounds with the same speed normal to the cushion as shown. If the impact time is 0.0003 s, find the average force acting on the ball by the cushion during collision.



A. 0 N
B. 640 N
C. 1280 N
D. 2560 N

(2024) *12.



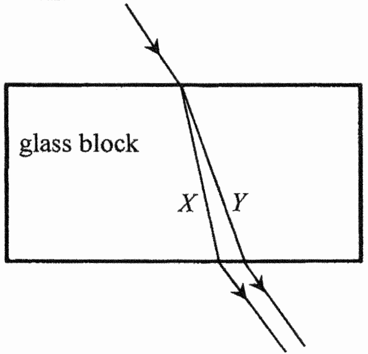
The figure shows a particle performing uniform circular motion. Which of the following correctly indicates the directions of its velocity and acceleration at the instant shown ?

	velocity	**acceleration**
A.	↑	←
B.	↖	←
C.	↖	→
D.	↑	→

(2024) *13. A satellite orbits the Earth in a circular orbit of radius r with a period of 24 hours. The period of another satellite which orbits the Earth in a circular orbit of radius $2r$ is

A. less than 24 hours.
B. 24 hours.
C. between 24 hours and 48 hours.
D. more than 48 hours.

(2024) 14. A light beam consisting of monochromatic light beam X and Y is incident on a rectangular glass block. It splits into two beams as shown.



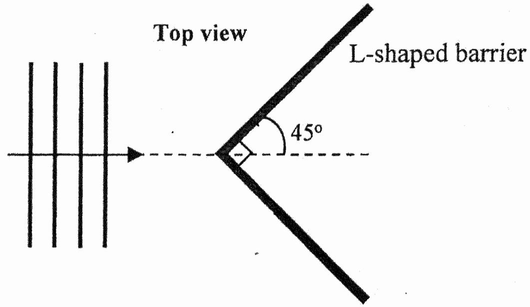
Which of the following comparisons of beam X and Y is/are correct ?

(1) X has a higher frequency.
(2) X has a larger speed in glass.
(3) In the glass block, the critical angle for X is greater.

A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

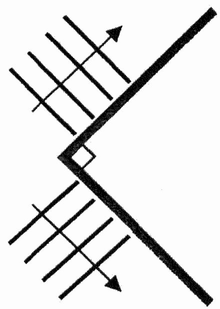
(2024)

15.

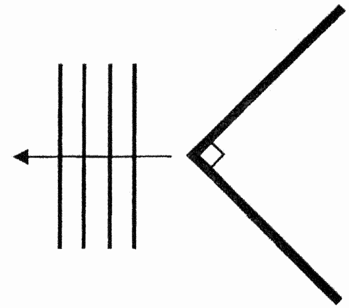


In a ripple tank, a series of plane water waves approach an L-shaped barrier as shown above. Which diagram below best shows the wave pattern reflected from the barrier ?

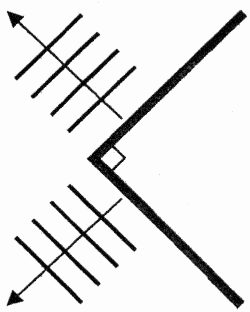
A.



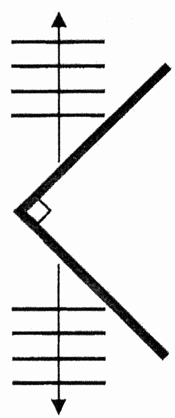
B.



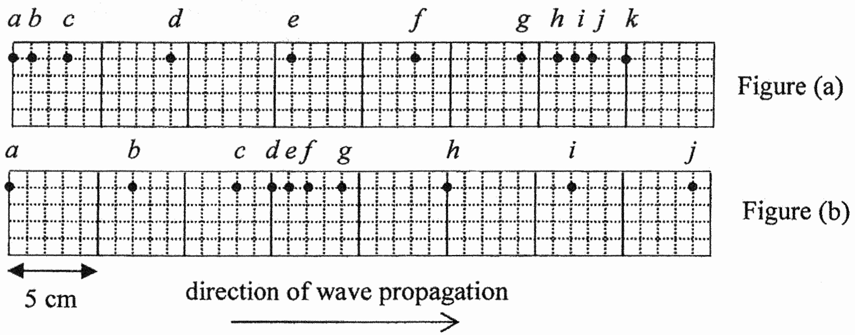
C.



D.



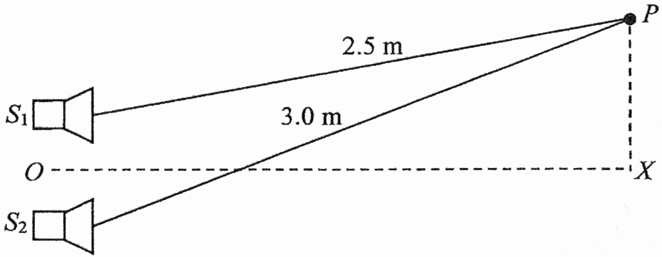
(2024) 16.



A series of particles is uniformly distributed along a slinky spring initially. Figure (a) and (b) show their positions at two instants when a travelling wave of frequency 10 Hz propagates along the spring from left to right. Which statement below is correct ?

A. Particle a is always stationary.
B. Particle b and c vibrate with different amplitudes.
C. The wavelength of the wave is 16 cm.
D. The smallest time interval for obtaining these two figures is 0.05 s.

(2024) 17. Two identical loudspeakers S_1 and S_2 connected to a signal generator produce sounds of wavelength 0.10 m which are in **anti-phase**.

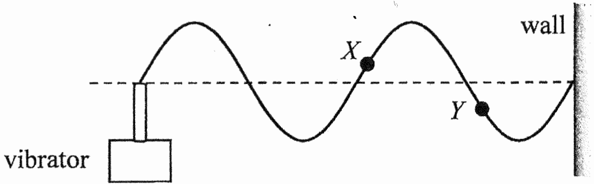


OX is the perpendicular bisector of the line joining S_1 and S_2. What type of interference occurs at X and P respectively ?

	X	P
A.	destructive	constructive
B.	constructive	constructive
C.	destructive	destructive
D.	constructive	destructive

(2024)

18. A string is tied to a vibrator at one end while the other end is fixed to a wall. The figure shows the stationary wave formed at a certain frequency of the vibrator.

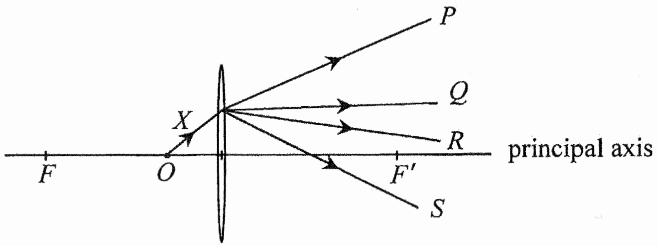


Which statement is correct when the frequency is doubled ?

A. The amplitude of the wave will double.
B. The wavelength will double.
C. The speed of the wave will double.
D. Particle X and Y will become in phase.

(2024)

19. X is a light ray from an object O placed on the principal axis of a convex lens as shown. F and F' are the foci of the lens. Which light ray best represents the emergent ray ?



A. P
B. Q
C. R
D. S

(2024) *20. An object of height 10 cm placed at 20 cm in front of a concave lens forms an image of height 8 cm. Find the focal length of the concave lens.

A. 4.4 cm
B. 8.9 cm
C. 40 cm
D. 80 cm

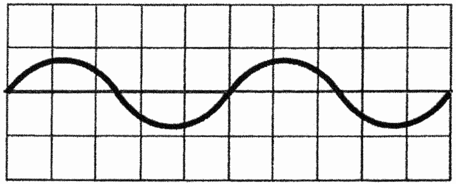
(2024)

21. Which description below is **INCORRECT** ?

A. Microwaves are a kind of electromagnetic waves.
B. Microwaves can be observed with the naked eye.
C. Microwaves propagate with the speed of light in a vacuum.
D. Microwaves are used in radar to detect the position of an aeroplane.

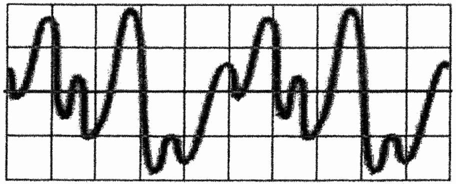
(2024) 22. Figure (a) and (b) are the CRO displays of musical note X and Y using the same time axis and intensity axis.

Figure (a)



note X

Figure (b)

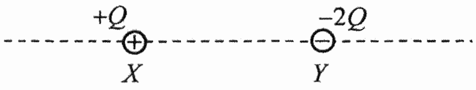


note Y

Which comparison of these two notes must be correct ?

	pitch	**loudness**
A.	the same	the same
B.	the same	not the same
C.	higher for Y	the same
D.	higher for Y	not the same

(2024) 23. Charges $+Q$ and $-2Q$ are fixed at X and Y as shown.

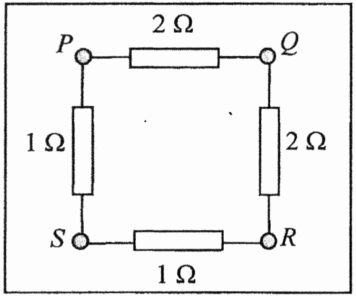


Which description(s) about the electric field along the dotted line is/are correct ?

(1) Between X and Y, the respective electric fields due to the two charges are in the same direction.
(2) To the right of Y, the resultant electric field is always directed to the left.
(3) On the dotted line, there exist two points where the resultant electric field is zero.

A. (2) only
B. (3) only
C. (1) and (2) only
D. (1) and (3) only

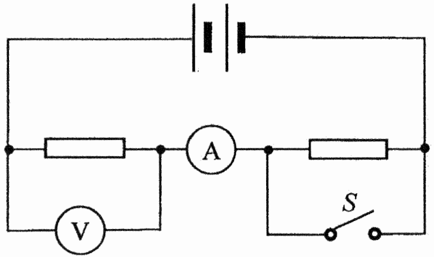
(2024) 24.



Four resistors are connected to terminals $PQRS$ of a circuit board as shown. Across which pair of terminals below is the equivalent resistance smallest ?

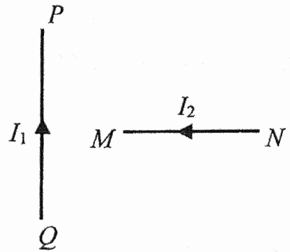
A. PQ
B. PR
C. PS
D. QS

(2024) 25. In the circuit below, the two resistors are identical. When S is closed, how would the readings of the ammeter and voltmeter change ?



	ammeter reading	**voltmeter reading**
A.	increase	increase
B.	increase	decrease
C.	decrease	increase
D.	decrease	decrease

(2024) 26.



PQ in the figure is a section of a fixed long straight wire carrying a current I_1. MN is a section of another long straight wire carrying a current I_2. MN is perpendicular to PQ and both wires are in the same plane. What is the direction of the magnetic force acting on MN due to PQ? Neglect the effects of the Earth's magnetic field.

A. pointing out of the paper

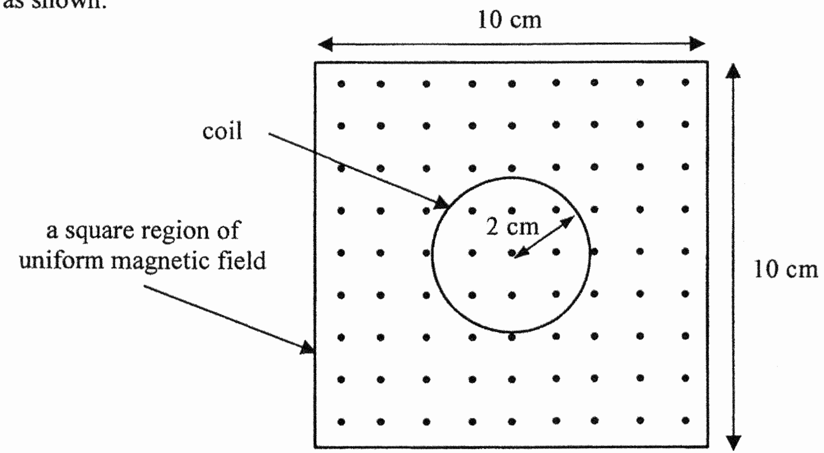
B. $\downarrow$

C. pointing into the paper

D. $\uparrow$

(2024)

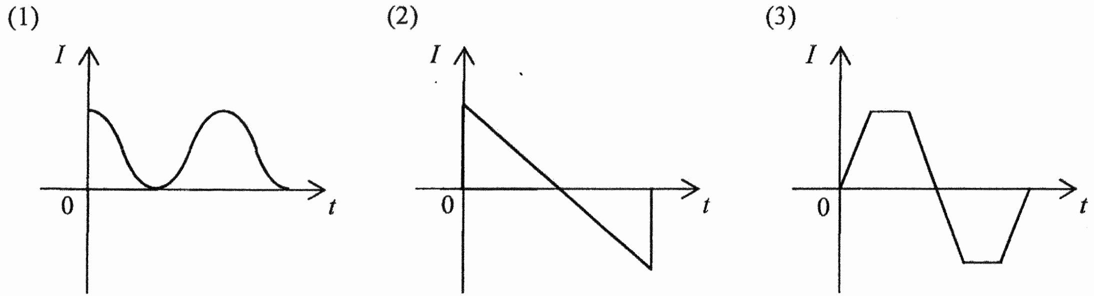
*27. A coil of radius 2 cm is placed inside a square region of uniform magnetic field pointing out of the paper as shown.



The magnetic field decreases uniformly from a strength of 5.0 T to 3.0 T in 0.8 s. What is the magnitude of the e.m.f. induced in the coil ?

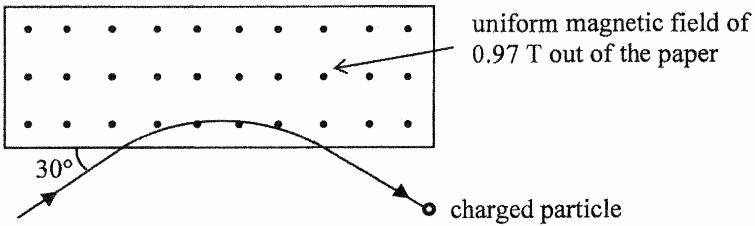
A. 2.5×10^{-2} V
B. 3.1×10^{-2} V
C. 3.1×10^{-3} V
D. 6.2×10^{-3} V

(2024) *28. Which of the following current against time (I- t) graphs represent an a.c. ?



A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

(2024) *29.



A charged particle enters and leaves a magnetic field of uniform flux density 0.97 T along the path as shown. It experiences a magnetic force of 5.9×10^{-12} N within the magnetic field. Given that the speed of the particle is 1.9×10^{7} m s^{-1}, find its charge.

A. -3.2×10^{-19} C
B. $+3.2 \times 10^{-19}$ C
C. -6.4×10^{-19} C
D. $+6.4 \times 10^{-19}$ C

(2024)

*30. A power station generates 217 MW of power for transmission to downtown under a voltage of 132 kV via a transmission cable which has an effective resistance of 0.02 Ω km^{-1}. The power loss for a total cable length of 40 km is about

A. 2.16 MW.
B. 6.49 MW.
C. 13.0 MW.
D. 21.6 MW.

(2024) 31. Which statements about domestic electricity are correct ?

(1) All fuses should be installed in the live wire.
(2) An electrical appliance can work even without an earth wire.
(3) An electrical appliance having a fuse in it can protect people from getting an electric shock.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

(2024)

*32. It takes 2.0 years for the activity of a sample containing a radioactive nuclide to drop from 800 Bq to 600 Bq. How long will it take for the activity of the sample to drop from 600 Bq to 300 Bq ?

A. 2.0 years
B. 2.5 years
C. 3.0 years
D. 4.8 years

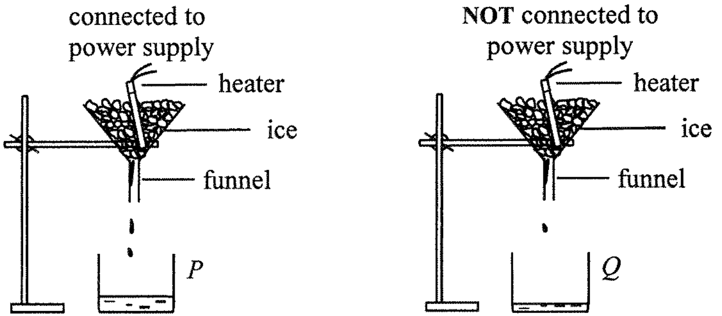
(2024) 33. In order to locate the cracks in an underground water pipe, a small amount of a radioisotope is added to the water in the pipe. The radioactive substance will leak through the crack to the soil nearby, thus radiation will be detected above ground. Which radioisotope (*P*, *Q*, *R* or *S*) below is the most suitable for this purpose ?

	isotope	**half-life**	**radiation emitted**
A.	*P*	66 hours	β
B.	*Q*	36 hours	γ
C.	*R*	7.2 hours	α
D.	*S*	45 minutes	α, β

(2025) 1. An equal amount of energy is supplied to each of the substances *P*, *Q*, *R* and *S* of mass and specific heat capacity listed below. Which substance would have the largest rise in temperature ?

	substance	mass / kg	specific heat capacity / J kg^{-1} $^{o}C^{-1}$
A.	*P*	2	4200
B.	*Q*	3	2100
C.	*R*	4	900
D.	*S*	5	840

(2025) 2. The set-up below can be used to find the specific latent heat of fusion of ice, l_f. The energy supplied by the heater is measured by a joulemeter. The mass of ice melted by the heater is given by the difference between the mass of water collected in beakers P and Q.



How would the calculated value of l_f be affected if

(I) the heater is not completely covered by the ice ?

(II) the mass of water collected in beaker P is taken as the mass of ice melted by the heater ?

	(I)	(II)
A.	decrease	increase
B.	decrease	decrease
C.	increase	increase
D.	increase	decrease

(2025) *3. Identical closed vessels X and Y contain carbon dioxide (CO_2) and oxygen (O_2) respectively.

vessel X vessel Y

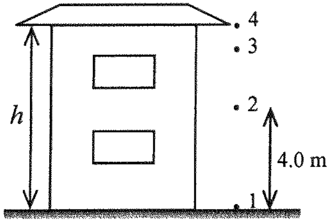
Both gases are at room temperature and atmospheric pressure. Which of the following physical quantities of the gas molecules in the two vessels is/are the same ?

(1) root-mean-square speed
(2) average kinetic energy
(3) frequency of collision with the walls of the vessel

A. (1) only
B. (2) only
C. (1) and (3) only
D. (2) and (3) only

(2025)

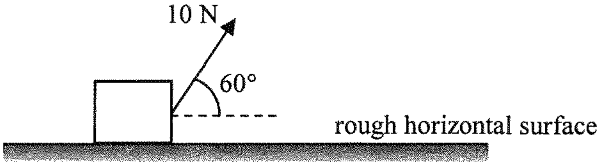
4.



Water drips regularly at a constant rate from a building's roof, which is at a height h above the ground. The droplets fall from rest such that the first droplet reaches the ground just when the fourth droplet leaves the roof as shown. If the second droplet is 4.0 m above the ground at this instant, find h. Neglect air resistance.

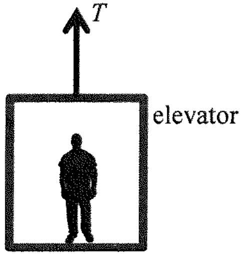
A. 5.8 m
B. 6.4 m
C. 7.2 m
D. 7.8 m

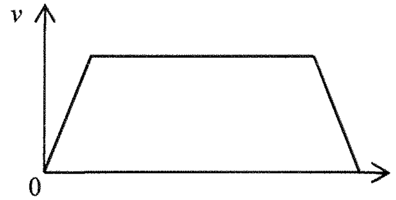
(2025) 5. A block, initially at rest, is pulled along a horizontal surface by a force of 10 N making an angle of 60° with the horizontal. There is a constant friction of 2 N between the block and the horizontal surface. What is the gain in kinetic energy of the block after moving 5 m ?



A. 33 J
B. 25 J
C. 15 J
D. 10 J

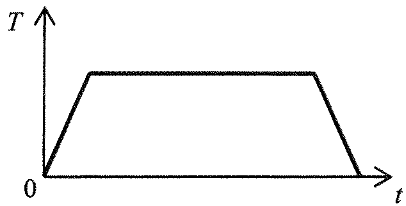
(2025) 6.



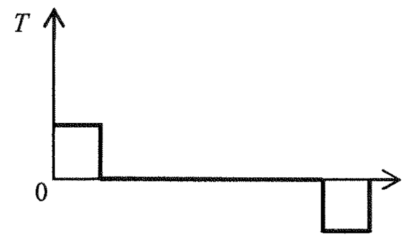


An elevator is pulled upward by a cable. The v-t graph above shows the variation of the elevator's velocity with time. Which of the following correctly indicates the variation of the tension T in the cable with time t ?

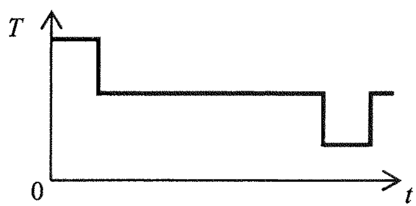
A.



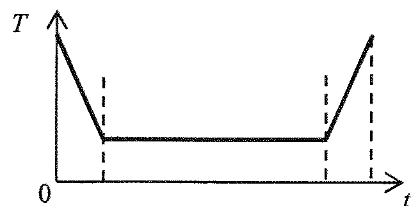
B.



C.

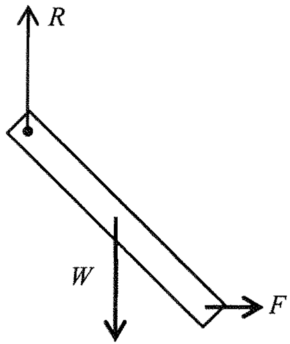


D.

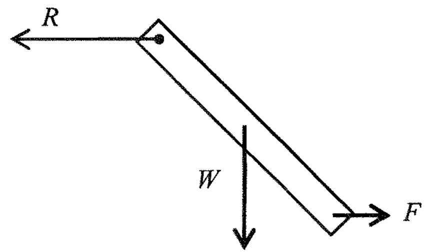


(2025) 7. A uniform rod of weight W hinged at its upper end is at rest under a horizontal force F applied at its lower end. Which figure correctly shows the force R acting on the rod at the hinge ?

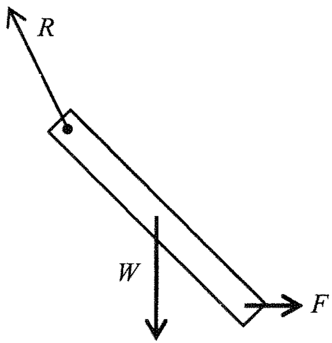
A.



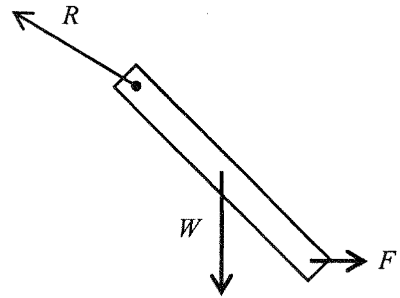
B.



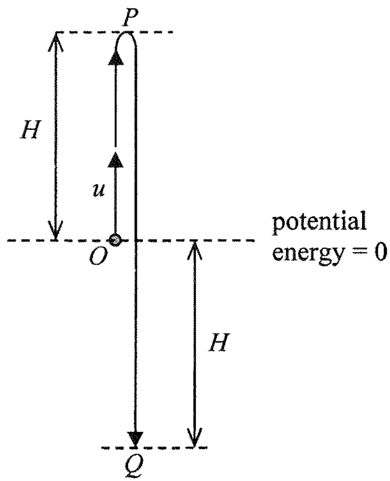
C.



D.



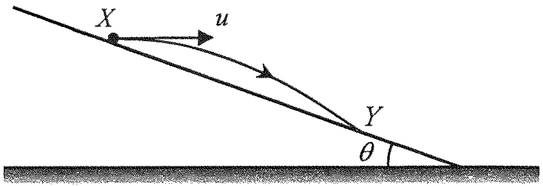
(2025) 8. A small ball of mass m is thrown vertically upward from point O with an initial speed u. It reaches the highest point P and then falls to point Q such that $OP = OQ = H$. The ball's potential energy at point O is taken to be zero.



What are the kinetic energy and potential energy of the ball at point Q ? Neglect air resistance.

	kinetic energy	**potential energy**
A.	mu^2	$-mgH$
B.	mu^2	mgH
C.	$\frac{1}{2}mu^2$	$-mgH$
D.	$\frac{1}{2}mu^2$	mgH

(2025) *9.



X and Y are two points on an incline making an angle θ with the horizontal. A particle projected horizontally from X with speed u lands at Y as shown. Which expression correctly gives its time of flight from X to Y? Neglect air resistance.

A. $\dfrac{u \sin \theta}{2g}$

B. $\dfrac{u \tan \theta}{2g}$

C. $\dfrac{2u \sin \theta}{g}$

D. $\dfrac{2u \tan \theta}{g}$

(2025) 10. Loosening a screw requires a pair of forces of 10 N acting tangentially as shown in Figure (1). A suitable screwdriver with a handle of diameter 0.025 m shown in Figure (2) is chosen. Estimate the magnitude of the pair of forces needed to act tangentially on the handle of the screwdriver.

Figure (1)

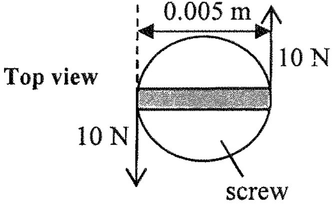


Figure (2)

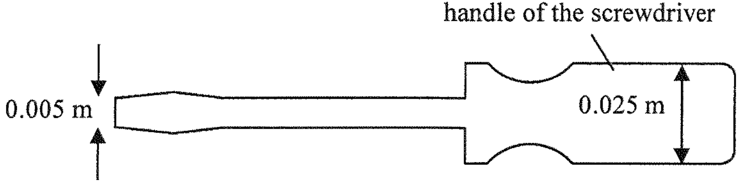
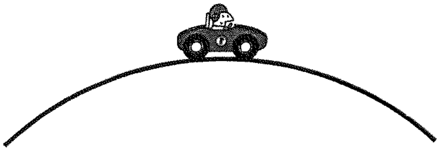


Diagram **NOT** drawn to scale

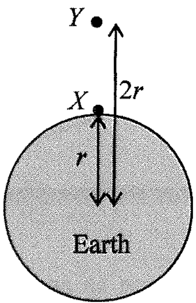
A. 2 N
B. 4 N
C. 5 N
D. 10 N

(2025) *11. A car of weight W travels with constant speed on a circular humpback bridge. At the moment when the car passes the top of the bridge, the normal reaction acting on it by the bridge is R. What is the centripetal force acting on the car at that moment ?



A. $W + R$
B. $W - R$
C. W
D. R

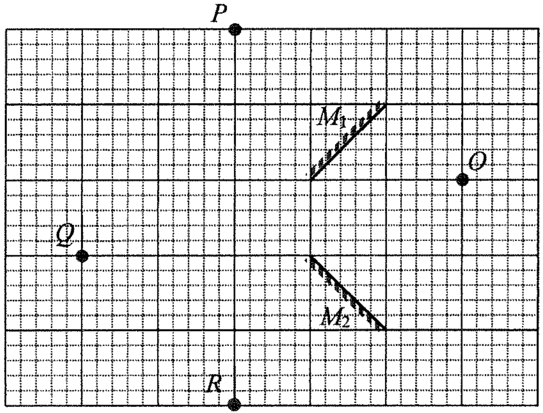
(2025) *12.



Two identical objects located at point X and point Y are at distances r and $2r$ from the Earth's centre. The weight of the object at point X is W. What is the weight of the object at point Y ?

A. $0.25W$
B. $0.5W$
C. $0.75W$
D. W

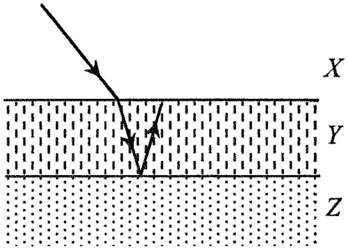
(2025) 13. A point object O is placed in front of two plane mirrors M_1 and M_2 as shown.



At which position(s) can an image be formed ?

A. P, Q and R
B. Q and R only
C. R only
D. P only

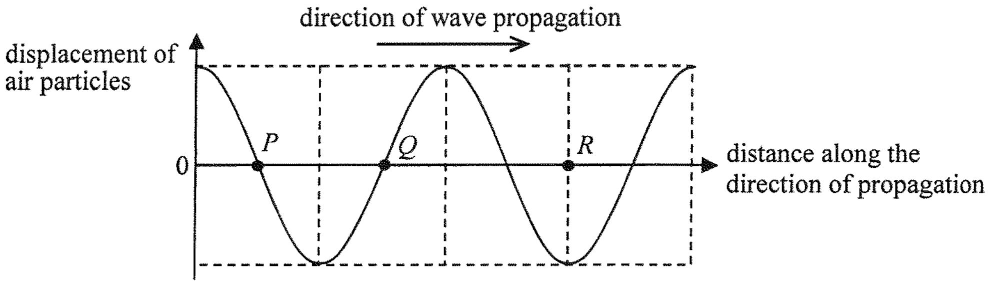
(2025) 14.



The figure shows three media *X*, *Y* and *Z* separated by parallel boundaries. A light ray travelling from *X* into *Y* then undergoes total internal reflection at the boundary between *Y* and *Z*. Arrange in descending order the speed of light in the respective media.

A. *X Y Z*
B. *X Z Y*
C. *Y Z X*
D. *Z X Y*

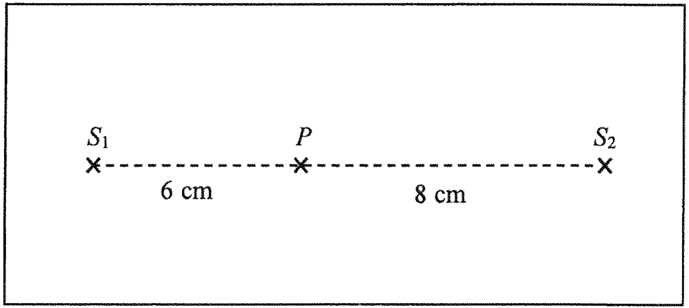
(2025) 15. The graph shows the variation of the displacement of air particles from their equilibrium positions with distance for a sound wave at a certain instant. Displacement along the direction of wave propagation is taken to be positive.



Which statement concerning the wave at that instant is correct ?

A. An air particle at P is momentarily at rest.
B. A rarefaction occurs at Q.
C. A compression occurs at R.
D. An air particle at R is moving with the greatest speed.

(2025) 16. In the ripple tank shown below, dippers S_1 and S_2 are vibrating in phase and generate water waves of wavelength 0.8 cm towards each other at the same speed. For a particle P at distances 6 cm and 8 cm from S_1 and S_2 respectively as shown, the amplitude of the waves reaching P from S_1 is 1 mm while that from S_2 is 2 mm.



Top view

What is the amplitude of oscillation of P ?

A. 0 mm
B. 1 mm
C. 2 mm
D. 3 mm

(2025) *17. The light from a sodium lamp consists of two wavelengths 589.00 nm and 589.59 nm. Find the angle between these two spectral lines in the second-order after the light passes through a diffraction grating of 600 lines per mm.

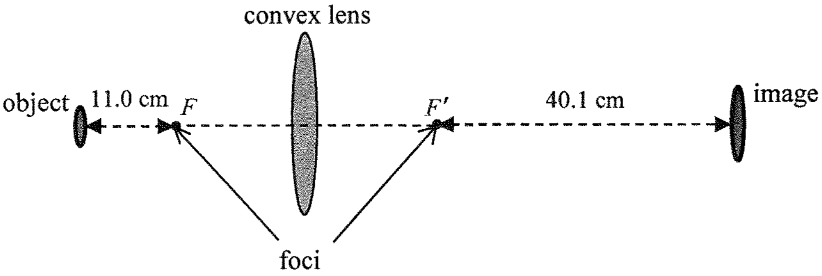
A. 0.058°
B. 0.041°
C. 0.022°
D. 0.021°

(2025) 18. In an experiment, monochromatic light is incident normally on a double slit and a diffraction grating placed in turn at the same position. The respective interference and diffraction patterns formed on the screen are observed. Which of the following statements about the patterns is/are correct ?

(1) The fringe separation of the interference pattern is wider.
(2) The fringes of the diffraction pattern are sharper.
(3) The width of the whole diffraction pattern is wider.

A. (1) only
B. (2) only
C. (1) and (3) only
D. (2) and (3) only

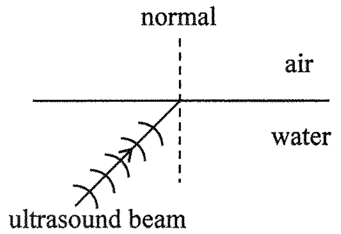
(2025) *19. An object is placed at a certain distance from a convex lens, the position of a screen is adjusted until a sharp image is formed.



Given that the distances of the object and the image to the respective foci F and F' of the convex lens are 11.0 cm and 40.1 cm as shown, the focal length of the lens is

A. 8.6 cm.
B. 17.3 cm.
C. 21.0 cm.
D. 25.9 cm.

(2025) 20. A source in water emits a beam of ultrasound towards the water-air interface as shown.



Which of the following statements is **INCORRECT** ?

A. The frequency of the refracted beam is the same as that of the incident beam.
B. The speed of the refracted beam becomes smaller.
C. The refracted beam leaving the interface bends towards the normal.
D. If the angle of incidence is large enough, total internal reflection may occur.

(2025) 21. Which of the following phenomena can be explained by the speed of waves being different in different media ?

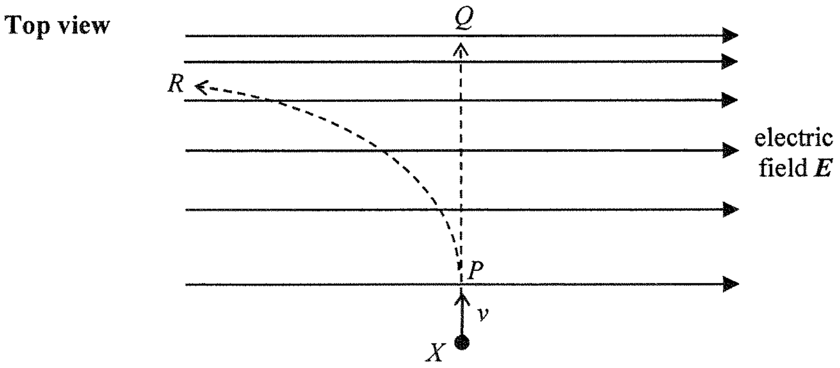
(1) echoes
(2) rainbows
(3) mirages

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

(2025) 22. Three light conducting spheres are suspended by nylon threads. Any two of them are found to attract each other when they are brought close to each other. Which conclusion below is correct ?

A. All three spheres are charged.
B. Only one sphere is charged.
C. Only one sphere is uncharged and the other two carry like charges.
D. Only one sphere is uncharged and the other two carry unlike charges.

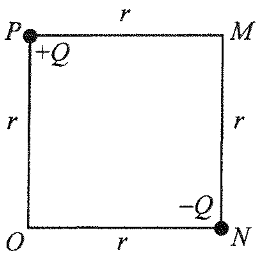
(2025) 23. A negatively charged particle X enters an electric field with initial velocity v perpendicular to the field lines as shown. Neglect the effects of gravity.



Particle X may move along

A. PQ with uniform speed.
B. PQ with increasing speed.
C. PR with decreasing speed.
D. PR with increasing speed.

(2025) *24.



Point charges $+Q$ and $-Q$ are fixed respectively at corners P and N of square $MNOP$ of side length r as shown above. What is the electric field at O ?

A. $\dfrac{\sqrt{2}Q}{4\pi\varepsilon_0 r^2}$ along the direction from P to N

B. $\dfrac{Q}{4\pi\varepsilon_0 r^2}$ along the direction from P to N

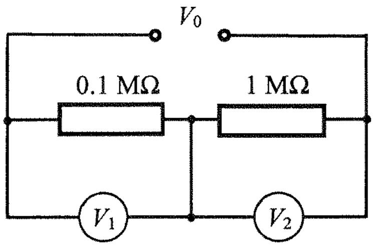
C. $\dfrac{\sqrt{2}Q}{4\pi\varepsilon_0 r^2}$ along the direction from N to P

D. $\dfrac{Q}{4\pi\varepsilon_0 r^2}$ along the direction from N to P

(2025) 25. A student tests an electric circuit by connecting a filament light bulb to a dry cell using two connecting wires. The bulb does not light up. Which of the following is certainly **NOT** a possible explanation ?

A. The dry cell may be flat.
B. The circuit may be open due to a bad connection.
C. The dry cell may have its polarities connected the wrong way round.
D. The bulb may have blown.

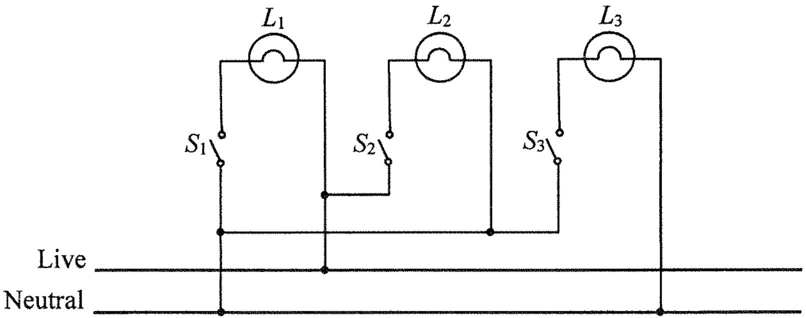
(2025) 26. In the circuit below, a voltage V_0 is applied across resistors of 0.1 MΩ and 1 MΩ connected in series. Voltmeter V_1 and V_2 are connected respectively across the two resistors as shown. V_1 is ideal while V_2 has an internal resistance of 1 MΩ.



Estimate the ratio of the voltmeter readings $V_1 : V_2$.

A. 5 : 1
B. 1 : 5
C. 10 : 1
D. 1 : 10

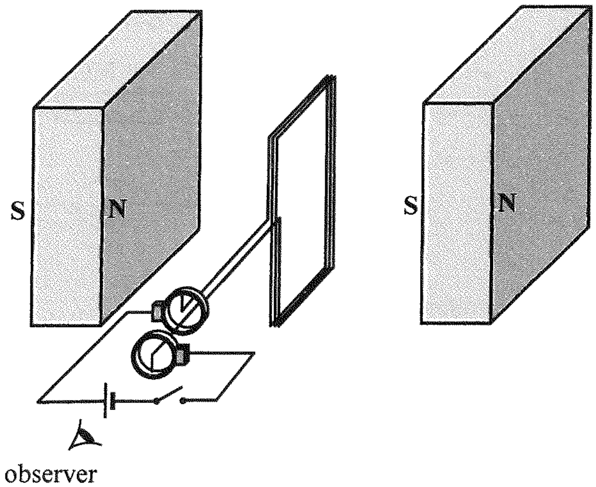
(2025) 27. The figure shows how lamps L_1, L_2 and L_3 are connected in the ring circuit of a house. There are some wrong connections in the circuit.



If switches S_2 and S_3 are closed while switch S_1 remains open, which of the lamps will light up ?

A. L_2 only
B. L_3 only
C. L_2 and L_3 only
D. none of them

(2025) 28. The figure shows a rectangular coil connected in series with a dry cell and a switch through a pair of slip rings. The magnetic field between the two magnets is uniform and is perpendicular to the plane of the coil at the moment shown.



When the switch is closed, the coil will

A. remain at rest.
B. rotate in a clockwise direction continuously.
C. rotate in an anticlockwise direction continuously.
D. change the direction of rotation after turning through 180°.

(2025) *29. A student uses a search coil to measure the magnetic field of a permanent magnet. However, no reading is observed on the CRO connected to the search coil. Which of the following is the possible reason ?

A. The magnet is not strong enough.
B. The magnetic field produced by the magnet is constant.
C. The sensitivity of the CRO is too low.
D. The size of the search coil is not large enough to pick up sufficient magnetic flux.

(2025) *30. An electric iron works at its rated power of 1100 W under a 50 Hz a.c. mains of root-mean-square voltage 220 V. What power will it deliver when working under a 100 Hz a.c. of root-mean-square voltage 110 V ?

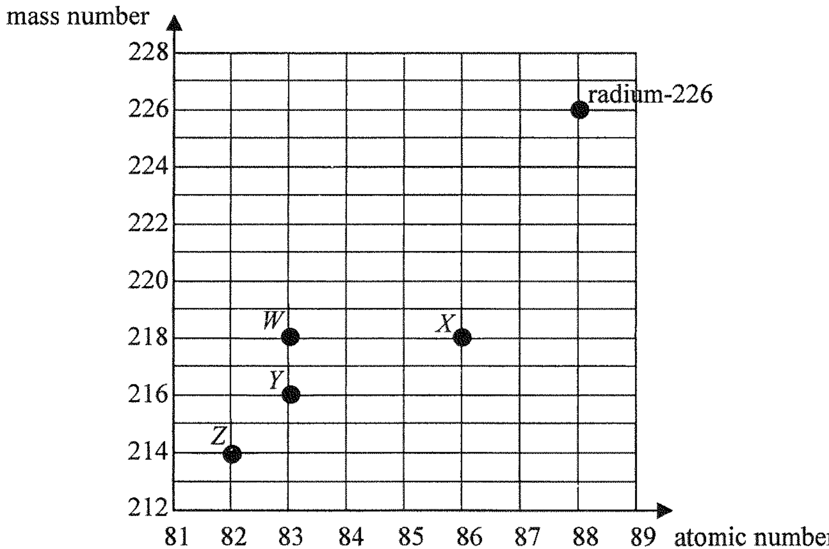
A. 275 W
B. 778 W
C. 550 W
D. 1100 W

(2025) 31. A radioactive source undergoes decay such that 300 nuclides have decayed in 1 minute. Its activity is

A. 0.2 Bq.
B. 5 Bq.
C. 300 Bq.
D. 18000 Bq.

(2025)

32. Radium-226 undergoes a series of α and β decays before finally becomes a stable nuclide. Referring to the plot of mass number against atomic number below, which nuclides, W, X, Y and Z, **cannot** be possible members in the decay series of radium-226 ?



A. W and X
B. W and Y
C. Z and X
D. Z and Y

(2025)

*33. In the following nuclear reaction, nuclide X and Y fuse to form nuclide Z under suitable conditions and the amount of energy released is 7.16 MeV.

$$X + Y \rightarrow Z$$

The total mass of X and Y is

A. 0.00385 u less than the mass of Z.
B. 0.00385 u more than the mass of Z.
C. 0.00769 u less than the mass of Z.
D. 0.00769 u more than the mass of Z.

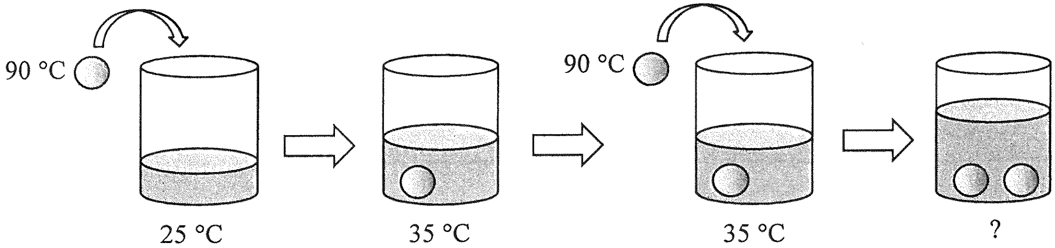
(2026) 1. Flasks X and Y are identical except that for their outer surfaces: X is black and Y is silvery. Each flask contains the same amount of water initially at 25 °C, sealed and equipped with a temperature sensor. The flasks are then placed in the same outdoor environment at 30 °C with sunlight. After 20 minutes,

A. the water in X is hotter as a black surface is a better absorber of radiation.
B. the water in X is hotter as a black surface is a better emitter of radiation.
C. the water in Y is hotter as a silvery surface is a poorer absorber of radiation.
D. the water in Y is hotter as a silvery surface is a poorer emitter of radiation.

(2026) 2. The resistance of a resistance thermometer is 100.0 Ω in pure melting ice. Its resistance increases linearly with temperature by 0.385 Ω $°C^{-1}$. Find the temperature when the resistance is 117.2 Ω.

A. 6.62 °C
B. 11.5 °C
C. 44.7 °C
D. 45.1 °C

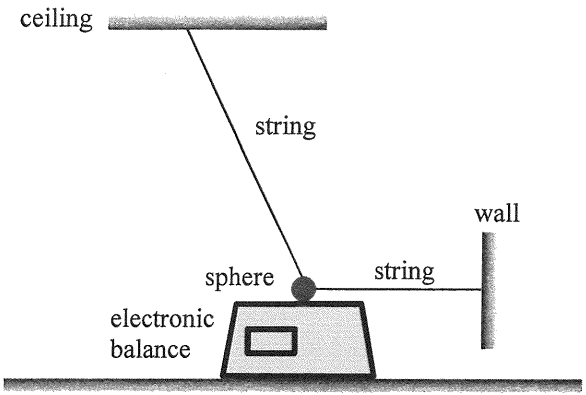
(2026) 3.



A metal sphere at 90 °C is put into a cup of water. The temperature of the water increases from 25 °C to 35 °C. If a second identical metal sphere, also at 90 °C, is added to the water, what is the final temperature of the water ? Assume there is no heat loss to the cup or the surroundings throughout the process.

A. 41.5 °C
B. 42.3 °C
C. 43.5 °C
D. 45.0 °C

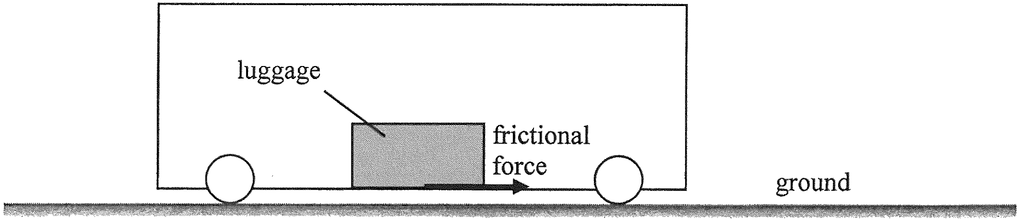
(2026) 4.



A sphere of weight 18 N is suspended by two light and inextensible strings. One string is attached horizontally to a wall, and the other is connected to the ceiling. The sphere rests on the smooth top surface of an electronic balance, which shows a reading of 10 N. Find the resultant force acting on the sphere by the two strings.

A. 0 N
B. 8 N
C. 10 N
D. 15 N

(2026) 5. A train is moving horizontally along a straight line. A piece of luggage rests on the floor of the train. The diagram shows the frictional force acting on the luggage, directed to the right. Neglect air resistance.

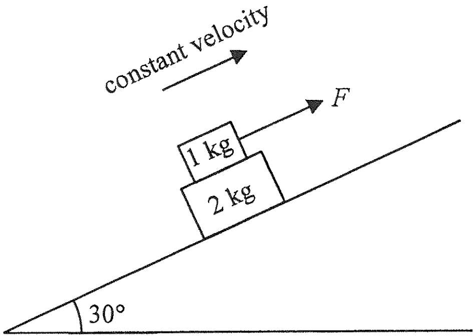


Observing from the ground, which of the following statements must be correct ?

(1) The train is travelling to the right.
(2) The train is accelerating.
(3) The direction of acceleration of the luggage is to the right.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

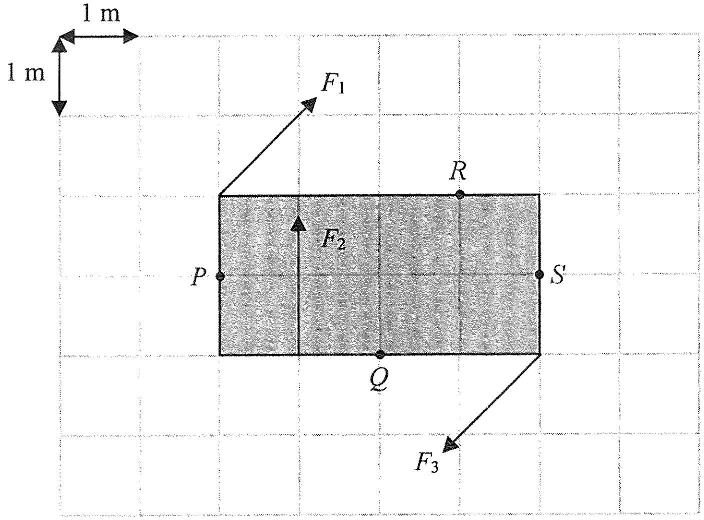
(2026) 6.



A 1-kg block is placed on a 2-kg block, and both are on a smooth incline at 30° to the horizontal. A constant force F is applied to the 1-kg block upward along the incline so that the two blocks move up the incline together with the same constant velocity as shown. What is the frictional force acting on the 1-kg block by the 2-kg block ? Neglect air resistance. ($g = 9.81 \text{ m s}^{-2}$)

A. 4.91 N upward along the incline
B. 9.81 N upward along the incline
C. 4.91 N downward along the incline
D. 9.81 N downward along the incline

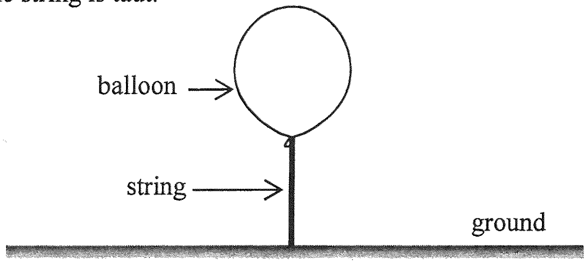
(2026) 7.



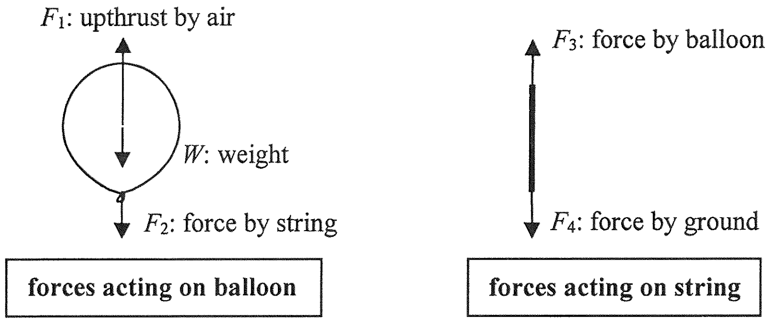
A rectangular plate of width 2 m and length 4 m is placed on a smooth horizontal surface. Three forces F_1, F_2 and F_3 of equal magnitude act on it as shown. F_1 and F_3 are parallel and opposite in direction. Taking the moment about which point below is the net moment the greatest ?

A. P
B. Q
C. R
D. S

(2026) 8. One end of a light and inextensible string is connected to a floating balloon and the other end is fixed to the ground as shown. The string is taut.



The free-body diagrams of the balloon and the string are shown below.



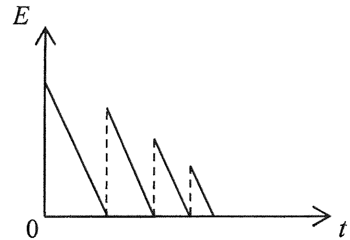
Which of the following is/are action and reaction pair(s) ?

(1) F_1 and W
(2) F_2 and F_3
(3) F_3 and F_4

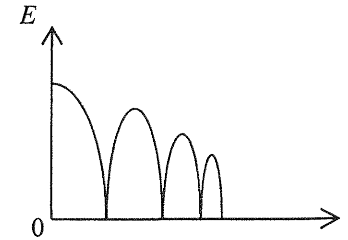
A. (1) only
B. (2) only
C. (1) and (3) only
D. (2) and (3) only

(2026) 9. A ball is released from a certain height. It rebounds from the ground several times before it comes to rest. Neglect air resistance. Which of the following graphs best represents the variation of the total mechanical energy E of the ball with time t ?

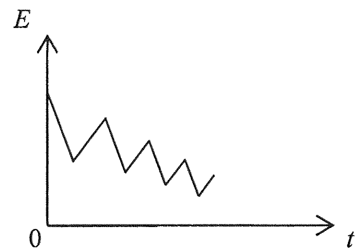
A.



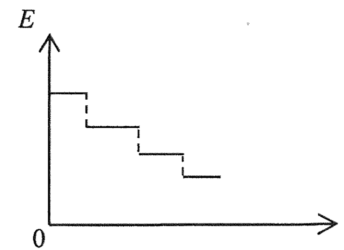
B.



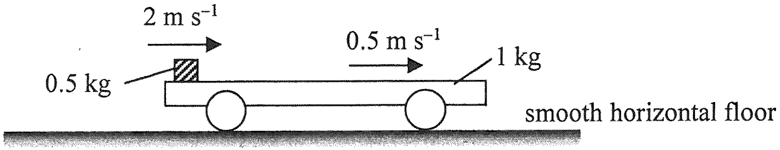
C.



D.



(2026) 10. A 1-kg trolley with a 0.5-kg block placed on its horizontal platform is resting on a smooth horizontal floor. The block is then given a sharp push to the right. At an instant after the push, the block and the trolley are moving to the right at 2 m s^{-1} and 0.5 m s^{-1} respectively as observed from the floor.



Finally, they travel together with the same speed v. Find v. Neglect air resistance.

A. 0.5 m s^{-1}
B. 1 m s^{-1}
C. 1.5 m s^{-1}
D. 2 m s^{-1}

(2026) *11. Satellites X and Y revolve around the Earth in circular orbits, in the same plane and in the same direction. The radii of their orbits are r and $4r$ respectively. The figure below shows an instant that X, Y and the Earth are on a straight line.

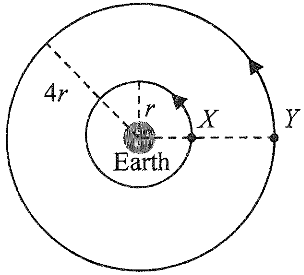
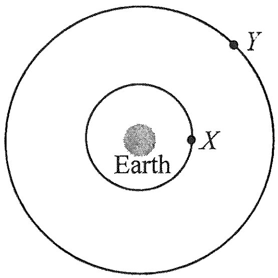


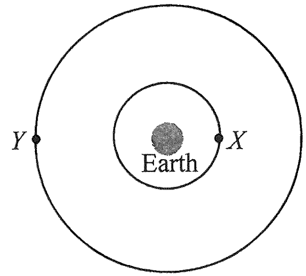
Diagram **NOT** drawn to scale

When X completes one cycle, which of the following diagrams best represents the position of Y ?

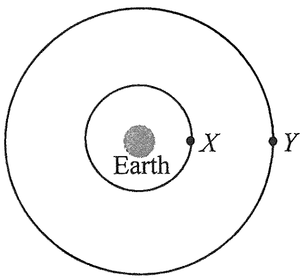
A.



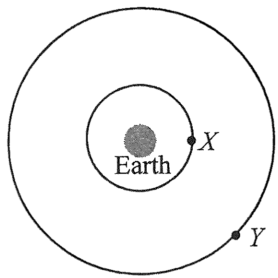
B.



C.



D.



(2026) *12. A rocket on the Earth's surface is launched by an upward thrust F, producing an upward acceleration g, where g is the acceleration due to gravity near the Earth's surface, as shown in Figure (a). R is the radius of the Earth.

Diagram **NOT** drawn to scale

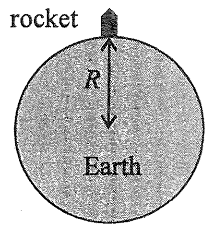


Figure (a)

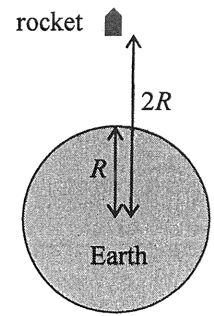


Figure (b)

Figure (b) shows an identical rocket at a distance $2R$ away from the Earth's centre. What is the upward thrust required for producing the same upward acceleration g ? Neglect air resistance.

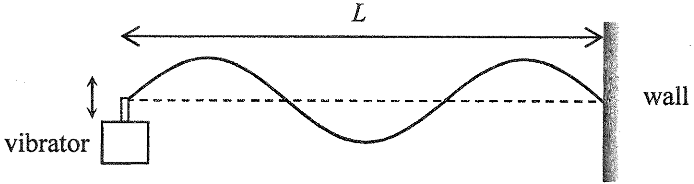
A. $\frac{F}{4}$

B. $\frac{F}{2}$

C. $\frac{5F}{8}$

D. $\frac{3F}{4}$

(2026) 13. The figure shows a string with one end fixed on the wall and the other end tied to a vibrator with frequency f. A stationary wave is formed as shown at a certain instant. The distance between the vibrator and the wall is L.



Which of the following statements are correct ?

(1) All particles on the string between two successive nodes are vibrating in phase.
(2) If the tension in the string increases while L remains unchanged, f must increase to maintain the same stationary wave pattern.
(3) The wavelength of the stationary wave is $\dfrac{L}{3}$.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

(2026) 14. Which of the following phenomena involve(s) the principle of superposition of waves ?

(1) Ripples in a pond overlap and create regions with different amplitudes.
(2) Stationary wave patterns are formed when two waves of the same frequency travel in opposite directions.
(3) The speed of waves changes when they travel from one medium to another.

A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

(2026) 15.

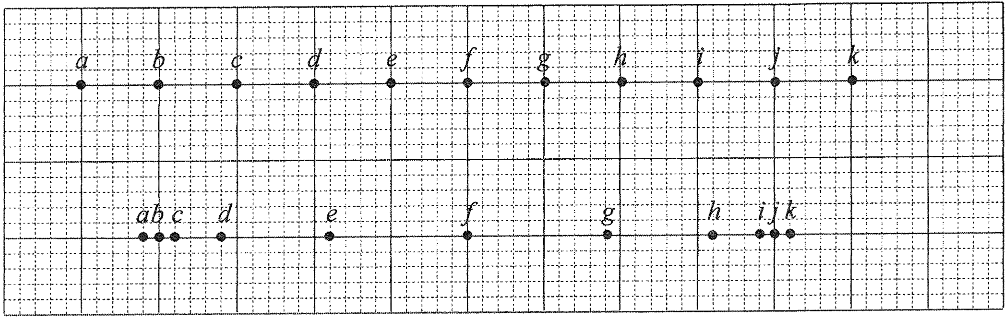
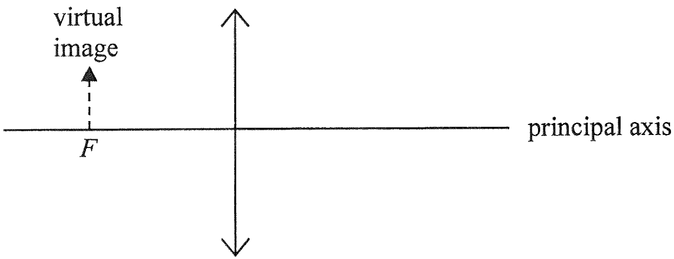


Figure (1) shows the equilibrium positions of particles *a* to *k* in a medium. A longitudinal wave is travelling through the medium and the positions of the particles at a certain instant are shown in Figure (2). Which of the following statements is/are correct at that instant ?

(1) The separation between *c* and *k* is one wavelength.
(2) *f* is momentarily at rest.
(3) *i* and *k* are moving in the same direction.

A. (2) only
B. (3) only
C. (1) and (2) only
D. (1) and (3) only

(2026) 16. The image of an object formed by a convex lens is virtual and is situated at the focus F as shown.

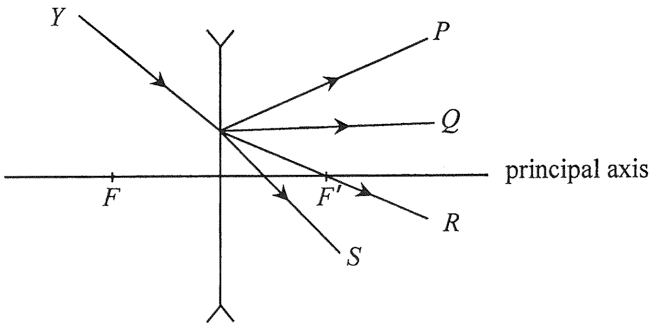


Which of the following statements is/are correct ?

(1) The object is at infinity.
(2) The object is larger than the image.
(3) The object and the image are on the same side of the lens.

A. (2) only
B. (3) only
C. (1) and (2) only
D. (1) and (3) only

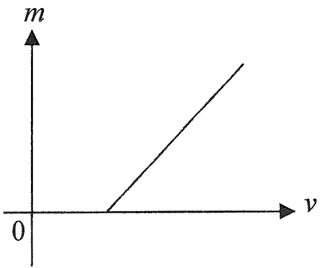
(2026) 17. Light ray Y is incident on a concave lens as shown. F and F' are the foci of the lens. Which light ray best represents the emergent ray ?



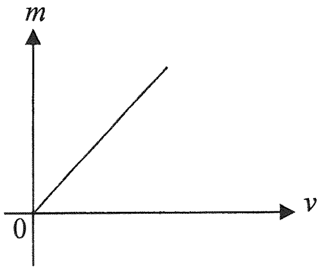
A. P
B. Q
C. R
D. S

(2026) *18. A convex lens is used to project sharp images of an object onto a screen at different object distances u. For each value of u, the corresponding image distance v and the linear magnification m are recorded. Which of the following best represents the graph of m against v ?

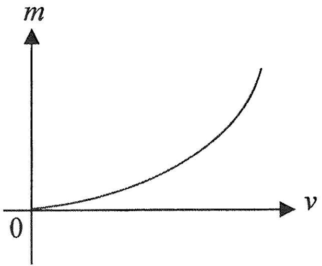
A.



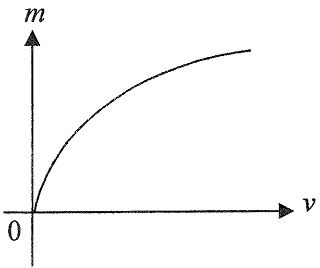
B.



C.



D.

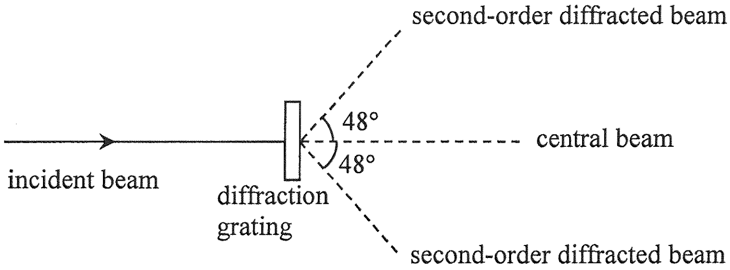


(2026) 19. A monochromatic light beam is directed to a double slit and a fringe pattern is formed. Which of the following physical phenomena is/are involved ?

(1) dispersion
(2) diffraction
(3) interference

A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

(2026) *20. A beam of monochromatic light is incident normally on a diffraction grating. Second-order diffracted beams are found at 48° to the incident direction as shown in the figure. Find the highest order of diffracted beam that can be observed.



A. 2nd order
B. 3rd order
C. 4th order
D. 5th order

(2026) 21. Two small charged spheres exert an electrostatic force of magnitude F on each other. If the amount of charge on each sphere is doubled and the distance between them is also doubled, the magnitude of the electrostatic force becomes

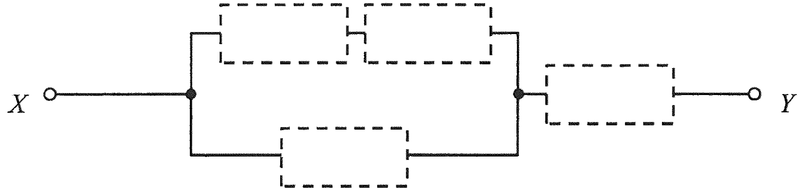
A. $\dfrac{F}{2}$.

B. F.

C. $2\,F$.

D. $4\,F$.

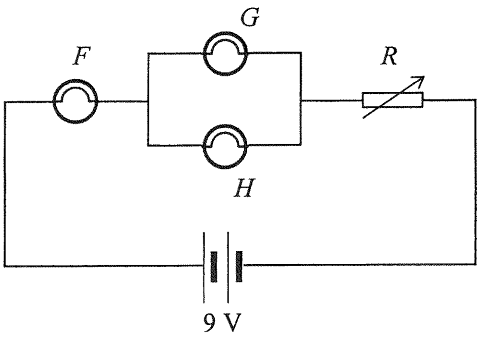
(2026) 22.



Four resistors of resistance 10 Ω, 20 Ω, 30 Ω and 40 Ω respectively are installed in the empty places in the above circuit. What is the minimum possible equivalent resistance across *X* and *Y* ?

A. 19.2 Ω
B. 25.6 Ω
C. 28.8 Ω
D. 32.2 Ω

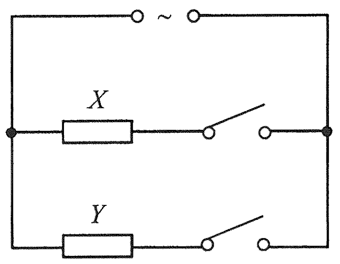
(2026) 23.



In the circuit above, light bulbs F, G and H are identical and the battery is ideal. Rheostat R is initially set to maximum resistance. The maximum voltage across each bulb without burning out is 2 V, and a burnt-out bulb becomes an open circuit. Initially all bulbs light up. If the resistance of R is gradually reduced to zero, which of the bulbs will burn out in the process ?

A. only bulb F
B. only bulbs G and H
C. none of them
D. all of them

(2026) 24.



The above shows a simplified diagram of the circuit in a heater. It consists of two heating elements X and Y. The time taken for the heater to heat a certain amount of water from room temperature to 100 °C is as follows.

only X is connected	180 s
only Y is connected	300 s

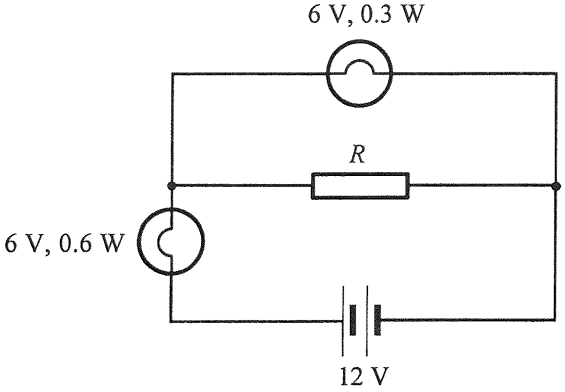
How long does it take to heat the same amount of water from room temperature to 100 °C if both X and Y are connected ?

A. 105 s
B. 113 s
C. 154 s
D. 240 s

(2026) 25. In household appliances, the function of the earth wire is

A. to provide a return path for the current during normal operations.
B. to provide a return path for the current when the neutral wire is disconnected.
C. to break the circuit when the live wire touches the metal case.
D. to provide a low-resistance path when the live wire touches the metal case.

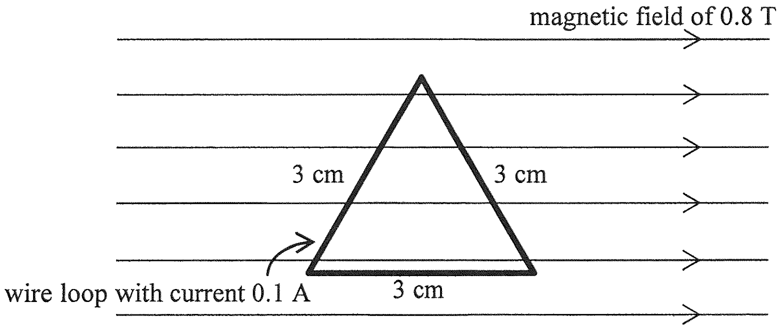
(2026) 26.



If both light bulbs work at their rated values as shown, find the resistance of R.

A. 30 Ω
B. 60 Ω
C. 90 Ω
D. 120 Ω

(2026) 27.



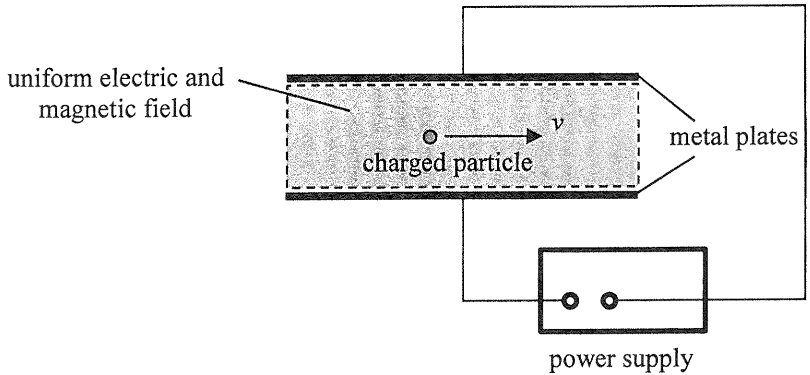
An equilateral triangular wire loop with sides of 3 cm is placed in a uniform magnetic field of 0.8 T. One of the sides of the loop is parallel to the field as shown. A current of 0.1 A flows through the loop. Find the net magnetic force acting on the loop.

A. 0 N
B. 2.8×10^{-3} N
C. 5.6×10^{-3} N
D. 7.6×10^{-3} N

(2026) *28. The input terminals of a transformer are connected to a 220 V mains supply. Its output power is 18 W and its output voltage is 5.0 V. If the efficiency of the transformer is 85%, what is the current drawn from the mains supply ?

A. 0.070 A
B. 0.082 A
C. 0.096 A
D. 0.55 A

(2026) *29.

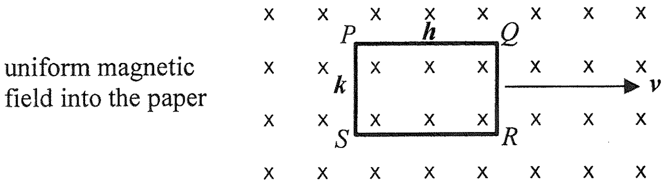


A particle with charge q travels horizontally at speed v without deflection through a region of uniform electric and magnetic field as shown. The electric field is set up by the potential difference across the two parallel metal plates. The magnetic field B is applied perpendicular to the paper. The whole set-up is in vacuum. Which of the following combinations allow the particle to travel horizontally without deflection through the region ? Neglect the effects of gravity.

	charge of particle	**speed of particle**	**magnetic field**
(1)	q	$2v$	$\frac{B}{2}$
(2)	$-q$	v	B
(3)	$2q$	v	B

A. (1) and (2) only
B. (2) and (3) only
C. (1) and (3) only
D. (1), (2) and (3)

(2026) *30. A rectangular copper loop *PQRS*, with its plane perpendicular to a uniform magnetic field of flux density $\boldsymbol{B}$, is moving with velocity $\boldsymbol{v}$ in a direction perpendicular to the field as shown. The length and width of the loop are $\boldsymbol{h}$ and $\boldsymbol{k}$ respectively.

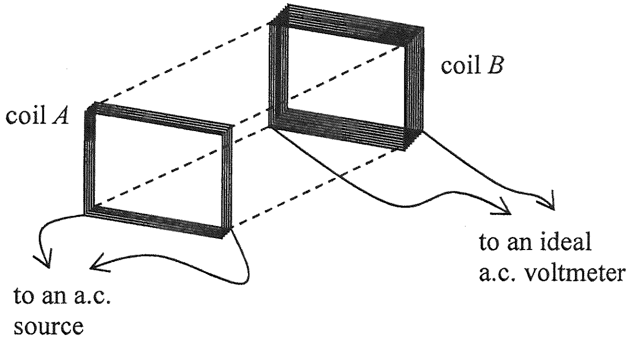


Which of the following statements is/are correct ?

(1) The potential difference across *PQ* is $\boldsymbol{Bhv}$.
(2) The potential difference across *QR* is $\boldsymbol{Bkv}$.
(3) There is **<u>no</u>** induced current flowing through the loop.

A. (1) only
B. (2) only
C. (1) and (3) only
D. (2) and (3) only

(2026) *31. Coil A and coil B are placed close and facing each other with the number of turns of coil B being doubled that of coil A. Coil A is connected to an a.c. source while coil B is connected to an ideal a.c. voltmeter as shown. The voltmeter shows a reading after the a.c. source is turned on.



Which of the following will increase the voltmeter reading ?

(1) The voltage of the a.c. source is increased.
(2) A thicker wire is used in coil B.
(3) A thin and large plastic sheet is placed between the two coils and parallel to them.

A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

(2026) 32. Which of the following units is used for measuring the radiation equivalent dose of a patient receiving radiotherapy treatment ?

A. sievert (Sv)
B. becquerel (Bq)
C. joule (J)
D. watt per square metre (W m^{-2})

(2026) *33. Consider the following nuclear reaction:

$$^{2}_{1}H + ^{3}_{1}H \rightarrow ^{4}_{2}He + ^{1}_{0}n$$

Using the data below, estimate the energy released when 5 g of $^{2}_{1}H$ completely reacts with $^{3}_{1}H$ through the reaction.

mass of $^{2}_{1}H$ = 2.0136 u

mass of $^{3}_{1}H$ = 3.0155 u

mass of $^{4}_{2}He$ = 4.0015 u

mass of $^{1}_{0}n$ = 1.0087 u

A. 3×10^{-12} J
B. 3×10^{-10} J
C. 4×10^{12} J
D. 2×10^{18} J